

THE CAFFARELLI-KOHN-NIRENBERG PARTIAL REGULARITY THEOREM: A SELF-CONTAINED PROOF WRITTEN FOR FORMALIZATION

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ABSTRACT. We give a complete and self-contained proof of the Caffarelli-Kohn-Nirenberg partial regularity theorem for suitable weak solutions of the three-dimensional incompressible Navier-Stokes equations with a force $f \in L^q$, $q > 5/2$: the singular set has vanishing one-dimensional parabolic Hausdorff measure. The exposition is written so that each statement can be transcribed into a single formal declaration. Every constant is named and its dependence is displayed, every cut-off function is written out, every passage to a limit is justified, and the external analytic inputs are isolated in an appendix with exact statements. We also construct, for every divergence-free L^2 datum, a global Leray–Hopf solution that is suitable, identify its associated pressure, and prove the corresponding forced results.

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Date: September 29, 2026.

Scott Armstrong was supported by the European Research Council (ERC) under the European Union’s Horizon Europe research and innovation programme, grant agreement No. 101200828.

Vlad Vicol was partially supported by the Collaborative NSF grant DMS-2307681 and by a Simons Investigator Award.

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INTRODUCTION

Leray's weak solutions of the incompressible Navier-Stokes equations in three space dimensions are global in time but are not known to be regular or unique. Partial regularity theory replaces the (open) question of full regularity by a quantitative bound on the size of the hypothetical singular set. Scheffer [20] initiated this programme, and Caffarelli, Kohn and Nirenberg [3] obtained the sharpest result known to date: for a *suitable* weak solution (one that additionally satisfies a localized form of the energy inequality) the set S of space-time points at which the velocity fails to be locally bounded satisfies $\mathcal{P}^1(S) = 0$, where $\mathcal{P}^1$ is the one-dimensional Hausdorff measure of the parabolic metric. Their hypothesis on the force was $f \in L^q$ with $q > 5/2$.

The proof in [3] rests on a two-step scheme that has since become standard. The first step is an ε -regularity theorem: if a suitable weak solution has small scale-invariant L^3 velocity energy and small scale-invariant $L^{3/2}$ pressure energy, together with a small force norm, on one parabolic cylinder, then the velocity has a Hölder representative on the half-cylinder with the same top centre. The second step is a covering argument: at every singular point, Theorem 2.9 gives $\limsup_{r \downarrow 0} r^{-1} \iint_{Q_r(z)} |\nabla u|^2 \geq \varepsilon_1^2$, and the finiteness of $\iint |\nabla u|^2$ then forces the singular set to be $\mathcal{P}^1$ -null. Two later treatments simplify the first step in different directions. Lin [15] replaced the explicit iteration of [3] by a blow-up (compactness plus linear decay) argument; Ladyzhenskaya and Seregin [10] and Colombo, De Lellis and Massaccesi [4] completed and corrected that argument. Kukavica [9] instead kept the direct iteration but organized it around a single monotone quantity. His regularity criterion uses local L^5 integrability and applies to divergence-free forces in L^2 , with an extension to L^q , $q > 5/3$, stated in his Theorem 2. This is a weaker notion of regularity than the Hölder conclusion considered here. A direct proof of the theorem, with a smallness criterion involving the vorticity component perpendicular to the velocity, is given by Wolf [27].

The present notes are written with a single purpose: to serve as the mathematical source of a machine-checked formalization. Accordingly, every constant is given a name ($C_1, C_2, \dots$) and is never reused; when a constant is absolute (that is, independent of the solution, of the cylinder, and of every parameter in the statement) we say so explicitly. The estimates used at later steps are stated explicitly and cross-referenced.

The last part of the manuscript (Section 7 onwards) shows that the class of solutions to which Theorem 2.10 applies is non-empty and rich. For every divergence-free initial datum $a \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ it constructs a global Leray–Hopf solution, in the sense of Escauriaza, Seregin and Šverák [6], that is a suitable weak solution in the sense of Definition 1.5; it identifies the canonical pressure of every Leray–Hopf solution; and it carries out the same construction with a force. Consequently the singular set of such a solution has vanishing one-dimensional parabolic Hausdorff measure. The construction follows Leray [14] in the form of Ożański and Pooley [18], with the Fourier– L^2 regularization and compactness arguments written out in full. The endpoint regularity theorem of Escauriaza, Seregin and Šverák, which uses Theorem 2.10 and these solutions, is the subject of the companion manuscript [2].

For background on suitable weak solutions, partial regularity and the harmonic-analysis tools used below we refer to [1, 3, 4, 5, 7, 8, 9, 10, 11, 12, 13, 15, 16, 17, 19, 20, 22, 23, 24, 25, 26].

Organization of the manuscript. Section 1 fixes the notation, defines suitable weak solutions with a force, and proves the two facts about the time variable that the definition does not literally contain: that $\partial_t u$ is a $L^{3/2}$ -in-time distribution of a definite order (Lemma 1.10), and that consequently u has a representative that is weakly continuous in time with values in L^2 (Lemma 1.11), as an optional strengthening. The energy estimates below use only the

almost-everywhere-in-time local energy inequality (Lemma 1.13), obtained directly by time differentiation. Section 2 states the three theorems. Section 3 records the exact scaling and translation invariance and proves the interpolation inequalities. Section 4 decomposes the pressure and proves the pressure decay lemma. Section 5 contains the ε -regularity core, in the direct-iteration form of [9] and [13, §13.8–13.9], and proves Theorems 2.7 and 2.9. Section 6 derives Theorem 2.10 from Theorem 2.9 by a Vitali covering argument. Section 7 and the sections after it form a separate part: they state and prove the existence of suitable Leray–Hopf solutions and of their associated pressures, with and without a force (Section 8 constructs the pressure and Sections 9–13 the solutions). Appendix A lists the external analytic inputs, and Appendix B records corrections to the sources of the existence part.

1. SETTING AND NOTATION

1.1. Ambient conventions. Throughout, space is $\mathbb{R}^3$ with the Euclidean norm $|\cdot|$, time is $\mathbb{R}$, and a space-time point is written $z = (x, t) \in \mathbb{R}^3 \times \mathbb{R}$. The viscosity is 1. Latin indices i, j, k run over $\{1, 2, 3\}$ and repeated indices are summed. We write $\partial_i = \partial/\partial x_i$, $\nabla = (\partial_1, \partial_2, \partial_3)$, $\Delta = \partial_i \partial_i$ (the spatial Laplacian), and $\partial_t = \partial/\partial t$. For a vector field u we write $\operatorname{div} u = \partial_i u_i$ and $(u \otimes u)_{ij} = u_i u_j$. Lebesgue measure on $\mathbb{R}^3$ and on $\mathbb{R}^3 \times \mathbb{R}$ is denoted dx and $dz = dx dt$; the measure of a measurable set E is $|E|$.

Convention 1.1. Constants are numbered $C_1, C_2, \dots$ in order of first appearance and are never reused with a different value. A constant is called *absolute* if it depends on nothing at all; since the space dimension is fixed at 3 and the viscosity at 1, dimensional constants are absolute here. Any other dependence is displayed in the notation, as in $C_{17}(q)$.

Convention 1.2 (Representatives and finite local quantities). Spatial balls and vector norms in statements are Euclidean. A displayed gradient of a Sobolev function is its weak gradient, determined almost everywhere. Smooth compactly supported tests mean C^∞ tests, not real-analytic tests. Integrals of nonnegative functions may first be read in $[0, \infty]$; real-valued scale quantities are used only on cylinders whose closures are compactly contained in the solution domain. Local finiteness then follows from (S1) and the local force assumption. No infinite integral is assigned a finite numerical value.

Distributional identities are tested against every compactly supported smooth test. A slice identity initially obtained for each test outside its own null set is promoted, when needed, to one common null set by a countable dense test family and continuity of the pairings. Convolutions of nonsmooth data, and identities between their measurable representatives, are understood almost everywhere. Pointwise regularity estimates concern the continuous representative subsequently constructed, not arbitrary values on a null set.

1.2. Balls, cylinders, and the parabolic metric. For $x \in \mathbb{R}^3$ and $r > 0$ put $B_r(x) = \{y \in \mathbb{R}^3 : |y - x| < r\}$, so that $|B_r(x)| = \frac{4}{3}\pi r^3$. For $z = (x, t)$ the *parabolic cylinder of radius r with top centre z* is

$$Q_r(z) = B_r(x) \times (t - r^2, t], \quad \text{so} \quad |Q_r(z)| = \frac{4}{3}\pi r^5. \quad (1)$$

We abbreviate $B_r = B_r(0)$ and $Q_r = Q_r((0, 0))$. The interval $(t - r^2, t]$ is open at the bottom and closed at the top so that $z \in Q_r(z)$; since the top face is a null set, every integral below is unaffected by this choice.

The *parabolic metric* on $\mathbb{R}^3 \times \mathbb{R}$ is

$$d_{\text{par}}(z, z') = \max\{|x - x'|, |t - t'|^{1/2}\}, \quad z = (x, t), \quad z' = (x', t'). \quad (2)$$

Lemma 1.3. d_{par} is a metric on $\mathbb{R}^3 \times \mathbb{R}$, and the open d_{par} -ball of radius r about $z = (x, t)$ is

$$\mathfrak{B}_r(z) = \{z' : d_{\text{par}}(z', z) < r\} = B_r(x) \times (t - r^2, t + r^2). \quad (3)$$

Moreover $Q_r(z) \subset \mathfrak{B}_r(z)$ and $\mathfrak{B}_r(z) \subset Q_{2r}((x, t + r^2))$ for every z and every $r > 0$.

Proof. Symmetry and the vanishing of $d_{\text{par}}(z, z')$ exactly when $z = z'$ are immediate from the definition. For the triangle inequality, let z, z', z'' be given. Since $|\cdot|$ is a metric on $\mathbb{R}^3$, $|x - x''| \leq |x - x'| + |x' - x''| \leq d_{\text{par}}(z, z') + d_{\text{par}}(z', z'')$. For the time entry, the elementary inequality $(a + b)^{1/2} \leq a^{1/2} + b^{1/2}$, valid for $a, b \geq 0$ because $a + b \leq a + 2a^{1/2}b^{1/2} + b$, gives

$$|t - t''|^{1/2} \leq (|t - t'| + |t' - t''|)^{1/2} \leq |t - t'|^{1/2} + |t' - t''|^{1/2} \leq d_{\text{par}}(z, z') + d_{\text{par}}(z', z'').$$

Taking the maximum of the two bounds yields $d_{\text{par}}(z, z'') \leq d_{\text{par}}(z, z') + d_{\text{par}}(z', z'')$.

Identity (3) is a restatement of the definition: $d_{\text{par}}(z', z) < r$ holds if and only if $|x' - x| < r$ and $|t' - t| < r^2$. The inclusion $Q_r(z) \subset \mathfrak{B}_r(z)$ holds because $(t - r^2, t] \subset (t - r^2, t + r^2)$. Finally $Q_{2r}((x, t + r^2)) = B_{2r}(x) \times (t - 3r^2, t + r^2]$ contains $B_r(x) \times (t - r^2, t + r^2)$. $\square$

1.3. Averages. Two distinct averaging operations occur, and they must not be confused. For a locally integrable g on a space-time set, the *spatial average* over $B_r(x)$ is the function of time

$$[g]_{B_r(x)}(t) = \frac{1}{|B_r(x)|} \int_{B_r(x)} g(y, t) dy, \quad (4)$$

defined for almost every t ; and the *space-time average* over $Q_r(z)$ is the number

$$(g)_{Q_r(z)} = \frac{1}{|Q_r(z)|} \iint_{Q_r(z)} g(y, s) dy ds. \quad (5)$$

For a vector field these are applied componentwise. We also write $[g]_{B_r(x)}$ for the function $t \mapsto [g]_{B_r(x)}(t)$ when no confusion is possible.

1.4. Function spaces. Let $U \subset \mathbb{R}^3$ be open and bounded. We write $L^s(U)$, $1 \leq s \leq \infty$, for the usual Lebesgue spaces, $H^1(U) = W^{1,2}(U)$, and $H_0^2(U)$ for the closure of $C_c^\infty(U)$ in the norm $\|\varphi\|_{H^2(U)} = (\sum_{|a| \leq 2} \|\partial^a \varphi\|_{L^2(U)}^2)^{1/2}$. For vector fields we write $H_0^2(U; \mathbb{R}^3)$ and we set

$$Z_U = (H_0^2(U; \mathbb{R}^3))^*, \quad (6)$$

the topological dual, with the dual norm. For a time interval J and a Banach space X , $L^s(J; X)$ is the usual Bochner space. We abbreviate

$$\|g\|_{L^s(E)} = \left(\iint_E |g|^s \right)^{1/s}$$

for a space-time set E and $1 \leq s < \infty$.

We record the two Sobolev embeddings that will be used on H_0^2 . Because a function in $H_0^2(U)$ extends by zero to a function in $H^2(\mathbb{R}^3)$ of the same norm, the constants are absolute and do not depend on U ; see External Input A.4.

Lemma 1.4. *There is an absolute constant C_1 such that for every bounded open $U \subset \mathbb{R}^3$ and every $\varphi \in H_0^2(U; \mathbb{R}^3)$,*

$$\|\varphi\|_{L^\infty(U)} \leq C_1 \|\varphi\|_{H^2(U)}, \quad \|\nabla \varphi\|_{L^6(U)} \leq C_1 \|\varphi\|_{H^2(U)}. \quad (7)$$

In particular, for every $s \in [1, \infty]$, $\|\varphi\|_{L^s(U)} \leq C_1 |U|^{1/s} \|\varphi\|_{H^2(U)}$, and for every $s \in [1, 6]$, $\|\nabla \varphi\|_{L^s(U)} \leq C_1 |U|^{1/s-1/6} \|\varphi\|_{H^2(U)}$.

Proof. Let $\tilde{\varphi}$ denote the extension of φ to $\mathbb{R}^3$ by zero. Since $\varphi \in H_0^2(U; \mathbb{R}^3)$ is a limit in H^2 of elements of $C_c^\infty(U; \mathbb{R}^3)$, and the zero extension of such an element belongs to $C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ with unchanged H^2 norm, the map $\varphi \mapsto \tilde{\varphi}$ is an isometry of $H_0^2(U; \mathbb{R}^3)$ into $H^2(\mathbb{R}^3; \mathbb{R}^3)$. By External Input A.4(iii) there is an absolute C with $\|\tilde{\varphi}\|_{L^\infty(\mathbb{R}^3)} \leq C \|\tilde{\varphi}\|_{H^2(\mathbb{R}^3)}$, and by External Input A.4(i) applied to each $\partial_i \tilde{\varphi} \in H^1(\mathbb{R}^3)$ there is an absolute C' with $\|\nabla \tilde{\varphi}\|_{L^6(\mathbb{R}^3)} \leq C' \|\nabla \tilde{\varphi}\|_{H^1(\mathbb{R}^3)} \leq C' \|\tilde{\varphi}\|_{H^2(\mathbb{R}^3)}$. Take $C_1 = \max\{C, C'\}$ and restrict to U . The two consequences follow from Hölder's inequality on U : $\|\varphi\|_{L^s(U)} \leq |U|^{1/s} \|\varphi\|_{L^\infty(U)}$ and, for $s \leq 6$, $\|\nabla \varphi\|_{L^s(U)} \leq |U|^{1/s-1/6} \|\nabla \varphi\|_{L^6(U)}$. $\square$

1.5. Suitable weak solutions with a force. Throughout the paper $\Omega \subset \mathbb{R}^3$ is open, $I \subset \mathbb{R}$ is an open interval, and we write

$$\mathcal{O} = \Omega \times I \subset \mathbb{R}^3 \times \mathbb{R}.$$

Only local statements are made: no initial condition, no boundary condition, and no global energy inequality is imposed. That the class below contains, for every divergence-free L^2 datum, a global Leray–Hopf solution is proved in Section 7 and the sections after it.

Definition 1.5 (Suitable weak solution). Let $q > 5/2$ and let $f \in L_{\text{loc}}^q(\mathcal{O}; \mathbb{R}^3)$. A pair (u, p) is a *suitable weak solution of the Navier-Stokes equations on $\mathcal{O}$ with force f* if the following four conditions hold.

(S1) (*Regularity class.*) For every open Ω' with $\overline{\Omega'} \subset \Omega$ compact and every interval J with $\overline{J} \subset I$ compact,

$$u \in L^\infty(J; L^2(\Omega'; \mathbb{R}^3)) \cap L^2(J; H^1(\Omega'; \mathbb{R}^3)), \quad p \in L^{3/2}(\Omega' \times J).$$

(S2) (*Incompressibility.*) $\operatorname{div} u = 0$ in $\mathcal{D}'(\mathcal{O})$, that is, $\iint_{\mathcal{O}} u \cdot \nabla \psi \, dz = 0$ for every $\psi \in C_c^\infty(\mathcal{O})$. No condition is imposed on $\operatorname{div} f$.

(S3) (*Momentum equation.*) For every $\varphi \in C_c^\infty(\mathcal{O}; \mathbb{R}^3)$,

$$\iint_{\mathcal{O}} \left[-u \cdot \partial_t \varphi - u_i u_j \partial_j \varphi_i + \nabla u : \nabla \varphi - p \operatorname{div} \varphi - f \cdot \varphi \right] dz = 0. \quad (8)$$

Here $\nabla u : \nabla \varphi = \partial_j u_i \partial_j \varphi_i$. Equation (8) is the distributional form of

$$\partial_t u + \operatorname{div}(u \otimes u) - \Delta u + \nabla p = f, \quad \operatorname{div} u = 0. \quad (9)$$

(S4) (*Local energy inequality.*) For every $\phi \in C_c^\infty(\mathcal{O})$ with $\phi \geq 0$,

$$2 \iint_{\mathcal{O}} |\nabla u|^2 \phi \, dz \leq \iint_{\mathcal{O}} \left[|u|^2 (\partial_t \phi + \Delta \phi) + (|u|^2 + 2p) u \cdot \nabla \phi + 2(f \cdot u) \phi \right] dz. \quad (10)$$

Remark 1.6 (The gradient is a datum). Condition (S1) asserts $u(\cdot, t) \in H^1(\Omega')$ for a.e. t , so the weak spatial gradient ∇u exists and is uniquely determined almost everywhere on $\mathcal{O}$. In a formalization it is convenient to carry ∇u as an explicit datum $\mathbf{G}$ satisfying $\iint \mathbf{G}_{ij} \chi = -\iint u_i \partial_j \chi$ for all $\chi \in C_c^\infty(\mathcal{O})$, rather than as a derived object; the a.e. uniqueness of $\mathbf{G}$ makes the two formulations equivalent. With that convention, all the quantities of §1.7 (in particular β , and hence θ of (112)) are functions of the tuple $(u, \mathbf{G}, p, f)$ and of (z, r) , and every statement below that mentions ∇u is to be read with $\mathbf{G}$ in its place. No mathematical content changes.

The sign conventions in (10) are those of [3], [15] and [9]: every term on the right carries the sign it acquires when $|u|^2 \phi$ is formally differentiated in time using (9) and integrated by parts, and the dissipation appears on the left with the factor 2. Formally, (10) is what one gets by multiplying (9) by $2u\phi$ and discarding the (nonnegative) defect measure.

Remark 1.7 (The force class). Following [3] we assume $f \in L^q$ with $q > 5/2$; unlike [9] and [13, Definition 13.4] we do *not* assume $\operatorname{div} f = 0$. Two comments.

(i) The exponent q enters only through the scale-invariant force functional λ of (24), for which $\lambda(z, r) \leq \|f\|_{L^q} r^\sigma$ with

$$\sigma = 3 - \frac{5}{q} > 1, \quad (11)$$

so that $\lambda(z, r) \rightarrow 0$ as $r \rightarrow 0$ at a rate strictly better than linear. The strict inequality $\sigma > 1$ is used in Lemma 4.6(d)–(e) and, through $\sigma > \varepsilon$, in Proposition 5.19; the threshold $q > 5/2$ reappears in Theorem 5.30 as the condition $\theta_0 > 5/2$ for the force slot. Kukavica [9, Theorem 2] shows that the decay statements of Sections 3 and 4 already hold for $q > 5/3$, where $\sigma > 0$ but possibly $\sigma \leq 1$. The endgame here uses $\sigma > 1$. Remark 1.8 shows that an L^2 force hypothesis does not suffice for this Hölder notion of regularity; the example does not establish sharpness of the exponent $5/2$.

(ii) We do *not* assume $\operatorname{div} f = 0$, in contrast with [9]. The hypothesis would matter at exactly two places: it would make the sum $p_7 + p_8$ vanish of the pressure decomposition of Proposition 4.4, and hence the term $C_{15}\kappa^{1/2}\lambda(z, \rho)^{1/2}$ in the pressure decay estimate (71) and the term $C_{32}(q)(r/\rho)^{3/2}\lambda^{3/2}$ in (82). Both terms are estimated in Lemma 4.6(d)–(e) and are carried through Section 5; in the scale iteration they contribute the term $C_{28}\kappa^{-3}\lambda(z, \rho)$ of (113), which is harmless because $\lambda(z, \rho) \leq \|f\|_{L^q} \rho^\sigma$ with $\sigma > 1 > \varepsilon$. In the endgame the force contribution to the pressure gradient is estimated in Lemma 5.26: the near force term is a completed second Riesz transform of the localized force, and the annular term has a fixed-scale $L^{3/2}$ time majorant. Neither estimate requires $\operatorname{div} f = 0$.

Remark 1.8 (Why an L^2 force does not suffice: a shear flow). A planar shear flow gives a counterexample at force exponent 2. Write $x = (x', x_3)$ with $x' = (x_1, x_2)$, and let $w \in W_2^{2,1}(\mathbb{R}^2 \times (-1, 1))$ be a solution of a two-dimensional heat equation that is unbounded near the origin; such w exist because $W_2^{2,1}$ is the borderline Sobolev case in parabolic dimension 4, for instance a compactly supported cutoff of $(\log(1/\varrho))^{1/4}$ near the origin, with $\varrho = (|x'|^2 + |t|)^{1/2}$. Its second spatial derivatives and first time derivative are bounded by $C\varrho^{-2}(\log(1/\varrho))^{-3/4}$ near the origin. Their squares are integrable in parabolic dimension 4, since $\int_0^{1/2} r^{-1}(\log(1/r))^{-3/2} dr < \infty$. Put

$$u(x, t) = (0, 0, w(x', t)), \quad p \equiv 0, \quad f = (0, 0, \partial_t w - \Delta_{x'} w).$$

Then $\operatorname{div} u = 0$ and $\operatorname{div} f = 0$ (both fields depend only on (x', t) and point in the x_3 direction), $u \cdot \nabla u = u_3 \partial_3 u = 0$, $f \in L_{loc}^2$, $p \in L^{3/2}$, and the local energy inequality holds with equality. Since $\nabla w \in L_{loc}^4$ by the parabolic Sobolev embedding in dimension 4, one computes $\iint_{Q_r(0)} |\nabla u|^2 = o(r^3)$, so $\beta(0, r)^2 = r^{-1} \iint_{Q_r(0)} |\nabla u|^2 \rightarrow 0$ and the hypothesis (34) of Theorem 2.9 holds for every threshold. Yet u is unbounded at the origin, so the origin is not a regular point in the sense of Definition 2.2. Hence the conclusion of Theorem 2.9 is *false* for $f \in L^2$ with the Hölder (indeed with the L^∞) notion of regularity. Kukavica's Theorem 1 for $f \in L^2$ is stated for the weaker notion “ $u \in L^5$ on a neighbourhood” [9, p. 719], which this example does not contradict.

Remark 1.9 (Comparison with the sources). Definition 1.5 is the class used by Lin [15, §2] and by Ladyzhenskaya–Seregin [10, Definition 2.1], with a force added and with the domain restricted to a product $\Omega \times I$. Two features of our formulation should be noted. First, $p \in L_{loc}^{3/2}$ is an explicit hypothesis. We use the modern suitable-solution class just cited, without claiming that its pressure hypothesis is identical to the original definition in [3] or deriving it from weaker local data. Second, [10] and [9] state the local energy inequality directly in the pointwise-in-time form (19) below; we take the distributional form (10) as the definition and *prove* the pointwise

form in Lemma 1.13, since the two are not literally the same statement and the derivation uses a time cutoff and Lebesgue differentiation.

1.6. The time derivative and weak continuity in time. We first derive a weakly continuous L^2 representative from (S3). This is a useful additional property of suitable solutions. The almost-every-time local energy inequality proved afterwards uses Lebesgue differentiation directly and does not require this representative. Fix, for the whole subsection, a ball $B = B_\rho(x_0)$ and a bounded open interval J with

$$\overline{B} \subset \Omega, \quad \overline{J} \subset I. \quad (12)$$

Lemma 1.10 (The time derivative lies in $L^{3/2}(J; Z_B)$). *Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force $f \in L^q_{loc}$, $q > 5/2$, and let B, J be as in (12). Then the distribution $\partial_t u$ on $B \times J$ belongs to $L^{3/2}(J; Z_B)$, where $Z_B = (H_0^2(B; \mathbb{R}^3))^*$ as in (6): there is a function $G \in L^{3/2}(J; Z_B)$ such that*

$$-\int_J \int_B u \cdot \varphi \theta'(t) dx dt = \int_J \langle G(t), \varphi \rangle \theta(t) dt \quad \text{for all } \varphi \in H_0^2(B; \mathbb{R}^3), \theta \in C_c^\infty(J), \quad (13)$$

and, with C_1 the constant of Lemma 1.4,

$$\begin{aligned} \|G\|_{L^{3/2}(J; Z_B)} &\leq \sqrt{3}C_1 \left[|J|^{1/6} |B|^{1/3} \|\nabla u\|_{L^2(B \times J)} + |J|^{1/15} |B|^{7/30} \|u\|_{L^{10/3}(B \times J)}^2 \right. \\ &\quad \left. + |B|^{1/6} \|p\|_{L^{3/2}(B \times J)} + |J|^{\frac{2}{3}-\frac{1}{q}} |B|^{1-\frac{1}{q}} \|f\|_{L^q(B \times J)} \right]. \quad (14) \end{aligned}$$

In particular, if $|B| \leq 1$ and $|J| \leq 1$ then

$$\|G\|_{L^{3/2}(J; Z_B)} \leq C_2 \left[\|\nabla u\|_{L^2(B \times J)} + \|u\|_{L^{10/3}(B \times J)}^2 + \|p\|_{L^{3/2}(B \times J)} + \|f\|_{L^q(B \times J)} \right], \quad (15)$$

where $C_2 = \sqrt{3}C_1$ is absolute.

Proof. Step 1: the candidate for G . For $t \in J$ define a linear functional on $H_0^2(B; \mathbb{R}^3)$ by

$$\langle G(t), \varphi \rangle = \int_B \left[u_i(x, t) u_j(x, t) \partial_j \varphi_i(x) - \nabla u(x, t) : \nabla \varphi(x) + p(x, t) \operatorname{div} \varphi(x) + f(x, t) \cdot \varphi(x) \right] dx. \quad (16)$$

By (S1) and $f \in L^q_{loc}$ each of the four integrands is in $L^1(B)$ for almost every $t \in J$, so $\langle G(t), \varphi \rangle$ is defined for a.e. t , and it is measurable in t for each fixed φ . Since $H_0^2(B; \mathbb{R}^3)$ is a separable Hilbert space, so is its dual Z_B ; hence, by Pettis' measurability theorem (External Input A.13), this weak measurability implies that $t \mapsto G(t)$ is strongly (Bochner) measurable, which is what membership in the Bochner space $L^{3/2}(J; Z_B)$ requires.

Step 2: the four bounds. Fix t for which all four integrands are integrable, and let $\varphi \in H_0^2(B; \mathbb{R}^3)$ with $\|\varphi\|_{H^2(B)} \leq 1$.

(a) *Dissipation.* By Cauchy-Schwarz and Lemma 1.4 (with $s = 2 \leq 6$, so $|B|^{1/2-1/6} \leq |B|^{1/3}$),

$$\left| \int_B \nabla u : \nabla \varphi \right| \leq \|\nabla u(\cdot, t)\|_{L^2(B)} \|\nabla \varphi\|_{L^2(B)} \leq C_1 |B|^{1/3} \|\nabla u(\cdot, t)\|_{L^2(B)}.$$

(b) *Transport.* Since $H_0^2(B) \not\hookrightarrow W^{1,\infty}(B)$ in three dimensions, the convective term must be paired against $\nabla \varphi \in L^6$ rather than against $\nabla \varphi \in L^\infty$. By Hölder with $\frac{5}{6} + \frac{1}{6} = 1$ and Lemma 1.4,

$$\left| \int_B u_i u_j \partial_j \varphi_i \right| \leq \|u(\cdot, t)\|_{L^{12/5}(B)}^2 \|\nabla \varphi\|_{L^6(B)} \leq C_1 \|u(\cdot, t)\|_{L^{12/5}(B)}^2,$$

where we used $\|u_i u_j\|_{L^{6/5}(B)} \leq \|u\|_{L^{12/5}(B)}^2$.

(c) *Pressure.* By Hölder with $\frac{2}{3} + \frac{1}{3} = 1$ and Lemma 1.4 with $s = 3 \leq 6$, using $|\operatorname{div} \varphi| \leq \sqrt{3} |\nabla \varphi|$,

$$\left| \int_B p \operatorname{div} \varphi \right| \leq \sqrt{3} \|p(\cdot, t)\|_{L^{3/2}(B)} \|\nabla \varphi\|_{L^3(B)} \leq \sqrt{3} C_1 |B|^{1/6} \|p(\cdot, t)\|_{L^{3/2}(B)}.$$

(d) *Force.* By Hölder with $\frac{1}{q} + \frac{1}{q'} = 1$ and Lemma 1.4,

$$\left| \int_B f \cdot \varphi \right| \leq \|f(\cdot, t)\|_{L^q(B)} \|\varphi\|_{L^{q'}(B)} \leq C_1 |B|^{1/q'} \|f(\cdot, t)\|_{L^q(B)}.$$

Adding,

$$\begin{aligned} \|G(t)\|_{Z_B} &\leq \sqrt{3} C_1 [|B|^{1/3} \|\nabla u(\cdot, t)\|_{L^2(B)} \\ &\quad + \|u(\cdot, t)\|_{L^{12/5}(B)}^2 + |B|^{1/6} \|p(\cdot, t)\|_{L^{3/2}(B)} \\ &\quad + |B|^{1/q'} \|f(\cdot, t)\|_{L^q(B)}]. \end{aligned}$$

Step 3: time integrability of each of the four terms. We show that each of the four functions of t on the right lies in $L^{3/2}(J)$.

(a') By (S1), $t \mapsto \|\nabla u(\cdot, t)\|_{L^2(B)}$ belongs to $L^2(J)$, and therefore to $L^{3/2}(J)$, with $\|\cdot\|_{L^{3/2}(J)} \leq |J|^{1/6} \|\cdot\|_{L^2(J)}$ by Hölder, since $\frac{1}{3/2} - \frac{1}{2} = \frac{1}{6}$.

(b) Interpolating $L^{12/5}$ between L^2 and L^6 with $\frac{5}{12} = \frac{3/4}{2} + \frac{1/4}{6}$ gives

$$\|u\|_{L^{12/5}(B)} \leq \|u\|_{L^2(B)}^{3/4} \|u\|_{L^6(B)}^{1/4},$$

so

$$\|u(\cdot, t)\|_{L^{12/5}(B)}^2 \leq \|u(\cdot, t)\|_{L^2(B)}^{3/2} \|u(\cdot, t)\|_{L^6(B)}^{1/2}.$$

By (S1) and External Input A.4(ii), $t \mapsto \|u(\cdot, t)\|_{L^6(B)}$ lies in $L^2(J)$ and $t \mapsto \|u(\cdot, t)\|_{L^2(B)}$ lies in $L^\infty(J)$; hence the product lies in $L^4(J) \subset L^{3/2}(J)$. Equivalently, and this is the form recorded in (14), the interpolation $L^\infty(J; L^2) \cap L^2(J; L^6) \subset L^{10/3}(B \times J)$ (External Input A.4(iv)) gives

$$\|u\|_{L^{12/5}(B)}^2 \leq \|u\|_{L^{10/3}(B)}^2 |B|^{2(5/12 - 3/10)}$$

and $t \mapsto \|u(\cdot, t)\|_{L^{10/3}(B)}^2 \in L^{5/3}(J) \subset L^{3/2}(J)$.

(c) $t \mapsto \|p(\cdot, t)\|_{L^{3/2}(B)}$ lies in $L^{3/2}(J)$ by (S1), with $L^{3/2}(J)$ -norm equal to $\|p\|_{L^{3/2}(B \times J)}$. This is the term that fixes the exponent $3/2$ in the statement: none of the other three terms forces an exponent below $3/2$, and this one is exactly $3/2$.

(d) $t \mapsto \|f(\cdot, t)\|_{L^q(B)}$ lies in $L^q(J)$ with norm $\|f\|_{L^q(B \times J)}$, and $q > 5/2 > 3/2$, so it lies in $L^{3/2}(J)$ with $\|\cdot\|_{L^{3/2}(J)} \leq |J|^{2/3 - 1/q} \|f\|_{L^q(B \times J)}$.

Collecting the four contributions gives (14): the first is $C_1 |B|^{1/3} |J|^{1/6} \|\nabla u\|_{L^2(B \times J)}$ by (a) and (a'); the second is $C_1 |B|^{7/30} |J|^{1/15} \|u\|_{L^{10/3}(B \times J)}^2$, since

$$\|u(\cdot, t)\|_{L^{12/5}(B)}^2 \leq |B|^{2(\frac{5}{12} - \frac{3}{10})} \|u(\cdot, t)\|_{L^{10/3}(B)}^2$$

with $2(\frac{5}{12} - \frac{3}{10}) = \frac{7}{30}$ and

$$\|\cdot\|_{L^{3/2}(J)} \leq |J|^{\frac{2}{3} - \frac{3}{5}} \|\cdot\|_{L^{5/3}(J)}$$

with $\frac{2}{3} - \frac{3}{5} = \frac{1}{15}$; the third is

$$\sqrt{3} C_1 |B|^{1/6} \|p\|_{L^{3/2}(B \times J)}$$

by (c); and the fourth is

$$C_1 |B|^{1/q'} |J|^{\frac{2}{3}-\frac{1}{q}} \|f\|_{L^q(B \times J)}$$

by (d), with $1/q' = 1 - 1/q$. In particular $G \in L^{3/2}(J; Z_B)$, and (15) follows by discarding the factors $|B|^a, |J|^b \leq 1$.

Step 4: G represents $\partial_t u$. Let $\varphi \in C_c^\infty(B; \mathbb{R}^3)$ and $\theta \in C_c^\infty(J)$, and apply (S3) with test field $(x, t) \mapsto \theta(t)\varphi(x)$, which belongs to $C_c^\infty(\mathcal{O}; \mathbb{R}^3)$ by (12). Since $\partial_t(\theta\varphi) = \theta'\varphi$ and all spatial derivatives fall on φ , (8) reads

$$-\int_J \int_B u \cdot \varphi \theta' dx dt = \int_J \theta(t) \int_B \left[u_i u_j \partial_j \varphi_i - \nabla u : \nabla \varphi + p \operatorname{div} \varphi + f \cdot \varphi \right] dx dt = \int_J \langle G(t), \varphi \rangle \theta(t) dt,$$

which is (13) for $\varphi \in C_c^\infty(B; \mathbb{R}^3)$. Both sides are continuous in φ with respect to the $H^2(B)$ norm (the left side because

$$\begin{aligned} & \left| \int_J \int_B u \cdot \varphi \theta' \right| \\ & \leq \|\theta'\|_{L^\infty} \|u\|_{L^1(B \times J)} \|\varphi\|_{L^\infty(B)} \\ & \leq C_1 \|\theta'\|_{L^\infty} \|u\|_{L^1(B \times J)} \|\varphi\|_{H^2(B)} \end{aligned}$$

the right side by Step 2) and $C_c^\infty(B; \mathbb{R}^3)$ is dense in $H_0^2(B; \mathbb{R}^3)$ by definition. Hence (13) holds for all $\varphi \in H_0^2(B; \mathbb{R}^3)$. $\square$

Lemma 1.11 (Weak continuity in time). *Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force $f \in L_{loc}^q$, $q > 5/2$, and let B, J be as in (12). Then there is a set $N \subset J$ of Lebesgue measure zero and a map $\bar{u} : \bar{J} \rightarrow L^2(B; \mathbb{R}^3)$ such that*

- (i) $\bar{u}(t) = u(\cdot, t)$ in $L^2(B; \mathbb{R}^3)$ for every $t \in J \setminus N$;
- (ii) $\sup_{t \in \bar{J}} \|\bar{u}(t)\|_{L^2(B)} \leq \|u\|_{L^\infty(J; L^2(B))} =: M_{B,J}$;
- (iii) for every $w \in L^2(B; \mathbb{R}^3)$ the function $t \mapsto \int_B \bar{u}(t) \cdot w dx$ is continuous on $\bar{J}$.

We say that u belongs to $C(\bar{J}; L_w^2(B))$ and we always work with the representative $\bar{u}$.

Proof. Step 1: a continuous Z_B -valued representative. By Lemma 1.10, $G \in L^{3/2}(J; Z_B) \subset L^1(J; Z_B)$, so $t \mapsto \int_{t_1}^t G(s) ds$ is a continuous Z_B -valued function on $\bar{J}$ for any fixed $t_1 \in J$. Set $v(t) = \int_{t_1}^t G(s) ds \in Z_B$. Regarding $u(\cdot, t) \in L^2(B)$ as an element of Z_B through $\langle u(\cdot, t), \varphi \rangle = \int_B u(\cdot, t) \cdot \varphi$ (a bounded map by Lemma 1.4), identity (13) says that the Z_B -valued distribution $u - v$ on J has vanishing time derivative, i.e. $\int_J (u(\cdot, t) - v(t)) \theta'(t) dt = 0$ in Z_B for every $\theta \in C_c^\infty(J)$. By the vector-valued du Bois-Reymond lemma (External Input A.14) applied in the Banach space Z_B , there is $\xi \in Z_B$ with $u(\cdot, t) = \xi + v(t)$ in Z_B for a.e. $t \in J$; let N contain the null set where this fails and the null set where the L^2 bound $\|u(\cdot, t)\|_{L^2(B)} \leq M_{B,J}$ fails. Define $\bar{u}(t) = \xi + v(t) \in Z_B$ for all $t \in \bar{J}$; then $\bar{u} : \bar{J} \rightarrow Z_B$ is continuous and agrees with $u(\cdot, t)$ for $t \in J \setminus N$. This is (i), once we know $\bar{u}(t) \in L^2$.

Step 2: $\bar{u}(t) \in L^2(B)$ with the bound (ii). Let $t_0 \in \bar{J}$ and choose $t_k \in J \setminus N$ with $t_k \rightarrow t_0$; this is possible because $J \setminus N$ has full measure in J and $\bar{J}$ is the closure of J . By (S1), $\|\bar{u}(t_k)\|_{L^2(B)} = \|u(\cdot, t_k)\|_{L^2(B)} \leq M_{B,J}$ for every k , by the choice of N , and $\bar{u}(t_k) \rightarrow \bar{u}(t_0)$ in Z_B by Step 1. We claim that a sequence $g_k \in L^2(B; \mathbb{R}^3)$ with $\|g_k\|_{L^2} \leq M$ that converges in Z_B to some $g \in Z_B$ satisfies $g \in L^2(B; \mathbb{R}^3)$, $\|g\|_{L^2} \leq M$, and $g_k \rightharpoonup g$ weakly in L^2 . Indeed, for $\varphi \in C_c^\infty(B; \mathbb{R}^3)$, $|\langle g, \varphi \rangle| = \lim_k |\int_B g_k \cdot \varphi| \leq M \|\varphi\|_{L^2(B)}$, so g extends to a bounded functional on the L^2 -closure of $C_c^\infty(B; \mathbb{R}^3)$, namely $L^2(B; \mathbb{R}^3)$, of norm at most M ; by the

Riesz representation theorem $g \in L^2(B; \mathbb{R}^3)$ with $\|g\|_{L^2} \leq M$. For weak convergence, let $w \in L^2(B; \mathbb{R}^3)$ and $\varepsilon > 0$, and choose $\varphi \in C_c^\infty(B; \mathbb{R}^3)$ with $\|w - \varphi\|_{L^2(B)} \leq \varepsilon/(4M + 1)$. Then

$$\left| \int_B (g_k - g) \cdot w \right| \leq \left| \int_B (g_k - g) \cdot (w - \varphi) \right| + \left| \int_B (g_k - g) \cdot \varphi \right| \leq 2M \frac{\varepsilon}{4M + 1} + |\langle g_k - g, \varphi \rangle|,$$

and the last term tends to 0 because $g_k \rightarrow g$ in Z_B and $\varphi \in H_0^2(B; \mathbb{R}^3)$. Hence

$$\limsup_k \left| \int_B (g_k - g) \cdot w \right| \leq \varepsilon/2 < \varepsilon.$$

Applying the claim with $g_k = \bar{u}(t_k)$, $g = \bar{u}(t_0)$, $M = M_{B,J}$ gives $\bar{u}(t_0) \in L^2(B; \mathbb{R}^3)$ with $\|\bar{u}(t_0)\|_{L^2(B)} \leq M_{B,J}$, which is (ii).

Step 3: weak continuity (iii). Let $t_0 \in \bar{J}$ and let $t_k \rightarrow t_0$ in $\bar{J}$ be arbitrary. By Step 2 each $\bar{u}(t_k) \in L^2(B)$ with norm at most $M_{B,J}$, and $\bar{u}(t_k) \rightarrow \bar{u}(t_0)$ in Z_B by Step 1. The claim of Step 2 gives $\bar{u}(t_k) \rightharpoonup \bar{u}(t_0)$ weakly in $L^2(B; \mathbb{R}^3)$, i.e. $\int_B \bar{u}(t_k) \cdot w \rightarrow \int_B \bar{u}(t_0) \cdot w$ for every $w \in L^2(B; \mathbb{R}^3)$. Since $t_k \rightarrow t_0$ was arbitrary, (iii) follows. $\square$

Lemma 1.12 (A common null set for the divergence-free condition). *Let $u \in L_{loc}^1(\mathcal{O}; \mathbb{R}^3)$ satisfy $\operatorname{div} u = 0$ in $\mathcal{D}'(\mathcal{O})$, let $B = B_\rho(x_0)$ with $\bar{B} \subset \Omega$, let J be an interval with $\bar{J} \subset I$, and let $E \subset B$ be compact. Then there is a Lebesgue-null set $N \subset J$ such that*

$$\int_B u(y, s) \cdot \nabla \psi(y) dy = 0 \quad \text{for every } s \in J \setminus N \text{ and every } \psi \in C_c^1(B) \text{ with } \operatorname{supp} \psi \subset E. \quad (17)$$

Proof. Let E' be a compact neighbourhood of E in B . Restriction of a function and its three first derivatives embeds $C_E^1 = \{\psi \in C^1(\mathbb{R}^3) : \operatorname{supp} \psi \subset E\}$ isometrically into $C(E')^4$. Since E' is compact metrizable, this space and its subspaces are separable. Let $\{\psi_k\}_{k \in \mathbb{N}} \subset C_c^\infty(B)$ with $\operatorname{supp} \psi_k \subset E'$ be a family whose C^1 closure contains C_E^1 . To construct it, mollify a countable dense subset of C_E^1 at a countable sequence of sufficiently small radii. The supports then lie in E' , and the mollifications converge in C^1 (External Input A.15). Testing (S2) with $\psi_k(x)\theta(t)$, $\theta \in C_c^\infty(J)$, gives $\int_J \theta(s) (\int_B u(\cdot, s) \cdot \nabla \psi_k) ds = 0$ for all θ , hence a null set $N_k \subset J$ off which $\int_B u(\cdot, s) \cdot \nabla \psi_k = 0$. Let $N = (\bigcup_k N_k) \cup \{s : \|u(\cdot, s)\|_{L^1(B)} = \infty\}$, a null set by Fubini. For $s \notin N$ and $\psi \in C_E^1$, choose $\psi_k \rightarrow \psi$ in C^1 ; then $|\int_B u(\cdot, s) \cdot \nabla(\psi - \psi_k)| \leq \|u(\cdot, s)\|_{L^1(B)} \|\nabla \psi - \nabla \psi_k\|_{L^\infty} \rightarrow 0$, which gives (17). $\square$

Lemma 1.13 (Pointwise-in-time local energy inequality). *Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force $f \in L_{loc}^q$, $q > 5/2$. Let $\phi \in C_c^\infty(\mathcal{O})$ with $\phi \geq 0$, and let $J = (t_-, t_+)$ be a bounded open interval with $\bar{J} \subset I$ and*

$$\phi(\cdot, s) = 0 \quad \text{for all } s \leq t_- + \varepsilon' \quad \text{for some } \varepsilon' > 0, \quad (18)$$

so that in particular $\operatorname{supp} \phi \subset \Omega \times (t_-, \infty)$. Then for almost every $t \in J$,

$$\begin{aligned} \int_\Omega |u(x, t)|^2 \phi(x, t) dx + 2 \int_{t_-}^t \int_\Omega |\nabla u|^2 \phi dx ds \\ \leq \int_{t_-}^t \int_\Omega \left[|u|^2 (\partial_s \phi + \Delta \phi) + (|u|^2 + 2p) u \cdot \nabla \phi + 2(f \cdot u) \phi \right] dx ds. \end{aligned} \quad (19)$$

Proof. Fix $t \in J$ and let $K = \operatorname{supp} \phi$, a compact subset of $\mathcal{O}$. Cover its spatial projection by finitely many balls whose closures lie in Ω , and let B denote their union. Choose an open interval J' with $\bar{J}' \subset I$, $K \subset B \times J'$ and $\bar{J} \subset J'$. The local integrability bounds used below hold on each ball and hence on their finite union. All spatial integrals below may be restricted to B , since the test function vanishes outside it.

Step 1: a Lipschitz cut-off in time. For $0 < h < t - t_-$ let $\chi_h : \mathbb{R} \rightarrow [0, 1]$ be the piecewise affine function

$$\chi_h(s) = \begin{cases} 1, & s \leq t - h, \\ \frac{t-s}{h}, & t - h < s < t, \\ 0, & s \geq t. \end{cases} \quad (20)$$

Then χ_h is Lipschitz with $\chi'_h = -h^{-1}\mathbf{1}_{(t-h,t)}$ almost everywhere. Fix $\zeta \in C_c^\infty((-1, 1))$ with $\zeta \geq 0$ and $\int_{\mathbb{R}} \zeta = 1$, put $\zeta_\varepsilon(s) = \varepsilon^{-1}\zeta(s/\varepsilon)$, and set $\chi_{h,\varepsilon} = \chi_h * \zeta_\varepsilon$, which is smooth, takes values in $[0, 1]$, equals 1 for $s \leq t - h - \varepsilon$ and 0 for $s \geq t + \varepsilon$. For ε small enough that $t + \varepsilon < t_+$, the function $\phi \chi_{h,\varepsilon}$ belongs to $C_c^\infty(\mathcal{O})$ and is nonnegative, so (10) applies to it. Using $\partial_s(\phi \chi_{h,\varepsilon}) = (\partial_s \phi) \chi_{h,\varepsilon} + \phi \chi'_{h,\varepsilon}$ and $\Delta(\phi \chi_{h,\varepsilon}) = (\Delta \phi) \chi_{h,\varepsilon}$, inequality (10) becomes

$$\begin{aligned} 2 \iint |\nabla u|^2 \phi \chi_{h,\varepsilon} - \iint |u|^2 \phi \chi'_{h,\varepsilon} \\ \leq \iint \left[|u|^2 (\partial_s \phi + \Delta \phi) + (|u|^2 + 2p) u \cdot \nabla \phi + 2(f \cdot u) \phi \right] \chi_{h,\varepsilon}. \end{aligned} \quad (21)$$

Step 2: $\varepsilon \rightarrow 0$ at fixed h . Each integrand in (21) other than the one containing χ' is in $L^1(B \times J')$: indeed $|\nabla u|^2 \phi \in L^1$ by (S1), $|u|^2 (\partial_s \phi + \Delta \phi) \in L^1$ because $u \in L^{10/3}(B \times J') \subset L^2$, $|u|^2 u \cdot \nabla \phi \in L^1$ because $u \in L^{10/3} \subset L^3$, $p u \cdot \nabla \phi \in L^1$ by Hölder with $\frac{2}{3} + \frac{1}{3} = 1$ and $p \in L^{3/2}$, $u \in L^3$, and $(f \cdot u) \phi \in L^1$ by Hölder with $f \in L^q$, $q > 5/2$, and $u \in L^{10/3}$ (note $\frac{1}{q} + \frac{3}{10} < \frac{2}{5} + \frac{3}{10} < 1$). Since $0 \leq \chi_{h,\varepsilon} \leq 1$ and $\chi_{h,\varepsilon} \rightarrow \chi_h$ pointwise away from the two corner points as $\varepsilon \rightarrow 0$, dominated convergence applies to all these terms. For the remaining term, $\chi'_{h,\varepsilon} = \chi'_h * \zeta_\varepsilon$ is bounded by h^{-1} and supported in $(t-h-\varepsilon, t+\varepsilon)$, and $\chi'_{h,\varepsilon} \rightarrow \chi'_h$ in $L^1(\mathbb{R})$; since $s \mapsto \int |u(x, s)|^2 \phi(x, s) dx$ is in $L^\infty(J')$ by (S1), we get $\iint |u|^2 \phi \chi'_{h,\varepsilon} \rightarrow -h^{-1} \int_{t-h}^t \int |u|^2 \phi dx ds$. Hence

$$\begin{aligned} 2 \int_{t_-}^t \int |\nabla u|^2 \phi \chi_h + \frac{1}{h} \int_{t-h}^t \int |u|^2 \phi dx ds \\ \leq \int_{t_-}^t \int \left[|u|^2 (\partial_s \phi + \Delta \phi) + (|u|^2 + 2p) u \cdot \nabla \phi + 2(f \cdot u) \phi \right] \chi_h. \end{aligned} \quad (22)$$

Step 3: $h \rightarrow 0$. On the right of (22) we have $0 \leq \chi_h \leq \mathbf{1}_{(-\infty, t)}$ and $\chi_h \rightarrow \mathbf{1}_{(-\infty, t)}$ pointwise, so dominated convergence gives the right side of (19). In the first term on the left, χ_h increases to $\mathbf{1}_{(-\infty, t)}$ and the integrand is nonnegative, so monotone convergence gives $2 \int_{t_-}^t \int |\nabla u|^2 \phi$. It remains to prove

$$\int |u(x, t)|^2 \phi(x, t) dx \leq \liminf_{h \rightarrow 0^+} \frac{1}{h} \int_{t-h}^t \int |u(x, s)|^2 \phi(x, s) dx ds. \quad (23)$$

Write $\Lambda(s) = \int |u(x, s)|^2 \phi(x, s) dx$. This function is locally integrable by (S1). At every one-sided Lebesgue point t of Λ ,

$$\lim_{h \downarrow 0} \frac{1}{h} \int_{t-h}^t \Lambda(s) ds = \Lambda(t).$$

Almost every t has this property. This gives (23) (in fact equality) at precisely the times required by the statement. The remaining terms converge by the preceding integrability and monotone-convergence arguments. $\square$

Remark 1.14. At the Lebesgue times used above the average converges to the slice energy. Only the corresponding lower bound is needed for the left side of (19). Every subsequent energy estimate uses an essential supremum or an integral in time, so it needs no assertion at an exceptional time. The earlier weak-time-continuity discussion provides a stronger optional representative convention, not an input to this route.

The local product-domain formulation loses no generality for local regularity on an open spacetime set: cover that set by relatively compact products $\Omega' \times J$. All weak-equation and energy pairings are integrable on the support of a test, by (S1), local interpolation, and the force assumption.

1.7. Scale-invariant quantities. Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force f , let $z = (x, t) \in \mathcal{O}$ and let $r > 0$ be such that $\overline{Q_r(z)} \subset \mathcal{O}$. The five quantities we work with are those of Kukavica [9, §2]:

$$\alpha(z, r) = \left(\frac{1}{r} \operatorname{ess\,sup}_{s \in (t-r^2, t)} \int_{B_r(x)} |u(y, s)|^2 dy \right)^{1/2}, \quad \beta(z, r) = \left(\frac{1}{r} \iint_{Q_r(z)} |\nabla u|^2 dz' \right)^{1/2}, \quad (24)$$

$$\gamma(z, r) = \left(\frac{1}{r^2} \iint_{Q_r(z)} |u|^3 dz' \right)^{1/3}, \quad \delta(z, r) = \left(\frac{1}{r^2} \iint_{Q_r(z)} |p|^{3/2} dz' \right)^{1/3},$$

$$\lambda(z, r) = r^\sigma \left(\iint_{Q_r(z)} |f|^q dz' \right)^{1/q} = r^\sigma \|f\|_{L^q(Q_r(z))}, \quad \sigma = 3 - \frac{5}{q} > 1. \quad (25)$$

When the base point is the origin we write $\alpha(r) = \alpha((0, 0), r)$, and so on. All five are dimensionless, in the precise sense of Lemma 3.3: each is unchanged when (u, p, f) is replaced by the rescaled triple (35) and r by r/μ . For $q = 2$ one has $\sigma = 1/2$ and $\lambda(z, r) = (r \iint_{Q_r(z)} |f|^2)^{1/2}$, which is the force functional of [9, §2].

Lemma 1.15 (Dictionary to the Lin, CKN and Ladyzhenskaya–Seregin notation). *Put*

$$\begin{aligned} A(z, r) &= \frac{1}{r} \operatorname{ess\,sup}_{s \in (t-r^2, t)} \int_{B_r(x)} |u|^2, & B(z, r) &= \frac{1}{r} \iint_{Q_r(z)} |\nabla u|^2, & C(z, r) &= \frac{1}{r^2} \iint_{Q_r(z)} |u|^3, \\ D(z, r) &= \frac{1}{r^2} \iint_{Q_r(z)} |p|^{3/2}, & E_q(z, r) &= \lambda(z, r). \end{aligned} \quad (26)$$

Then

$$\alpha = A^{1/2}, \quad \beta = B^{1/2}, \quad \gamma = C^{1/3}, \quad \delta = D^{1/3}, \quad \lambda = E_q. \quad (27)$$

Proof. Each identity is the definition (24) with the outer exponent removed, compared with (26). $\square$

Remark 1.16 (Letters used differently by different authors). A, B, C, D are Lin's notation [15, §3], which is also that of [3]. Ladyzhenskaya–Seregin [10, §5] and Tsai [25, §6.3] write $E(r)$ for what is here $B(z, r)$, i.e. for the scale-invariant Dirichlet energy; we never use the letter E alone, precisely to avoid this collision, and we write E_q only as an alias for λ when comparing with [25]. Tsai [25, §6.3] also uses $\tilde{C}$ for the velocity excess, which agrees with (28) below.

We shall also use the two *excess* (oscillation) quantities, in which the velocity is measured against its space-time mean and the pressure against its spatial mean at each time,

$$\tilde{C}(z, r) = \frac{1}{r^2} \iint_{Q_r(z)} |u - (u)_{Q_r(z)}|^3 dz', \quad \tilde{D}(z, r) = \frac{1}{r^2} \iint_{Q_r(z)} |p(y, s) - [p]_{B_r(x)}(s)|^{3/2} dy ds, \quad (28)$$

the *drift functional*

$$\Psi(z, r) = r |(u)_{Q_r(z)}|, \quad (29)$$

and the *excess* of (u, p) on $Q_r(z)$, in the normalization of [25, (6.14)],

$$\begin{aligned} \varphi(z, r) &= \tilde{C}(z, r)^{1/3} + \tilde{D}(z, r)^{2/3} \\ &= \left(\frac{1}{r^2} \iint_{Q_r(z)} |u - (u)_{Q_r(z)}|^3 \right)^{1/3} + \left(\frac{1}{r^2} \iint_{Q_r(z)} |p - [p]_{B_r(x)}|^{3/2} \right)^{2/3}. \end{aligned} \quad (30)$$

Both φ and Ψ are dimensionless (Lemma 3.3).

Remark 1.17 (Why the two means are different). In (30) the velocity is compared with a *space-time* mean and the pressure with a *spatial* mean at each time. These choices serve different purposes.

The pressure of a suitable weak solution is determined only up to an additive function of time: if (u, p) is a suitable weak solution with force f and $c \in L_{loc}^{3/2}(I)$, then $(u, p + c)$ is one as well, since $\nabla c = 0$ and $c u \cdot \nabla \phi$ integrates to 0 against $\operatorname{div} u = 0$ after an integration by parts. Hence no statement about p can be invariant unless a spatial mean is subtracted at each time.

The velocity excess uses a space-time mean because the parabolic Campanato characterization of Hölder continuity (External Input A.9) measures oscillation in both space and time. A spatial mean alone removes all information about a spatially constant, time-dependent velocity. This observation explains the choice of excess; it is not a counterexample to regularity under the full suitable-solution hypotheses.

Lemma 1.18 (Comparison of the excess with γ, δ). *For every z and every admissible r ,*

$$\tilde{C}(z, r) \leq 8C(z, r), \quad \tilde{D}(z, r) \leq 2^{3/2}D(z, r), \quad \varphi(z, r) \leq 2\gamma(z, r) + 2\delta(z, r)^2, \quad (31)$$

and $\Psi(z, r) \leq C_3 \gamma(z, r)$ with $C_3 = (4\pi/3)^{-1/3}$, an absolute constant.

Proof. Jensen's inequality, applied to $\tau \mapsto \tau^3$ with the normalized Lebesgue measure on $Q_r(z)$, gives

$$|(u)_{Q_r(z)}|^3 \leq (|Q_r(z)|^{-1} \iint |u|)^3 \leq |Q_r(z)|^{-1} \iint |u|^3,$$

hence

$$\iint_{Q_r(z)} |(u)_{Q_r(z)}|^3 \leq \iint_{Q_r(z)} |u|^3.$$

Combining with $|a - b|^3 \leq 4(|a|^3 + |b|^3)$ gives $\tilde{C} \leq 4(C + C) = 8C$. The same argument with $\tau \mapsto \tau^{3/2}$, applied for each fixed s on $B_r(x)$, gives

$$\int_{B_r(x)} |[p]_{B_r(x)}(s)|^{3/2} dy \leq \int_{B_r(x)} |p(\cdot, s)|^{3/2} dy,$$

and with $|a - b|^{3/2} \leq 2^{1/2}(|a|^{3/2} + |b|^{3/2})$ this gives $\tilde{D} \leq 2^{1/2} \cdot 2D = 2^{3/2}D$. Taking the $1/3$ and $2/3$ powers, $\tilde{C}^{1/3} \leq 8^{1/3}C^{1/3} = 2\gamma$ and $\tilde{D}^{2/3} \leq (2^{3/2})^{2/3}D^{2/3} = 2\delta^2$, which is the third inequality. Finally, by Jensen again,

$$\begin{aligned} & |(u)_{Q_r(z)}| \\ & \leq (|Q_r(z)|^{-1} \iint |u|^3)^{1/3}, \\ & = \left(\frac{4\pi}{3} \right)^{-1/3} r^{-5/3} (r^2 \gamma^3)^{1/3} \\ & = \left(\frac{4\pi}{3} \right)^{-1/3} r^{-1} \gamma \end{aligned}$$

so $\Psi = r \left| (u)_{Q_r(z)} \right| \leq (\frac{4\pi}{3})^{-1/3} \gamma$. $\square$

Remark 1.19 (Normalizations that are *not* used). Lin [15] writes $v_\theta = \theta^{-5} \iint_{Q_\theta} v$ and $P_\theta(t) = \theta^{-3} \int_{B_\theta} P$. These differ from the genuine averages (4), (5) by the fixed factors $|Q_1| = \frac{4\pi}{3}$ and $|B_1| = \frac{4\pi}{3}$. We use only genuine averages: every mean in this paper divides by the measure of the set it is taken over. This matters for the Campanato step (External Input A.9), which is stated for genuine averages, and for the Poincaré and Jensen inequalities, which are false with a mismatched normalizing factor.

2. STATEMENTS

2.1. Regular and singular points.

Definition 2.1 (Parabolic Hölder spaces). Let $E \subset \mathbb{R}^3 \times \mathbb{R}$ and $\gamma \in (0, 1]$. A function $w : E \rightarrow \mathbb{R}^m$ belongs to $C^{\gamma, \gamma/2}(E)$ if it is bounded and

$$[w]_{C^{\gamma, \gamma/2}(E)} = \sup_{z \neq z' \in E} \frac{|w(z) - w(z')|}{d_{\text{par}}(z, z')^\gamma} < \infty,$$

with d_{par} as in (2). We set $\|w\|_{C^{\gamma, \gamma/2}(E)} = \|w\|_{L^\infty(E)} + [w]_{C^{\gamma, \gamma/2}(E)}$.

Definition 2.2 (Regular and singular points). Let (u, p) be a suitable weak solution on $\mathcal{O} = \Omega \times I$ with force f . A point $z_0 \in \mathcal{O}$ is a *regular point* of u if there exist an open set $\mathcal{N}$ with $z_0 \in \mathcal{N} \subset \mathcal{O}$, an exponent $\gamma \in (0, 1]$, and a function $w \in C^{\gamma, \gamma/2}(\mathcal{N})$ such that $u = w$ almost everywhere in $\mathcal{N}$. Otherwise z_0 is a *singular point*. The *singular set* is

$$S = S(u) = \{z_0 \in \mathcal{O} : z_0 \text{ is a singular point of } u\}.$$

Lemma 2.3. S is relatively closed in $\mathcal{O}$.

Proof. Let $R = \mathcal{O} \setminus S$ be the set of regular points and let $z_0 \in R$, with $\mathcal{N}, \gamma, w$ as in Definition 2.2. Every $z_1 \in \mathcal{N}$ is then regular, witnessed by the same $\mathcal{N}, \gamma$ and w . Hence $\mathcal{N} \subset R$, so R is open in $\mathcal{O}$ and S is relatively closed. $\square$

Definition 2.4 (Parabolic Hausdorff measure). For $\alpha \geq 0$, $\delta > 0$ and $E \subset \mathbb{R}^3 \times \mathbb{R}$ put

$$\begin{aligned} \mathcal{P}_\delta^\alpha(E) &= \inf \left\{ \sum_{j \in J} r_j^\alpha : J \subset \mathbb{N}, \quad E \subset \bigcup_{j \in J} \mathfrak{B}_{r_j}(z_j), \quad 0 < r_j \leq \delta \quad (j \in J) \right\}, \\ \mathcal{P}^\alpha(E) &= \lim_{\delta \rightarrow 0^+} \mathcal{P}_\delta^\alpha(E) = \sup_{\delta > 0} \mathcal{P}_\delta^\alpha(E), \end{aligned}$$

where empty and finite J are allowed, an empty sum is zero, and $\mathfrak{B}_r(z)$ is the d_{par} -ball (3). For $\alpha = 0$ each nonempty covering ball has cost 1. The same index convention is used for the cylinder gauge below. Thus $\mathcal{P}^\alpha$ is the *spherical* α -dimensional Hausdorff measure of the metric space $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}})$: the gauge is the radius of a covering ball rather than the diameter of an arbitrary covering set (External Input A.8). The two differ by a factor between 1 and 2^α and in particular have the same null sets, which is all that is used.

Remark 2.5 (Gauge independence). Definition 2.4 uses the *radius* of a covering ball as gauge. The Hausdorff measure of $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}})$ built, as in the usual measure-theoretic convention, from covers by arbitrary sets E_j of small d_{par} -diameter and the gauge $(\text{diam}_{d_{\text{par}}} E_j)^\alpha$, denoted by $\mathcal{P}_{\text{diam}}^\alpha$, satisfies

$$\mathcal{P}_{\text{diam}}^\alpha(E) \leq 2^\alpha \mathcal{P}^\alpha(E) \quad \text{and} \quad \mathcal{P}^\alpha(E) \leq 2^\alpha \mathcal{P}_{\text{diam}}^\alpha(E) \quad (32)$$

for every E . Indeed, a ball $\mathfrak{B}_r(z)$ has $\text{diam}_{d_{\text{par}}} \leq 2r$, which gives the first inequality; conversely a non-empty set E_j of diameter d_j containing a point z_j satisfies $E_j \subset \mathfrak{B}_{r_j}(z_j)$ for every $r_j > d_j$, and letting $r_j \downarrow d_j$ gives the second, in fact with factor 1 for $\alpha > 0$: choose the radii with arbitrarily small summable excess in $\sum_j r_j^\alpha$, including arbitrarily small positive radii when $d_j = 0$. At $\alpha = 0$ both measures are counting measure: a finite set of n points requires n sufficiently small covering sets, and every infinite set contains finite subsets of arbitrarily large cardinality. Empty covering sets are discarded at zero cost in the diameter convention. Thus both comparisons include $\alpha = 0$ and $E = \emptyset$. In particular the two have exactly the same null sets, and Theorem 2.10 (an assertion that a set is $\mathcal{P}^1$ -null) is unaffected by the choice of gauge. The same applies to the cylinder gauge of Lemma 2.6.

Lemma 2.6 (Cylinders and parabolic balls give the same null sets). *Define $\widehat{\mathcal{P}}^\alpha$ exactly as in Definition 2.4 but with the covering sets $\mathfrak{B}_{r_j}(z_j)$ replaced by cylinders $Q_{r_j}(z_j)$. Then for every E and every $\alpha \geq 0$,*

$$\mathcal{P}^\alpha(E) \leq \widehat{\mathcal{P}}^\alpha(E) \leq 2^\alpha \mathcal{P}^\alpha(E).$$

In particular $\mathcal{P}^\alpha(E) = 0$ if and only if $\widehat{\mathcal{P}}^\alpha(E) = 0$.

Proof. If $E \subset \bigcup_j Q_{r_j}(z_j)$ with $r_j \leq \delta$ then $E \subset \bigcup_j \mathfrak{B}_{r_j}(z_j)$ by Lemma 1.3, with the same radii; hence $\mathcal{P}_\delta^\alpha(E) \leq \widehat{\mathcal{P}}_\delta^\alpha(E)$. Conversely, if $E \subset \bigcup_j \mathfrak{B}_{r_j}(z_j)$ with $z_j = (x_j, t_j)$ and $r_j \leq \delta$, then by Lemma 1.3 $\mathfrak{B}_{r_j}(z_j) \subset Q_{2r_j}((x_j, t_j + r_j^2))$, so E is covered by cylinders of radii $2r_j \leq 2\delta$ and $\widehat{\mathcal{P}}_{2\delta}^\alpha(E) \leq \sum_j (2r_j)^\alpha = 2^\alpha \sum_j r_j^\alpha$. Taking the infimum and then $\delta \rightarrow 0^+$ gives the second inequality. $\square$

2.2. The three theorems.

Theorem 2.7 (ε -regularity, L^3 form). *For every $q > 5/2$ there exist $\varepsilon_0 = \varepsilon_0(q) > 0$, $\gamma_0 = \gamma_0(q) \in (0, \frac{2}{3}]$ and $C_4 = C_4(q) < \infty$ with the following property. Let (u, p) be a suitable weak solution on $\mathcal{O} = \Omega \times I$, with $\overline{Q_1} \subset \mathcal{O}$, and force $f \in L^q(Q_1; \mathbb{R}^3)$, and suppose that, with the notation (24),*

$$\iint_{Q_1} (|u|^3 + |p|^{3/2} + |f|^q) dz \leq \varepsilon_0, \quad \text{equivalently} \quad \gamma(1)^3 + \delta(1)^3 + \lambda(1)^q \leq \varepsilon_0. \quad (33)$$

Then u agrees almost everywhere on $Q_{1/2}$ with a function $w \in C^{\gamma_0, \gamma_0/2}(\overline{Q_{1/2}}; \mathbb{R}^3)$ satisfying $\|w\|_{C^{\gamma_0, \gamma_0/2}(\overline{Q_{1/2}})} \leq C_4$. In particular every point of the open cylinder $B_{1/2}(0) \times (-\frac{1}{4}, 0)$ is a regular point of u .

Remark 2.8 (The top face). Definition 2.2 requires an *open* neighbourhood, whereas $Q_{1/2}$ contains its top face $\{t = 0\}$; Theorem 2.7 therefore does not by itself declare the points of that face regular. This costs nothing, because Theorem 2.9 is not deduced from Theorem 2.7 by applying it at z_0 : it is proved in §5.6 from Proposition 5.22, whose conclusion (123) holds at every centre of the *open* d_{par} -ball $\mathfrak{B}_{r_2}(z_0)$, and from Theorem 5.30, which produces a Hölder representative on an open d_{par} -ball. The upper-semicontinuity Lemma 5.20 is exactly what makes the passage to later base points legitimate.

Theorem 2.9 (ε -regularity, gradient form). *For every $q > 5/2$ there exists $\varepsilon_1 = \varepsilon_1(q) > 0$ with the following property. Let (u, p) be a suitable weak solution on $\mathcal{O} = \Omega \times I$ with force $f \in L_{\text{loc}}^q(\mathcal{O}; \mathbb{R}^3)$, and let $z_0 \in \mathcal{O}$. Suppose that the following inequality holds, with both limsups*

taken in $[0, \infty]$:¹

$$\limsup_{r \rightarrow 0^+} \beta(z_0, r)^2 = \limsup_{r \rightarrow 0^+} \frac{1}{r} \iint_{Q_r(z_0)} |\nabla u|^2 dz < \varepsilon_1^2. \quad (34)$$

Then z_0 is a regular point of u .

Theorem 2.10 (Caffarelli-Kohn-Nirenberg). *Let (u, p) be a suitable weak solution on $\mathcal{O} = \Omega \times I$ with force $f \in L_{loc}^q(\mathcal{O}; \mathbb{R}^3)$ for some $q > 5/2$. Then the singular set S satisfies $\mathcal{P}^1(S) = 0$.*

Theorems 2.7 and 2.9 are proved in §5.6; Theorem 2.10 is deduced from Theorem 2.9 in Section 6.

Remark 2.11 (Strictness in (34)). The inequality in (34) is strict, as in [10, Theorem 2.2] and [9, Theorem 1]; the proof produces a threshold ε_1 and needs $B(z_0, r) < \varepsilon_1^2$ for all sufficiently small r , which is what the strict limsup inequality supplies. For $0 < \varepsilon'_1 < \varepsilon_1$, the nonstrict hypothesis $\limsup_{r \downarrow 0} B(z_0, r) \leq (\varepsilon'_1)^2 < \varepsilon_1^2$ suffices.

3. INVARIANCES AND INTERPOLATION

3.1. Translation and parabolic rescaling.

Definition 3.1. Let $z_0 = (x_0, t_0) \in \mathbb{R}^3 \times \mathbb{R}$ and $\mu > 0$. For functions u, p, f defined on $\mathcal{O} = \Omega \times I$ set

$$\begin{aligned} u^{\mu, z_0}(y, s) &= \mu u(x_0 + \mu y, t_0 + \mu^2 s), \\ p^{\mu, z_0}(y, s) &= \mu^2 p(x_0 + \mu y, t_0 + \mu^2 s), \\ f^{\mu, z_0}(y, s) &= \mu^3 f(x_0 + \mu y, t_0 + \mu^2 s), \end{aligned} \quad (35)$$

defined on $\mathcal{O}^{\mu, z_0} = (\mu^{-1}(\Omega - x_0)) \times (\mu^{-2}(I - t_0))$.

Lemma 3.2 (The class is invariant). *Let $q > 5/2$. If (u, p) is a suitable weak solution on $\mathcal{O}$ with force $f \in L_{loc}^q(\mathcal{O}; \mathbb{R}^3)$, then for every z_0 and every $\mu > 0$ the pair $(u^{\mu, z_0}, p^{\mu, z_0})$ is a suitable weak solution on $\mathcal{O}^{\mu, z_0}$ with force f^{μ, z_0} .*

Proof. Throughout write $u^\mu = u^{\mu, z_0}$ and let $T(y, s) = (x_0 + \mu y, t_0 + \mu^2 s)$, so that $u^\mu = \mu(u \circ T)$, and T maps $\mathcal{O}^{\mu, z_0}$ onto $\mathcal{O}$ with Jacobian μ^5 in (y, s) .

(S1). If $\Omega'' \times J''$ has compact closure in $\mathcal{O}^{\mu, z_0}$ then $T(\Omega'' \times J'')$ has compact closure in $\mathcal{O}$, and

$$\int_{\Omega''} |u^\mu(y, s)|^2 dy = \mu^2 \mu^{-3} \int_{T_x(\Omega'')} |u(\cdot, t_0 + \mu^2 s)|^2 dx = \mu^{-1} \int_{T_x(\Omega'')} |u|^2 dx,$$

where $T_x(y) = x_0 + \mu y$; by the same change of variables,

$$\iint |\nabla_y u^\mu|^2 dy ds = \mu^2 \mu^2 \mu^{-5} \iint |\nabla u|^2 dz = \mu^{-1} \iint |\nabla u|^2,$$

$$\iint |u^\mu|^2 = \mu^{-3} \iint |u|^2,$$

$$\iint |p^\mu|^{3/2} = \mu^3 \mu^{-5} \iint |p|^{3/2} = \mu^{-2} \iint |p|^{3/2}$$

and $\iint |f^\mu|^q = \mu^{3q-5} \iint |f|^q$. All are finite, so (S1) holds for (u^μ, p^μ) .

(S2) and (S3). Given $\psi \in C_c^\infty(\mathcal{O}^{\mu, z_0})$, put $\widehat{\psi} = \psi \circ T^{-1} \in C_c^\infty(\mathcal{O})$. Then $\nabla_y \psi = \mu(\nabla \widehat{\psi}) \circ T$ and $\iint u^\mu \cdot \nabla_y \psi dy ds = \mu \cdot \mu \cdot \mu^{-5} \iint u \cdot \nabla \widehat{\psi} dz = \mu^{-3} \iint u \cdot \nabla \widehat{\psi} dz = 0$, which is (S2). For (S3),

¹Suitability and interior geometry make each sufficiently small cylinder mass finite, so the squared real β expression agrees with the nonnegative integral expression.

let $\varphi \in C_c^\infty(\mathcal{O}^{\mu, z_0}; \mathbb{R}^3)$ and put $\widehat{\varphi} = \varphi \circ T^{-1}$. Using $\partial_s \varphi = \mu^2(\partial_t \widehat{\varphi}) \circ T$, $\nabla_y \varphi = \mu(\nabla \widehat{\varphi}) \circ T$, one computes the five terms of (8) for $(u^\mu, p^\mu, f^\mu, \varphi)$:

$$\begin{aligned} \iint u^\mu \cdot \partial_s \varphi &= \mu \cdot \mu^2 \cdot \mu^{-5} \iint u \cdot \partial_t \widehat{\varphi} = \mu^{-2} \iint u \cdot \partial_t \widehat{\varphi}, \\ \iint u_i^\mu u_j^\mu \partial_j \varphi_i &= \mu^2 \cdot \mu \cdot \mu^{-5} \iint u_i u_j \partial_j \widehat{\varphi}_i = \mu^{-2} \iint u_i u_j \partial_j \widehat{\varphi}_i, \\ \iint \nabla_y u^\mu : \nabla_y \varphi &= \mu^2 \cdot \mu \cdot \mu^{-5} \iint \nabla u : \nabla \widehat{\varphi} = \mu^{-2} \iint \nabla u : \nabla \widehat{\varphi}, \\ \iint p^\mu \operatorname{div}_y \varphi &= \mu^2 \cdot \mu \cdot \mu^{-5} \iint p \operatorname{div} \widehat{\varphi} = \mu^{-2} \iint p \operatorname{div} \widehat{\varphi}, \\ \iint f^\mu \cdot \varphi &= \mu^3 \cdot \mu^{-5} \iint f \cdot \widehat{\varphi} = \mu^{-2} \iint f \cdot \widehat{\varphi}. \end{aligned}$$

Every term carries the same factor μ^{-2} , so (8) for (u^μ, p^μ, f^μ) is μ^{-2} times (8) for (u, p, f) with test field $\widehat{\varphi}$, hence holds.

(S4). Let $\phi \in C_c^\infty(\mathcal{O}^{\mu, z_0})$, $\phi \geq 0$, and $\widehat{\phi} = \phi \circ T^{-1} \geq 0$. Then $\partial_s \phi + \Delta_y \phi = \mu^2((\partial_t \widehat{\phi} + \Delta \widehat{\phi}) \circ T)$, so

$$\begin{aligned} \iint |\nabla_y u^\mu|^2 \phi &= \mu^{-1} \iint |\nabla u|^2 \widehat{\phi}, \\ \iint |u^\mu|^2 (\partial_s \phi + \Delta_y \phi) &= \mu^2 \mu^2 \mu^{-5} \iint |u|^2 (\partial_t \widehat{\phi} + \Delta \widehat{\phi}), \\ \iint |u^\mu|^2 u^\mu \cdot \nabla_y \phi &= \mu^3 \mu \mu^{-5} \iint |u|^2 u \cdot \nabla \widehat{\phi}, \\ \iint p^\mu u^\mu \cdot \nabla_y \phi &= \mu^2 \mu \mu \mu^{-5} \iint p u \cdot \nabla \widehat{\phi}, \\ \iint (f^\mu \cdot u^\mu) \phi &= \mu^3 \mu \mu^{-5} \iint (f \cdot u) \widehat{\phi}. \end{aligned}$$

Each of the five quantities equals μ^{-1} times the corresponding quantity for $(u, p, f, \widehat{\phi})$. Since $\mu^{-1} > 0$, inequality (10) for (u, p, f) with test function $\widehat{\phi}$ is equivalent to (10) for (u^μ, p^μ, f^μ) with test function ϕ . $\square$

Lemma 3.3 (Transformation of the scale-invariant quantities). *Let (u, p, f) , z_0 , μ be as in Lemma 3.2, let $z = (x, t)$, and let $\zeta = (\mu^{-1}(x - x_0), \mu^{-2}(t - t_0))$ be the corresponding point for the rescaled solution. Then for every $r > 0$ with $\overline{Q_{\mu r}(z)} \subset \mathcal{O}$,*

$$\begin{aligned} A(\zeta, r; u^{\mu, z_0}) &= A(z, \mu r; u), & B(\zeta, r; u^{\mu, z_0}) &= B(z, \mu r; u), & C(\zeta, r; u^{\mu, z_0}) &= C(z, \mu r; u), \\ D(\zeta, r; p^{\mu, z_0}) &= D(z, \mu r; p), & E_q(\zeta, r; f^{\mu, z_0}) &= E_q(z, \mu r; f), & \Psi(\zeta, r; u^{\mu, z_0}) &= \Psi(z, \mu r; u), \end{aligned} \tag{36}$$

and likewise $\widetilde{C}(\zeta, r) = \widetilde{C}(z, \mu r)$, $\widetilde{D}(\zeta, r) = \widetilde{D}(z, \mu r)$ and $\varphi(\zeta, r) = \varphi(z, \mu r)$.

Proof. All six identities are the same computation. The map $T(y, s) = (x_0 + \mu y, t_0 + \mu^2 s)$ carries $Q_r(\zeta)$ onto $Q_{\mu r}(z)$ with $dz = \mu^5 dy ds$ and carries $B_r(\cdot)$ onto $B_{\mu r}(\cdot)$ with $dx = \mu^3 dy$.

Hence, writing $v = u^{\mu, z_0}$,

$$\begin{aligned}
A(\zeta, r; v) &= \frac{1}{r} \operatorname{ess\,sup}_s \int_{B_r(\cdot)} |v|^2 dy \\
&= \frac{1}{r} \mu^2 \mu^{-3} \operatorname{ess\,sup} \int_{B_{\mu r}(\cdot)} |u|^2 dx \\
&= \frac{1}{\mu r} \operatorname{ess\,sup} \int_{B_{\mu r}(\cdot)} |u|^2 dx = A(z, \mu r; u), \\
B(\zeta, r; v) &= \frac{1}{r} \mu^2 \mu^2 \mu^{-5} \iint_{Q_{\mu r}(z)} |\nabla u|^2 = \frac{1}{\mu r} \iint_{Q_{\mu r}(z)} |\nabla u|^2 = B(z, \mu r; u), \\
C(\zeta, r; v) &= \frac{1}{r^2} \mu^3 \mu^{-5} \iint_{Q_{\mu r}(z)} |u|^3 = \frac{1}{(\mu r)^2} \iint_{Q_{\mu r}(z)} |u|^3 = C(z, \mu r; u), \\
D(\zeta, r; p^{\mu, z_0}) &= \frac{1}{r^2} \mu^3 \mu^{-5} \iint_{Q_{\mu r}(z)} |p|^{3/2} = D(z, \mu r; p), \\
E_q(\zeta, r; f^{\mu, z_0}) &= r^{3-\frac{5}{q}} (\mu^{3q} \mu^{-5})^{1/q} \left(\iint_{Q_{\mu r}(z)} |f|^q \right)^{1/q} = (\mu r)^{3-\frac{5}{q}} \|f\|_{L^q(Q_{\mu r}(z))} = E_q(z, \mu r; f).
\end{aligned}$$

For Ψ , the space-time mean transforms as $(v)_{Q_r(\zeta)} = \mu (u)_{Q_{\mu r}(z)}$ (the Jacobian cancels against the normalization), so $\Psi(\zeta, r; v) = r\mu \left| (u)_{Q_{\mu r}(z)} \right| = \Psi(z, \mu r; u)$. The same transformation of the mean, together with the computation for C , gives $\tilde{C}(\zeta, r) = \tilde{C}(z, \mu r)$; and since $[p^{\mu, z_0}]_{B_r(\cdot)}(s) = \mu^2 [p]_{B_{\mu r}(\cdot)}(t_0 + \mu^2 s)$, the computation for D gives $\tilde{D}(\zeta, r) = \tilde{D}(z, \mu r)$. Adding the appropriate powers yields the statement for φ . $\square$

3.2. Interpolation on balls and cylinders. We use the Sobolev and Poincaré inequalities on a ball in the following scale-explicit form; see External Input A.4 and External Input A.5.

Lemma 3.4 (Sobolev and Poincaré on a ball). *There is an absolute constant C_5 such that for every $r > 0$, every $x \in \mathbb{R}^3$ and every $v \in H^1(B_r(x))$,*

$$\|v - [v]_{B_r(x)}\|_{L^6(B_r(x))} \leq C_5 \|\nabla v\|_{L^2(B_r(x))}, \quad (37)$$

$$\|v\|_{L^6(B_r(x))} \leq C_5 \left(\|\nabla v\|_{L^2(B_r(x))} + r^{-1} \|v\|_{L^2(B_r(x))} \right), \quad (38)$$

and, for every $g \in W^{1,1}(B_r(x))$,

$$\|g - [g]_{B_r(x)}\|_{L^1(B_r(x))} \leq C_5 r \|\nabla g\|_{L^1(B_r(x))}. \quad (39)$$

Proof. Inequality (37) for $r = 1$, $x = 0$ is External Input A.5(ii); the general case follows from the change of variables $v(y) = \psi((y - x)/r)$, under which $\|v - [v]_{B_r(x)}\|_{L^6(B_r(x))} = r^{1/2} \|\psi - [\psi]_{B_1}\|_{L^6(B_1)}$ and $\|\nabla v\|_{L^2(B_r(x))} = r^{1/2} \|\nabla \psi\|_{L^2(B_1)}$, so that both sides carry the same power $r^{1/2}$ and the constant is unchanged. Inequality (39) is External Input A.5(i) rescaled in the same way: both sides carry the factor r^3 after the substitution, once the factor r on the right is accounted for. For (38), write $v = (v - [v]_{B_r(x)}) + [v]_{B_r(x)}$ and use (37) together with

$$\begin{aligned}
\|[v]_{B_r(x)}\|_{L^6(B_r(x))} &= |B_r(x)|^{1/6} |[v]_{B_r(x)}| \\
&\leq |B_r(x)|^{1/6} |B_r(x)|^{-1/2} \|v\|_{L^2(B_r(x))} = \left(\frac{4\pi}{3}\right)^{-1/3} r^{-1} \|v\|_{L^2(B_r(x))},
\end{aligned}$$

where the middle step is the Cauchy–Schwarz inequality. Enlarging the constant to cover all three inequalities gives C_5 . $\square$

Lemma 3.5 (Interpolation on a ball). *Let $2 \leq \ell \leq 6$ and set*

$$a = a(\ell) = \frac{3}{4}(\ell - 2) \in [0, 3], \quad \text{so that} \quad \frac{\ell}{2} - a = \frac{6-\ell}{4} \in [0, 1]. \quad (40)$$

There is an absolute constant C_6 such that for every $r > 0$, every $x \in \mathbb{R}^3$ and every $v \in H^1(B_r(x))$,

$$\int_{B_r(x)} |v|^\ell dy \leq C_6 \left(\int_{B_r(x)} |\nabla v|^2 \right)^a \left(\int_{B_r(x)} |v|^2 \right)^{\frac{\ell}{2}-a} + \frac{C_6}{r^{2a}} \left(\int_{B_r(x)} |v|^2 \right)^{\frac{\ell}{2}}. \quad (41)$$

Proof. Write $B = B_r(x)$. Let $\theta \in [0, 1]$ be defined by $\frac{1}{\ell} = \frac{1-\theta}{2} + \frac{\theta}{6}$, that is $\theta = 3(\frac{1}{2} - \frac{1}{\ell}) = \frac{3(\ell-2)}{2\ell}$; this is in $[0, 1]$ precisely because $2 \leq \ell \leq 6$. The Hölder interpolation inequality between L^2 and L^6 gives $\|v\|_{L^\ell(B)} \leq \|v\|_{L^2(B)}^{1-\theta} \|v\|_{L^6(B)}^\theta$, hence

$$\int_B |v|^\ell = \|v\|_{L^\ell(B)}^\ell \leq \|v\|_{L^2(B)}^{\ell(1-\theta)} \|v\|_{L^6(B)}^{\ell\theta}. \quad (42)$$

The two exponents are

$$\ell\theta = \frac{3(\ell-2)}{2} = 2a, \quad \ell(1-\theta) = \ell - \frac{3(\ell-2)}{2} = \frac{6-\ell}{2} = 2\left(\frac{\ell}{2} - a\right).$$

Inserting (38) into (42) and using $(s_1 + s_2)^{2a} \leq 2^{2a}(s_1^{2a} + s_2^{2a}) \leq 64(s_1^{2a} + s_2^{2a})$ for $s_1, s_2 \geq 0$ and $0 \leq a \leq 3$,

$$\begin{aligned} \int_B |v|^\ell &\leq 64 C_5^{2a} \|v\|_{L^2(B)}^{2(\frac{\ell}{2}-a)} \left(\|\nabla v\|_{L^2(B)}^{2a} + r^{-2a} \|v\|_{L^2(B)}^{2a} \right) \\ &= 64 C_5^{2a} \left[\left(\int_B |\nabla v|^2 \right)^a \left(\int_B |v|^2 \right)^{\frac{\ell}{2}-a} + r^{-2a} \left(\int_B |v|^2 \right)^{\frac{\ell}{2}} \right]. \end{aligned}$$

Since $0 \leq a \leq 3$, the factor C_5^{2a} is bounded by $\max\{1, C_5^6\}$; take $C_6 = 64 \max\{1, C_5^6\}$, which is absolute. $\square$

Remark 3.6 (Two instances, and a defect in the source). The two cases used below are

$$\ell = 3, \quad a = \frac{3}{4} : \quad \int_{B_r(x)} |v|^3 \leq C_6 \left(\int |\nabla v|^2 \right)^{3/4} \left(\int |v|^2 \right)^{3/4} + \frac{C_6}{r^{3/2}} \left(\int |v|^2 \right)^{3/2}, \quad (43)$$

$$\ell = \frac{10}{3}, \quad a = 1 : \quad \int_{B_r(x)} |v|^{10/3} \leq C_6 \left(\int |\nabla v|^2 \right) \left(\int |v|^2 \right)^{2/3} + \frac{C_6}{r^2} \left(\int |v|^2 \right)^{5/3}. \quad (44)$$

Integrating (44) in time over $(t-r^2, t)$ and using (S1) gives the embedding $L^\infty L^2 \cap L^2 H^1 \subset L^{10/3}$ of External Input A.4(iv).

The inequality [15, (2.1)] is printed with the exponents a and $\ell/2 - a$ interchanged, i.e. with $(\int |\nabla v|^2)^{\ell/2-a} (\int |v|^2)^a$ in place of the first term of (41). That form agrees with (41) when $\ell = 3$, the only case Lin uses, but it is false for every $\ell \in (3, 6]$. Indeed, take $r = 1$, fix $\psi \in C_c^\infty(B_1)$ with $\psi \not\equiv 0$, and set $v_\varepsilon(y) = \varepsilon^{-3/2} \psi(y/\varepsilon)$ for $0 < \varepsilon < 1$. Then

$$\int_{B_1} |v_\varepsilon|^2 = \int_{B_1} |\psi|^2, \quad \int_{B_1} |\nabla v_\varepsilon|^2 = \varepsilon^{-2} \int_{B_1} |\nabla \psi|^2, \quad \int_{B_1} |v_\varepsilon|^\ell = \varepsilon^{3-\frac{3\ell}{2}} \int_{B_1} |\psi|^\ell,$$

so the printed right side is $O(\varepsilon^{-2(\ell/2-a)}) + O(1) = O(\varepsilon^{(\ell-6)/2}) + O(1)$ while the left side is of exact order $\varepsilon^{3-3\ell/2}$; since $3 - \frac{3\ell}{2} < \min\{\frac{\ell-6}{2}, 0\}$ for $\ell > 3$, the printed inequality fails as $\varepsilon \rightarrow 0$. With the exponents as in (41) the first term is of order $\varepsilon^{-2a} = \varepsilon^{3-3\ell/2}$, matching the left side exactly.

Lemma 3.7 (Interpolation on cylinders). *There is an absolute constant C_7 with the following property. Let (u, p) satisfy (S1), let $z = (x, t)$ and let $0 < r \leq \rho$ with $\overline{Q_\rho(z)} \subset \mathcal{O}$. Then*

$$C(z, r) \leq C_7 \left[\left(\frac{r}{\rho} \right)^3 A(z, \rho)^{3/2} + \left(\frac{\rho}{r} \right)^3 A(z, \rho)^{3/4} B(z, \rho)^{3/4} \right]. \quad (45)$$

Equivalently, in Kukavica's variables, $\gamma(z, r)^3 \leq C_7 [(r/\rho)^3 \alpha(z, \rho)^3 + (\rho/r)^3 \alpha(z, \rho)^{3/2} \beta(z, \rho)^{3/2}]$.

Proof. Write $A = A(z, \rho)$, $B = B(z, \rho)$, $B_r = B_r(x)$, $B_\rho = B_\rho(x)$, and for $s \in (t - \rho^2, t)$ put

$$Y(s) = \int_{B_r} |u(y, s)|^2 dy, \quad G(s) = \int_{B_\rho} |\nabla u(y, s)|^2 dy.$$

All statements below hold for almost every such s ; we suppress “a.e.”.

Step 1: a bound for $Y(s)$. Split

$$\int_{B_r} |u|^2 = \int_{B_r} \left(|u|^2 - [u^2]_{B_\rho}(s) \right) + |B_r| [u^2]_{B_\rho}(s) \leq \int_{B_\rho} \left| |u|^2 - [u^2]_{B_\rho}(s) \right| + \left(\frac{r}{\rho} \right)^3 \int_{B_\rho} |u|^2,$$

where we used $B_r \subset B_\rho$ in the first integral and $|B_r| / |B_\rho| = (r/\rho)^3$ in the second. Apply (39) to $g = |u(\cdot, s)|^2$, which lies in $W^{1,1}(B_\rho)$ with $|\nabla g| \leq 2|u||\nabla u|$ because $u(\cdot, s) \in H^1(B_\rho)$ and $u(\cdot, s) \in L^4(B_\rho)$ by (41) with $q = 4$:

$$\int_{B_\rho} \left| |u|^2 - [u^2]_{B_\rho} \right| \leq 2C_5 \rho \int_{B_\rho} |u| |\nabla u| \leq 2C_5 \rho \|u(\cdot, s)\|_{L^2(B_\rho)} G(s)^{1/2}.$$

Since $\|u(\cdot, s)\|_{L^2(B_\rho)}^2 \leq \rho A$ by definition of A ,

$$Y(s) \leq 2C_5 \rho^{3/2} A^{1/2} G(s)^{1/2} + \frac{r^3}{\rho^2} A. \quad (46)$$

Step 2: the pointwise-in-time L^3 bound. Apply (43) on B_r :

$$\int_{B_r} |u|^3 \leq C_6 \left(\int_{B_r} |\nabla u|^2 \right)^{3/4} Y(s)^{3/4} + \frac{C_6}{r^{3/2}} Y(s)^{3/2}.$$

In the first term use

$$\int_{B_r} |\nabla u|^2 \leq G(s), \quad Y(s) \leq \int_{B_\rho} |u|^2 \leq \rho A,$$

obtaining $C_6 \rho^{3/4} A^{3/4} G(s)^{3/4}$. In the second term insert (46) and use $(s_1 + s_2)^{3/2} \leq 2^{3/2} (s_1^{3/2} + s_2^{3/2})$:

$$\frac{C_6}{r^{3/2}} Y(s)^{3/2} \leq \frac{2^{3/2} C_6}{r^{3/2}} \left[(2C_5)^{3/2} \rho^{9/4} A^{3/4} G(s)^{3/4} + \frac{r^{9/2}}{\rho^3} A^{3/2} \right].$$

Hence, with $C_8 = C_6(1 + 2^{3/2}(2C_5)^{3/2} + 2^{3/2})$,

$$\int_{B_r} |u(\cdot, s)|^3 \leq C_8 \left(\rho^{3/4} + \frac{\rho^{9/4}}{r^{3/2}} \right) A^{3/4} G(s)^{3/4} + C_8 \frac{r^3}{\rho^3} A^{3/2}. \quad (47)$$

Step 3: integration in time. Integrate (47) over $s \in (t - r^2, t)$. For the first term use Hölder with exponents $\frac{4}{3}$ and 4 and then $\int_{t-r^2}^t G(s) ds = \rho B$:

$$\int_{t-r^2}^t G(s)^{3/4} ds \leq (r^2)^{1/4} \left(\int_{t-r^2}^t G(s) ds \right)^{3/4} \leq r^{1/2} (\rho B)^{3/4}.$$

The second term contributes $C_8 r^5 \rho^{-3} A^{3/2}$. Therefore

$$\begin{aligned} \iint_{Q_r(z)} |u|^3 &\leq C_8 \left(\rho^{3/4} + \frac{\rho^{9/4}}{r^{3/2}} \right) A^{3/4} r^{1/2} \rho^{3/4} B^{3/4} \\ &+ C_8 \frac{r^5}{\rho^3} A^{3/2} \\ &= C_8 \left(\rho^{3/2} r^{1/2} + \frac{\rho^3}{r} \right) A^{3/4} B^{3/4} + C_8 \frac{r^5}{\rho^3} A^{3/2}. \end{aligned}$$

Dividing by r^2 ,

$$C(z, r) \leq C_8 \left[\left(\frac{\rho}{r} \right)^{3/2} + \left(\frac{\rho}{r} \right)^3 \right] A^{3/4} B^{3/4} + C_8 \left(\frac{r}{\rho} \right)^3 A^{3/2},$$

and since $r \leq \rho$ gives $(\rho/r)^{3/2} \leq (\rho/r)^3$, the claim follows with $C_7 = 2C_8$. $\square$

Corollary 3.8 (Gagliardo–Nirenberg in Kukavica’s variables). *With C_7 as in Lemma 3.7, for every z and every $\rho > 0$ with $\overline{Q_\rho(z)} \subset \mathcal{O}$,*

$$\gamma(z, \rho) \leq C_9 \alpha(z, \rho)^{1/2} \beta(z, \rho)^{1/2} + C_9 \alpha(z, \rho), \quad C_9 = (2C_7)^{1/3}, \quad (48)$$

The same cylinder interpolation estimate also gives

$$C(z, \rho) \leq 2C_7 [A(z, \rho)^{3/2} + A(z, \rho)^{3/4} B(z, \rho)^{3/4}].$$

Proof. Take $r = \rho$ in (45): $C(z, \rho) \leq C_7 [A^{3/2} + A^{3/4} B^{3/4}]$. Using $\gamma = C^{1/3}$, $\alpha = A^{1/2}$, $\beta = B^{1/2}$ from Lemma 1.15 and $(s_1 + s_2)^{1/3} \leq s_1^{1/3} + s_2^{1/3}$,

$$\gamma(z, \rho) = C(z, \rho)^{1/3} \leq C_7^{1/3} (A^{1/2} + A^{1/4} B^{1/4}) = C_7^{1/3} (\alpha + \alpha^{1/2} \beta^{1/2}).$$

Since $C_7^{1/3} \leq (2C_7)^{1/3} = C_9$, (48) follows. $\square$

Lemma 3.9 (Monotonicity in the radius). *Let $z \in \mathcal{O}$ and $0 < \rho_1 < \rho_2$ with $\overline{Q_{\rho_2}(z)} \subset \mathcal{O}$. Then*

$$\begin{aligned} \alpha(z, \rho_1) &\leq \left(\frac{\rho_2}{\rho_1} \right)^{1/2} \alpha(z, \rho_2), & \beta(z, \rho_1) &\leq \left(\frac{\rho_2}{\rho_1} \right)^{1/2} \beta(z, \rho_2), \\ \gamma(z, \rho_1) &\leq \left(\frac{\rho_2}{\rho_1} \right)^{2/3} \gamma(z, \rho_2), & \delta(z, \rho_1)^2 &\leq \left(\frac{\rho_2}{\rho_1} \right)^{4/3} \delta(z, \rho_2)^2, \end{aligned} \quad (49)$$

and $\lambda(z, \rho_1) \leq (\rho_1/\rho_2)^\sigma \lambda(z, \rho_2)$.

Proof. All five follow from $Q_{\rho_1}(z) \subset Q_{\rho_2}(z)$, $B_{\rho_1}(x) \subset B_{\rho_2}(x)$ and $(t - \rho_1^2, t) \subset (t - \rho_2^2, t)$, which make every integral in (24) monotone in the radius, combined with the explicit power of r in the prefactor. For instance

$$\begin{aligned} \alpha(z, \rho_1)^2 &= \rho_1^{-1} \operatorname{ess\,sup}_{(t-\rho_1^2, t)} \int_{B_{\rho_1}(x)} |u|^2 \\ &\leq \rho_1^{-1} \operatorname{ess\,sup}_{(t-\rho_2^2, t)} \int_{B_{\rho_2}(x)} |u|^2; \\ &= (\rho_2/\rho_1) \alpha(z, \rho_2)^2 \end{aligned}$$

enlarging the domain of integration in the same way in the three remaining integrals gives $\beta(z, \rho_1)^2 \leq (\rho_2/\rho_1) \beta(z, \rho_2)^2$, $\gamma(z, \rho_1)^3 \leq (\rho_2/\rho_1)^2 \gamma(z, \rho_2)^3$ and $\delta(z, \rho_1)^3 \leq (\rho_2/\rho_1)^2 \delta(z, \rho_2)^3$;

raising the last two to the powers $1/3$ and $2/3$ gives the stated forms. For λ , $\|f\|_{L^q(Q_{\rho_1}(z))} \leq \|f\|_{L^q(Q_{\rho_2}(z))}$ gives

$$\lambda(z, \rho_1) \leq \rho_1^\sigma \|f\|_{L^q(Q_{\rho_2}(z))} = (\rho_1/\rho_2)^\sigma \lambda(z, \rho_2).$$

□

Remark 3.10. Kukavica [9, (2.23) and the line above it] states the fourth inequality of (49) as $\delta(\rho_1) \leq (\rho_2/\rho_1)^{4/3} \delta(\rho_2)$. The exponent $4/3$ belongs to δ^2 , not to δ : the correct statements are $\delta(\rho_1) \leq (\rho_2/\rho_1)^{2/3} \delta(\rho_2)$ and $\delta(\rho_1)^2 \leq (\rho_2/\rho_1)^{4/3} \delta(\rho_2)^2$. Since $\rho_2/\rho_1 \geq 1$ the printed inequality is weaker than the truth, so the argument of [9] is unaffected.

4. THE PRESSURE

Remark 4.1 (Operator and source conventions). The pressure identities below use weak derivatives and completed L^p Riesz operators. One may first prove the Newtonian derivative adjoint identity on compact smooth tests by heat-kernel subordination and then pass to nonsmooth compact sources by density. Exterior kernel representations are justified on sets separated from the source support. Local potential integrability and growth at infinity identify the harmonic residual by Liouville's theorem; they do not turn a singular Hessian kernel into an everywhere absolutely integrable convolution.

The singly centred tensor $-u_i(u_j - c_j)$ used in the pressure-decay estimate is distinct from the doubly centred tensor $-(u_i - c_i)(u_j - c_j)$ used in Proposition 4.11. For the latter, subtract the two distributional pairings and use incompressibility and Hessian symmetry to cancel the constant terms. The nine scalar Riesz estimates give the factor $9C_{11}$. A source obtained by differentiating the velocity tensor contains no force term: the force potentials p_7, p_8 are added separately.

Throughout this section (u, p) is a suitable weak solution on $\mathcal{O}$ with force $f \in L^q_{loc}$, $q > 5/2$, and $z_0 = (x_0, t_0) \in \mathcal{O}$ and $\rho > 0$ are such that $\overline{Q_\rho(z_0)} \subset \mathcal{O}$. We write $B_\rho = B_\rho(x_0)$, $J_\rho = (t_0 - \rho^2, t_0)$, and

$$\kappa = \frac{r}{\rho} \in (0, \frac{1}{2}] \quad (50)$$

for the two radii $0 < r \leq \rho/2$ appearing in the estimates. The symbol κ keeps this meaning up to and including Lemma 5.17, where it is still a free parameter; from Convention 5.18 onwards it denotes the *fixed* absolute constant defined there, and the functional θ of (112) is understood with that fixed value.

4.1. The spatial Poisson equation for the pressure.

Lemma 4.2 (Δp at almost every time). *For almost every $t \in J_\rho$ and every $\psi \in C_c^\infty(B_\rho)$,*

$$\int_{B_\rho} p(y, t) \Delta \psi(y) dy = \int_{B_\rho} U_{ij}(y, t) \partial_i \partial_j \psi(y) dy - \int_{B_\rho} f(y, t) \cdot \nabla \psi(y) dy, \quad (51)$$

where

$$U_{ij}(y, t) = -u_i(y, t)(u_j(y, t) - [u_j]_{B_\rho}(t)). \quad (52)$$

Equivalently, $\Delta p = \partial_i \partial_j U_{ij} + \operatorname{div} f$ in $\mathcal{D}'(B_\rho)$ for a.e. $t \in J_\rho$.

Proof. Step 1: a null set that works for all ψ . Let $\mathcal{S} \subset C_c^\infty(B_\rho)$ be a countable set that is dense in $C_c^\infty(B_\rho)$ for the C^2 norm together with uniform control of supports, for instance the set of finite rational-coefficient combinations of a fixed countable family of bump functions. It suffices to prove (51) for each fixed $\psi \in \mathcal{S}$ outside a null set N_ψ depending on ψ , since then

(51) holds for all $\psi \in \mathcal{S}$ outside $N = \bigcup_{\psi \in \mathcal{S}} N_\psi$, and both sides of (51) are continuous in ψ for the C^2 norm on a fixed compact support (each integrand is in $L^1(B_\rho)$ for a.e. t).

Step 2: the mean of u against a gradient. Applying (S2) to $\psi(x)\theta(t)$ with $\theta \in C_c^\infty(J_\rho)$ gives $\int_{J_\rho} \theta(t) \left(\int_{B_\rho} u(\cdot, t) \cdot \nabla \psi \right) dt = 0$ for all such θ , hence

$$\int_{B_\rho} u(y, t) \cdot \nabla \psi(y) dy = 0 \quad \text{for a.e. } t \in J_\rho. \quad (53)$$

Step 3: testing the momentum equation with a gradient. Fix $\psi \in \mathcal{S}$ and $\theta \in C_c^\infty(J_\rho)$, and apply (S3) with $\varphi(x, t) = \theta(t)\nabla \psi(x)$, which lies in $C_c^\infty(\mathcal{O}; \mathbb{R}^3)$. The five terms of (8) are:

$$\iint u \cdot \partial_t \varphi = \int \theta'(t) \left(\int_{B_\rho} u \cdot \nabla \psi \right) dt = 0$$

by (53);

$$\iint \nabla u : \nabla \varphi = \int \theta(t) \left(\int_{B_\rho} \partial_j u_i \partial_j \partial_i \psi \right) dt = \int \theta(t) \left(- \int_{B_\rho} u_i \partial_i (\Delta \psi) \right) dt = 0,$$

where the middle equality is an integration by parts in x_j , licit because $u(\cdot, t) \in H^1(B_\rho)$ and $\psi \in C_c^\infty(B_\rho)$, and the last equality is (53) applied to $\Delta \psi \in C_c^\infty(B_\rho)$; and

$$\begin{aligned} \iint u_i u_j \partial_j \varphi_i &= \int \theta \int u_i u_j \partial_i \partial_j \psi, \\ \iint p \operatorname{div} \varphi &= \int \theta \int p \Delta \psi, \quad \iint f \cdot \varphi = \int \theta \int f \cdot \nabla \psi. \end{aligned}$$

Thus (8) reads $\int \theta(t) \left[- \int u_i u_j \partial_i \partial_j \psi - \int p \Delta \psi - \int f \cdot \nabla \psi \right] dt = 0$ for every $\theta \in C_c^\infty(J_\rho)$, whence for a.e. t

$$\int_{B_\rho} p \Delta \psi = - \int_{B_\rho} u_i u_j \partial_i \partial_j \psi - \int_{B_\rho} f \cdot \nabla \psi. \quad (54)$$

Step 4: replacing u_j by $u_j - [u_j]_{B_\rho}(t)$. Let $c \in \mathbb{R}^3$ be a constant vector. Then, for a.e. t ,

$$\int_{B_\rho} u_i c_j \partial_i \partial_j \psi = c_j \int_{B_\rho} u_i \partial_i (\partial_j \psi) = 0$$

by (53) applied to $\partial_j \psi \in C_c^\infty(B_\rho)$. Taking $c = [u]_{B_\rho}(t)$, which is a constant vector for each fixed t , turns $-\int u_i u_j \partial_i \partial_j \psi$ into $-\int u_i (u_j - c_j) \partial_i \partial_j \psi = \int U_{ij} \partial_i \partial_j \psi$. Substituting into (54) gives (51). $\square$

4.2. The cut-off and the decomposition.

Lemma 4.3 (The spatial cut-off). *There is a family of constants $C_{10}(k)$, $k \in \mathbb{N}_0$, absolute in the sense that they depend on k only, with the following property. For every x_0 and every $\rho > 0$ there is $\eta = \eta_{x_0, \rho} \in C_c^\infty(\mathbb{R}^3)$ with*

$$0 \leq \eta \leq 1, \quad \eta \equiv 1 \text{ on } B_{13\rho/20}(x_0), \quad \operatorname{supp} \eta \subset B_{3\rho/4}(x_0), \quad |\partial^a \eta| \leq C_{10}(|a|)\rho^{-|a|} \text{ on } \mathbb{R}^3. \quad (55)$$

Moreover $\nabla \eta$ and all higher derivatives of η vanish outside the annulus

$$\mathcal{A}_\rho = B_{3\rho/4}(x_0) \setminus \overline{B_{13\rho/20}(x_0)}, \quad (56)$$

and for every $x \in B_r(x_0)$ with $r \leq \rho/2$ and every $y \in \mathcal{A}_\rho$ one has $|x - y| \geq \frac{3\rho}{20}$.

Proof. Let $\zeta \in C_c^\infty(B_1)$ be a fixed nonnegative radial mollifier with $\int_{\mathbb{R}^3} \zeta = 1$, and for $\varepsilon > 0$ put $\zeta_\varepsilon(y) = \varepsilon^{-3} \zeta(y/\varepsilon)$. Since $\text{supp } \zeta$ is compact in B_1 , choose $a_* < 1$ with $\text{supp } \zeta \subset \overline{B_{a_*}}$. The convolution below then has support in $\overline{B_{(7/10+a_*/20)\rho}(x_0)}}$, which is a compact subset of $B_{3\rho/4}(x_0)$. Set $\eta = \mathbf{1}_{B_{7\rho/10}(x_0)} * \zeta_{\rho/20}$. Then $0 \leq \eta \leq 1$; if $|y - x_0| \leq \frac{7\rho}{10} - \frac{\rho}{20} = \frac{13\rho}{20}$ then the whole ball $B_{\rho/20}(y)$ lies in $B_{7\rho/10}(x_0)$ and $\eta(y) = 1$; if $|y - x_0| \geq \frac{7\rho}{10} + \frac{\rho}{20} = \frac{3\rho}{4}$ then $B_{\rho/20}(y) \cap B_{7\rho/10}(x_0) = \emptyset$ and $\eta(y) = 0$. Also $\partial^a \eta = \mathbf{1}_{B_{7\rho/10}(x_0)} * \partial^a \zeta_{\rho/20}$, so $|\partial^a \eta| \leq \|\partial^a \zeta_{\rho/20}\|_{L^1} = (20/\rho)^{|a|} \|\partial^a \zeta\|_{L^1(\mathbb{R}^3)}$, giving (55) with $C_{10}(k) = 20^k \max_{|a|=k} \|\partial^a \zeta\|_{L^1(\mathbb{R}^3)}$. Since η is constant on $B_{13\rho/20}(x_0)$ and on the complement of $\overline{B_{3\rho/4}(x_0)}$, all its derivatives of order ≥ 1 vanish off $\mathcal{A}_\rho$. Finally if $|x - x_0| < r \leq \rho/2$ and $|y - x_0| \geq \frac{13\rho}{20}$ then $|x - y| \geq \frac{13\rho}{20} - \frac{\rho}{2} = \frac{3\rho}{20}$. $\square$

Proposition 4.4 (Local decomposition of the pressure). *Let η be as in Lemma 4.3 and let $N(y) = -\frac{1}{4\pi|y|}$ be the Newtonian kernel of $\mathbb{R}^3$, so that $\Delta N = \delta_0$. Then for almost every $t \in J_\rho$ the identity*

$$\Delta(\eta p) = \partial_i \partial_j (\eta U_{ij}) + (\partial_i \partial_j \eta) U_{ij} - \partial_j (U_{ij} \partial_i \eta) - \partial_i (U_{ij} \partial_j \eta) - p \Delta \eta + 2 \partial_j ((\partial_j \eta) p) + \eta \operatorname{div} f \quad (57)$$

holds in $\mathcal{D}'(\mathbb{R}^3)$, all functions being extended by zero outside B_ρ , and consequently

$$\eta p = p_1 + p_2 + p_3 + p_4 + p_5 + p_6 + p_7 + p_8 \quad \text{a.e. in } \mathbb{R}^3, \quad (58)$$

where, with R_i the i -th Riesz transform of $\mathbb{R}^3$ (External Input A.1),

$$\begin{aligned} p_1 &= -R_i R_j (\eta U_{ij}), & p_2 &= N * ((\partial_i \partial_j \eta) U_{ij}), \\ p_3 &= -\partial_j N * (U_{ij} \partial_i \eta), & p_4 &= -\partial_i N * (U_{ij} \partial_j \eta), \\ p_5 &= -N * (p \Delta \eta), & p_6 &= 2 \partial_j N * ((\partial_j \eta) p), \\ p_7 &= \partial_j N * (\eta f_j), & p_8 &= -N * ((\partial_j \eta) f_j). \end{aligned} \quad (59)$$

If, in addition, $\operatorname{div} f = 0$ locally in $\mathcal{D}'(\mathcal{O})$, then $p_7 + p_8 = 0$ and the last two terms may be omitted. This scalar-test condition is not part of Definition 1.5; it imposes no whole-space divergence condition on a zero extension of f .

Proof. Step 1: the identity (57). Fix t for which Lemma 4.2 holds. For $\psi \in C_c^\infty(\mathbb{R}^3)$ we have $\eta \psi \in C_c^\infty(B_\rho)$ by (55), so all pairings below are legitimate. Three elementary distributional identities are used. First, for a scalar distribution T and smooth η ,

$$\Delta(\eta T) = \eta \Delta T + 2 \partial_j \eta \partial_j T + T \Delta \eta. \quad (60)$$

Second, for a matrix field U ,

$$\eta \partial_i \partial_j U_{ij} = \partial_i \partial_j (\eta U_{ij}) + (\partial_i \partial_j \eta) U_{ij} - \partial_j (U_{ij} \partial_i \eta) - \partial_i (U_{ij} \partial_j \eta), \quad (61)$$

which is verified by expanding the right side: $\partial_i \partial_j (\eta U_{ij}) = \eta \partial_i \partial_j U_{ij} + \partial_i \eta \partial_j U_{ij} + \partial_j \eta \partial_i U_{ij} + (\partial_i \partial_j \eta) U_{ij}$, while $\partial_j (U_{ij} \partial_i \eta) = \partial_j U_{ij} \partial_i \eta + U_{ij} \partial_i \partial_j \eta$ and $\partial_i (U_{ij} \partial_j \eta) = \partial_i U_{ij} \partial_j \eta + U_{ij} \partial_i \partial_j \eta$; substituting and cancelling gives (61). Third, $2 \partial_j \eta \partial_j p = 2 \partial_j ((\partial_j \eta) p) - 2 p \Delta \eta$. Now combine: by (60) and Lemma 4.2,

$$\Delta(\eta p) = \eta (\partial_i \partial_j U_{ij} + \operatorname{div} f) + 2 \partial_j ((\partial_j \eta) p) - 2 p \Delta \eta + p \Delta \eta,$$

and inserting (61) yields exactly (57).

Step 2: the representation (58). The function $\eta p(\cdot, t)$ belongs to $L^{3/2}(\mathbb{R}^3)$ and has compact support in $\overline{B_{3\rho/4}(x_0)}$. We apply External Input A.3 with

$$G_{ij} = \eta U_{ij}, \quad H_j = -U_{ij} \partial_i \eta - U_{ji} \partial_i \eta + 2(\partial_j \eta) p + \eta f_j, \quad K = (\partial_i \partial_j \eta) U_{ij} - p \Delta \eta - (\partial_j \eta) f_j.$$

All three have compact support in $\overline{B_{3\rho/4}(x_0)}$; $G \in L^{3/2}$ because $U(\cdot, t) \in L^{3/2}(B_\rho)$ by Hölder ($|U| \leq |u| \left| u - [u]_{B_\rho} \right|$ with $u(\cdot, t) \in L^2 \cap L^6$), $H \in L^m$ with $m = \frac{5}{4} \in (1, 3)$, and $K \in L^{m'}$ with $m' = \frac{5}{4} \in (1, 3/2)$: indeed each of $U(\cdot, t)$, $p(\cdot, t)$ and $f(\cdot, t)$ lies in $L^{3/2}(B_\rho)$ (for f because $q > 5/2 > 3/2$), all the densities are supported in the bounded set $\overline{B_{3\rho/4}(x_0)}$, and $L^{3/2}(B_\rho) \subset L^{5/4}(B_\rho)$ by Hölder. The exponent $5/4$ lies strictly inside the admissible range of External Input A.3. The identity $\Delta(\eta p) = \partial_i \partial_j G_{ij} + \partial_j H_j + K$ is (57), and External Input A.3 then gives $\eta p = -R_i R_j G_{ij} + \partial_j N * H_j + N * K$, which is (58) with the terms grouped as in (59).

Step 3: the solenoidal case. If $\operatorname{div} f = 0$ in $\mathcal{D}'(\mathcal{O})$ then, arguing as in Step 2 of the proof of Lemma 4.2, $\int_{B_\rho} f(\cdot, t) \cdot \nabla \psi = 0$ for a.e. t and every $\psi \in C_c^\infty(B_\rho)$, i.e. $\operatorname{div} f(\cdot, t) = 0$ in $\mathcal{D}'(B_\rho)$. Hence the last term of (57) vanishes, and since $\eta \operatorname{div} f = \operatorname{div}(\eta f) - (\nabla \eta) \cdot f$ we get $\partial_j N * (\eta f_j) - N * ((\partial_j \eta) f_j) = N * (\eta \operatorname{div} f) = 0$, i.e. $p_7 + p_8 = 0$. $\square$

4.3. Bounds for the eight terms. Throughout this subsection $0 < r \leq \rho/2$, $\kappa = r/\rho$, $B_r = B_r(x_0)$, and for a.e. $t \in J_\rho$ we abbreviate

$$\begin{aligned} \mathbf{u}(t) &= \|u(\cdot, t)\|_{L^2(B_\rho)}, & \mathbf{g}(t) &= \|\nabla u(\cdot, t)\|_{L^2(B_\rho)}, \\ \mathbf{p}(t) &= \|p(\cdot, t)\|_{L^{3/2}(B_\rho)}, & \mathbf{f}(t) &= \|f(\cdot, t)\|_{L^q(B_\rho)}. \end{aligned} \quad (62)$$

By (24),

$$\operatorname{ess\,sup}_{J_\rho} \mathbf{u} \leq \rho^{1/2} \alpha(\rho), \quad \|\mathbf{g}\|_{L^2(J_\rho)} = \rho^{1/2} \beta(\rho), \quad \|\mathbf{p}\|_{L^{3/2}(J_\rho)} = \rho^{4/3} \delta(\rho)^2, \quad \|\mathbf{f}\|_{L^q(J_\rho)} = \rho^{-\sigma} \lambda(\rho), \quad (63)$$

where $\alpha(\rho) = \alpha(z_0, \rho)$ and so on. We use three instances of Hölder's inequality in time, valid for any measurable $g \geq 0$ on J_ρ and any $0 < r \leq \rho/2$:

$$\left(\int_{t_0-r^2}^{t_0} g^{3/2} \right)^{2/3} \leq r^{1/3} \|g\|_{L^2(J_\rho)}, \quad \left(\int_{t_0-r^2}^{t_0} g^{3/2} \right)^{2/3} \leq \|g\|_{L^{3/2}(J_\rho)}, \quad (64)$$

$$\left(\int_{t_0-r^2}^{t_0} g^{3/2} \right)^{2/3} \leq r^{2(\frac{2}{3}-\frac{1}{q})} \|g\|_{L^q(J_\rho)} \quad (q \geq \frac{3}{2}). \quad (65)$$

The first follows from Hölder with exponents $\frac{4}{3}$ and 4 (so that the measure factor is $(r^2)^{2/3-1/2} = r^{1/3}$), the third from Hölder with exponents $\frac{2q}{3}$ and its conjugate; the second is monotonicity of the integral.

Lemma 4.5 (Algebraic bounds on U). *Let U be as in (52), with its Euclidean matrix norm $|U| = (\sum_{i,j} U_{ij}^2)^{1/2}$. For almost every $t \in J_\rho$,*

$$\|U(\cdot, t)\|_{L^{3/2}(B_\rho)} \leq C_5 \mathbf{u}(t) \mathbf{g}(t), \quad \|U(\cdot, t)\|_{L^1(B_\rho)} \leq \left(\frac{4\pi}{3}\right)^{1/3} C_5 \rho \mathbf{u}(t) \mathbf{g}(t). \quad (66)$$

Proof. Pointwise $|U| \leq |u| \left| u - [u]_{B_\rho} \right|$. By Hölder with $\frac{2}{3} = \frac{1}{2} + \frac{1}{6}$ and (37),

$$\|U(\cdot, t)\|_{L^{3/2}(B_\rho)} \leq \|u(\cdot, t)\|_{L^2(B_\rho)} \|u(\cdot, t) - [u]_{B_\rho}(t)\|_{L^6(B_\rho)} \leq C_5 \mathbf{u}(t) \mathbf{g}(t).$$

For the L^1 bound, Hölder with $1 = \frac{2}{3} + \frac{1}{3}$ gives

$$\|U\|_{L^1(B_\rho)} \leq |B_\rho|^{1/3} \|U\|_{L^{3/2}(B_\rho)} = \left(\frac{4\pi}{3}\right)^{1/3} \rho \|U\|_{L^{3/2}(B_\rho)}.$$

$\square$

All space–time norms of the pressure terms below use jointly measurable representatives. For the singular term, the localized tensor is a strongly measurable function of time with values in the separable space $L^{3/2}$; the bounded spatial Riesz operator preserves this measurability and admits a jointly measurable representative. The remaining localized potentials are handled by measurable truncated kernel integrals and their a.e. limits. In particular, their spatial power integrals and the slice norms and majorants used below are measurable functions of time, so that Fubini and the time Hölder inequalities apply.

Lemma 4.6 (Term-by-term bounds). *There is an absolute constant C_{12} , and a constant $C_{13} = C_{13}(q)$, such that for every $0 < r \leq \rho/2$, with $\kappa = r/\rho$,*

$$\frac{1}{r^{4/3}} \|p_1\|_{L^{3/2}(Q_r(z_0))} \leq C_{12} \kappa^{-1} \alpha(\rho) \beta(\rho), \quad (67)$$

$$\frac{1}{r^{4/3}} \sum_{k=2}^4 \|p_k\|_{L^{3/2}(Q_r(z_0))} \leq C_{12} \kappa \alpha(\rho) \beta(\rho), \quad (68)$$

$$\frac{1}{r^{4/3}} \sum_{k=5}^6 \|p_k\|_{L^{3/2}(Q_r(z_0))} \leq C_{12} \kappa^{2/3} \delta(\rho)^2, \quad (69)$$

$$\frac{1}{r^{4/3}} \sum_{k=7}^8 \|p_k\|_{L^{3/2}(Q_r(z_0))} \leq C_{13} \kappa \lambda(\rho). \quad (70)$$

Proof. Throughout, c denotes a quantity bounded by an absolute constant times $1 + C_5 + C_{10}(1) + C_{10}(2) + C_{11}$, where C_{11} is the constant of External Input A.1; at the end we let C_{12} be the largest of the finitely many such quantities.

(a) *The term p_1 .* By External Input A.1 applied for each fixed t to each of the nine functions $\eta U_{ij}(\cdot, t) \in L^{3/2}(\mathbb{R}^3)$,

$$\|p_1(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq C_{11} \sum_{i,j} \|\eta U_{ij}(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq 9C_{11} \|U(\cdot, t)\|_{L^{3/2}(B_\rho)} \leq 9C_{11} C_5 \mathbf{u}(t) \mathbf{g}(t),$$

using $0 \leq \eta \leq 1$, $\text{supp } \eta \subset B_\rho$ and Lemma 4.5. Restricting to B_r , raising to the power $3/2$, integrating over $(t_0 - r^2, t_0)$ and using the first inequality of (64) with $g = \mathbf{u} \mathbf{g}$ together with (63),

$$\begin{aligned} \|p_1\|_{L^{3/2}(Q_r(z_0))} &\leq 9C_{11} C_5 r^{1/3} \|\mathbf{u} \mathbf{g}\|_{L^2(J_\rho)} \\ &\leq 9C_{11} C_5 r^{1/3} (\text{ess sup } \mathbf{u}) \|\mathbf{g}\|_{L^2(J_\rho)} \leq 9C_{11} C_5 r^{1/3} \rho \alpha(\rho) \beta(\rho). \end{aligned}$$

Dividing by $r^{4/3}$ gives (67).

(b) *The terms p_2, p_3, p_4 .* The densities $(\partial_i \partial_j \eta) U_{ij}$, $U_{ij} \partial_i \eta$ and $U_{ij} \partial_j \eta$ vanish outside the annulus $\mathcal{A}_\rho$ of (56), and for $x \in B_r$, $y \in \mathcal{A}_\rho$ we have $|x - y| \geq \frac{3\rho}{20}$ by Lemma 4.3, hence

$$|N(x - y)| = \frac{1}{4\pi |x - y|} \leq \frac{5}{3\pi \rho} \leq \frac{2}{\rho}, \quad |\nabla N(x - y)| = \frac{1}{4\pi |x - y|^2} \leq \frac{100}{9\pi \rho^2} \leq \frac{4}{\rho^2}.$$

By Cauchy–Schwarz over the indices, $\sum_{i,j} |\partial_i \partial_j \eta| |U_{ij}| \leq 3C_{10}(2) \rho^{-2} |U|$ and $\sum_j |\sum_i U_{ij} \partial_i \eta| \leq 3C_{10}(1) \rho^{-1} |U|$. Therefore, for $x \in B_r$,

$$|p_2(x, t)| \leq \frac{2}{\rho} \cdot \frac{3C_{10}(2)}{\rho^2} \|U(\cdot, t)\|_{L^1(B_\rho)}, \quad |p_3(x, t)| + |p_4(x, t)| \leq 2 \cdot \frac{4}{\rho^2} \cdot \frac{3C_{10}(1)}{\rho} \|U(\cdot, t)\|_{L^1(B_\rho)},$$

so by Lemma 4.5

$$\sum_{k=2}^4 |p_k(x, t)| \leq c \rho^{-2} u(t) g(t).$$

Hence

$$\|p_k(\cdot, t)\|_{L^{3/2}(B_r)} \leq |B_r|^{2/3} \|p_k(\cdot, t)\|_{L^\infty(B_r)} \leq c r^2 \rho^{-2} u(t) g(t),$$

and by (64) and (63),

$$\sum_{k=2}^4 \|p_k\|_{L^{3/2}(Q_r(z_0))} \leq c \frac{r^2}{\rho^2} r^{1/3} \rho \alpha(\rho) \beta(\rho) = c \frac{r^{7/3}}{\rho} \alpha(\rho) \beta(\rho),$$

and dividing by $r^{4/3}$ gives (68).

(c) *The terms p_5, p_6 .* The densities $p \Delta \eta$ and $(\partial_j \eta) p$ also vanish outside $\mathcal{A}_\rho$, so the same kernel bounds give, for $x \in B_r$,

$$\begin{aligned} & |p_5(x, t)| + |p_6(x, t)| \\ & \leq \left(\frac{2}{\rho} \cdot \frac{3C_{10}(2)}{\rho^2} + 2 \cdot \frac{4}{\rho^2} \cdot \frac{3C_{10}(1)}{\rho} \right) \|p(\cdot, t)\|_{L^1(B_\rho)} \\ & \leq c \rho^{-3} \cdot \left(\frac{4\pi}{3} \right)^{1/3} \rho \mathfrak{p}(t) = c \rho^{-2} \mathfrak{p}(t), \end{aligned}$$

where $\|p\|_{L^1(B_\rho)} \leq |B_\rho|^{1/3} \mathfrak{p}$ by Hölder. Then $\|p_5(\cdot, t) + p_6(\cdot, t)\|_{L^{3/2}(B_r)} \leq c r^2 \rho^{-2} \mathfrak{p}(t)$, and by the second inequality of (64) and (63),

$$\sum_{k=5}^6 \|p_k\|_{L^{3/2}(Q_r(z_0))} \leq c \frac{r^2}{\rho^2} \|\mathfrak{p}\|_{L^{3/2}(J_\rho)} = c \frac{r^2}{\rho^2} \rho^{4/3} \delta(\rho)^2 = c r^2 \rho^{-2/3} \delta(\rho)^2,$$

and dividing by $r^{4/3}$ gives $c (r/\rho)^{2/3} \delta(\rho)^2$, which is (69).

(d) *The term p_7 .* By External Input A.2 with $\frac{1}{15} = \frac{2}{5} - \frac{1}{3}$, for a.e. t , $\|p_7(\cdot, t)\|_{L^{15}(\mathbb{R}^3)} \leq C'_{11} \sum_{j=1}^3 \|\eta f_j(\cdot, t)\|_{L^{5/2}(\mathbb{R}^3)} \leq 3C'_{11} \|f(\cdot, t)\|_{L^{5/2}(B_\rho)}$. By Hölder on B_ρ (legitimate since $q > 5/2$), $\|f(\cdot, t)\|_{L^{5/2}(B_\rho)} \leq |B_\rho|^{\frac{2}{5}-\frac{1}{q}} \mathfrak{f}(t) = c \rho^{\frac{6}{5}-\frac{3}{q}} \mathfrak{f}(t)$, and by Hölder on B_r ,

$$\|p_7(\cdot, t)\|_{L^{3/2}(B_r)} \leq |B_r|^{\frac{2}{3}-\frac{1}{15}} \|p_7(\cdot, t)\|_{L^{15}(\mathbb{R}^3)} \leq c r^{9/5} \rho^{\frac{6}{5}-\frac{3}{q}} \mathfrak{f}(t).$$

Applying (65) and (63),

$$\|p_7\|_{L^{3/2}(Q_r(z_0))} \leq c r^{9/5} \rho^{\frac{6}{5}-\frac{3}{q}} r^{\frac{4}{3}-\frac{2}{q}} \rho^{-\sigma} \lambda(\rho) = c r^{\frac{47}{15}-\frac{2}{q}} \rho^{-\frac{9}{5}+\frac{2}{q}-\frac{4}{3}+\frac{4}{3}} \lambda(\rho),$$

where we used $\frac{6}{5}-\frac{3}{q}-\sigma = \frac{6}{5}-\frac{3}{q}-3+\frac{5}{q} = -\frac{9}{5}+\frac{2}{q}$. Dividing by $r^{4/3}$ and recalling $\frac{47}{15}-\frac{4}{3} = \frac{27}{15} = \frac{9}{5}$,

$$\frac{1}{r^{4/3}} \|p_7\|_{L^{3/2}(Q_r(z_0))} \leq c \kappa^{\frac{9}{5}-\frac{2}{q}} \lambda(\rho) \leq c \kappa \lambda(\rho),$$

the last step because $q > 5/2$ gives $\frac{9}{5}-\frac{2}{q} > \frac{9}{5}-\frac{4}{5} = 1$ and $\kappa \leq \frac{1}{2} < 1$.

(e) *The term p_8 .* Its density $(\partial_j \eta) f_j$ is supported in $\mathcal{A}_\rho$, so for $x \in B_r$,

$$|p_8(x, t)| \leq \frac{2}{\rho} \cdot \frac{3C_{10}(1)}{\rho} \|f(\cdot, t)\|_{L^1(B_\rho)} \leq c \rho^{-2} |B_\rho|^{1-\frac{1}{q}} \mathfrak{f}(t) = c \rho^{1-\frac{3}{q}} \mathfrak{f}(t).$$

Hence $\|p_8(\cdot, t)\|_{L^{3/2}(B_r)} \leq c r^2 \rho^{1-\frac{3}{q}} \mathfrak{f}(t)$ and, by (65),

$$\frac{1}{r^{4/3}} \|p_8\|_{L^{3/2}(Q_r(z_0))} \leq c r^{2-\frac{4}{3}+\frac{4}{3}-\frac{2}{q}} \rho^{1-\frac{3}{q}-\sigma} \lambda(\rho) = c r^{2-\frac{2}{q}} \rho^{-2+\frac{2}{q}} \lambda(\rho) = c \kappa^{2-\frac{2}{q}} \lambda(\rho) \leq c \kappa \lambda(\rho),$$

because $q > 5/2$ gives $2 - \frac{2}{q} > \frac{6}{5} > 1$. Adding (d) and (e) and taking $C_{13}(q)$ to be the resulting constant proves (70). $\square$

4.4. The pressure decay lemma.

Theorem 4.7 (Pressure decay). *There is an absolute constant C_{14} and a constant $C_{15} = C_{15}(q)$ such that for every suitable weak solution as above, every z_0 , every ρ with $Q_\rho(z_0) \subset \mathcal{O}$, and every $0 < r \leq \rho/2$, writing $\kappa = r/\rho$,*

$$\delta(z_0, r) \leq C_{14} \kappa^{-1/2} \alpha(z_0, \rho)^{1/2} \beta(z_0, \rho)^{1/2} + C_{14} \kappa^{1/3} \delta(z_0, \rho) + C_{15} \kappa^{1/2} \lambda(z_0, \rho)^{1/2}. \quad (71)$$

If moreover $\operatorname{div} f = 0$ then the last term may be omitted; that hypothesis is never used below.

Proof. Since $r \leq \rho/2 < \frac{13\rho}{20}$, Lemma 4.3 gives $\eta \equiv 1$ on B_r , so $p = \eta p$ on $Q_r(z_0)$ and (58) gives $p = \sum_{k=1}^8 p_k$ there. By definition of δ ,

$$\delta(z_0, r)^2 = \left(\frac{1}{r^2} \iint_{Q_r(z_0)} |p|^{3/2} \right)^{2/3} = \frac{1}{r^{4/3}} \|p\|_{L^{3/2}(Q_r(z_0))} \leq \frac{1}{r^{4/3}} \sum_{k=1}^8 \|p_k\|_{L^{3/2}(Q_r(z_0))},$$

by the triangle inequality in $L^{3/2}(Q_r(z_0))$. Lemma 4.6 and $\kappa \leq \frac{1}{2}$, which gives $\kappa \leq \kappa^{-1}$, yield

$$\delta(z_0, r)^2 \leq 2C_{12} \kappa^{-1} \alpha(\rho) \beta(\rho) + C_{12} \kappa^{2/3} \delta(\rho)^2 + C_{13} \kappa \lambda(\rho).$$

Taking square roots and using $(s_1 + s_2 + s_3)^{1/2} \leq s_1^{1/2} + s_2^{1/2} + s_3^{1/2}$ for $s_i \geq 0$ gives (71) with $C_{14} = (2C_{12})^{1/2}$ and $C_{15} = C_{13}^{1/2}$. Under the additional hypothesis $\operatorname{div} f = 0$ one has $p_7 + p_8 = 0$ by Proposition 4.4 and the term with C_{13} is absent. $\square$

4.5. The harmonic part and its interior estimates.

Corollary 4.8 (Calderón–Zygmund part, harmonic part, force part). *Set*

$$p_{\text{CZ}} = p_1 = -R_i R_j (\eta U_{ij}), \quad p_{\text{har}} = p_2 + p_3 + p_4 + p_5 + p_6, \quad p_{\text{f}} = p_7 + p_8, \quad (72)$$

so that $p = p_{\text{CZ}} + p_{\text{har}} + p_{\text{f}}$ on $B_{13\rho/20}(x_0)$ for a.e. $t \in J_\rho$, with $p_{\text{f}} = 0$ when $\operatorname{div} f = 0$ (which Definition 1.5 does not assume). Then for a.e. $t \in J_\rho$ the function $p_{\text{har}}(\cdot, t)$ is harmonic on $B_{13\rho/20}(x_0)$, and for every multi-index a there is a constant $C_{16}(|a|)$, depending on $|a|$ and on $C_{10}(1), C_{10}(2)$ only, with

$$\sup_{x \in B_{\rho/2}(x_0)} |\partial^a p_{\text{har}}(x, t)| \leq C_{16}(|a|) \rho^{-2-|a|} \left(\alpha(z_0, \rho)^2 + \|p(\cdot, t)\|_{L^{3/2}(B_\rho)} \right). \quad (73)$$

The exponent depends on $|a|$ and not only on its bound: replacing $\rho^{-2-|a|}$ by ρ^{-2-k} for all $|a| \leq k$ would be false for $\rho > 1$, as the suitable weak solution $u \equiv 0$, $p \equiv 1$, $f \equiv 0$ shows, for which $p_{\text{har}} \equiv 1$ on $B_{\rho/2}(x_0)$ while the right side with $k = 1$ is $O(\rho^{-1})$.

Proof. Each of $p_2, \dots, p_6$ is of the form $N * g$ or $\partial_j N * g$ with g supported in $\overline{\mathcal{A}_\rho} \subset \mathbb{R}^3 \setminus B_{13\rho/20}(x_0)$ and $g(\cdot, t) \in L^1(\mathbb{R}^3)$. For x in the open set $B_{13\rho/20}(x_0)$ the kernel $y \mapsto N(x - y)$ is smooth and harmonic in x on $\operatorname{supp} g$, and $|\partial_x^a N(x - y)| \leq c_a |x - y|^{-1-|a|}$ is integrable against $g(\cdot, t) dy$ locally uniformly in x ; hence differentiation under the integral sign is licit and $\Delta_x(N * g)(x) = \int \Delta_x N(x - y) g(y) dy = 0$ there. The same applies to $\partial_j N * g$. So $p_{\text{har}}(\cdot, t)$ is harmonic on $B_{13\rho/20}(x_0)$.

For (73), let $x \in B_{\rho/2}(x_0)$ and $y \in \mathcal{A}_\rho$, so $|x - y| \geq \frac{3\rho}{20}$ by Lemma 4.3. Differentiating under the integral, $|\partial^a p_{\text{har}}(x, t)|$ is bounded by $c_a \rho^{-1-|a|}$ times the total mass of the densities, one extra

factor ρ^{-1} being present for p_3, p_4, p_6 because the kernel is $\partial_j N$; together with $|\partial\eta| \leq C_{10}(1)\rho^{-1}$, $|\partial^2\eta| \leq C_{10}(2)\rho^{-2}$ this gives

$$|\partial^a p_{\text{har}}(x, t)| \leq c_a (C_{10}(1) + C_{10}(2)) \rho^{-3-|a|} \left(\|U(\cdot, t)\|_{L^1(B_\rho)} + \|p(\cdot, t)\|_{L^1(B_\rho)} \right).$$

Now $\|p(\cdot, t)\|_{L^1(B_\rho)} \leq (\frac{4\pi}{3})^{1/3} \rho \|p(\cdot, t)\|_{L^{3/2}(B_\rho)}$ by Hölder, and

$$\|U(\cdot, t)\|_{L^1(B_\rho)} \leq \|u(\cdot, t)\|_{L^2(B_\rho)} \|u(\cdot, t) - [u]_{B_\rho}(t)\|_{L^2(B_\rho)} \leq 2 \|u(\cdot, t)\|_{L^2(B_\rho)}^2 \leq 2\rho \alpha(z_0, \rho)^2,$$

where the middle step uses that $g \mapsto g - [g]_{B_\rho}$ has operator norm at most 2 on $L^2(B_\rho)$ (indeed at most 1, by orthogonality), and the last is (63). Substituting and absorbing the numeric factors into $C_{16}(|a|)$ gives (73). $\square$

Lemma 4.9 (Interior estimates for harmonic functions in $L^{3/2}$). *There is an absolute constant C_{17} with the following property. Let h be harmonic on $B_\rho(x_0)$ with $h \in L^{3/2}(B_\rho(x_0))$. Then*

$$\|h\|_{L^\infty(B_{3\rho/4}(x_0))} \leq C_{17} \rho^{-2} \|h\|_{L^{3/2}(B_\rho(x_0))}, \quad (74)$$

$$\|\nabla h\|_{L^\infty(B_{\rho/2}(x_0))} \leq C_{17} \rho^{-3} \|h\|_{L^{3/2}(B_\rho(x_0))}, \quad (75)$$

and for every $0 < r \leq \rho/2$,

$$\frac{1}{r^2} \int_{B_r(x_0)} |h|^{3/2} \leq C_{17} \frac{r}{\rho} \cdot \frac{1}{\rho^2} \int_{B_\rho(x_0)} |h|^{3/2}, \quad (76)$$

$$\frac{1}{r^2} \int_{B_r(x_0)} |h - [h]_{B_r(x_0)}|^{3/2} \leq C_{17} \left(\frac{r}{\rho}\right)^{5/2} \cdot \frac{1}{\rho^2} \int_{B_\rho(x_0)} |h|^{3/2}. \quad (77)$$

Proof. We use the classical smooth harmonic representative of h . If h is initially given as an $L^{3/2}$ function with $\Delta h = 0$ distributionally, Weyl's lemma gives an a.e.-equal smooth harmonic representative on the ball. All supremum and gradient estimates below refer to this representative; its $L^{3/2}$ norm agrees with that of h .

(74). Let $x \in B_{3\rho/4}(x_0)$. Then $B_{\rho/4}(x) \subset B_\rho(x_0)$, and the mean value property (External Input A.6) gives $h(x) = |B_{\rho/4}(x)|^{-1} \int_{B_{\rho/4}(x)} h$. Hence, by Hölder with $1 = \frac{2}{3} + \frac{1}{3}$,

$$|h(x)| \leq |B_{\rho/4}(x)|^{-1} |B_\rho(x_0)|^{1/3} \|h\|_{L^{3/2}(B_\rho(x_0))} = \left(\frac{4\pi}{3}\right)^{-2/3} 4^3 \rho^{-2} \|h\|_{L^{3/2}(B_\rho(x_0))}.$$

(75). Each $\partial_i h$ is harmonic on $B_\rho(x_0)$, and for $x \in B_{\rho/2}(x_0)$ the gradient estimate (External Input A.6) on the ball $B_{\rho/4}(x) \subset B_{3\rho/4}(x_0)$ gives $|\nabla h(x)| \leq \frac{3}{\rho/4} \|h\|_{L^\infty(B_{3\rho/4}(x_0))}$; combining with (74) proves (75).

(76). For $r \leq \rho/2$,

$$\begin{aligned} & \int_{B_r(x_0)} |h|^{3/2} \\ & \leq |B_r(x_0)| \|h\|_{L^\infty(B_{3\rho/4}(x_0))}^{3/2} \\ & \leq \frac{4\pi}{3} r^3 (C'_{17} \rho^{-2})^{3/2} \left(\int_{B_\rho(x_0)} |h|^{3/2} \right) \end{aligned}$$

by (74), where C'_{17} is the constant there. Dividing by r^2 gives $r\rho^{-3}$ times $\int_{B_\rho(x_0)} |h|^{3/2}$, i.e. (76).

(77). For $x \in B_r(x_0)$, $|h(x) - [h]_{B_r(x_0)}| \leq \sup_{y, y' \in B_r(x_0)} |h(y) - h(y')| \leq 2r \|\nabla h\|_{L^\infty(B_{\rho/2}(x_0))}$, since $B_r(x_0) \subset B_{\rho/2}(x_0)$ is convex. Hence, by (75),

$$\int_{B_r(x_0)} |h - [h]_{B_r(x_0)}|^{3/2} \leq \frac{4\pi}{3} r^3 (2r C'_{17} \rho^{-3})^{3/2} \int_{B_\rho(x_0)} |h|^{3/2} = c r^{9/2} \rho^{-9/2} \int_{B_\rho(x_0)} |h|^{3/2},$$

and dividing by r^2 gives $c (r/\rho)^{5/2} \rho^{-2} \int_{B_\rho(x_0)} |h|^{3/2}$. $\square$

4.6. The oscillation form of the pressure estimate. The estimates of Lemma 4.6 measure the velocity through $\alpha\beta$. For the scale iteration one also needs a form in which the velocity enters only through its oscillation. Define, for $Q_\rho(z_0) \subset \mathcal{O}$,

$$\widehat{C}(z_0, \rho) = \frac{1}{\rho^2} \iint_{Q_\rho(z_0)} |u(y, s) - [u]_{B_\rho}(s)|^3 dy ds. \quad (78)$$

Lemma 4.10 (Δp with the doubly centred nonlinearity). *For a.e. $t \in J_\rho$, $\Delta p = \partial_i \partial_j \widehat{U}_{ij} + \operatorname{div} f$ in $\mathcal{D}'(B_\rho)$, where*

$$\widehat{U}_{ij}(y, t) = -(u_i(y, t) - [u_i]_{B_\rho}(t))(u_j(y, t) - [u_j]_{B_\rho}(t)). \quad (79)$$

Proof. Write $c = [u]_{B_\rho}(t) \in \mathbb{R}^3$, a constant vector at fixed t . Expanding, $(u_i - c_i)(u_j - c_j) = u_i u_j - c_i u_j - c_j u_i + c_i c_j$. For $\psi \in C_c^\infty(B_\rho)$ and a.e. t , $\int c_i u_j \partial_i \partial_j \psi = c_i \int u_j \partial_j (\partial_i \psi) = 0$ and $\int c_j u_i \partial_i \partial_j \psi = c_j \int u_i \partial_i (\partial_j \psi) = 0$ by (53), while $\int c_i c_j \partial_i \partial_j \psi = 0$ because ψ has compact support. Hence $\int u_i u_j \partial_i \partial_j \psi = \int (u_i - c_i)(u_j - c_j) \partial_i \partial_j \psi$, and substituting into (54) gives the claim. $\square$

Proposition 4.11 (Oscillation form of the pressure estimate). *There is an absolute constant C_{18} such that for every z_0 and $\rho > 0$ with $\overline{Q_\rho(z_0)} \subset \mathcal{O}$ and every $0 < r \leq \rho/2$, the following hold.*

(i) (Pointwise in time, when $\operatorname{div} f = 0$.) *For a.e. $t \in J_\rho$,*

$$\frac{1}{r^2} \int_{B_r(x_0)} |p(y, t)|^{3/2} dy \leq C_{18} \left(\frac{\rho}{r}\right)^2 \frac{1}{\rho^2} \int_{B_\rho} |u(y, t) - [u]_{B_\rho}(t)|^3 dy + C_{18} \frac{r}{\rho} \cdot \frac{1}{\rho^2} \int_{B_\rho} |p(y, t)|^{3/2} dy. \quad (80)$$

(ii-a) (Integrated in time, when $\operatorname{div} f = 0$.) *If $\operatorname{div} f = 0$ in $\mathcal{D}'(\mathcal{O})$ then*

$$D(z_0, r) \leq C_{18} \left(\frac{\rho}{r}\right)^2 \widehat{C}(z_0, \rho) + C_{18} \frac{r}{\rho} D(z_0, \rho). \quad (81)$$

(ii-b) (Integrated in time, general force.) *Without any hypothesis on $\operatorname{div} f$, with $C_{32}(q) = 2^{1/2}(C_{18} + C_{13}(q)^{3/2})$,*

$$D(z_0, r) \leq C_{32}(q) \left[\left(\frac{\rho}{r}\right)^2 \widehat{C}(z_0, \rho) + \frac{r}{\rho} D(z_0, \rho) + \left(\frac{r}{\rho}\right)^{3/2} \lambda(z_0, \rho)^{3/2} \right]. \quad (82)$$

Proof. Run the decomposition of Proposition 4.4 with $\widehat{U}$ of (79) in place of U , which is legitimate by Lemma 4.10. In part (i) we assume $\operatorname{div} f = 0$, so that $p_7 + p_8 = 0$ and $p = \sum_{k=1}^6 p_k$ on $B_{13\rho/20}(x_0) \supset B_r(x_0)$; part (ii) adds the contribution of $p_7 + p_8$ at the end. Write $w(\cdot, t) = u(\cdot, t) - [u]_{B_\rho}(t)$, so that $|\widehat{U}| \leq |w|^2$ and $\|\widehat{U}(\cdot, t)\|_{L^{3/2}(B_\rho)} \leq \|w(\cdot, t)\|_{L^3(B_\rho)}^2$.

(i). For p_1 , External Input A.1 gives $\|p_1(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq 9C_{11} \|w(\cdot, t)\|_{L^3(B_\rho)}^2$, hence

$$\int_{B_r(x_0)} |p_1(\cdot, t)|^{3/2} \leq (9C_{11})^{3/2} \int_{B_\rho} |w(\cdot, t)|^3,$$

and dividing by r^2 yields the first term of (80) with the factor $(\rho/r)^2$. For p_2, p_3, p_4 , the far-field bounds of Lemma 4.6(b) with U replaced by $\widehat{U}$ give, for $x \in B_r(x_0)$,

$$\sum_{k=2}^4 |p_k(x, t)| \leq c \rho^{-3} \left\| \widehat{U}(\cdot, t) \right\|_{L^1(B_\rho)} \leq c \rho^{-3} |B_\rho|^{1/3} \|w(\cdot, t)\|_{L^3(B_\rho)}^2 = c \rho^{-2} \|w(\cdot, t)\|_{L^3(B_\rho)}^2,$$

so $\int_{B_r(x_0)} (\sum_{k=2}^4 |p_k|)^{3/2} \leq c r^3 \rho^{-3} \int_{B_\rho} |w(\cdot, t)|^3$, and dividing by r^2 gives $c (r/\rho) \rho^{-2} \int_{B_\rho} |w|^3 \leq c (\rho/r)^2 \rho^{-2} \int_{B_\rho} |w|^3$. For p_5, p_6 , Lemma 4.6(c) gives $|p_5| + |p_6| \leq c \rho^{-2} \|p(\cdot, t)\|_{L^{3/2}(B_\rho)}$ on $B_r(x_0)$, so

$$\int_{B_r(x_0)} (|p_5| + |p_6|)^{3/2} \leq c r^3 \rho^{-3} \int_{B_\rho} |p(\cdot, t)|^{3/2},$$

and dividing by r^2 gives $c (r/\rho) \rho^{-2} \int_{B_\rho} |p|^{3/2}$, the second term of (80). Summing the three groups, using $(\sum_k |p_k|)^{3/2} \leq 3^{1/2} \sum_k |p_k|^{3/2}$ for three groups, and absorbing numeric factors into C_{18} proves (i).

(ii-a). Integrate (80) over $t \in (t_0 - r^2, t_0)$ and enlarge the two time integrals on the right from $(t_0 - r^2, t_0)$ to $J_\rho = (t_0 - \rho^2, t_0)$, which is legitimate since both integrands are nonnegative. This gives exactly (81).

(ii-b). For a general force write $p = (\sum_{k=1}^6 p_k) + (p_7 + p_8)$ on $B_{13\rho/20}(x_0)$ and use $|a + b|^{3/2} \leq 2^{1/2}(|a|^{3/2} + |b|^{3/2})$. The first group is bounded exactly as in (ii-a), and by Lemma 4.6(d)–(e)

$$\frac{1}{r^2} \iint_{Q_{r(z_0)}} |p_7 + p_8|^{3/2} = \left(\frac{1}{r^{4/3}} \|p_7 + p_8\|_{L^{3/2}(Q_{r(z_0)})} \right)^{3/2} \leq (C_{13}(q) \kappa \lambda(z_0, \rho))^{3/2},$$

which gives (82) with $C_{32}(q) = 2^{1/2}(C_{18} + C_{13}(q)^{3/2})$. $\square$

Remark 4.12 (The pressure-decay step in Lin's proof). Lin [15, Lemma 3.4] uses a coefficient C_{θ_0} depending on the lower scale ratio θ_0 , not the small factor $C\theta_0$. His Corollary 3.5 restricts $r > \theta_0 \rho$, so the factor $(\rho/r)^2$ multiplying the velocity oscillation in (80) can be absorbed into C_{θ_0} . The velocity term of the corollary is therefore consistent with that estimate.

The gap in the deduction from the printed lemma concerns the harmonic pressure term. The useful unnormalized estimate retains the factor $(r/\rho)^3$ in that term, supplied by the harmonic interior estimate. After division by r^2 this becomes $(r/\rho) \rho^{-2} \int_{B_\rho} |p|^{3/2}$, as in the corollary. Without the cubic factor, direct rescaling gives $(\rho/r)^2 \rho^{-2} \int_{B_\rho} |p|^{3/2}$ instead. Thus the lemma as printed is too weak to imply its corollary; the missing cubic decay belongs to the harmonic term. We use (81), retaining the explicit scale ratios throughout.

5. THE ε -REGULARITY CORE

This section proves Theorem 2.7 and Theorem 2.9. The architecture is that of Lemarié-Rieusset [13, §13.8–13.9], which combines the scale iteration of Kukavica [9] with the parabolic Morrey/potential endgame of O'Leary [17]. There are four steps.

Step 1 (§5.2) The local energy inequality and the pressure decomposition of Section 4 give a pair of coupled decay estimates for α, β, δ at two scales. This is [13, Lemma 13.3, (13.30)–(13.31), pp. 430–434]; we obtain it with Kukavica's backward-Gaussian test function.

Step 2 (§5.3) Iterating these gives, on a neighbourhood of a point at which $\limsup_{r \rightarrow 0} \beta(z_0, r)$ is small, the parabolic Morrey bounds $\mathbf{1}_Q u \in \mathcal{M}^{3, \tau_2}$, $\mathbf{1}_Q \nabla u \in \mathcal{M}^{2, \tau_3}$, $\mathbf{1}_Q p \in \mathcal{M}^{3/2, \tau_p}$ with $\tau_2 > 5$. We use Kukavica's Lemma 3; compare [13, Lemma 13.4, pp. 435–439]. The pressure exponents differ, as explained in Remark 5.24.

Step 3 (§5.4) A representation of ϕu by parabolic Riesz potentials, together with the Adams–Hedberg inequality, improves the Morrey exponent of u in finitely many rounds. This is [13, Lemma 13.5, pp. 439–441].

Step 4 (§5.5) Once the exponents are large enough, the same representation and the parabolic Campanato lemma give Hölder continuity. This is [13, Lemma 13.6 and Proposition 13.4].

Convention 5.1 (Standing choices in this section). Throughout Section 5:

$$\begin{aligned} q > \frac{5}{2}, \quad \sigma = 3 - \frac{5}{q} > 1, \quad \varepsilon = \frac{2}{5}, \quad \tau_2 = \frac{5}{1 - \varepsilon} = \frac{25}{3}, \\ \frac{1}{\tau_3} = \frac{1}{\tau_2} + \frac{1}{5}, \quad \tau_3 = \frac{25}{8}, \quad \tau_p = \frac{25}{8}. \end{aligned} \quad (83)$$

The value $\varepsilon = 2/5$ is admissible because $\varepsilon < \min\{2/3, \sigma\}$ (Proposition 5.22), and it gives $\tau_2 = 25/3$ and $\tau_3 = 25/8$. Steps 3 and 4 use these actual values in the product and potential exponent identities, not merely $\tau_2 > 5$ and $\tau_3 > 5/2$. The pressure exponent $\tau_p = 25/8$ is recorded for completeness but is *not* consumed downstream: the pressure enters Steps 3 and 4 through fixed-scale $L^{3/2}$ bounds for the harmonic remainder in Lemma 5.26. See Remark 5.24. The pressure is taken in $L_{loc}^{3/2}$, i.e. the exponent called q_0 in [13] is $q_0 = 3/2$; see Remark 5.24.

5.1. Parabolic Morrey spaces and Riesz potentials.

Definition 5.2 (Parabolic Morrey space). For $1 \leq P < \infty$ and $P \leq \tau < \infty$ and $g \in L_{loc}^P(\mathbb{R}^3 \times \mathbb{R})$ put

$$\|g\|_{\mathcal{M}^{P,\tau}} = \sup_{z \in \mathbb{R}^3 \times \mathbb{R}} \sup_{r > 0} r^{-5(\frac{1}{P} - \frac{1}{\tau})} \left(\iint_{\mathfrak{B}_r(z)} |g|^P dz' \right)^{1/P}, \quad (84)$$

with $\mathfrak{B}_r(z)$ the d_{par} -ball (3), and $\mathcal{M}^{P,\tau} = \{g : \|g\|_{\mathcal{M}^{P,\tau}} < \infty\}$.

The number 5 is the homogeneous dimension of $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}})$: by Lemma 1.3, $|\mathfrak{B}_r(z)| = \frac{8\pi}{3} r^5$. Note $L^\tau \subset \mathcal{M}^{P,\tau}$ and $\mathcal{M}^{\tau,\tau} = L^\tau$.

Lemma 5.3 (Cylinders, one-sided or two-sided, give the same space). *Let $1 \leq P \leq \tau < \infty$ and put, for g measurable,*

$$\begin{aligned} N_1(g) &= \sup_{z,r} r^{-5(\frac{1}{P} - \frac{1}{\tau})} \left(\iint_{Q_r(z)} |g|^P \right)^{1/P}, \\ N_2(g) &= \sup_{z,r} r^{-5(\frac{1}{P} - \frac{1}{\tau})} \left(\iint_{\mathfrak{B}_r(z)} |g|^P \right)^{1/P} = \|g\|_{\mathcal{M}^{P,\tau}}. \end{aligned}$$

Then

$$N_1(g) \leq N_2(g) \leq 2^{5(\frac{1}{P} - \frac{1}{\tau})} N_1(g). \quad (85)$$

Proof. The first inequality is $Q_r(z) \subset \mathfrak{B}_r(z)$ (Lemma 1.3). For the second, fix $z = (x, t)$ and $r > 0$ and set $z' = (x, t + r^2)$. By Lemma 1.3, $\mathfrak{B}_r(z) \subset Q_{2r}(z')$, so $\iint_{\mathfrak{B}_r(z)} |g|^P \leq \iint_{Q_{2r}(z')} |g|^P \leq N_1(g)^P (2r)^{5P(\frac{1}{P} - \frac{1}{\tau})}$. Taking P -th roots and dividing by $r^{5(\frac{1}{P} - \frac{1}{\tau})}$ gives the claim. $\square$

Remark 5.4 (One-sided cylinders are what the energy inequality produces). The local energy inequality is naturally an estimate on the one-sided cylinders $Q_r(z) = B_r(x) \times (t - r^2, t]$, and the hypothesis of Theorem 2.9 is stated on these cylinders. The original [3, Proposition 2] instead uses a shifted backward cylinder; we do not identify its cylinder convention with this one. Lemarié-Rieusset [13, §13.8] uses the two-sided cylinders $(t - r^2, t + r^2) \times B_r(x)$ throughout, so his hypothesis is formally stronger than ours. Lemma 5.3 shows the difference is immaterial

for Morrey norms, and Lemma 5.23 below converts the one-sided bounds produced by Step 2 into the two-sided form used by Steps 3 and 4: the point is that Step 2 produces its bounds simultaneously at *every* centre in a neighbourhood, and $\mathfrak{B}_r(z) \subset Q_{\sqrt{2}r}((x, t + r^2))$.

Remark 5.5 (Normalization of local norms). The metric ball has volume $(8\pi/3)r^5$; the backward cylinder has volume $(4\pi/3)r^5$. Thus lowering the local integrability exponent from P to P' contributes the volume of the unit domain to the power $1/P' - 1/P$, using the domain of integration. These factors, and the comparison in (85), cannot be replaced by literal equality of ball and cylinder norms.

Definition 5.6 (Parabolic Riesz potential). For $a \in (0, 5)$ and $g \in L^1_{loc}(\mathbb{R}^3 \times \mathbb{R})$ set

$$(I_a g)(z) = \iint_{\mathbb{R}^3 \times \mathbb{R}} \frac{|g(w)|}{d_{\text{par}}(z, w)^{5-a}} dw, \quad z \in \mathbb{R}^3 \times \mathbb{R}. \quad (86)$$

External input 5.7 (Parabolic Hardy–Littlewood maximal theorem). For $g \in L^1_{loc}(\mathbb{R}^3 \times \mathbb{R})$ put

$$(\mathcal{M}g)(z) = \sup_{r>0} \frac{1}{|\mathfrak{B}_r(z)|} \iint_{\mathfrak{B}_r(z)} |g| dw.$$

Then $\mathcal{M}g$ is lower semicontinuous, hence measurable. The doubling property $|\mathfrak{B}_{2r}(z)| = 2^5 |\mathfrak{B}_r(z)|$ and the Vitali covering lemma give the weak $(1, 1)$ bound $|\{\mathcal{M}g > \varsigma\}| \leq C\varsigma^{-1} \|g\|_{L^1}$; the trivial L^∞ bound and Marcinkiewicz interpolation then give, for every $P \in (1, \infty)$, a constant $C_{19}(P)$ with $\|\mathcal{M}g\|_{L^P} \leq C_{19}(P) \|g\|_{L^P}$. The same construction yields the Lebesgue differentiation theorem on $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}}, dz)$, used in Lemma 5.11. Reference: [22, Ch. I]; the metric measure space $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}}, dz)$ is of homogeneous type.

Lemma 5.8 (Hedberg’s inequality). Let $1 \leq P \leq \tau < \infty$ and $0 < a < 5/\tau$. Set

$$\varkappa = \frac{a\tau}{5} \in (0, 1). \quad (87)$$

There is a constant $C_{20}(a, \tau)$ such that for every $g \in \mathcal{M}^{P, \tau}$ and every z ,

$$(I_a g)(z) \leq C_{20}(a, \tau) (\mathcal{M}g(z))^{1-\varkappa} \|g\|_{\mathcal{M}^{P, \tau}}^\varkappa. \quad (88)$$

Proof. Fix z and $R > 0$ and split $I_a g(z) = A_R + B_R$ with A_R the integral over $\{d_{\text{par}}(z, w) < R\} = \mathfrak{B}_R(z)$ and B_R the integral over the complement.

Near part. Decompose $\mathfrak{B}_R(z)$ into the dyadic shells $S_j = \mathfrak{B}_{2^{-j}R}(z) \setminus \mathfrak{B}_{2^{-j-1}R}(z)$, $j \geq 0$. On S_j we have $d_{\text{par}}(z, w)^{-(5-a)} \leq (2^{-j-1}R)^{-(5-a)}$, so

$$\begin{aligned} A_R &\leq \sum_{j \geq 0} (2^{-j-1}R)^{-(5-a)} \iint_{\mathfrak{B}_{2^{-j}R}(z)} |g| \\ &\leq \sum_{j \geq 0} (2^{-j-1}R)^{-(5-a)} \frac{8\pi}{3} (2^{-j}R)^5 \mathcal{M}g(z) \\ &= \frac{8\pi}{3} 2^{5-a} R^a \mathcal{M}g(z) \sum_{j \geq 0} 2^{-ja}, \end{aligned}$$

so $A_R \leq CR^a \mathcal{M}g(z)$ with $C = \frac{8\pi}{3} 2^{5-a} (1 - 2^{-a})^{-1}$.

Far part. Decompose the complement into $T_j = \mathfrak{B}_{2^{j+1}R}(z) \setminus \mathfrak{B}_{2^jR}(z)$, $j \geq 0$. On T_j , $d_{\text{par}}(z, w)^{-(5-a)} \leq (2^jR)^{-(5-a)}$. By Hölder’s inequality on $\mathfrak{B}_{2^{j+1}R}(z)$ with exponents P and

P' , and Definition 5.2,

$$\begin{aligned} \iint_{\mathfrak{B}_{2^{j+1}R}(z)} |g| &\leq |\mathfrak{B}_{2^{j+1}R}|^{1-\frac{1}{P}} \left(\iint_{\mathfrak{B}_{2^{j+1}R}} |g|^P \right)^{1/P} \\ &\leq C' (2^{j+1}R)^{5(1-\frac{1}{P})} (2^{j+1}R)^{5(\frac{1}{P}-\frac{1}{\tau})} \|g\|_{\mathcal{M}^{P,\tau}} \\ &= C' (2^{j+1}R)^{5(1-\frac{1}{\tau})} \|g\|_{\mathcal{M}^{P,\tau}}, \end{aligned}$$

with $C' = (\frac{8\pi}{3})^{1-1/P}$. Hence

$$B_R \leq C' 2^5 \|g\|_{\mathcal{M}^{P,\tau}} \sum_{j \geq 0} (2^j R)^{a-\frac{5}{\tau}} = C'' R^{a-\frac{5}{\tau}} \|g\|_{\mathcal{M}^{P,\tau}},$$

the geometric series converging because $a - \frac{5}{\tau} < 0$ by hypothesis.

Optimisation. If $\mathcal{M}g(z) = 0$ then $g = 0$ a.e. and there is nothing to prove; if $\mathcal{M}g(z) = \infty$ the bound is trivial. Otherwise choose R with $R^{5/\tau} = \|g\|_{\mathcal{M}^{P,\tau}} / \mathcal{M}g(z)$, that is $R = (\|g\|_{\mathcal{M}^{P,\tau}} / \mathcal{M}g(z))^{\tau/5}$. Then $R^a \mathcal{M}g(z) = \|g\|_{\mathcal{M}^{P,\tau}}^{a\tau/5} \mathcal{M}g(z)^{1-a\tau/5}$ and $R^{a-5/\tau} \|g\|_{\mathcal{M}^{P,\tau}}$ equals the same quantity. Adding the two bounds gives (88) with $C_{20} = C + C''$. $\square$

Corollary 5.9 (Adams' theorem). *Let $1 < P \leq \tau < \infty$ and $0 < a < 5/\tau$, and set $\varkappa = a\tau/5$, $\lambda = 1 - \varkappa$, so that*

$$\frac{1}{\tau/\lambda} = \frac{1}{\tau} - \frac{a}{5}. \quad (89)$$

There is $C_{21}(a, P, \tau)$ such that

$$\|I_a g\|_{\mathcal{M}^{P/\lambda, \tau/\lambda}} \leq C_{21}(a, P, \tau) \|g\|_{\mathcal{M}^{P,\tau}} \quad \text{for all } g \in \mathcal{M}^{P,\tau}. \quad (90)$$

Proof. Write $\Lambda = \|g\|_{\mathcal{M}^{P,\tau}}$ and let $z_0, r > 0$. By Lemma 5.8, $|I_a g|^{P/\lambda} \leq C_{20}^{P/\lambda} (\mathcal{M}g)^P \Lambda^{\varkappa P/\lambda}$, since $(1 - \varkappa)P/\lambda = P$. Therefore

$$\iint_{\mathfrak{B}_r(z_0)} |I_a g|^{P/\lambda} \leq C_{20}^{P/\lambda} \Lambda^{\varkappa P/\lambda} \iint_{\mathfrak{B}_r(z_0)} (\mathcal{M}g)^P.$$

It therefore suffices to prove the Morrey bound

$$\iint_{\mathfrak{B}_r(z_0)} (\mathcal{M}g)^P \leq C r^{5(1-\frac{P}{\tau})} \Lambda^P. \quad (91)$$

Split $g = g_1 + g_2$ with $g_1 = g \mathbf{1}_{\mathfrak{B}_{2r}(z_0)}$. For g_1 , External Input 5.7 gives $\iint (\mathcal{M}g_1)^P \leq C_{19}(P)^P \iint |g_1|^P \leq C_{19}(P)^P (2r)^{5(1-\frac{P}{\tau})} \Lambda^P$. For g_2 and $z \in \mathfrak{B}_r(z_0)$, any ball $\mathfrak{B}_\varrho(z)$ meeting $\text{supp } g_2$ has $\varrho > r$, and then $\mathfrak{B}_\varrho(z) \subset \mathfrak{B}_{2\varrho}(z_0)$, so

$$\mathcal{M}g_2(z) \leq \sup_{\varrho > r} \frac{1}{\frac{8\pi}{3}\varrho^5} \iint_{\mathfrak{B}_{2\varrho}(z_0)} |g| \leq \sup_{\varrho > r} \frac{C}{\varrho^5} (2\varrho)^{5(1-\frac{1}{\tau})} \Lambda = C' r^{-5/\tau} \Lambda,$$

using the Hölder step of the previous proof. Hence $\iint_{\mathfrak{B}_r(z_0)} (\mathcal{M}g_2)^P \leq \frac{8\pi}{3} r^5 (C' r^{-5/\tau} \Lambda)^P = C'' r^{5(1-\frac{P}{\tau})} \Lambda^P$. Adding the two contributions, with the factor 2^{P-1} from $(s_1 + s_2)^P \leq 2^{P-1}(s_1^P + s_2^P)$ absorbed into C , gives (91), and therefore $\iint_{\mathfrak{B}_r(z_0)} |I_a g|^{P/\lambda} \leq C r^{5(1-\frac{P}{\tau})} \Lambda^{P/\lambda}$. Since $5(1 - \frac{P/\lambda}{\tau/\lambda}) = 5(1 - \frac{P}{\tau})$, this is exactly (90). $\square$

Remark 5.10. Lemma 5.8 and Corollary 5.9 are [13, Lemma 5.3 and Corollary 5.1, p. 93–94], stated there for a general space of homogeneous type of homogeneous dimension Q ; we have specialized to $Q = 5$ and written the proof out. Corollary 5.9 is due to Adams [1]. Note that

only the *Morrey* exponent τ enters the condition $a\tau < 5$ and the gain (89); the integrability exponent P is carried along.

Lemma 5.11 (Parabolic Campanato lemma). *Let $P \in [1, \infty)$, $\gamma \in (0, 1)$ and $h \in L_{loc}^P(\mathbb{R}^3 \times \mathbb{R})$, and suppose*

$$H = \sup_z \sup_{r>0} \frac{1}{r^\gamma} \left(\frac{1}{|\mathfrak{B}_r(z)|} \iint_{\mathfrak{B}_r(z)} |h - (h)_{\mathfrak{B}_r(z)}|^P \right)^{1/P} < \infty,$$

where $(h)_{\mathfrak{B}_r(z)}$ is the average of h over $\mathfrak{B}_r(z)$. Then h agrees almost everywhere with a function $\bar{h}$ with $|\bar{h}(z) - \bar{h}(z')| \leq C_{22}(\gamma, P) H d_{\text{par}}(z, z')^\gamma$ for all z, z' , and moreover, for every z and every $R > 0$,

$$\|\bar{h}\|_{L^\infty(\mathfrak{B}_R(z))} \leq 2^\gamma C_{22}(\gamma, P) H R^\gamma + \left(\frac{1}{|\mathfrak{B}_R(z)|} \iint_{\mathfrak{B}_R(z)} |h|^P \right)^{1/P}. \quad (92)$$

Conversely, if h satisfies such a Hölder bound with constant $[h]$ then $H \leq 2^\gamma [h]$.

Proof. The converse is immediate: for $w \in \mathfrak{B}_r(z)$,

$$|h(w) - (h)_{\mathfrak{B}_r(z)}| \leq |\mathfrak{B}_r(z)|^{-1} \iint_{\mathfrak{B}_r(z)} |h(w) - h(w')| dw' \leq [h] (2r)^\gamma,$$

since $d_{\text{par}}(w, w') < 2r$ for $w, w' \in \mathfrak{B}_r(z)$.

For the direct statement, fix $\Phi \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R})$ supported in $\mathfrak{B}_1(0)$ with $\iint \Phi = 1$, and set $\Phi_\epsilon(x, t) = \epsilon^{-5} \Phi(x/\epsilon, t/\epsilon^2)$, so that Φ_ϵ is supported in $\mathfrak{B}_\epsilon(0)$, $\iint \Phi_\epsilon = 1$, $\|\Phi_\epsilon\|_\infty \leq \epsilon^{-5} \|\Phi\|_\infty$, $\|\nabla_x \Phi_\epsilon\|_\infty \leq \epsilon^{-6} \|\nabla \Phi\|_\infty$ and $\|\partial_t \Phi_\epsilon\|_\infty \leq \epsilon^{-7} \|\partial_t \Phi\|_\infty$.

(a) *Successive scales.* Since $\iint (\Phi_\epsilon - \Phi_{\epsilon/2}) = 0$ and both are supported in $\mathfrak{B}_\epsilon(0)$,

$$\Phi_\epsilon * h(z) - \Phi_{\epsilon/2} * h(z) = \iint_{\mathfrak{B}_{\epsilon/2}(z)} (\Phi_\epsilon(z - w) - \Phi_{\epsilon/2}(z - w)) (h(w) - (h)_{\mathfrak{B}_{\epsilon/2}(z)}) dw,$$

so, bounding the kernel by $C\epsilon^{-5} \leq C' |\mathfrak{B}_\epsilon(z)|^{-1}$ and using Hölder's inequality and the hypothesis,

$$|\Phi_\epsilon * h(z) - \Phi_{\epsilon/2} * h(z)| \leq CH\epsilon^\gamma. \quad (93)$$

(b) *Two points at one scale.* Let z, z' with $\varrho = d_{\text{par}}(z, z') \leq \epsilon$. Since $\iint (\Phi_\epsilon(z - \cdot) - \Phi_\epsilon(z' - \cdot)) = 0$ and both kernels are supported in $\mathfrak{B}_{\epsilon+\varrho}(z) \subset \mathfrak{B}_{2\epsilon}(z)$,

$$\Phi_\epsilon * h(z) - \Phi_\epsilon * h(z') = \iint_{\mathfrak{B}_{2\epsilon}(z)} (\Phi_\epsilon(z - w) - \Phi_\epsilon(z' - w)) (h(w) - (h)_{\mathfrak{B}_{2\epsilon}(z)}) dw.$$

Writing $z = (x, t)$, $z' = (x', t')$ and using the mean value theorem separately in x and in t ,

$$|\Phi_\epsilon(z - w) - \Phi_\epsilon(z' - w)| \leq \frac{|x - x'|}{\epsilon^6} \|\nabla \Phi\|_\infty + \frac{|t - t'|}{\epsilon^7} \|\partial_t \Phi\|_\infty \leq \frac{C}{\epsilon^5} \cdot \frac{\varrho}{\epsilon},$$

because $|x - x'| \leq \varrho$ and $|t - t'| \leq \varrho^2 \leq \varrho\epsilon$. Hence

$$|\Phi_\epsilon * h(z) - \Phi_\epsilon * h(z')| \leq CH\epsilon^\gamma \frac{\varrho}{\epsilon}. \quad (94)$$

(c) *A continuous representative.* By (93), for $0 < \epsilon' \leq \epsilon$ with $\epsilon' = 2^{-m}\epsilon$,

$$|\Phi_\epsilon * h(z) - \Phi_{\epsilon'} * h(z)| \leq CH \sum_{k=0}^{m-1} (2^{-k}\epsilon)^\gamma \leq \frac{CH}{1 - 2^{-\gamma}} \epsilon^\gamma, \quad (95)$$

uniformly in z , so $(\Phi_{2^{-k}} * h)_{k \geq 0}$ has a uniform limit $\bar{h}$. Repeat part (a) with $\epsilon' \in [\epsilon/2, \epsilon]$ in place of $\epsilon/2$. Both kernels still have integral one, support in $\mathfrak{B}_\epsilon(0)$, and supremum bounded by $C\epsilon^{-5}$. Consequently

$$\sup_z |\Phi_\epsilon * h(z) - \Phi_{\epsilon'} * h(z)| \leq CH\epsilon^\gamma \quad (\epsilon/2 \leq \epsilon' \leq \epsilon).$$

For any small $s > 0$, choose k with $2^{-k-1} \leq s \leq 2^{-k}$ and compare $\Phi_s * h$ to $\Phi_{2^{-k}} * h$. This comparison tends to zero uniformly as $s \downarrow 0$, proving $\bar{h} = \lim_{s \downarrow 0} \Phi_s * h$. Telescoping from an arbitrary initial scale ϵ now gives $\sup_z |\bar{h}(z) - \Phi_\epsilon * h(z)| \leq C'H\epsilon^\gamma$. The limit is continuous because each mollification is continuous. By the Lebesgue differentiation theorem on the homogeneous space $(\mathbb{R}^3 \times \mathbb{R}, d_{\text{par}}, dz)$ (a consequence of External Input 5.7), $\Phi_\epsilon * h(z) \rightarrow h(z)$ at every Lebesgue point of h , hence $\bar{h} = h$ almost everywhere.

(d) *The Hölder bound.* Let $z \neq z'$ and $\varrho = d_{\text{par}}(z, z')$. By (95) with $\epsilon = \varrho$ and $\epsilon' \rightarrow 0$, $|\bar{h}(z) - \Phi_\varrho * h(z)| \leq C'H\varrho^\gamma$ and likewise at z' ; by (94) with $\epsilon = \varrho$, $|\Phi_\varrho * h(z) - \Phi_\varrho * h(z')| \leq CH\varrho^\gamma$. Adding the three gives $|\bar{h}(z) - \bar{h}(z')| \leq C_{22}(\gamma, P)H\varrho^\gamma$.

(e) *The supremum bound* (92). For $w \in \mathfrak{B}_R(z)$,

$$|\bar{h}(w) - (h)_{\mathfrak{B}_R(z)}| = \left| \frac{1}{|\mathfrak{B}_R(z)|} \iint_{\mathfrak{B}_R(z)} (\bar{h}(w) - \bar{h}(w')) dw' \right| \leq C_{22}(\gamma, P)H(2R)^\gamma,$$

because $d_{\text{par}}(w, w') < 2R$, and $|(h)_{\mathfrak{B}_R(z)}| \leq (|\mathfrak{B}_R(z)|^{-1} \iint_{\mathfrak{B}_R(z)} |h|^P)^{1/P}$ by Jensen's inequality. $\square$

Proposition 5.12 (Heat potentials of Morrey sources are Hölder). *Let $\gamma \in (0, 1)$ and define*

$$\frac{1}{\theta_0} = \frac{2-\gamma}{5}, \quad \frac{1}{\theta_1} = \frac{1-\gamma}{5}, \quad \text{so} \quad \theta_0 = \frac{5}{2-\gamma} > \frac{5}{2}, \quad \theta_1 = \frac{5}{1-\gamma} > 5. \quad (96)$$

Let $1 \leq P \leq \theta_0 < \theta_1 < \infty$, let $K \in \mathbb{N}$, let $\varsigma_1, \dots, \varsigma_K$ be smooth on $\mathbb{R}^3 \setminus \{0\}$ and homogeneous of degree 1, and let $\varsigma_k(D)$ be the associated Fourier multipliers, acting in the space variable only. Let $F \in \mathcal{M}^{P, \theta_0}$ and $G_1, \dots, G_K \in \mathcal{M}^{P, \theta_1}$, all supported in a fixed bounded set, and put

$$h = W_+ * \left(F + \sum_{k=1}^K \varsigma_k(D) G_k \right), \quad W_+(x, t) = \mathbf{1}_{t>0} (4\pi t)^{-3/2} e^{-|x|^2/(4t)}, \quad (97)$$

the convolution being taken in $\mathbb{R}^3 \times \mathbb{R}$ in the distributional, equivalently almost-everywhere kernel, sense. In particular, its slice formula is

$$\begin{aligned} h(x, t) &= \int_{\mathbb{R}} (W_+(\cdot, t-s) *_x F(\cdot, s))(x) ds \\ &\quad + \sum_k \int_{\mathbb{R}} (\varsigma_k(D) (W_+(\cdot, t-s) *_x G_k(\cdot, s)))(x) ds \end{aligned} \quad \text{a.e.}$$

The translated kernel products are absolutely integrable almost everywhere; this is a conclusion, not an additional source hypothesis. The distributional pairing remains an identity for every compactly supported smooth test. The finite constant C_{23} below is chosen from γ, P and the finite symbol family before the sources or their support. Then h agrees a.e. with a function $\bar{h}$ that is Hölder of exponent γ with respect to d_{par} , with

$$[\bar{h}]_{C^{\gamma, \gamma/2}} \leq C_{23}(\gamma, P, \varsigma_1, \dots, \varsigma_K) \left(\|F\|_{\mathcal{M}^{P, \theta_0}} + \sum_k \|G_k\|_{\mathcal{M}^{P, \theta_1}} \right) =: C_{23}\mathcal{N}, \quad (98)$$

and, for every z and $R > 0$,

$$\|\bar{h}\|_{L^\infty(\mathfrak{B}_R(z))} \leq 2^\gamma C_{23} \mathcal{N} R^\gamma + \left(\frac{1}{|\mathfrak{B}_R(z)|} \iint_{\mathfrak{B}_R(z)} |h|^P \right)^{1/P}. \quad (99)$$

Proof. By External Input A.11 the kernels satisfy, for $w = (y, s)$ with $s > 0$ and $\varrho = d_{\text{par}}(w, 0)$,

$$\begin{aligned} |W_+| &\leq C\varrho^{-3}, \quad |\nabla_y W_+| \leq C\varrho^{-4}, \quad |\partial_s W_+| \leq C\varrho^{-5}, \\ |\varsigma(D)W_+| &\leq C\varrho^{-4}, \quad |\nabla_y \varsigma(D)W_+| \leq C\varrho^{-5}, \quad |\partial_s \varsigma(D)W_+| \leq C\varrho^{-6}. \end{aligned} \quad (100)$$

The causal kernels vanish for $s < 0$; their values on $s = 0$ are set to zero and do not affect convolution integrals. A general multiplier kernel need not have zero right trace at $s = 0$, even when $y \neq 0$. Accordingly no classical time derivative or cross-zero mean-value estimate is asserted there. By linearity it suffices to treat $K = 1$ and to sum the resulting bounds, so we write $\varsigma = \varsigma_1$, $G = G_1$. Note that $\varsigma(D)W_+$ is a locally integrable function by (100) ($\iint_{\{\varrho < R\}} \varrho^{-4} = \frac{40\pi}{3} R < \infty$), so $W_+ * \varsigma(D)G = (\varsigma(D)W_+) * G$.

Fix z and $r > 0$. Split $h = h^{\text{near}} + \sum_{j \geq 6} h_j$, where

$$h^{\text{near}} = W_+ * (F^{\text{near}} + \varsigma(D)G^{\text{near}}), \quad F^{\text{near}} = \mathbf{1}_{\mathfrak{B}_{2^6 r}(z)} F, \quad G^{\text{near}} = \mathbf{1}_{\mathfrak{B}_{2^6 r}(z)} G,$$

and, for $j \geq 6$, $\Gamma_j = \mathfrak{B}_{2^{j+1}r}(z) \setminus \mathfrak{B}_{2^j r}(z)$, $F_j = \mathbf{1}_{\Gamma_j} F$, $G_j = \mathbf{1}_{\Gamma_j} G$, $h_j = W_+ * (F_j + \varsigma(D)G_j)$. The sum over $j \geq 6$ is finite because F and G have bounded support. We estimate

$$\mathcal{E}_j(z, r) = \left(\frac{1}{|\mathfrak{B}_r(z)|^2} \iint_{\mathfrak{B}_r(z)} \iint_{\mathfrak{B}_r(z)} |h_j(w) - h_j(w')|^P dw dw' \right)^{1/P}. \quad (101)$$

Near part, treated as one block. No cancellation is used. For $w \in \mathfrak{B}_r(z)$ and $v \in \mathfrak{B}_{2^6 r}(z)$ we have $d_{\text{par}}(w, v) < 2^7 r$, so Young's inequality with the truncated kernels gives

$$\|h^{\text{near}}\|_{L^P(\mathfrak{B}_r(z))} \leq \|\mathbf{1}_{\varrho < 2^7 r} \varrho^{-3}\|_{L^1} \|F^{\text{near}}\|_{L^P} + C_\varsigma \|\mathbf{1}_{\varrho < 2^7 r} \varrho^{-4}\|_{L^1} \|G^{\text{near}}\|_{L^P}.$$

Since $|\mathfrak{B}_s| = \frac{8\pi}{3} s^5$, we have $\iint_{\{\varrho < R\}} \varrho^{-3} = \frac{40\pi}{3} \int_0^R s ds = \frac{20\pi}{3} R^2$ and $\iint_{\{\varrho < R\}} \varrho^{-4} = \frac{40\pi}{3} R$, so the two kernel norms are $C r^2$ and $C r$. By Definition 5.2, $\|F^{\text{near}}\|_{L^P} \leq (2^6 r)^{5(\frac{1}{P} - \frac{1}{\theta_0})} \|F\|_{\mathcal{M}^{P, \theta_0}}$ and $\|G^{\text{near}}\|_{L^P} \leq (2^6 r)^{5(\frac{1}{P} - \frac{1}{\theta_1})} \|G\|_{\mathcal{M}^{P, \theta_1}}$. Using $\mathcal{E}^{\text{near}} \leq 2 |\mathfrak{B}_r|^{-1/P} \|h^{\text{near}}\|_{L^P(\mathfrak{B}_r(z))}$, $|\mathfrak{B}_r|^{-1/P} = C r^{-5/P}$, and $2 - 5/\theta_0 = 1 - 5/\theta_1 = \gamma$ from (96),

$$\mathcal{E}^{\text{near}}(z, r) \leq C r^\gamma (\|F\|_{\mathcal{M}^{P, \theta_0}} + \|G\|_{\mathcal{M}^{P, \theta_1}}).$$

No series and no constraint beyond $P \leq \theta_0$ is involved.

Far shells, $j \geq 6$. Write $L = 2^j r$, let $w, w' \in \mathfrak{B}_r(z)$ and split Γ_j into the regions where $t_w - s$ and $t_{w'} - s$ are both positive, both nonpositive, or have opposite signs. Spatial separation along the intervening segments, or time separation when it is the larger coordinate, is comparable to L . On the first region the separate space and time mean-value estimates give

$$\begin{aligned} |W_+(w - v) - W_+(w' - v)| &\leq C d_{\text{par}}(w, w') L^{-4}, \\ |(\varsigma(D)W_+)(w - v) - (\varsigma(D)W_+)(w' - v)| &\leq C d_{\text{par}}(w, w') L^{-5}. \end{aligned}$$

The contribution from the second region vanishes. By Hölder and the Morrey bounds, the first region contributes at most

$$C r L^{\gamma-1} (\|F\|_{\mathcal{M}^{P, \theta_0}} + \|G\|_{\mathcal{M}^{P, \theta_1}}).$$

Here $1 - 5/\theta_0 = \gamma - 1$ and $-5/\theta_1 = \gamma - 1$.

On the crossing region the source time s lies between t_w and $t_{w'}$, an interval of length at most $2r^2$. Since $j \geq 6$, membership in Γ_j then forces spatial separation comparable to L . Cover

this spatial shell and time interval by at most $C(L/r)^3$ parabolic balls of radius Cr . Hölder on each such ball gives

$$\iint_{\text{crossing region}} |G| \leq C(L/r)^3 r^{5-5/\theta_1} \|G\|_{\mathcal{M}^{P,\theta_1}} = CL^3 r^{1+\gamma} \|G\|_{\mathcal{M}^{P,\theta_1}}.$$

The kernel size bound CL^{-4} therefore bounds its crossing contribution by $C(r/L)r^\gamma \|G\|_{\mathcal{M}^{P,\theta_1}}$. This estimate does not differentiate through time zero. For the Gaussian part, the zero extension and its time derivative vanish at $t = 0$ at nonzero spatial separation. Alternatively, $W_+(x, \tau) \leq C\tau L^{-5}$ for $|x| \asymp L$, $0 < \tau \leq 2r^2$, by the exponential formula. The same cover yields

$$\iint_{\text{crossing region}} |F| \leq CL^3 r^\gamma \|F\|_{\mathcal{M}^{P,\theta_0}},$$

so this contribution is at most $C(r/L)^2 r^\gamma \|F\|_{\mathcal{M}^{P,\theta_0}}$. Consequently

$$\mathcal{E}_j(z, r) \leq Cr^\gamma (2^{j(\gamma-1)} + 2^{-j} + 2^{-2j}) (\|F\|_{\mathcal{M}^{P,\theta_0}} + \|G\|_{\mathcal{M}^{P,\theta_1}}).$$

All three shell series converge for $0 < \gamma < 1$.

For comparison with a pointwise shell-difference estimate, set $\delta = d_{\text{par}}(w, w')$. The case $\delta = 0$ is immediate. Otherwise repeat the crossing cover at radius $C\delta$ instead of Cr ; the time strip has length at most δ^2 . The two crossing contributions are bounded by $C\delta^{1+\gamma} L^{-1} \|G\|_{\mathcal{M}^{P,\theta_1}}$ and $C\delta^{2+\gamma} L^{-2} \|F\|_{\mathcal{M}^{P,\theta_0}}$. Since $\delta \leq 2r \leq L$, each is at most the corresponding source norm times $C\delta L^{\gamma-1}$. Together with the positive-time region, this gives

$$|h_j(w) - h_j(w')| \leq C\delta L^{\gamma-1} (\|F\|_{\mathcal{M}^{P,\theta_0}} + \|G\|_{\mathcal{M}^{P,\theta_1}}).$$

Conclusion. By the triangle inequality in $L^P(\mathfrak{B}_r \times \mathfrak{B}_r)$ and Jensen's inequality,

$$\left(\frac{1}{|\mathfrak{B}_r(z)|} \iint_{\mathfrak{B}_r(z)} |h - (h)_{\mathfrak{B}_r(z)}|^P \right)^{1/P} \leq \mathcal{E}^{\text{near}}(z, r) + \sum_{j \geq 6} \mathcal{E}_j(z, r) \leq Cr^\gamma (\|F\|_{\mathcal{M}^{P,\theta_0}} + \|G\|_{\mathcal{M}^{P,\theta_1}}),$$

uniformly in z and r . Lemma 5.11 now gives (98) with $C_{23} = C_{22}(\gamma, P) \cdot C$, and (99) follows by averaging the resulting Hölder bound over $\mathfrak{B}_R(z)$, as in part (e) of that lemma. Its coefficient is $2^\gamma C_{23}$, since C_{23} already includes C_{22} . For $K > 1$ apply the above to each pair $(0, G_k)$ and to $(F, 0)$ and add the $K + 1$ resulting seminorm bounds. $\square$

Remark 5.13 (Averages and the continuous representative). There is also a direct dyadic proof of Lemma 5.11, avoiding mollification. Set $a_r(z) = (h)_{\mathfrak{B}_r(z)}$. Hölder's inequality and the volume ratio give

$$|a_{r/2}(z) - a_r(z)| \leq 2^{5/P} H r^\gamma.$$

Telescoping at dyadic radii gives a locally uniform limit of the averages, equal to h almost everywhere by differentiation. To compare two centres at distance r , compare both radius- r averages to the average on a radius- $3r$ ball containing both balls. This gives a bound CHr^γ ; the two telescoping tails give the same bound. The resulting single representative is Hölder everywhere. Averaging its Hölder estimate gives (92). This is an alternative to the mollification proof above; its constants may depend on (γ, P) .

5.2. Step 1: the local energy and pressure estimates. All two-scale estimates in this section are applied with the closure of the outer cylinder inside the solution domain. Cutoff coefficients are fixed before the solution. The absolute energy constants and the q -dependent force constants may be enlarged to dominate the finitely many explicit cutoff and norm-comparison constants, without changing their stated dependence. The iteration uses one fixed contraction

κ chosen in (115); a bound proved only at that choice is not an assertion uniform over every contraction.

Lemma 5.14 (The backward Gaussian test function). *Let $G(x, \vartheta) = (4\pi\vartheta)^{-3/2}e^{-|x|^2/(4\vartheta)}$ and, for $r > 0$ and $z_0 = (x_0, t_0)$, set*

$$\psi_r(x, t) = r^2 G(x - x_0, r^2 - (t - t_0)), \quad (x, t) \in \mathbb{R}^3 \times (-\infty, t_0 + r^2). \quad (102)$$

Then ψ_r is smooth and positive there, and there is an absolute constant $C_{24} \geq 1$ such that, for all $0 < r \leq \rho/2$,

$$\partial_t \psi_r + \Delta \psi_r = 0 \quad \text{on } \mathbb{R}^3 \times (-\infty, t_0 + r^2), \quad (103)$$

$$\psi_r \geq C_{24}^{-1} r^{-1} \quad \text{on } Q_r(z_0), \quad (104)$$

$$\psi_r \leq C_{24} r^{-1}, \quad |\nabla \psi_r| \leq C_{24} r^{-2} \quad \text{on } \mathbb{R}^3 \times (-\infty, t_0], \quad (105)$$

$$\psi_r \leq C_{24} \frac{r^2}{\rho^3}, \quad |\nabla \psi_r| \leq C_{24} \frac{r^2}{\rho^4} \quad \text{on } Q_\rho(z_0) \setminus Q_{\rho/2}(z_0). \quad (106)$$

Proof. We may take $z_0 = (0, 0)$. Write $\vartheta = r^2 - t > 0$ for $t < r^2$ and $s = |x|^2/(4\vartheta)$. By External Input A.10, $\partial_\vartheta G = \Delta G$, so $\partial_t \psi_r = -r^2(\partial_\vartheta G)(x, \vartheta) = -r^2(\Delta G)(x, \vartheta) = -\Delta \psi_r$, which is (103).

(104). On Q_r we have $|x| < r$ and $t \in (-r^2, 0]$, so $\vartheta \in [r^2, 2r^2]$ and $s \leq r^2/(4r^2) = 1/4$. Hence $\psi_r \geq r^2(8\pi r^2)^{-3/2}e^{-1/4} = (8\pi)^{-3/2}e^{-1/4}r^{-1}$.

(105). For $t \leq 0$, $\vartheta \geq r^2$, so $\psi_r \leq r^2(4\pi\vartheta)^{-3/2} \leq (4\pi)^{-3/2}r^{-1}$. For the gradient, $\nabla \psi_r = -r^2 \frac{x}{2\vartheta} G(x, \vartheta)$, and $\frac{|x|}{2\vartheta} = \vartheta^{-1/2}s^{1/2}$, so

$$|\nabla \psi_r| = r^2(4\pi)^{-3/2}\vartheta^{-2}s^{1/2}e^{-s} \leq r^2(4\pi)^{-3/2}(2e)^{-1/2}\vartheta^{-2}, \quad (107)$$

using $\sup_{s \geq 0} s^{1/2}e^{-s} = (2e)^{-1/2}$. With $\vartheta \geq r^2$ this gives $|\nabla \psi_r| \leq (4\pi)^{-3/2}(2e)^{-1/2}r^{-2}$.

(106). Let $(x, t) \in Q_\rho \setminus Q_{\rho/2}$, so $t \in (-\rho^2, 0]$ and either (A) $t \leq -\rho^2/4$, or (B) $|x| \geq \rho/2$.

In case (A), $\vartheta = r^2 - t \geq \rho^2/4$, hence $\psi_r \leq r^2(4\pi\vartheta)^{-3/2} \leq r^2(\pi\rho^2)^{-3/2} = \pi^{-3/2}r^2\rho^{-3}$, and by (107) $|\nabla \psi_r| \leq (4\pi)^{-3/2}(2e)^{-1/2}r^2(4/\rho^2)^2 = 16(4\pi)^{-3/2}(2e)^{-1/2}r^2\rho^{-4}$.

In case (B), put $\varsigma = \rho^2/(16\vartheta)$, so that $s = |x|^2/(4\vartheta) \geq \varsigma$ and $\vartheta = \rho^2/(16\varsigma)$. Then

$$\begin{aligned} \psi_r &\leq r^2(4\pi\vartheta)^{-3/2}e^{-\varsigma} = r^2(4\pi)^{-3/2} \left(\frac{16\varsigma}{\rho^2} \right)^{3/2} e^{-\varsigma} \\ &= 64(4\pi)^{-3/2} \frac{r^2}{\rho^3} \varsigma^{3/2} e^{-\varsigma} \leq 64(4\pi)^{-3/2} \left(\frac{3}{2} \right)^{3/2} e^{-3/2} \frac{r^2}{\rho^3}, \end{aligned}$$

using $\sup_{\varsigma \geq 0} \varsigma^{3/2}e^{-\varsigma} = (3/2)^{3/2}e^{-3/2}$. For the gradient, (107) and $s^{1/2}e^{-s} \leq e^{-1/2}e^{-s/2} \leq e^{-1/2}e^{-\varsigma/2}$ (valid because $\sup_s s^{1/2}e^{-s/2} = e^{-1/2}$) give

$$|\nabla \psi_r| \leq r^2(4\pi)^{-3/2}e^{-1/2} \left(\frac{16\varsigma}{\rho^2} \right)^2 e^{-\varsigma/2} = 256(4\pi)^{-3/2}e^{-1/2} \frac{r^2}{\rho^4} \varsigma^2 e^{-\varsigma/2} \leq C \frac{r^2}{\rho^4}.$$

Taking C_{24} to be the largest of the finitely many constants produced in (105)–(106) and of $((8\pi)^{-3/2}e^{-1/4})^{-1}$, which is what makes (104) read $\psi_r \geq C_{24}^{-1}r^{-1}$, completes the proof. $\square$

Lemma 5.15 (Caccioppoli inequality; [13, (13.30)]). *There is an absolute constant C_{25} and a constant $C_{26} = C_{26}(q)$ such that for every suitable weak solution (u, p) on $\mathcal{O}$ with force $f \in L_{loc}^q$, $q > 5/2$, every z_0 and $\rho > 0$ with $Q_\rho(z_0) \subset \mathcal{O}$, and every $0 < r \leq \rho/2$, writing $\kappa = r/\rho$ and*

suppressing the base point z_0 ,

$$\begin{aligned} \alpha(r) + \beta(r) &\leq C_{25}\kappa\alpha(\rho) \\ &+ C_{25}\kappa^{-1}\alpha(\rho)^{1/2}\beta(\rho)^{1/2}\gamma(\rho)^{1/2} \\ &+ C_{25}\kappa^{-1}\delta(\rho)\gamma(\rho)^{1/2} + C_{26}\kappa^{-1/2}\gamma(\rho)^{1/2}\lambda(\rho)^{1/2}. \end{aligned} \quad (108)$$

Proof. Write $B_\varrho = B_\varrho(x_0)$ and $J_\rho = (t_0 - \rho^2, t_0)$. Since $t_0 \in I$ and I is open, there is $\epsilon_0 > 0$ with $[t_0 - \rho^2 + \frac{1}{2}\epsilon_0, t_0 + \epsilon_0] \subset I$; shrinking ϵ_0 we may and do also require

$$\epsilon_0 < \frac{1}{2}r^2, \quad (109)$$

which is legitimate because $r > 0$ is fixed before ϵ_0 is chosen and only smallness of ϵ_0 is used. By (109) the function ψ_r of Lemma 5.14 is smooth on a neighbourhood of $\overline{B_\rho(x_0)} \times [t_0 - \rho^2, t_0 + \epsilon_0]$, since that set lies in $\mathbb{R}^3 \times (-\infty, t_0 + r^2)$.

Step 1: the test function. Let $\eta_1 \in C_c^\infty(B_\rho)$ with $0 \leq \eta_1 \leq 1$, $\eta_1 \equiv 1$ on $B_{\rho/2}$, $\text{supp } \eta_1 \subset B_{3\rho/4}$ and $|\partial^a \eta_1| \leq C_{10}(|a|)\rho^{-|a|}$; this is Lemma 4.3 verbatim, up to renaming the two radii $13\rho/20 \rightsquigarrow \rho/2$ and $3\rho/4 \rightsquigarrow 3\rho/4$, which only changes the numerical constants. Let $\eta_2 \in C^\infty(\mathbb{R})$ with $0 \leq \eta_2 \leq 1$, $\eta_2 \equiv 0$ on $(-\infty, t_0 - \rho^2 + \frac{1}{4}\epsilon_0]$, $\eta_2 \equiv 1$ on $[t_0 - \rho^2/4, t_0 + \frac{1}{2}\epsilon_0]$, $\text{supp } \eta_2 \subset (t_0 - \rho^2 + \frac{1}{8}\epsilon_0, t_0 + \epsilon_0)$ and $|\eta_2'| + \rho^2 |\eta_2''| \leq C\rho^{-2}$ on $(-\infty, t_0]$, where the transition occupies the interval $[t_0 - \rho^2 + \frac{1}{4}\epsilon_0, t_0 - \rho^2/4]$ of length $\geq \rho^2/2$; such η_2 exists by mollifying a piecewise affine function as in Lemma 4.3. Put $\eta = \eta_1 \eta_2$ and

$$\phi = \psi_r \eta \in C_c^\infty(\mathcal{O}), \quad \phi \geq 0,$$

with ψ_r as in Lemma 5.14. Then $\eta \equiv 1$ on $Q_{\rho/2}(z_0)$, and $\nabla \eta$, $\partial_t \eta$, $\Delta \eta$ all vanish on $Q_{\rho/2}(z_0)$ and, for $t \leq t_0$, are supported in $Q_\rho(z_0) \setminus Q_{\rho/2}(z_0)$.

Step 2: the four terms. Apply Lemma 1.13 with this ϕ and $t \in (t_0 - r^2, t_0]$. Since $\partial_t \psi_r + \Delta \psi_r = 0$,

$$\partial_t \phi + \Delta \phi = (\partial_t \eta + \Delta \eta) \psi_r + 2 \nabla \eta \cdot \nabla \psi_r, \quad (110)$$

which vanishes on $Q_{\rho/2}(z_0)$ and, by (106) and the bounds on η , satisfies $|\partial_t \phi + \Delta \phi| \leq C\rho^{-2} \cdot C_{24}r^2\rho^{-3} + 2C\rho^{-1}C_{24}r^2\rho^{-4} \leq C'r^2\rho^{-5}$ on $Q_\rho(z_0)$, $t \leq t_0$. Also, by (105) and $\rho \geq 2r$,

$$|\nabla \phi| \leq \eta |\nabla \psi_r| + \psi_r |\nabla \eta| \leq C_{24}r^{-2} + C_{24}r^{-1}C\rho^{-1} \leq C'r^{-2} \quad \text{on } Q_\rho(z_0), \quad (111)$$

and $\phi \leq C_{24}r^{-1}$. Since $\text{supp } \phi(\cdot, s) \subset \overline{B_{3\rho/4}} =: E$ for every s , Lemma 1.12 supplies a single null set $N \subset J_\rho$ off which $\int_{B_\rho} u(\cdot, s) \cdot \nabla \psi = 0$ for all $\psi \in C_c^1(B_\rho)$ supported in E ; applying it to $\psi = \phi(\cdot, s)$ gives $\int_{B_\rho} \Gamma(s) u(\cdot, s) \cdot \nabla \phi(\cdot, s) dy = 0$ for $s \notin N$ and any function Γ of time alone. We use this with $\Gamma(s) = [|u|^2]_{B_\rho}(s)$. Therefore Lemma 1.13 gives, for almost every $t \in (t_0 - r^2, t_0]$,

$$\int |u(\cdot, t)|^2 \phi(\cdot, t) + 2 \int_{t_0 - \rho^2}^t \int |\nabla u|^2 \phi \leq I_1 + I_2 + I_3 + I_4,$$

where

$$\begin{aligned} I_1 &= \iint_{Q_\rho(z_0)} |u|^2 |\partial_s \phi + \Delta \phi|, \\ I_2 &= \iint_{Q_\rho(z_0)} \left| |u|^2 - [u^2]_{B_\rho} \right| |u| |\nabla \phi|, \\ I_3 &= 2 \iint_{Q_\rho(z_0)} |p| |u| |\nabla \phi|, \\ I_4 &= 2 \iint_{Q_\rho(z_0)} |f| |u| \phi. \end{aligned}$$

I_1 . By the bound on $\partial_s \phi + \Delta \phi$,

$$I_1 \leq C \frac{r^2}{\rho^5} \iint_{Q_\rho(z_0)} |u|^2 \leq C \frac{r^2}{\rho^5} \rho^2 \operatorname{ess\,sup}_{J_\rho} \int_{B_\rho} |u|^2 = C \frac{r^2}{\rho^3} \rho \alpha(\rho)^2 = C \kappa^2 \alpha(\rho)^2.$$

I_2 . By (111) and Hölder's inequality with $\frac{2}{3} + \frac{1}{3} = 1$,

$$I_2 \leq \frac{C}{r^2} \left\| |u|^2 - [u^2]_{B_\rho} \right\|_{L^{3/2}(Q_\rho(z_0))} \|u\|_{L^3(Q_\rho(z_0))}.$$

For a.e. s , apply the squared-velocity Sobolev–Poincaré estimate recorded after External Input A.5. The closed outer-cylinder containment supplies a slightly larger spatial ball on which the slice has its H^1 weak gradient. The estimate gives $\left\| |u|^2 - [u^2]_{B_\rho}(s) \right\|_{L^{3/2}(B_\rho)} \leq 2C \|u(\cdot, s)\|_{L^2(B_\rho)} \|\nabla u(\cdot, s)\|_{L^2(B_\rho)}$. Taking the $L^{3/2}$ norm in s over J_ρ and using Hölder in time as in (64),

$$\begin{aligned} &\left\| |u|^2 - [u^2]_{B_\rho} \right\|_{L^{3/2}(Q_\rho(z_0))} \\ &\leq C (\operatorname{ess\,sup} u) |J_\rho|^{1/6} \|\mathbf{g}\|_{L^2(J_\rho)} \\ &\leq C \rho^{1/2} \alpha(\rho) \cdot \rho^{1/3} \cdot \rho^{1/2} \beta(\rho) = C \rho^{4/3} \alpha(\rho) \beta(\rho), \end{aligned}$$

with $\mathbf{u}, \mathbf{g}$ as in (62). Since $\|u\|_{L^3(Q_\rho(z_0))} = \rho^{2/3} \gamma(\rho)$,

$$I_2 \leq \frac{C}{r^2} \rho^{4/3} \alpha(\rho) \beta(\rho) \rho^{2/3} \gamma(\rho) = C \frac{\rho^2}{r^2} \alpha(\rho) \beta(\rho) \gamma(\rho) = C \kappa^{-2} \alpha(\rho) \beta(\rho) \gamma(\rho).$$

I_3 . By (111) and Hölder, $I_3 \leq C r^{-2} \|p\|_{L^{3/2}(Q_\rho(z_0))} \|u\|_{L^3(Q_\rho(z_0))} = C r^{-2} \rho^{4/3} \delta(\rho)^2 \rho^{2/3} \gamma(\rho) = C \kappa^{-2} \delta(\rho)^2 \gamma(\rho)$.

I_4 . Since $\phi \leq C_{24} r^{-1}$ and $q > 5/2$, Hölder with $\frac{1}{q} + \frac{1}{q'} = 1$ and then Hölder on $Q_\rho(z_0)$ (using $q' < \frac{5}{3} < 3$) give

$$\begin{aligned} I_4 &\leq \frac{C}{r} \|f\|_{L^q(Q_\rho(z_0))} \|u\|_{L^{q'}(Q_\rho(z_0))} \\ &\leq \frac{C}{r} \rho^{-\sigma} \lambda(\rho) \cdot |Q_\rho(z_0)|^{\frac{1}{q'} - \frac{1}{3}} \rho^{2/3} \gamma(\rho) \\ &\leq \frac{C'}{r} \rho^{\frac{5}{q} - 3} \lambda(\rho) \rho^{\frac{10}{3} - \frac{5}{q}} \rho^{2/3} \gamma(\rho) = C' \kappa^{-1} \gamma(\rho) \lambda(\rho), \end{aligned}$$

where we used $|Q_\rho(z_0)|^{\frac{1}{q'} - \frac{1}{3}} = C \rho^{5(\frac{2}{3} - \frac{1}{q})} = C \rho^{\frac{10}{3} - \frac{5}{q}}$ and $\frac{5}{q} - 3 - \sigma$ cancels.

Step 3: the left side. On $Q_r(z_0)$ we have $\eta = 1$ and $\psi_r \geq C_{24}^{-1} r^{-1}$, so $\int |u(\cdot, t)|^2 \phi(\cdot, t) \geq C_{24}^{-1} r^{-1} \int_{B_r} |u(\cdot, t)|^2$ for $t \in (t_0 - r^2, t_0]$, and $2 \int_{t_0 - \rho^2}^{t_0} \int |\nabla u|^2 \phi \geq 2 C_{24}^{-1} r^{-1} \iint_{Q_r(z_0)} |\nabla u|^2 = 2 C_{24}^{-1} \beta(r)^2$. Taking the essential supremum over the full-measure set of good times $t \in (t_0 - r^2, t_0)$ bounds the terminal energy term. For dissipation, choose good times $t_n \uparrow t_0$, discard the

nonnegative terminal energy, and use monotone convergence on $\mathbf{1}_{(t_0-\rho^2, t_n)} |\nabla u|^2 \phi$. This gives the integral up to t_0 without assuming that t_0 is a good time. Since $I_1, \dots, I_4$ do not depend on the terminal time,

$$\alpha(r)^2 + \beta(r)^2 \leq \frac{3}{2} C_{24} (I_1 + I_2 + I_3 + I_4) \leq C [\kappa^2 \alpha(\rho)^2 + \kappa^{-2} \alpha \beta \gamma + \kappa^{-2} \delta^2 \gamma + \kappa^{-1} \gamma \lambda] (\rho).$$

Taking square roots and using $\alpha(r) + \beta(r) \leq \sqrt{2}(\alpha(r)^2 + \beta(r)^2)^{1/2}$ and $(s_1 + s_2 + s_3 + s_4)^{1/2} \leq \sum s_i^{1/2}$ gives (108). $\square$

Remark 5.16 (Comparison with the sources). Inequality (108) is [9, (2.18)] with the force term in the L^q form of [9, Theorem 2], and it is [13, (13.30)] after the substitutions $U_\rho = \rho \alpha(\rho)^2$, $V_\rho = \rho \beta(\rho)^2$, $W_\rho = \rho^2 \gamma(\rho)^3$, $P_\rho = \rho^2 \delta(\rho)^3$. We have used Kukavica's test function $\psi_r \eta$ rather than Lemarié-Rieusset's $r^3 \omega \theta H(4r^2 + t - s, x - y)$; the two differ by the normalizing power of r (his ψ is $\approx r$ times ours, matching his un-normalized U_r, V_r) and by the auxiliary cut-off θ in time, which is not needed here because Lemma 1.13 already provides the pointwise-in-time inequality. The four terms and their exponents agree.

5.3. Step 2: the scale iteration.

Lemma 5.17 (Combined decay inequality). *For $\kappa \in (0, \frac{1}{2}]$ and $Q_\rho(z) \subset \mathcal{O}$ set*

$$\theta(z, \rho) = \alpha(z, \rho) + \beta(z, \rho) + \kappa^{-4} \delta(z, \rho)^2. \quad (112)$$

There are an absolute constant C_{27} and a constant $C_{28} = C_{28}(q)$ such that for every such z, ρ, κ , suppressing z ,

$$\theta(\kappa \rho) \leq C_{27} \kappa^{2/3} \theta(\rho) + C_{27} \kappa^{-5} (\beta(\rho)^{1/2} + \beta(\rho)) \theta(\rho) + C_{28} \kappa^{-1/2} \theta(\rho)^{1/2} \lambda(\rho)^{1/2} + C_{28} \kappa^{-3} \lambda(\rho), \quad (113)$$

and, if in addition $\theta(\rho) \leq 1$,

$$\theta(\kappa \rho) \leq C_{27} \kappa^{2/3} \theta(\rho) + 2C_{27} \kappa^{-5} \theta(\rho)^{1/2} \theta(\rho) + C_{28} \kappa^{-1/2} \theta(\rho)^{1/2} \lambda(\rho)^{1/2} + C_{28} \kappa^{-3} \lambda(\rho). \quad (114)$$

Proof. Write $r = \kappa \rho$ and suppress ρ on the right. All of α, β and $\kappa^{-4} \delta^2$ are $\leq \theta$ by definition.

Pressure part. By Theorem 4.7 and $(s_1 + s_2 + s_3)^2 \leq 3(s_1^2 + s_2^2 + s_3^2)$,

$$\kappa^{-4} \delta(r)^2 \leq 3C_{14}^2 \kappa^{-5} \alpha \beta + 3C_{14}^2 \kappa^{2/3} (\kappa^{-4} \delta^2) + 3C_{15}^2 \kappa^{-3} \lambda \leq 3C_{14}^2 \kappa^{-5} \beta \theta + 3C_{14}^2 \kappa^{2/3} \theta + 3C_{15}^2 \kappa^{-3} \lambda.$$

Energy part. By Corollary 3.8, $\gamma \leq C_9(\alpha^{1/2} \beta^{1/2} + \alpha) \leq C_9(\theta^{1/2} \beta^{1/2} + \theta)$ and hence $\gamma^{1/2} \leq C_9^{1/2}(\theta^{1/4} \beta^{1/4} + \theta^{1/2})$. Insert this into (108), term by term, using $\beta \leq \theta$ and $\delta \leq \kappa^2 \theta^{1/2}$ (from $\kappa^{-4} \delta^2 \leq \theta$):

$$\begin{aligned} \kappa \alpha &\leq \kappa \theta; \\ \kappa^{-1} \alpha^{1/2} \beta^{1/2} \gamma^{1/2} &\leq C_9^{1/2} \kappa^{-1} \theta^{1/2} \beta^{1/2} (\theta^{1/4} \beta^{1/4} + \theta^{1/2}) \leq 2C_9^{1/2} \kappa^{-1} \beta^{1/2} \theta; \\ \kappa^{-1} \delta \gamma^{1/2} &\leq \kappa \theta^{1/2} C_9^{1/2} (\theta^{1/4} \beta^{1/4} + \theta^{1/2}) \leq 2C_9^{1/2} \kappa \theta; \\ \kappa^{-1/2} \gamma^{1/2} \lambda^{1/2} &\leq C_9^{1/2} \kappa^{-1/2} (\theta^{1/4} \beta^{1/4} + \theta^{1/2}) \lambda^{1/2} \leq 2C_9^{1/2} \kappa^{-1/2} \theta^{1/2} \lambda^{1/2}, \end{aligned}$$

where in the second line we used $\beta^{1/4} \theta^{1/4} \leq \theta^{1/2}$ and in the third and fourth $\beta^{1/4} \theta^{1/4} \leq \theta^{1/2}$ again. Hence

$$\alpha(r) + \beta(r) \leq C_{25} (1 + 2C_9^{1/2}) \kappa \theta + 2C_{25} C_9^{1/2} \kappa^{-1} \beta^{1/2} \theta + 2C_{26} C_9^{1/2} \kappa^{-1/2} \theta^{1/2} \lambda^{1/2}.$$

Adding the two parts and using $\kappa \leq \kappa^{2/3}$ and $\kappa^{-1} \leq \kappa^{-5}$, valid for $\kappa \leq \frac{1}{2}$, gives (113) with $C_{27} = 3C_{14}^2 + C_{25}(1 + 2C_9^{1/2}) + 2C_{25} C_9^{1/2}$ and $C_{28} = 3C_{15}^2 + 2C_{26} C_9^{1/2}$ (sums, not maxima, so that nothing is lost when the two groups are added). If $\theta(\rho) \leq 1$ then $\beta \leq \theta \leq \theta^{1/2}$, so $\beta^{1/2} + \beta \leq 2\theta^{1/2}$, which gives (114). $\square$

Convention 5.18 (The constants $\kappa, \eta, \varepsilon_*$). Recall $\varepsilon = 2/5$. Define, once and for all,

$$\kappa = \min\left\{\frac{1}{2}, (8C_{27})^{-\frac{1}{2/3-\varepsilon}}\right\}, \quad \eta = \min\left\{1, \left(\frac{\kappa^{5+\varepsilon}}{16C_{27}}\right)^2\right\}, \quad \varepsilon_* = \min\left\{1, \left(\frac{\kappa^{5+\varepsilon}}{32C_{27}}\right)^2\right\}, \quad (115)$$

so that

$$C_{27}\kappa^{2/3-\varepsilon} \leq \frac{1}{8}, \quad 2C_{27}\kappa^{-5-\varepsilon}\eta^{1/2} \leq \frac{1}{8}, \quad 2C_{27}\kappa^{-5-\varepsilon}(\varepsilon_*^{1/2} + \varepsilon_*) \leq \frac{1}{8}. \quad (116)$$

Since $\varepsilon = 2/5$ is fixed and C_{27} is absolute, κ, η and ε_* are *absolute constants*. Finally set

$$C_{29} = C_{29}(q) = 2C_{28}^2\kappa^{-1-2\varepsilon} + C_{28}\kappa^{-3-\varepsilon}, \quad \Lambda_0 = \Lambda_0(q) = \frac{\eta}{2C_{29}}. \quad (117)$$

Proposition 5.19 (The iteration). *Let $z \in \mathcal{O}$ and $r_5 > 0$ with $Q_{r_5}(z) \subset \mathcal{O}$, and suppose*

$$\theta(z, r_5) \leq \eta \quad \text{and} \quad \lambda(z, r_5) \leq \Lambda_0. \quad (118)$$

Then $\theta(z, \kappa^n r_5) \leq \eta \kappa^{n\varepsilon}$ for every $n \in \mathbb{N}_0$, and consequently

$$\max\{\alpha(z, r), \beta(z, r), \delta(z, r)^2\} \leq \theta(z, r) \leq M r^\varepsilon, \quad 0 < r \leq r_5, \quad M := \kappa^{-4/3-\varepsilon} \eta r_5^{-\varepsilon}. \quad (119)$$

Proof. Put $\rho_n = \kappa^n r_5$, $\theta_n = \theta(z, \rho_n)$, $\lambda_n = \lambda(z, \rho_n)$, $T_n = \theta_n \kappa^{-n\varepsilon}$ and $L_n = \lambda_n \kappa^{-n\varepsilon}$. By Lemma 3.9, $\lambda_n \leq \kappa^{n\sigma} \lambda_0$, and $\sigma > 1 > \varepsilon$ gives $L_n \leq \kappa^{n(\sigma-\varepsilon)} \lambda_0 \leq \Lambda_0$ for all n .

We prove $T_n \leq \eta$ by induction; $T_0 = \theta(z, r_5) \leq \eta$ by hypothesis. Assume $T_n \leq \eta$; then $\theta_n \leq \eta \leq 1$, so (114) applies with $\rho = \rho_n$. Dividing it by $\kappa^{(n+1)\varepsilon}$,

$$T_{n+1} \leq C_{27}\kappa^{2/3-\varepsilon}T_n + 2C_{27}\kappa^{-5-\varepsilon}\theta_n^{1/2}T_n + C_{28}\kappa^{-1/2-\varepsilon}T_n^{1/2}L_n^{1/2} + C_{28}\kappa^{-3-\varepsilon}L_n.$$

By (116) the first term is $\leq \frac{1}{8}T_n$ and, since $\theta_n^{1/2} \leq \eta^{1/2}$, the second is $\leq \frac{1}{8}T_n$. For the third, Young's inequality $ab \leq \frac{1}{8}a^2 + 2b^2$ with $a = T_n^{1/2}$ and $b = C_{28}\kappa^{-1/2-\varepsilon}L_n^{1/2}$ gives $\leq \frac{1}{8}T_n + 2C_{28}^2\kappa^{-1-2\varepsilon}L_n$. Hence

$$T_{n+1} \leq \frac{3}{8}T_n + C_{29}L_n \leq \frac{3}{8}\eta + C_{29}\Lambda_0 = \frac{3}{8}\eta + \frac{1}{2}\eta = \frac{7}{8}\eta \leq \eta,$$

completing the induction.

For (119), let $0 < r \leq r_5$ and choose $n \in \mathbb{N}_0$ with $\kappa^{n+1}r_5 < r \leq \kappa^n r_5$. By Lemma 3.9 with $\rho_1 = r$, $\rho_2 = \rho_n$ and $\rho_n/r < \kappa^{-1}$,

$$\alpha(z, r) \leq \kappa^{-1/2}\alpha(z, \rho_n), \quad \beta(z, r) \leq \kappa^{-1/2}\beta(z, \rho_n), \quad \delta(z, r)^2 \leq \kappa^{-4/3}\delta(z, \rho_n)^2,$$

so $\theta(z, r) \leq \kappa^{-4/3}\theta(z, \rho_n) \leq \kappa^{-4/3}\eta\kappa^{n\varepsilon}$. Finally $\kappa^n < r/(\kappa r_5)$ gives $\kappa^{n\varepsilon} < \kappa^{-\varepsilon}r_5^{-\varepsilon}r^\varepsilon$, whence $\theta(z, r) \leq \kappa^{-4/3-\varepsilon}\eta r_5^{-\varepsilon}r^\varepsilon$. $\square$

Lemma 5.20 (Upper semicontinuity of α in the base point). *Let $0 < r < R$ and $t_1 < t_2$ be such that $\overline{B_R(x_0)} \times [t_1, t_2] \subset \mathcal{O}$ and $t_2 \in I$. Then*

$$\lim_{h \rightarrow 0^+} \operatorname{ess\,sup}_{s \in (t_1, t_2+h)} \int_{B_r(x_0)} |u(y, s)|^2 dy \leq \operatorname{ess\,sup}_{s \in (t_1, t_2)} \int_{B_R(x_0)} |u(y, s)|^2 dy. \quad (120)$$

Proof. Write S for the right side, which is finite by (S1). Choose $\epsilon_0 > 0$ with $[t_1, t_2 + \epsilon_0] \subset I$ and $\overline{B_R(x_0)} \times [t_1, t_2 + \epsilon_0] \subset \mathcal{O}$; this is possible because $\mathcal{O}$ is open and the left-hand set is compact. Let $\psi_1 \in C_c^\infty(B_R(x_0))$ with $0 \leq \psi_1 \leq 1$ and $\psi_1 \equiv 1$ on $B_r(x_0)$. Define

$$\Xi(a, b) = \int_a^b \int_{B_R(x_0)} \left[|u|^2 |\Delta \psi_1| + (|u|^2 + 2|p|) |u| |\nabla \psi_1| + 2|f| |u| \right] dy ds,$$

which is finite for $[a, b] \subset [t_1, t_2 + \epsilon_0]$ by the integrability checks in Step 2 of the proof of Lemma 1.13; by absolute continuity of the integral, $\Xi(a, b) \rightarrow 0$ as $b - a \rightarrow 0$.

Fix $0 < h < \min\{\epsilon_0/2, t_2 - t_1\}$ and let $T \in (t_2, t_2 + h)$. Let $T_0 \in (t_2 - h, t_2)$ and $h' \in (0, t_2 - T_0)$, and let $\chi = \chi_{T_0, h'}$ be a smooth nondecreasing function with $\chi \equiv 0$ on $(-\infty, T_0]$, $\chi \equiv 1$ on $[T_0 + h', \infty)$ and $0 \leq \chi' \leq 2/h'$ supported in $(T_0, T_0 + h')$; such χ is obtained by mollifying the piecewise affine function as in (20). Multiply χ by a smooth time cut-off equal to 1 on $(-\infty, t_2 + \epsilon_0/2]$ and vanishing near $t_2 + \epsilon_0$, which does not affect the computation below because we integrate only up to $T < t_2 + \epsilon_0/2$. Then $\phi(y, s) = \psi_1(y)\chi(s) \in C_c^\infty(\mathcal{O})$ is nonnegative and vanishes for s near t_1 , so Lemma 1.13 applies for almost every T (for each fixed cutoff). Since $\chi(T) = 1$, $\partial_s \phi = \psi_1 \chi'$, $\Delta \phi = \chi \Delta \psi_1$, $\nabla \phi = \chi \nabla \psi_1$ and $0 \leq \chi \leq 1$, discarding the (nonnegative) dissipation term gives

$$\int_{B_r(x_0)} |u(y, T)|^2 dy \leq \int |u(y, T)|^2 \psi_1 dy \leq \int_{T_0}^{T_0+h'} \int_{B_R(x_0)} |u|^2 \chi' dy ds + \Xi(T_0, T).$$

The first term on the right is at most $\frac{2}{h'} \int_{T_0}^{T_0+h'} (\int_{B_R(x_0)} |u(\cdot, s)|^2) ds \leq 2S$ provided $(T_0, T_0+h') \subset (t_1, t_2)$. To remove the factor 2, choose instead T_0 to be a Lebesgue point in $(t_2 - h, t_2)$ of the function $s \mapsto \int_{B_R(x_0)} |u(\cdot, s)|^2$ (almost every point is one) and let $h' \rightarrow 0^+$ through a fixed sequence, intersecting the countably many full-measure sets of terminal times: the first term then converges to $\int_{B_R(x_0)} |u(\cdot, T_0)|^2 \leq S$ if T_0 is moreover a point where the essential supremum bound holds, which is the case for a.e. $T_0 \in (t_1, t_2)$. Hence

$$\int_{B_r(x_0)} |u(y, T)|^2 dy \leq S + \Xi(t_2 - h, t_2 + h) \quad \text{for a.e. } T \in (t_2, t_2 + h).$$

The same bound holds trivially for a.e. $T \in (t_1, t_2)$ since $B_r(x_0) \subset B_R(x_0)$. Taking the essential supremum over $T \in (t_1, t_2 + h)$ and letting $h \rightarrow 0^+$, so that $\Xi(t_2 - h, t_2 + h) \rightarrow 0$, gives (120). $\square$

Corollary 5.21. *Let $z_0 \in \mathcal{O}$ and $0 < \varrho < R$ with $\overline{B_R(x_0)} \times [t_0 - R^2, t_0] \subset \mathcal{O}$ and $t_0 \in I$. Then*

$$\limsup_{z \rightarrow z_0} \alpha(z, \varrho)^2 \leq \frac{R}{\varrho} \alpha(z_0, R)^2, \quad \lim_{z \rightarrow z_0} \beta(z, \varrho) = \beta(z_0, \varrho), \quad \lim_{z \rightarrow z_0} \delta(z, \varrho) = \delta(z_0, \varrho). \quad (121)$$

Proof. Let $z = (x, t) \rightarrow z_0$. For z close enough to z_0 we have $B_\varrho(x) \subset B_R(x_0)$ and $(t - \varrho^2, t) \subset (t_0 - R^2, t_0 + h)$ for any prescribed $h > 0$. Hence

$$\alpha(z, \varrho)^2 = \frac{1}{\varrho} \operatorname{ess\,sup}_{(t-\varrho^2, t)} \int_{B_\varrho(x)} |u|^2 \leq \frac{1}{\varrho} \operatorname{ess\,sup}_{(t_0-R^2, t_0+h)} \int_{B_R(x_0)} |u|^2,$$

Since $\varrho < R$ strictly, fix an intermediate radius r with $\varrho < r < R$. For z with $|x - x_0| < r - \varrho$ we have $B_\varrho(x) \subset B_r(x_0)$, so

$$\alpha(z, \varrho)^2 \leq \frac{1}{\varrho} \operatorname{ess\,sup}_{s \in (t_0-R^2, t_0+h)} \int_{B_r(x_0)} |u(y, s)|^2 dy.$$

Letting $z \rightarrow z_0$ and then $h \rightarrow 0^+$ and applying Lemma 5.20 with $t_1 = t_0 - R^2$, $t_2 = t_0$, inner radius r and outer radius R (so that $r < R$ as the lemma requires) gives $\limsup_{z \rightarrow z_0} \alpha(z, \varrho)^2 \leq \varrho^{-1} \operatorname{ess\,sup}_{(t_0-R^2, t_0)} \int_{B_R(x_0)} |u|^2 = (R/\varrho) \alpha(z_0, R)^2$. For β and δ , the maps $z \mapsto \iint_{Q_\varrho(z)} |\nabla u|^2$ and $z \mapsto \iint_{Q_\varrho(z)} |p|^{3/2}$ are continuous, because $|Q_\varrho(z) \triangle Q_\varrho(z_0)| \rightarrow 0$ as $z \rightarrow z_0$ and $|\nabla u|^2, |p|^{3/2} \in L^1$ near z_0 , so absolute continuity of the integral applies. $\square$

Proposition 5.22 (Step 2: Morrey decay on a neighbourhood). *Let $q > 5/2$, $\varepsilon = 2/5$, and let $\kappa, \eta, \varepsilon_*, \Lambda_0$ be as in Convention 5.18. Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force $f \in L^q_{loc}$, and let $z_0 \in \mathcal{O}$ satisfy*

$$\limsup_{r \rightarrow 0^+} \beta(z_0, r) < \varepsilon_*. \quad (122)$$

Then there are $r_2 > 0$ with $\mathfrak{B}_{2r_2}(z_0) \subset \mathcal{O}$ and $M \geq 1$ such that

$$\max\{\alpha(z, r), \beta(z, r), \delta(z, r)^2\} \leq Mr^\varepsilon \quad \text{for all } z \in \mathfrak{B}_{r_2}(z_0) \text{ and } 0 < r < r_2. \quad (123)$$

Proof. Step (a): descent at z_0 . Choose $r_4 > 0$ with $\overline{\mathfrak{B}_{4r_4}(z_0)} \subset \mathcal{O}$ and $\beta(z_0, \rho) \leq \varepsilon_*$ for all $\rho \leq r_4$; this is possible by (122). Put $M_f = \|f\|_{L^q(\mathfrak{B}_{4r_4}(z_0))}$, so that $\lambda(z, \rho) \leq M_f \rho^\sigma$ whenever $Q_\rho(z) \subset \mathfrak{B}_{4r_4}(z_0)$. Put $T_n = \theta(z_0, \kappa^n r_4) \kappa^{-n\varepsilon}$ and $L_n = \lambda(z_0, \kappa^n r_4) \kappa^{-n\varepsilon}$, so that $L_n \leq M_f r_4^\sigma \kappa^{n(\sigma-\varepsilon)}$. Apply (113) (the hypothesis $\theta(z_0, \cdot) \leq \eta$ of (114) is not yet known) at z_0 with $\rho = \kappa^n r_4$ and divide by $\kappa^{(n+1)\varepsilon}$:

$$\begin{aligned} T_{n+1} &\leq C_{27} \kappa^{2/3-\varepsilon} T_n \\ &\quad + C_{27} \kappa^{-5-\varepsilon} (\beta(z_0, \kappa^n r_4)^{1/2} + \beta(z_0, \kappa^n r_4)) T_n \\ &\quad + C_{28} \kappa^{-1/2-\varepsilon} T_n^{1/2} L_n^{1/2} + C_{28} \kappa^{-3-\varepsilon} L_n. \end{aligned}$$

Since $\kappa^n r_4 \leq r_4$ we have $\beta(z_0, \kappa^n r_4) \leq \varepsilon_*$ for every n , so the second term is at most

$$C_{27} \kappa^{-5-\varepsilon} (\varepsilon_*^{1/2} + \varepsilon_*) T_n \leq \frac{1}{16} T_n$$

by the third inequality of (116); the first is $\leq \frac{1}{8} T_n$ by the first inequality there; and the third is $\leq \frac{1}{8} T_n + 2C_{28}^2 \kappa^{-1-2\varepsilon} L_n$ by Young's inequality $ab \leq \frac{1}{8} a^2 + 2b^2$. Hence $T_{n+1} \leq \frac{5}{16} T_n + C_{29} L_n \leq \frac{3}{8} T_n + C_{29} L_n$, and

$$T_n \leq \left(\frac{3}{8}\right)^n T_0 + C_{29} M_f r_4^\sigma \sum_{k=0}^{n-1} \left(\frac{3}{8}\right)^{n-1-k} \kappa^{k(\sigma-\varepsilon)} \leq \left(\frac{3}{8}\right)^n T_0 + C M_f r_4^\sigma, \quad C := \frac{8}{5} C_{29},$$

because $\sum_{k=0}^{n-1} (3/8)^{n-1-k} \kappa^{k(\sigma-\varepsilon)} \leq \sum_{m \geq 0} (3/8)^m = \frac{8}{5}$ (each factor $\kappa^{k(\sigma-\varepsilon)} \leq 1$). Shrinking r_4 so that $C M_f r_4^\sigma \leq \eta/8$ and choosing n_0 with $(3/8)^{n_0} T_0 \leq \eta/8$ (possible since $T_0 = \theta(z_0, r_4) < \infty$ by (S1)), we obtain

$$\theta(z_0, R_0) \leq \frac{1}{4} \eta \kappa^{n_0 \varepsilon} \leq \frac{1}{4} \eta, \quad R_0 := \kappa^{n_0} r_4. \quad (124)$$

Step (b): transfer to nearby centres. Let $\varrho \in (0, R_0)$ with $(R_0/\varrho)^{4/3+\varepsilon} \leq 2$. By Lemma 3.9, $\beta(z_0, \varrho) \leq (R_0/\varrho)^{1/2} \beta(z_0, R_0)$ and $\delta(z_0, \varrho)^2 \leq (R_0/\varrho)^{4/3} \delta(z_0, R_0)^2$, and by Corollary 5.21 applied with $R = R_0$,

$$\limsup_{z \rightarrow z_0} \theta(z, \varrho) \leq \left(\frac{R_0}{\varrho}\right)^{4/3} \theta(z_0, R_0) \leq 2 \cdot \frac{1}{4} \eta = \frac{1}{2} \eta.$$

Hence there is $r_2 \in (0, \varrho/4)$ with $\mathfrak{B}_{2r_2}(z_0) \subset \mathcal{O}$ and

$$\theta(z, \varrho) \leq \eta \quad \text{for every } z \in \mathfrak{B}_{r_2}(z_0). \quad (125)$$

Shrinking r_4 further at the start if necessary, we may also assume $\lambda(z, \varrho) \leq M_f \varrho^\sigma \leq \Lambda_0$ for all such z .

Step (c): iterate at every centre. For each $z \in \mathfrak{B}_{r_2}(z_0)$, Proposition 5.19 with $r_5 = \varrho$ gives $\theta(z, r) \leq Mr^\varepsilon$ for $0 < r \leq \varrho$, with coefficient $\kappa^{-4/3-\varepsilon} \eta \varrho^{-\varepsilon}$ independent of z . Set $M = \max\{1, \kappa^{-4/3-\varepsilon} \eta \varrho^{-\varepsilon}\}$; the estimate remains valid and $M \geq 1$ as asserted. Since $r_2 < \varrho$, this is (123). $\square$

Lemma 5.23 (Step 2 in Morrey form). *Let z_0, r_2, M be as in Proposition 5.22 and put $Q_2 = \mathfrak{B}_{r_2/4}(z_0)$. Then, with τ_2, τ_3, τ_p as in (83),*

$$\mathbf{1}_{Q_2} u \in \mathcal{M}^{3, \tau_2}, \quad \mathbf{1}_{Q_2} \nabla u \in \mathcal{M}^{2, \tau_3}, \quad \mathbf{1}_{Q_2} p \in \mathcal{M}^{3/2, \tau_p}, \quad (126)$$

with norms bounded by $C(M, r_2)$. Explicitly $\tau_2 = 25/3 > 5$ and $\tau_3 = \tau_p = 25/8 > 5/2$, and $\frac{1}{\tau_3} = \frac{1}{\tau_2} + \frac{1}{5}$.

Proof. By Lemma 5.3 it suffices to bound the integrals over one-sided cylinders $Q_\varrho(z)$, $z \in \mathbb{R}^3 \times \mathbb{R}$, $\varrho > 0$. If $Q_\varrho(z) \cap Q_2 = \emptyset$ there is nothing to prove.

Small radii, $\varrho \leq r_2/8$. Pick $z' = (x', t') \in Q_\varrho(z) \cap Q_2$, so that $Q_\varrho(z) \subset Q_{2\varrho}((x', t' + \varrho^2))$ by the triangle inequality for d_{par} and Lemma 1.3. The centre $z'' = (x', t' + \varrho^2)$ satisfies $d_{\text{par}}(z'', z_0) \leq d_{\text{par}}(z', z_0) + \varrho < r_2/4 + r_2/8 < r_2$, so $z'' \in \mathfrak{B}_{r_2}(z_0)$, and $\varrho' = 2\varrho < r_2$. By Proposition 5.22 and Corollary 3.8,

$$\iint_{Q_\varrho(z)} |u|^3 \leq \iint_{Q_{\varrho'}(z'')} |u|^3 = \varrho'^2 \gamma(z'', \varrho')^3 \leq \varrho'^2 (2C_9 M \varrho'^\varepsilon)^3 = C M^3 \varrho'^{2+3\varepsilon},$$

and $2 + 3\varepsilon = 16/5 = 5(1 - 3/\tau_2)$ with $\tau_2 = 25/3$, which is the Morrey condition for $\mathcal{M}^{3, \tau_2}$. Similarly

$$\iint_{Q_\varrho(z)} |\nabla u|^2 \leq \varrho' \beta(z'', \varrho')^2 \leq M^2 \varrho'^{1+2\varepsilon}, \quad \iint_{Q_\varrho(z)} |p|^{3/2} \leq \varrho'^2 \delta(z'', \varrho')^3 \leq M^{3/2} \varrho'^{2+\frac{3}{2}\varepsilon},$$

and $1 + 2\varepsilon = 9/5 = 5(1 - 2/\tau_3)$ with $\tau_3 = 25/8$, while $2 + \frac{3}{2}\varepsilon = 13/5 = 5(1 - \frac{3/2}{\tau_p})$ with $\tau_p = 25/8$. Finally $\frac{1}{\tau_3} = \frac{8}{25} = \frac{3}{25} + \frac{5}{25} = \frac{1}{\tau_2} + \frac{1}{5}$.

Large radii, $\varrho > r_2/8$. Cover the bounded set Q_2 by finitely many cylinders of radius $r_2/8$ and apply the small-radius case to each: this gives $M_0 := \iint_{Q_2} |u|^3 < \infty$, and likewise $M'_0 := \iint_{Q_2} |\nabla u|^2 < \infty$ and $M''_0 := \iint_{Q_2} |p|^{3/2} < \infty$ (all three are finite already by (S1)). For $\varrho > r_2/8$,

$$\iint_{Q_\varrho(z)} |\mathbf{1}_{Q_2} u|^3 \leq M_0 = M_0 \left(\frac{r_2}{8}\right)^{-16/5} \left(\frac{r_2}{8}\right)^{16/5} \leq M_0 \left(\frac{r_2}{8}\right)^{-16/5} \varrho^{16/5},$$

which is the required Morrey bound with the constant enlarged by $M_0(r_2/8)^{-16/5}$; identically for ∇u with the exponent $9/5$ and for p with the exponent $13/5$. $\square$

Remark 5.24 (Comparison with [13, Lemma 13.4]). Lemarié-Rieusset states the conclusion of Step 2 as $\mathbf{1}_{Q_2} u \in \mathcal{M}^{3, \tau_2}$ and $\mathbf{1}_{Q_2} p \in \mathcal{M}^{q_0, \tau_2/2}$ under the restriction $1 < \tau_2/5 < \min\{q_0, 2\}$, equivalently $5 < \tau_2 < 5 \min\{q_0, 2\}$. This is the condition printed in his Lemma 13.4. In particular $\tau_2 < 5q_0$, so both terms in $\lambda = \min\{10/\tau_2, 10q_0/\tau_2 - 2\}$ are positive, as required by his iteration on p. 437. With $q_0 = 3/2$ that restriction forces $\tau_2 < 15/2$, i.e. $\varepsilon < 1/3$. We instead iterate Kukavica's functional $\theta = \alpha + \beta + \kappa^{-4}\delta^2$, in which the pressure is normalized by δ^2 rather than by P_+ ; this allows $\varepsilon = 2/5$ and hence $\tau_2 = 25/3$ for the velocity, at the price of the weaker pressure exponent $\tau_p = 25/8$ instead of $\tau_2/2 = 25/6$. The weaker exponent is sufficient: the pressure Morrey norm is used below only through fixed-scale $L^{3/2}$ integrability in Lemma 5.26, never through its Morrey exponent. The gradient bound $\mathbf{1}_{Q_2} \nabla u \in \mathcal{M}^{2, \tau_3}$, $\frac{1}{\tau_3} = \frac{1}{\tau_2} + \frac{1}{5}$, is the one extracted in [13, p. 439] from the same iteration. We use this gradient information explicitly in the pressure-gradient estimate and the velocity bootstrap below.

5.4. Step 3: the local representation and the Morrey bootstrap. Throughout §5.4–§5.5, z_0, r_2, M are as in Proposition 5.22, $Q_2 = \mathfrak{B}_{r_2/4}(z_0)$, and $\varphi \in C_c^\infty(\mathbb{R}^3 \times \mathbb{R})$ satisfies

$$\varphi \equiv 1 \text{ on } Q_3 := \mathfrak{B}_{r_3}(z_0), \quad \text{supp } \varphi \subset Q_2. \quad (127)$$

with $0 < r_3 < r_2/4$. We write $N(y) = -\frac{1}{4\pi|y|}$ as in Proposition 4.4, W_+ for the forward heat kernel of (97) (unit viscosity, as throughout), and I_1, I_2 for the parabolic Riesz potentials of Definition 5.6.

We keep the pressure gradient in the undifferentiated source of the localized heat equation. The pressure estimate below uses a fixed spatial localization and one measurable weak gradient; it does not require a commutator expansion.

Lemma 5.25 (The localized equation). *Set $v = \varphi u$. Then, in $\mathcal{D}'(\mathbb{R}^3 \times \mathbb{R})$,*

$$\partial_t v - \Delta v = \mathbf{g} + \sum_{i=1}^3 \partial_i \mathbf{h}_i - \varphi \nabla p, \quad \mathbf{g} = (\partial_t \varphi)u + (\Delta \varphi)u - \varphi u \cdot \nabla u + \varphi f, \quad \mathbf{h}_i = -2(\partial_i \varphi)u, \quad (128)$$

and the compact-support adjoint identity proved below gives

$$v = W_+ * \left(\mathbf{g} + \sum_i \partial_i \mathbf{h}_i - \varphi \nabla p \right). \quad (129)$$

Proof. By (S3), $\partial_t u - \Delta u = -u \cdot \nabla u - \nabla p + f$ in $\mathcal{D}'(\mathcal{O})$, where $u \cdot \nabla u = \text{div}(u \otimes u)$ makes sense as a distribution because $u \otimes u \in L_{loc}^{5/3}$. Multiplying by φ and using the Leibniz rules $\partial_t(\varphi u) = (\partial_t \varphi)u + \varphi \partial_t u$ and $\Delta(\varphi u) = (\Delta \varphi)u + 2\nabla \varphi \cdot \nabla u + \varphi \Delta u$, together with $2\nabla \varphi \cdot \nabla u = \sum_i \partial_i(2(\partial_i \varphi)u) - 2(\Delta \varphi)u$, gives (128). For (129), write $\mathcal{F}$ for the compactly supported distribution on the right-hand side of (128). Given a compactly supported smooth vector test ψ , let

$$\Psi(y, s) = \int_s^\infty \int_{\mathbb{R}^3} W_+(x - y, t - s) \psi(x, t) dx dt.$$

Then $-\partial_s \Psi - \Delta_y \Psi = \psi$. Choose a smooth compact cutoff equal to one on a neighbourhood of the supports of v and $\mathcal{F}$. Testing the localized equation with this cutoff times Ψ gives

$$\langle v, \psi \rangle = \langle \mathcal{F}, \Psi \rangle = \langle W_+ * \mathcal{F}, \psi \rangle.$$

The last equality is the defining adjoint convolution identity; for the locally integrable source terms it also follows from Fubini. The source support makes the cutoff immaterial in the middle pairing. Thus (129) holds distributionally, and almost everywhere whenever the source potential is represented by a locally integrable function. This proof requires neither an algebra of arbitrary causal distributions nor unrestricted uniqueness for the heat equation. $\square$

Lemma 5.26 (Localized pressure-gradient Morrey bound). *Let $\mathfrak{B}_{2R}(z) \subset \mathcal{O}$, $R > 0$, and $25/3 \leq \tau \leq 25$. If $\mathbf{1}_{\mathfrak{B}_R(z)} u \in \mathcal{M}^{3, \tau}$ and $\mathbf{1}_{\mathfrak{B}_R(z)} \nabla u \in \mathcal{M}^{2, 25/8}$, then the distributional spatial gradient of p has a jointly measurable representative Dp on $\mathfrak{B}_{R/2}(z)$, and*

$$\mathbf{1}_{\mathfrak{B}_{R/2}(z)} Dp \in \mathcal{M}^{6/5, \varkappa}, \quad \varkappa = \min \left\{ q, \left(\frac{1}{\tau} + \frac{8}{25} \right)^{-1} \right\}. \quad (130)$$

In particular, for every smooth test ψ compactly supported there, $\iint p \partial_i \psi = -\iint D_i p \psi$. The bounds are finite functions of the incoming Morrey norms, the local energy, pressure and force norms on a fixed surrounding cylinder, q, τ , and the localization radii. No bound is required on a shrinking-cell norm of the unnormalized pressure.

Proof. Write $z = (x, t)$ and fix the cylinder $Q = B_R(x) \times (t - 3R^2/4, t + R^2/4]$, which lies in $\mathfrak{B}_R(z)$ and whose closure lies in $\mathfrak{B}_{2R}(z)$. Let η be the spatial cutoff of Lemma 4.3, and let $c(s) = [u(\cdot, s)]_{B_R(x)}$. In the pressure decomposition of Proposition 4.4 use the singly centred tensor $U_{ij} = -u_i(u_j - c_j)$. Its differentiated, localized source is

$$V_i = \sum_j [\eta(u_j - c_j) \partial_j u_i + (\partial_j \eta) u_i (u_j - c_j)].$$

All sources in this proof are extended by zero outside the fixed time window of Q . The weak product rule, incompressibility and the completed second Riesz transforms give, on the inner cylinder,

$$D_i p = - \sum_j \partial_i \partial_j \Delta^{-1} (\mathbf{1}_Q V_j) + \sum_j \partial_i \partial_j \Delta^{-1} (\mathbf{1}_Q \eta f_j) + H_i, \quad H = \nabla(p_{\text{har}} + p_8). \quad (131)$$

Here $p_{\text{har}} = p_2 + \dots + p_6$ is the smooth harmonic remainder on the inner ball; it agrees almost everywhere there with $p - p_1 - p_7 - p_8$. The gradient in H is taken of this smooth representative and the smooth annular potential p_8 , not of an arbitrary representative of p . The operators are their $L^{6/5}$ extensions, not pointwise differentiation of a possibly undefined singular integral. Spatial mollification and almost-everywhere limits select jointly measurable representatives; the slice weak-derivative identities, followed by Fubini, give the asserted space-time pairing. Uniqueness of locally integrable weak derivatives identifies these representatives on overlapping cylinders.

We give the estimates on the three terms. The fixed spatial mean c is essentially bounded in time by the local energy bound. Thus its restriction to Q belongs to every Morrey class used below. Hölder's inequality gives $u \nabla u \in \mathcal{M}^{6/5, (1/\tau + 8/25)^{-1}}$. The remaining products in V have at least this regularity: for example, $u(u - c)$ has integrability $3/2$ and Morrey exponent $\tau/2$ before lowering the two exponents. The bounded cutoff derivatives do not change membership. Since $q > 5/2$, the force term also belongs to $\mathcal{M}^{6/5, \kappa}$. These facts use only the fixed support and Hölder's inequality, with its volume factors.

For completeness, the spatial Riesz operators preserve this parabolic Morrey bound for the spatially supported sources just used. Split each source into $B_{2r}(y)$ and the spatial annuli of radii $2^k r$, on the time interval of a testing cylinder. The near term is bounded by the $L^{6/5}$ Calderón–Zygmund estimate. For the annuli use the kernel bound $C|x - y|^{-3}$, Hölder in space and time, and the source Morrey bound on the larger cylinders. After normalization, the series is bounded by a constant times

$$\sum_{k \geq 0} 2^{k(5/3 - 5/\kappa)} \leq \sum_{k \geq 0} 2^{-2k/15} < \infty, \quad \kappa \leq 25/9.$$

The constants here depend only on κ and the fixed-exponent Calderón–Zygmund constant, not the number of annuli or the source. The Newtonian kernel derivatives used in these estimates include, for $y \neq 0$,

$$4\pi \partial_i \partial_j \partial_l N(y) = -3\delta_{jl} \frac{y_i}{|y|^5} - 3\delta_{ij} \frac{y_l}{|y|^5} - 3\delta_{il} \frac{y_j}{|y|^5} + 15 \frac{y_i y_j y_l}{|y|^7}, \quad (132)$$

obtained by differentiating $N(y) = -(4\pi|y|)^{-1}$.

For the remainder, harmonic interior estimates on the fixed ball and the separated support of the density of p_8 give, for almost every s ,

$$\sup_{B_{R/2}(x)} |H(\cdot, s)| \leq M_R(s), \quad \int_{t-3R^2/4}^{t+R^2/4} M_R(s)^{3/2} ds < \infty.$$

More explicitly one may take, with absolute constants,

$$M_R(s) = CR^{-3} \left(\alpha((x, t + R^2/4), R)^2 + \|p(\cdot, s)\|_{L^{3/2}(B_R(x))} \right) + CR^{-2} \|f(\cdot, s)\|_{L^{3/2}(B_R(x))}.$$

The pressure term is integrated at this fixed scale; finiteness follows from (S1) and $q > 5/2$. Hölder in time shows that on any testing cylinder of radius r the $6/5$ -power integral of the restricted remainder is at most $Cr^{17/5} \|M_R\|_{L^{3/2}}^{6/5}$. For small r this is stronger than the required power $r^{5-6/\varkappa}$, since $\varkappa \leq 25/9$; for large r use the finite integral on its bounded support. Adding the three bounds proves (130). $\square$

We record three elementary facts about parabolic Morrey spaces.

Lemma 5.27 (Hölder, embeddings, and Minkowski). *Let all functions be measurable on $\mathbb{R}^3 \times \mathbb{R}$.*

(i) (Hölder.) *If $g_1 \in \mathcal{M}^{P_1, \tau_1}$, $g_2 \in \mathcal{M}^{P_2, \tau_2}$, $\frac{1}{P} = \frac{1}{P_1} + \frac{1}{P_2}$ and $\frac{1}{\tau} = \frac{1}{\tau_1} + \frac{1}{\tau_2}$, then*

$$\|g_1 g_2\|_{\mathcal{M}^{P, \tau}} \leq \|g_1\|_{\mathcal{M}^{P_1, \tau_1}} \|g_2\|_{\mathcal{M}^{P_2, \tau_2}}. \quad (133)$$

(ii) (Lowering the integrability exponent.) *If $1 \leq P' \leq P \leq \tau$ then*

$$\|g\|_{\mathcal{M}^{P', \tau}} \leq \left(\frac{8\pi}{3}\right)^{\frac{1}{P'} - \frac{1}{P}} \|g\|_{\mathcal{M}^{P, \tau}},$$

with no support hypothesis.

(iii) (Lowering the Morrey exponent.) *If $P \leq \varkappa \leq \tau$ and $\text{supp } g \subset E$ with $d_{\text{par}}\text{-diam } E \leq D$, then*

$$\|g\|_{\mathcal{M}^{P, \varkappa}} \leq \max\{1, D^{5(\frac{1}{\varkappa} - \frac{1}{\tau})}\} \|g\|_{\mathcal{M}^{P, \tau}}. \quad (134)$$

(iv) (Minkowski.) *If $k \in L^1(\mathbb{R}^3)$ and $g \in \mathcal{M}^{P, \tau}$ then $\|\int_{\mathbb{R}^3} k(y) g(\cdot - y, \cdot) dy\|_{\mathcal{M}^{P, \tau}} \leq \|k\|_{L^1} \|g\|_{\mathcal{M}^{P, \tau}}$; the same holds for a convolution in all four space-time variables with $k \in L^1(\mathbb{R}^3 \times \mathbb{R})$.*

Proof. (i) Hölder's inequality on each ball $\mathfrak{B}_r(z)$ with exponents P_1/P and P_2/P , and $5(\frac{1}{P} - \frac{1}{\tau}) = 5(\frac{1}{P_1} - \frac{1}{\tau_1}) + 5(\frac{1}{P_2} - \frac{1}{\tau_2})$.

(ii) Hölder on $\mathfrak{B}_r(z)$ against $|\mathfrak{B}_r|^{\frac{1}{P'} - \frac{1}{P}} = (\frac{8\pi}{3})^{\frac{1}{P'} - \frac{1}{P}} r^{5(\frac{1}{P'} - \frac{1}{P})}$, and $5(\frac{1}{P'} - \frac{1}{P}) + 5(\frac{1}{P} - \frac{1}{\tau}) = 5(\frac{1}{P'} - \frac{1}{\tau})$.

(iii) If $D = 0$, then E contains at most one point and $g = 0$ almost everywhere, so the assertion is immediate. Assume $D > 0$. Let $r > 0$ and z . If $r \leq D$ then $r^{-5(\frac{1}{P} - \frac{1}{\varkappa})} = r^{5(\frac{1}{\varkappa} - \frac{1}{\tau})} r^{-5(\frac{1}{P} - \frac{1}{\tau})} \leq D^{5(\frac{1}{\varkappa} - \frac{1}{\tau})} r^{-5(\frac{1}{P} - \frac{1}{\tau})}$, giving the bound. If $r > D$ then E is contained in the closed ball of radius D centred at any point of E . Apply the Morrey bound on the open ball of radius $D + \delta$ and let $\delta \downarrow 0$ (the empty-set case is immediate). This gives $(\iint_{\mathfrak{B}_r(z)} |g|^P)^{1/P} \leq D^{5(\frac{1}{P} - \frac{1}{\tau})} \|g\|_{\mathcal{M}^{P, \tau}} \leq r^{5(\frac{1}{P} - \frac{1}{\varkappa})} D^{5(\frac{1}{\varkappa} - \frac{1}{\tau})} \|g\|_{\mathcal{M}^{P, \tau}}$, again as claimed.

(iv) Minkowski's integral inequality in $L^P(\mathfrak{B}_r(z))$ together with the translation invariance of $\|\cdot\|_{\mathcal{M}^{P, \tau}}$:

$$\begin{aligned} & \left(\iint_{\mathfrak{B}_r(z)} \left| \int k(y) g(\cdot - y) dy \right|^P \right)^{1/P} \\ & \leq \int |k(y)| \left(\iint_{\mathfrak{B}_r(z - (y, 0))} |g|^P \right)^{1/P} dy \\ & \leq \|k\|_{L^1} r^{5(\frac{1}{P} - \frac{1}{\tau})} \|g\|_{\mathcal{M}^{P, \tau}} \end{aligned}$$

$\square$

Proposition 5.28 (Step 3: one bootstrap round). *Assume $q > 5/2$ and the conclusions (126) of Step 2. Let $\tau > 5$ be such that $\mathbf{1}_{Q_2}u \in \mathcal{M}^{3,\tau}$ and*

$$\frac{1}{\tau_3} + \frac{1}{\tau} > \frac{2}{5}. \quad (135)$$

Since $\tau_3 = 25/8$, this last condition is equivalent, for $\tau > 0$, to $\tau < 25/2$. It is a hypothesis, not a consequence of $\tau > 5$ alone. Then $\mathbf{1}_{Q_3}u \in \mathcal{M}^{3,\varsigma}$ where

$$\frac{1}{\varsigma} = \frac{1}{\tau} - \varpi, \quad \varpi := \frac{1}{5} - \frac{1}{\tau_2} = \frac{\varepsilon}{5} = \frac{2}{25}. \quad (136)$$

Proof. First suppose $25/3 \leq \tau < 25/2$. We work on a small symmetric parabolic ball $\mathfrak{B}_R(z)$ with $\mathfrak{B}_{2R}(z) \subset Q_2$, and choose a smooth cutoff φ equal to one on $\mathfrak{B}_{R/4}(z)$, supported in $\mathfrak{B}_{R/2}(z)$. Its derivatives are bounded by constants depending only on the fixed radius. Set

$$\frac{1}{\kappa} = \frac{1}{\tau} + \frac{8}{25}, \quad \frac{1}{\theta} = \frac{1}{\tau} + \frac{3}{25}, \quad \frac{1}{\varsigma} = \frac{1}{\tau} - \frac{2}{25}.$$

Then $3/2 < \kappa < 5/2$, $3 < \theta < 5$ and $\varsigma > 5$. Since $\kappa < 5/2 < q$, Lemma 5.26 gives $\mathbf{1}_{\mathfrak{B}_{R/2}(z)}Dp \in \mathcal{M}^{6/5,\kappa}$.

The undifferentiated source. With the notation of Lemma 5.25, put

$$F = (\partial_t \varphi + \Delta \varphi)u - \varphi u \cdot \nabla u + \varphi f - \varphi Dp, \quad G_i = -2(\partial_i \varphi)u.$$

By Morrey Hölder, $\varphi u \cdot \nabla u \in \mathcal{M}^{6/5,\kappa}$ because $5/6 = 1/3 + 1/2$ and $1/\kappa = 1/\tau + 1/\tau_3$. The cutoff-linear terms have this same membership by lowering exponents on their bounded support. The force term does too, since $f \in L_{\text{loc}}^q$, $6/5 < \kappa < q$. Together with the pressure-gradient estimate, this proves $F \in \mathcal{M}^{6/5,\kappa}$.

The differentiated sources. The fields G_i initially belong to $\mathcal{M}^{3,\tau}$. Since $3 < \theta < \tau$ and they have bounded support, Lemma 5.27(iii) yields $G_i \in \mathcal{M}^{3,\theta}$.

The two potential estimates. The localized equation and its causal representation give

$$v = W_+ * F + \sum_i (\partial_i W_+) * G_i, \quad |v| \leq C \left(I_2 |F| + \sum_i I_1 |G_i| \right) \quad \text{almost everywhere.}$$

The second inequality uses only the heat kernel and its spatial derivatives, with the maximum gauge of Definition 5.6. Its constant is absolute and chosen before the sources. The same proof may first use the sum gauge $d_\Sigma = |x - y| + |t - s|^{1/2}$: for $a = 1, 2$,

$$I_a^\Sigma g \leq I_a g \leq 2^{5-a} I_a^\Sigma g.$$

These absolute comparison factors transfer both the pointwise bound and the Morrey estimates to the maximum gauge; the two potentials are not identified literally. Corollary 5.9 supplies both the almost-everywhere finiteness needed for these convolutions and the bounds

$$I_2 |F| \in \mathcal{M}^{P_F, \varsigma}, \quad P_F = \frac{6/5}{1 - 2\kappa/5}, \quad I_1 |G_i| \in \mathcal{M}^{P_G, \varsigma}, \quad P_G = \frac{3}{1 - \theta/5},$$

because

$$\frac{1}{\varsigma} = \frac{1}{\kappa} - \frac{2}{5} = \frac{1}{\theta} - \frac{1}{5} = \frac{1}{\tau} - \frac{2}{25}.$$

Both P_F and P_G are at least 3 and at most ς . Lower their integrability exponents to 3 using Lemma 5.27(ii), and add the finitely many terms. Since $v = u$ on $\mathfrak{B}_{R/4}(z)$, this proves the desired local velocity bound. All constants are finite functions of the incoming bounds and the cutoff radius; no uniform bound independent of the solution's incoming norms is asserted by this proposition.

Choose R from the positive gap $r_2/4 - r_3$ and cover $\overline{Q_3}$ by finitely many such inner balls with their doubled balls in Q_2 . Restriction and the triangle inequality for Morrey norms give the conclusion on Q_3 .

Finally, if $5 < \tau < 25/3$, use the Step 2 velocity exponent $25/3$ instead of τ . The case just proved gives exponent 25. Here $(1/\tau - 2/25)^{-1} < 25$, so lowering the Morrey exponent on the bounded set Q_3 gives exactly (136). $\square$

Corollary 5.29 (The bootstrap terminates after one round). *With $\varepsilon = 2/5$, Proposition 5.28 applies once with $\tau = \tau_2 = 25/3$ and yields $\mathbf{1}_{Q_3}u \in \mathcal{M}^{3,\varsigma}$ with $\varsigma = 25$; and then*

$$\frac{1}{\varsigma} + \frac{1}{\tau_2} = \frac{1}{25} + \frac{3}{25} = \frac{4}{25} < \frac{1}{5}. \quad (137)$$

A second round is not available, since $\frac{1}{\tau_3} + \frac{1}{\varsigma} = \frac{8}{25} + \frac{1}{25} = \frac{9}{25} < \frac{2}{5}$ violates (135).

Proof. Condition (135) for $\tau = \tau_2$ reads $\frac{8}{25} + \frac{3}{25} = \frac{11}{25} > \frac{10}{25}$, which holds. The gain is $\frac{1}{\varsigma} = \frac{3}{25} - \frac{2}{25} = \frac{1}{25}$. The two displayed computations are immediate. $\square$

5.5. Step 4: Hölder continuity.

Theorem 5.30 (Endgame). *Let $q > 5/2$ and let z_0, r_2, M be as in Proposition 5.22 with $\varepsilon = 2/5$. Then u agrees almost everywhere on $Q_3 = \mathfrak{B}_{r_3}(z_0)$, for every $r_3 < r_2/4$, with a function that is Hölder continuous with respect to d_{par} of exponent*

$$\gamma_0 = \min\left\{2 - \frac{5}{q}, \frac{1}{5}\right\} > 0, \quad (138)$$

with norm bounded in terms of q, M, r_2, r_3 and the norms $\|u\|_{L^{10/3}}, \|p\|_{L^{3/2}}, \|f\|_{L^q}$ on $\mathfrak{B}_{r_2}(z_0)$. In particular every point of Q_3 is a regular point of u in the sense of Definition 2.2.

Proof. Choose an intermediate symmetric ball Q'_2 with $\overline{Q_3} \subset Q'_2 \Subset Q_2$. Corollary 5.29, applied with Q'_2 as its inner set, gives $\mathbf{1}_{Q'_2}u \in \mathcal{M}^{3,25}$; Step 2 also gives $\mathbf{1}_{Q'_2}u \in \mathcal{M}^{3,25/3}$. Choose a new cutoff $\varphi \in C_c^\infty(Q'_2)$ with $\varphi = 1$ near $\overline{Q_3}$. Thus every localized source below is supported where the improved velocity exponent is available. Apply Lemma 5.26 on finitely many balls whose doubled balls lie in Q'_2 and whose inner balls cover $\text{supp } \varphi$. Weak-derivative uniqueness identifies the pressure gradients on overlaps. Since $1/25 + 8/25 = 9/25$, the resulting pressure gradient belongs locally to $\mathcal{M}^{6/5, \min\{q, 25/9\}}$. Use the same sources $F = \mathbf{g} - \varphi Dp$ and $G_i = \mathbf{h}_i$ as in the proof of Proposition 5.28. Morrey Hölder and bounded-support lowering now give

$$F \in \mathcal{M}^{6/5, \theta_0}, \quad G_i \in \mathcal{M}^{3, \theta_1}, \quad \theta_0 = \min\{q, 25/9\}, \quad \theta_1 = 25/3.$$

The pressure is retained in F , not converted into a differentiated quadratic source. The local equation therefore has the representation $v = W_+ * (F + \sum_i \partial_i G_i)$. Lower the integrability of the G_i to $6/5$, and, on the fixed bounded supports, lower the Morrey exponents to the values required by Proposition 5.12. The conditions are

$$\gamma \leq 2 - \frac{5}{\theta_0} = \min\{2 - 5/q, 1/5\}, \quad \gamma \leq 1 - \frac{5}{\theta_1} = \frac{2}{5}.$$

They hold for $\gamma = \gamma_0 > 0$ in (138). The resulting single representative agrees almost everywhere with u on Q_3 . On overlapping local balls the continuous representatives coincide, since they agree almost everywhere; a finite cover gives the claimed Hölder representative and norm bound.

For the quantitative bound, fix the cutoff family from the radii before the solution and retain the source-norm bounds in the preceding estimates. The fixed-scale harmonic remainder is controlled by the local energy and pressure bounds; the local energy inequalities bound these

quantities in terms of the Step 2 bound and the listed surrounding norms. The Calderón–Zygmund constants occur only at fixed exponents, and the finite cover and cutoff derivatives depend only on the radii. Together with the heat-potential estimate this gives exactly the dependence asserted in the theorem. $\square$

Lemma 5.31 (One-sided pressure-gradient estimate). *Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force f and exponent $q > 5/2$. Write $E_R = B_R(0) \times (-R^2, 0]$. Suppose $\overline{E_R} \subset \mathcal{O}$, $0 < r < R$, and $25/3 \leq \tau \leq 25$. If*

$$\mathbf{1}_{E_R} u \in \mathcal{M}^{3, \tau}, \quad \mathbf{1}_{E_R} \nabla u \in \mathcal{M}^{2, 25/8},$$

then the weak spatial gradient of p has a jointly measurable, locally integrable representative Dp near $\overline{E_r}$, and

$$\mathbf{1}_{E_r} Dp \in \mathcal{M}^{6/5, \varkappa}, \quad \varkappa = \min \left\{ q, \left(\frac{1}{\tau} + \frac{8}{25} \right)^{-1} \right\}.$$

Its Morrey norm is bounded in terms of the two displayed norms, q, τ , the radii, and the following past-time quantities only:

$$A_R = \operatorname{ess\,sup}_{-R^2 < s < 0} \int_{B_R} |u(y, s)|^2 dy, \quad \|p\|_{L^{3/2}(E_R)}, \quad \|f\|_{L^q(E_R)}.$$

No Morrey estimate or quantitative norm at positive times is assumed.

Proof. Choose $r < a < b < R$ and a fixed spatial cutoff equal to one on B_b and supported in B_R . At almost every time $s \in (-a^2, 0)$, apply the spatial pressure decomposition with $c(s) = [u]_{B_R}(s)$ and the singly centred tensor $-u_i(u_j - c_j)$. The differentiated source is

$$V_i = \sum_j [\eta(u_j - c_j) \partial_j u_i + (\partial_j \eta) u_i (u_j - c_j)].$$

Set all sources to zero outside the time interval $(-a^2, 0]$. The spatial operators do not change this time support. On E_r , the same slice weak-derivative identity used in Lemma 5.26 is

$$D_i p = - \sum_j \partial_i \partial_j \Delta^{-1} (\mathbf{1}_{(-a^2, 0]} V_j) + \sum_j \partial_i \partial_j \Delta^{-1} (\mathbf{1}_{(-a^2, 0]} \eta f_j) + H_i.$$

All cutoffs in this identity are spatial; no time derivative of the indicator is taken. The mean satisfies $|c(s)| \leq |B_R|^{-1/2} A_R^{1/2}$ for a.e. past time. Morrey Hölder gives $\mathbf{1}_{(-a^2, 0]} \eta u \nabla u \in \mathcal{M}^{6/5, (1/\tau + 8/25)^{-1}}$. The terms containing c or a cutoff derivative have at least this membership after bounded-support lowering: $u(u - c)$ first has integrability $3/2$ and Morrey exponent $\tau/2$. The restricted force belongs to $\mathcal{M}^{6/5, \varkappa}$ by Hölder and bounded-support lowering from L^q .

For clarity, the Riesz estimate remains valid across the truncation time. On a testing cylinder of radius ℓ , split a source spatially into the ball of radius 2ℓ and dyadic annuli of radii $L = 2^k \ell$, while keeping its time interval unchanged. At exponent $P = 6/5$, the near term follows from the spatial L^P estimate integrated in time. For an annulus the kernel bound and spatial Hölder give

$$\|Tg_k\|_{L^P(B_\ell \times J_\ell)} \leq C \ell^{3/P} L^{-3/P} \|g\|_{L^P(B_{CL} \times J_\ell)} \leq C \ell^{3/P} L^{2/P - 5/\varkappa} \|g\|_{\mathcal{M}^{P, \varkappa}}.$$

Here the last norm is estimated on a parabolic ball of radius CL containing the spatial annulus and the original time interval. After division by $\ell^{5/P - 5/\varkappa}$, the annular factor is $C 2^{k(5/3 - 5/\varkappa)}$. Since $\varkappa \leq 25/9$, it is summable, with exponent at most $-2/15$. The argument applies directly to the zero-extended past sources; it never asks for their future values.

Harmonic interior estimates and separation from the annular force source give a majorant on B_r of the form

$$\sup_{B_r} |H(\cdot, s)| \leq m(s), \quad m(s) \leq C_{r,R} (A_R + \|p(\cdot, s)\|_{L^{3/2}(B_R)} + \|f(\cdot, s)\|_{L^{3/2}(B_R)}).$$

Thus $m \in L^{3/2}(-a^2, 0)$ with a bound depending only on the stated past quantities. For a testing cylinder of radius ℓ , Hölder in time bounds the $6/5$ -power integral of the restricted remainder by $C\ell^{17/5}\|m\|_{L^{3/2}}^{6/5}$. This is sufficient at small radii, since $17/5 > 5 - 6/\varkappa$; bounded support handles large radii. This proves the asserted Morrey estimate.

To obtain a representative on an open neighbourhood as well, use the same spatial construction on a short time collar of $[-a^2, 0]$ lying inside the solution domain. Local suitability gives $u\nabla u \in L_{\text{loc}}^{5/4}$ there by $u \in L_{\text{loc}}^{10/3}$ and $\nabla u \in L_{\text{loc}}^2$. The spatial $L^{5/4}$ Riesz estimate and the harmonic remainder therefore give a locally integrable gradient on that collar. Spatial mollification and a.e. limits give joint measurability; the slice identities and Fubini give the spacetime weak-derivative identity. Weak-derivative uniqueness identifies it with the past representative already estimated. The collar is used only for this qualitative identity, not for a quantitative Morrey bound. $\square$

Proposition 5.32 (One-sided bootstrap and endgame). *Let (u, p) be a suitable weak solution on $\mathcal{O}$ with force f . Let $E_R = B_R(0) \times (-R^2, 0]$ with $\overline{E_R} \subset \mathcal{O}$, and assume the finite past bounds*

$$\begin{aligned} \mathbf{1}_{E_R} u &\in \mathcal{M}^{3,25/3}, & \mathbf{1}_{E_R} \nabla u &\in \mathcal{M}^{2,25/8}, \\ A_R &< \infty, & p &\in L^{3/2}(E_R), \quad f \in L^q(E_R), \quad q > 5/2. \end{aligned}$$

For every $0 < r < R$ one has $\mathbf{1}_{E_r} u \in \mathcal{M}^{3,25}$. For every $0 < \rho < r$, there is a globally defined parabolically Hölder function $\bar{v}$ of exponent $\gamma_0 = \min\{2 - 5/q, 1/5\} > 0$, agreeing a.e. with u on E_ρ . Its Hölder seminorm and its supremum on $\overline{E_\rho}$ are bounded in terms of the displayed past bounds, q , and R, r, ρ . No conclusion of regularity on the top face in the open-neighbourhood sense is asserted.

Proof. First choose $r < a < b < R$. By Lemma 5.31, $\mathbf{1}_{E_b} Dp \in \mathcal{M}^{6/5,25/11}$, because $1/(25/3) + 8/25 = 11/25$ and $25/11 < q$. Choose a smooth product cutoff φ equal to one near $\overline{E_r}$, whose past support lies in E_a and whose full support lies in the open solution domain and inside the neighbourhood where Dp was constructed. Explicitly, the spatial cutoff is supported in B_a and is one on B_r ; the temporal cutoff vanishes before $-a^2$ and is one on $[-r^2, 0]$, remaining one for a short future interval before vanishing inside the available time collar. All derivatives at times $t \leq 0$ are bounded in terms of the fixed radii. The future cutoff may depend on the collar, but its derivatives occur only at $t > 0$.

Define the full localized sources

$$\begin{aligned} F &= (\partial_t \varphi + \Delta \varphi)u - \varphi u \cdot \nabla u + \varphi f - \varphi Dp, & G_i &= -2(\partial_i \varphi)u, \\ F^- &= \mathbf{1}_{\{t \leq 0\}} F, & G_i^- &= \mathbf{1}_{\{t \leq 0\}} G_i. \end{aligned}$$

The local equation and its adjoint-test proof give the full distributional representation for φu . Splitting the source in that representation, rather than differentiating $\mathbf{1}_{\{t \leq 0\}} \varphi u$, introduces no terminal-time distribution. The kernels W_+ and $\partial_i W_+$ vanish at negative time differences. Hence

$$\varphi u = W_+ * F^- + \sum_i (\partial_i W_+) * G_i^- \quad \text{a.e. on } \{t \leq 0\}.$$

All future sources make zero contribution on this set; the single time slice $t = 0$ has spacetime measure zero.

Morrey Hölder and bounded-support lowering, applied only to the past sources, give

$$F^- \in \mathcal{M}^{6/5, 25/11}, \quad G_i^- \in \mathcal{M}^{3, 25/6}.$$

For the latter use $25/6 < 25/3$; for the former use the pressure estimate above and $u \nabla u$ with reciprocal Morrey exponent $11/25$. Adams' estimate with potential orders two and one gives respectively

$$I_2|F^-| \in \mathcal{M}^{66/5, 25}, \quad I_1|G_i^-| \in \mathcal{M}^{18, 25}.$$

Lower the integrability exponents to three. The heat-kernel pointwise bound and the a.e. identity on E_r now prove $\mathbf{1}_{E_r} u \in \mathcal{M}^{3, 25}$, with constants depending only on the past bounds and the chosen radii.

For the final step choose new intermediate radii $\rho < a' < b' < r$ and a new cutoff equal to one near $\overline{E}_\rho$ whose past support lies in $E_{a'}$. Its sources are therefore supported where the improved velocity bound is available. Apply Lemma 5.31 with outer radius r and inner radius b' at $\tau = 25$. The resulting past pressure-gradient exponent is $\theta_0 = \min\{q, 25/9\}$. The same source calculation gives

$$F^- \in \mathcal{M}^{6/5, \theta_0}, \quad G_i^- \in \mathcal{M}^{3, 25/3}.$$

Here the nonlinear term uses the improved exponent 25, not $25/3$; all other past bounds on E_r follow by restriction from E_R . Lower the integrability of G_i^- to $6/5$ and their Morrey exponents, on their bounded supports, to those prescribed by Proposition 5.12 for $\gamma = \gamma_0$. The requisite inequalities are

$$\gamma_0 \leq 2 - 5/\theta_0 = \min\{2 - 5/q, 1/5\}, \quad \gamma_0 \leq 1 - 5/(25/3) = 2/5.$$

That proposition gives a global Hölder representative of the potential of these past sources. By causality it agrees a.e. with u on E_ρ . Finally, for $z \in \overline{E}_\rho$ average its Hölder inequality over $y \in E_\rho$ to obtain

$$|\bar{v}(z)| \leq |E_\rho|^{-1/3} \|u\|_{L^3(E_\rho)} + [\bar{v}]_{C^{\gamma_0, \gamma_0/2}} \text{diam}_{d_{\text{par}}}(\overline{E}_\rho)^{\gamma_0}.$$

This controls the supremum using past data only and proves the proposition. $\square$

Remark 5.33 (The exponent delivered by this route). With the fixed choice $\varepsilon = 2/5$, the single bootstrap round gives velocity exponent 25, source exponents $\theta_0 = \min\{q, 25/9\}$ and $\theta_1 = 25/3$, and hence $\gamma_0 = \min\{2 - 5/q, 1/5\}$. No optimization over a variable ε is asserted here: the pressure-gradient estimates above were used only in the stated range $25/3 \leq \tau \leq 25$. The ceiling $2/3$ appearing in the blow-up route [10, Lemma 2.4], [25, Remark 6.6] is not attained by this argument; only the positivity of γ_0 is used in Theorems 2.7–2.10.

5.6. Proof of Theorems 2.7 and 2.9.

Proof of Theorem 2.9. Let $\varepsilon_1 = \varepsilon_*$, the absolute constant of Convention 5.18; note that the hypothesis (34) of Theorem 2.9 is stated for β^2 against ε_1^2 , whereas Proposition 5.22 is stated for β against ε_* , so the two thresholds are the same number and no squaring discrepancy arises. If $\limsup_{r \rightarrow 0^+} \beta(z_0, r)^2 < \varepsilon_1^2$ then $\limsup_{r \rightarrow 0^+} \beta(z_0, r) < \varepsilon_*$, so Proposition 5.22 provides r_2 and M , and Lemma 5.23, Proposition 5.28, Corollary 5.29 and Theorem 5.30 show that u agrees a.e. on the open set $\mathfrak{B}_{r_2/8}(z_0)$ with a function that is Hölder continuous of exponent γ_0 with respect to d_{par} . Since $z_0 \in \mathfrak{B}_{r_2/8}(z_0)$, Definition 2.2 is satisfied and z_0 is a regular point. $\square$

Lemma 5.34 (Caccioppoli inequality, γ -form). *Under the hypotheses of Lemma 5.15, and with the same notation,*

$$\alpha(r) + \beta(r) \leq C'_{25} \left[\kappa \gamma(\rho) + \kappa^{-1} \gamma(\rho)^{3/2} + \kappa^{-1} \delta(\rho) \gamma(\rho)^{1/2} \right] + C'_{26} \kappa^{-1/2} \gamma(\rho)^{1/2} \lambda(\rho)^{1/2}, \quad (139)$$

with C'_{25} absolute and $C'_{26} = C'_{26}(q)$.

Proof. Run the proof of Lemma 5.15 verbatim, changing only the bounds of I_1 and I_2 .

I_1 . By Hölder on $Q_\rho(z_0)$ with $1 = \frac{2}{3} + \frac{1}{3}$, $\iint_{Q_\rho(z_0)} |u|^2 \leq |Q_\rho(z_0)|^{1/3} \|u\|_{L^3(Q_\rho(z_0))}^2 = C\rho^{5/3}\rho^{4/3}\gamma(\rho)^2 = C\rho^3\gamma(\rho)^2$, so $I_1 \leq Cr^2\rho^{-5} \cdot \rho^3\gamma(\rho)^2 = C\kappa^2\gamma(\rho)^2$.

I_2 . Instead of the Poincaré step, use the spatial integral bound

$$\int_{B_\rho} ||u|^2 - [u^2]_{B_\rho}| |u| \leq 2 \int_{B_\rho} |u|^3,$$

valid at almost every time by Hölder's inequality on the mean term. Thus

$$I_2 \leq Cr^{-2} \iint_{Q_\rho(z_0)} |u|^3 = Cr^{-2}\rho^2\gamma(\rho)^3 = C\kappa^{-2}\gamma(\rho)^3.$$

The bounds on I_3 and I_4 are unchanged. Adding and taking square roots as in Step 3 of that proof gives (139). $\square$

Lemma 5.35 (Smallness at one scale propagates to all centres). *There is $\varepsilon_0 = \varepsilon_0(q) > 0$ such that: if (u, p) is a suitable weak solution on an open set containing $\overline{Q_1}$ with force $f \in L^q(Q_1)$ and (33) holds, then, with $r_5 := \kappa/4$,*

$$\theta(z, r_5) \leq \eta \quad \text{and} \quad \lambda(z, r_5) \leq \Lambda_0 \quad \text{for every } z \in Q_{3/4}.$$

Proof. Let $z = (x, t) \in Q_{3/4}$, so $|x| < 3/4$ and $t \in (-\frac{9}{16}, 0]$. Then $Q_{1/4}(z) \subset Q_1$: indeed $B_{1/4}(x) \subset B_1(0)$ and $(t - \frac{1}{16}, t] \subset (-\frac{9}{16} - \frac{1}{16}, 0] \subset (-1, 0]$. By (33),

$$\gamma(z, \tfrac{1}{4})^3 = 16 \iint_{Q_{1/4}(z)} |u|^3 \leq 16\varepsilon_0, \quad \delta(z, \tfrac{1}{4})^3 \leq 16\varepsilon_0, \quad \lambda(z, \tfrac{1}{4}) = \left(\tfrac{1}{4}\right)^\sigma \|f\|_{L^q(Q_{1/4}(z))} \leq \varepsilon_0^{1/q}.$$

Apply Lemma 5.34 with $\rho = 1/4$ and $r = r_5 = \kappa/4$, and Proposition 4.11(ii-b). For a.e. time s , write $c(s) = [u]_{B_{1/4}(x)}(s)$. The inequalities $|a - b|^3 \leq 4(|a|^3 + |b|^3)$ and Jensen's inequality give

$$\int_{B_{1/4}(x)} |u(y, s) - c(s)|^3 dy \leq 4 \int_{B_{1/4}(x)} |u(y, s)|^3 dy + 4|B_{1/4}| |c(s)|^3 \leq 8 \int_{B_{1/4}(x)} |u(y, s)|^3 dy.$$

Integrating in time and multiplying by $(1/4)^{-2}$ proves $\widehat{C}(z, \frac{1}{4}) \leq 8\gamma(z, \frac{1}{4})^3$ for the spatial mean used here. Taking cube roots in the pressure estimate gives

$$\delta(z, r_5) \leq C_{32}(q)^{1/3} [2\kappa^{-2/3}\gamma(z, \tfrac{1}{4}) + \kappa^{1/3}\delta(z, \tfrac{1}{4}) + \kappa^{1/2}\lambda(z, \tfrac{1}{4})^{1/2}].$$

Every term on the right of both displays tends to 0 as $\varepsilon_0 \rightarrow 0$, with κ fixed. Choose $\varepsilon_0(q)$ so small that $\theta(z, r_5) = \alpha(z, r_5) + \beta(z, r_5) + \kappa^{-4}\delta(z, r_5)^2 \leq \eta$; and $\lambda(z, r_5) \leq \lambda(z, \frac{1}{4}) \leq \varepsilon_0^{1/q} \leq \Lambda_0$ after shrinking ε_0 once more. $\square$

Proof of Theorem 2.7. Let $\varepsilon_0(q)$ be as in Lemma 5.35 and $r_5 = \kappa/4$. The set of admissible centres is the *one-sided* cylinder $Q_{3/4}$, so the centre-shift device of Lemma 5.23, which moves a centre forward in time by ϱ^2 , is not available as it stands; and $\overline{Q_{1/2}}$ contains its top face, so no cut-off supported in $\{t \leq 0\}$ can equal 1 there. Three modifications repair this.

Step 1: the scale iteration at every centre of $Q_{3/4}$. By Lemma 5.35, $\theta(z, r_5) \leq \eta$ and $\lambda(z, r_5) \leq \Lambda_0$ for every $z \in Q_{3/4}$, with r_5 fixed. Proposition 5.19 therefore gives

$$\max\{\alpha(z, r), \beta(z, r), \delta(z, r)^2\} \leq M r^\varepsilon \quad \text{for all } z \in Q_{3/4}, \quad 0 < r \leq r_5, \quad (140)$$

with $M = \kappa^{-4/3-\varepsilon}\eta r_5^{-\varepsilon}$ an absolute constant.

Step 2: a one-sided Morrey transfer. Put $Q_2^\sharp = B_{5/8}(0) \times (-\frac{25}{64}, 0]$, a one-sided cylinder with $\overline{Q_{1/2}} \cap \{t \leq 0\} \subset Q_2^\sharp \subset Q_{3/4}$. We claim that (126) holds with Q_2 replaced by $Q_2^\sharp$. By Lemma 5.3

it suffices to bound the integrals over one-sided cylinders $Q_\varrho(w)$. Since $Q_2^\sharp \subset \{t \leq 0\}$, only $Q_\varrho(w) \cap \{t \leq 0\}$ contributes. If that set meets $Q_2^\sharp$, pick $w' \in Q_\varrho(w) \cap Q_2^\sharp$ and set

$$w'' = (x_{w'}, \min\{t_w + \varrho^2, 0\}),$$

the *truncated* centre shift. Then $w'' \in Q_{3/4}$, because $|x_{w'}| < 5/8 < 3/4$ and $t_{w''} \leq 0$ while $t_{w''} \geq t_{w'} > -\frac{25}{64} > -\frac{9}{16}$. Moreover $Q_\varrho(w) \cap \{t \leq 0\} \subset Q_{2\varrho}(w'')$: for the spatial part, $|x - x_{w''}| \leq |x - x_w| + |x_w - x_{w'}| < 2\varrho$; for the upper time end, $s \leq \min\{t_w, 0\} \leq t_{w''}$; and for the lower end, $s > t_w - \varrho^2 \geq t_{w''} - 3\varrho^2 > t_{w''} - 4\varrho^2$. The computation of Lemma 5.23 now applies with w'' in place of z'' and (140) in place of (123), for $\varrho \leq r_5/2$; the large-radius case is unchanged. This gives $\mathbf{1}_{Q_2^\sharp} u \in \mathcal{M}^{3, \tau_2}$, $\mathbf{1}_{Q_2^\sharp} \nabla u \in \mathcal{M}^{2, \tau_3}$, $\mathbf{1}_{Q_2^\sharp} p \in \mathcal{M}^{3/2, \tau_p}$, with norms bounded in terms of M and r_5 , hence by absolute constants.

Step 3: the one-sided bootstrap and endgame. We apply Proposition 5.32 with $R = 5/8$, $r = 9/16$ and $\rho = 1/2$. The velocity and gradient Morrey hypotheses on $E_R = Q_2^\sharp$ are precisely those proved in Step 2. Its pressure Morrey bound also controls $\|p\|_{L^{3/2}(E_R)}$, and $\|f\|_{L^q(E_R)} \leq \varepsilon_0^{1/q}$ by the smallness assumption. For the remaining past energy bound, choose a fixed $0 < \ell < \min\{r_5, 3/4 - R\}/4$ and cover B_R by finitely many balls $B_\ell(x_j)$ with $x_j \in B_R$. Cover $(-R^2, 0)$ by finitely many time intervals $(t_k - \ell^2, t_k)$ with $-R^2 < t_k \leq 0$. Their cylinders have centres in $Q_{3/4}$, so Step 1 gives

$$\operatorname{ess\,sup}_{t_k - \ell^2 < s < t_k} \int_{B_\ell(x_j)} |u(y, s)|^2 dy \leq \ell \alpha((x_j, t_k), \ell)^2 \leq M^2 \ell^{1+2\varepsilon}.$$

Summing over the finite spatial cover and taking the maximum over the time intervals bounds A_R by a constant depending only on M, r_5 and the fixed radii. Thus every quantitative hypothesis of Proposition 5.32 is controlled by the past data.

The first conclusion of that proposition upgrades the velocity to $\mathbf{1}_{E_{9/16}} u \in \mathcal{M}^{3, 25}$ before its final pressure-gradient estimate and Hölder step are applied. It supplies a single function $\bar{v}$, of exponent $\gamma_0 = \min\{2 - 5/q, 1/5\}$, agreeing a.e. with u on $Q_{1/2}$ and satisfying $[\bar{v}]_{C^{\gamma_0, \gamma_0/2}} \leq C(q)$. The causal source splitting in that proposition requires no positive-time Morrey bound. To control its value as well as its oscillation, put $E = Q_{1/2}$ and average over this same past cylinder:

$$\sup_{z \in \bar{E}} |\bar{v}(z)| \leq |E|^{-1/3} \|u\|_{L^3(E)} + [\bar{v}]_{C^{\gamma_0, \gamma_0/2}} \operatorname{diam}_{d_{\text{par}}}(\bar{E})^{\gamma_0} \leq C\varepsilon_0^{1/3} + C(q).$$

Taking $w = \bar{v}|_{\bar{Q}_{1/2}}$ gives the required bound $\|w\|_{C^{\gamma_0, \gamma_0/2}(\bar{Q}_{1/2})} \leq C_4(q)$. Every point of the open interior has an open neighbourhood within $Q_{1/2}$ and is therefore regular. The argument makes no assertion of open-neighbourhood regularity on its top face. $\square$

6. FROM THE GRADIENT CRITERION TO THE PARTIAL REGULARITY THEOREM

This section proves Theorem 2.10 assuming Theorem 2.9.

Lemma 6.1 ($\mathcal{P}^1$ finite implies Lebesgue null). *Let $E \subset \mathbb{R}^3 \times \mathbb{R}$ with $\mathcal{P}^1(E) < \infty$. Then E has Lebesgue measure zero in $\mathbb{R}^4$.*

Proof. Let $M = \mathcal{P}^1(E)$ and let $\delta \in (0, 1]$. Since $\mathcal{P}_\delta^1(E) \leq \mathcal{P}^1(E) = M$, there are z_j and $r_j \in (0, \delta]$ with $E \subset \bigcup_j \mathfrak{B}_{r_j}(z_j)$ and $\sum_j r_j \leq M + 1$. By (3), $|\mathfrak{B}_r(z)| = |B_r(x)| \cdot 2r^2 = \frac{8\pi}{3} r^5$, so

$$|E| \leq \sum_j |\mathfrak{B}_{r_j}(z_j)| = \frac{8\pi}{3} \sum_j r_j^5 \leq \frac{8\pi}{3} \delta^4 \sum_j r_j \leq \frac{8\pi}{3} \delta^4 (M + 1).$$

Letting $\delta \rightarrow 0^+$ gives $|E| = 0$. Equivalently, the same cover gives zero five-dimensional parabolic Hausdorff measure, which dominates Lebesgue measure up to an absolute normalization. This comparison gives an alternative proof of the lemma. $\square$

Proof of Theorem 2.10. Let $\varepsilon_1 = \varepsilon_1(q)$ be the constant of Theorem 2.9 and set $\varepsilon_{\text{sing}} = \varepsilon_1^2 > 0$.

Step 1: the defect inequality at singular points. Let $z \in S$. By Theorem 2.9 (in contrapositive form), (34) must fail, so

$$\limsup_{r \rightarrow 0^+} \frac{1}{r} \iint_{Q_r(z)} |\nabla u|^2 dz' \geq \varepsilon_{\text{sing}}. \quad (141)$$

Consequently, for every $\tau > 0$ there exists $r \in (0, \tau)$ with $Q_r(z) \subset \mathcal{O}$ and

$$\iint_{Q_r(z)} |\nabla u|^2 dz' > \frac{1}{2} \varepsilon_{\text{sing}} r. \quad (142)$$

Step 2: reduction to relatively compact sets. Equivalently, one can cover the domain by countably many relatively compact local boxes, run the following argument on each closed smaller box, and use countable subadditivity. This uses only second countability and local finiteness of the Dirichlet energy. We give the compact-exhaustion version for comparison. $\mathcal{O}$ is open in $\mathbb{R}^4$, so there are compact sets $K_1 \subset K_2 \subset \dots$ with $\bigcup_n K_n = \mathcal{O}$ and $K_n \subset \text{int } K_{n+1}$; for instance $K_n = \{z \in \mathcal{O} : |z| \leq n, \text{dist}(z, \mathbb{R}^4 \setminus \mathcal{O}) \geq 1/n\}$ when $\mathcal{O} \neq \mathbb{R}^4$, and $K_n = \{|z| \leq n\}$ otherwise. Since $\mathcal{P}^1$ is an outer measure (External Input A.8) and $S = \bigcup_n (S \cap K_n)$, it suffices to prove $\mathcal{P}^1(S \cap K_n) = 0$ for each n . Fix n , write $K = K_n$ and $V_0 = \text{int } K_{n+1}$, so that $K \subset V_0$, $\overline{V_0} \subset K_{n+2} \subset \mathcal{O}$ is compact, and by (S1)

$$\mathcal{E}_0 := \iint_{V_0} |\nabla u|^2 dz' < \infty. \quad (143)$$

By Lemma 2.3, S is relatively closed in $\mathcal{O}$, so $S \cap K$ is compact.

Step 3: the covering. Let V be any open set with $S \cap K \subset V \subset V_0$, and let $\delta > 0$. For each $z \in S \cap K$ choose, by Step 1, a radius $r_z \in (0, \delta)$ such that $\mathfrak{B}_{r_z}(z) \subset V$ and (142) holds with $r = r_z$; this is possible because V is open and $z \in V$, so $\mathfrak{B}_r(z) \subset V$ for all small r . The family $\mathcal{F} = \{\mathfrak{B}_{r_z}(z) : z \in S \cap K\}$ consists of balls of the metric space $(\mathbb{R}^4, d_{\text{par}})$ with radii bounded by δ . That space is separable: $\mathbb{Q}^4$ is a countable d_{par} -dense subset, because $d_{\text{par}} \leq |x - x'| + |t - t'|^{1/2}$ and both terms can be made small by a rational approximation. By External Input A.7 there is a countable subfamily $\{\mathfrak{B}_{r_j}(z_j)\}_{j \in J}$, $J \subset \mathbb{N}$, $r_j = r_{z_j}$, which is pairwise disjoint and such that every member of $\mathcal{F}$ is contained in $\mathfrak{B}_{5r_j}(z_j)$ for some j . Since each $z \in S \cap K$ lies in $\mathfrak{B}_{r_z}(z) \in \mathcal{F}$,

$$S \cap K \subset \bigcup_{j \in J} \mathfrak{B}_{5r_j}(z_j), \quad 5r_j < 5\delta. \quad (144)$$

Step 4: the measure estimate. By Lemma 1.3, $Q_{r_j}(z_j) \subset \mathfrak{B}_{r_j}(z_j) \subset V$, and the cylinders $Q_{r_j}(z_j)$ are pairwise disjoint because the balls $\mathfrak{B}_{r_j}(z_j)$ are. Hence, by (142),

$$\sum_j r_j \leq \frac{2}{\varepsilon_{\text{sing}}} \sum_j \iint_{Q_{r_j}(z_j)} |\nabla u|^2 \leq \frac{2}{\varepsilon_{\text{sing}}} \iint_V |\nabla u|^2, \quad (145)$$

and therefore, by (144) and Definition 2.4,

$$\mathcal{P}_{5\delta}^1(S \cap K) \leq \sum_j 5r_j \leq \frac{10}{\varepsilon_{\text{sing}}} \iint_V |\nabla u|^2. \quad (146)$$

Step 5: $S \cap K$ is Lebesgue null. Apply (146) with $V = V_0$: for every $\delta > 0$, $\mathcal{P}_{5\delta}^1(S \cap K) \leq 10\varepsilon_{\text{sing}}^{-1}\mathcal{E}_0$, whence $\mathcal{P}^1(S \cap K) = \lim_{\delta \rightarrow 0} \mathcal{P}_{5\delta}^1(S \cap K) \leq 10\varepsilon_{\text{sing}}^{-1}\mathcal{E}_0 < \infty$ by (143). By Lemma 6.1, $|S \cap K| = 0$.

Step 6: conclusion. Let $\varepsilon > 0$. Since $|\nabla u|^2 \in L^1(V_0)$ by (143), absolute continuity of the integral gives $\tau > 0$ such that $\iint_W |\nabla u|^2 < \varepsilon$ for every measurable $W \subset V_0$ with $|W| < \tau$. By Step 5 and outer regularity of Lebesgue measure there is an open $V' \supset S \cap K$ with $|V'| < \tau$; replacing V' by $V = V' \cap V_0$, which is open and still contains $S \cap K$, we get $|V| < \tau$ and hence $\iint_V |\nabla u|^2 < \varepsilon$. Now (146) gives $\mathcal{P}_{5\delta}^1(S \cap K) \leq 10\varepsilon_{\text{sing}}^{-1}\varepsilon$ for every $\delta > 0$, so $\mathcal{P}^1(S \cap K) \leq 10\varepsilon_{\text{sing}}^{-1}\varepsilon$. As $\varepsilon > 0$ was arbitrary, $\mathcal{P}^1(S \cap K) = 0$. By Step 2, $\mathcal{P}^1(S) \leq \sum_n \mathcal{P}^1(S \cap K_n) = 0$. $\square$

Remark 6.2 (Why parabolic balls rather than cylinders in the covering). The Vitali lemma with the $5r$ enlargement is a statement about balls of a metric space, and the parabolic cylinders $Q_r(z)$ are not balls of d_{par} . Two intersecting backward cylinders need not be contained in one another's enlargements about the original top centres: a point later than one top time remains outside every cylinder with that top time. We therefore run the covering with the genuine d_{par} -balls $\mathfrak{B}_r(z)$, and recover the statement about cylinders through Lemma 2.6. The defect inequality (142) is still stated on cylinders, which is where the local energy inequality lives, and the inclusion $Q_r(z) \subset \mathfrak{B}_r(z)$ of Lemma 1.3 is what links the two. An alternative used in [25, §6.2, pp. 96–97] uses the shifted cylinders $Q^*(z, r) = B_r(x) \times (t - 7r^2/8, t + r^2/8)$, which contain both z and $Q_{r/2}(z)$.

Global existence of suitable Leray–Hopf solutions

The remaining sections construct, for every divergence-free L^2 datum, a global Leray–Hopf solution that is a suitable weak solution in the sense of Definition 1.5, together with its associated pressure, and their versions with a force. The Caffarelli–Kohn–Nirenberg theorem, Theorem 2.10, then applies to these solutions. The results are stated in Section 7, the pressure is constructed in Section 8, and the solutions in Sections 9–13.

7. SOLUTION CLASSES AND MAIN RESULTS

This section fixes the solution notions and states the results of this part. The suitable weak solutions are those of Definition 1.5; the Leray–Hopf solutions transcribe (1.3)–(1.7) of [6]. Velocity and gradient as data. In this part a velocity is a map $u: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^3$, and its weak spatial gradient is carried as separate data $Du: \mathbb{R}^3 \times \mathbb{R} \rightarrow \mathbb{R}^{3 \times 3}$ with $Du_{ij} = \partial_j u_i$, as in Remark 1.6; for open $\Omega \subset \mathbb{R}^3$ and an open interval $I \subset \mathbb{R}$, $C_c^\infty(\Omega \times I)$ denotes the smooth functions with compact support in $\Omega \times I$. Let $q > 5/2$ and $f \in L_{\text{loc}}^q(\Omega \times I; \mathbb{R}^3)$. The quadruple (u, Du, p, f) is a *suitable weak solution* on $\Omega \times I$ if (u, p) is a suitable weak solution with force f on $\Omega \times I$ in the sense of Definition 1.5, with Du in the role of ∇u . Explicitly, conditions (S1)–(S4) of Definition 1.5 then read as follows.

- (S1) For every open Ω' with compact closure $\overline{\Omega'} \subset \Omega$ and every interval J with compact closure $\overline{J} \subset I$: u, Du, p, f are measurable on $\Omega' \times J$; $\text{ess sup}_{t \in J} \int_{\Omega'} |u(x, t)|^2 dx < \infty$; $\iint_{\Omega' \times J} |u|^2 + |Du|^2 < \infty$; $p \in L^{3/2}(\Omega' \times J)$; $f \in L^q(\Omega' \times J)$; and for a.e. $t \in J$, $Du(\cdot, t)$ is the weak gradient of $u(\cdot, t)$ on Ω' .
- (S2) $\iint u \cdot \nabla \psi = 0$ for every $\psi \in C_c^\infty(\Omega \times I)$.
- (S3) For every $\varphi \in C_c^\infty(\Omega \times I; \mathbb{R}^3)$,

$$\iint \left[-u \cdot \partial_t \varphi - u_i u_j \partial_j \varphi_i + Du_{ij} \partial_j \varphi_i - p \operatorname{div} \varphi - f \cdot \varphi \right] = 0.$$

(S4) For every $\psi \in C_c^\infty(\Omega \times I)$ with $\psi \geq 0$,

$$2 \iint |Du|^2 \psi \leq \iint \left[|u|^2 (\partial_t \psi + \Delta \psi) + (|u|^2 + 2p) u \cdot \nabla \psi + 2(f \cdot u) \psi \right].$$

Regular points, the singular set and the parabolic Hausdorff measure are those of Definitions 2.2 and 2.4.

Divergence-free data. Following [6], J is the closure in $L^2(\mathbb{R}^3; \mathbb{R}^3)$ of the smooth compactly supported divergence-free fields. We use the equivalent description by the weak divergence.

Lemma 7.1. *A field $a \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ belongs to J if and only if $\int_{\mathbb{R}^3} a \cdot \nabla \psi = 0$ for every $\psi \in C_c^\infty(\mathbb{R}^3)$.*

Proof of Lemma 7.1. If $a \in J$, choose $a_k \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ with $\operatorname{div} a_k = 0$ and $a_k \rightarrow a$ in L^2 . For $\psi \in C_c^\infty(\mathbb{R}^3)$, integration by parts gives $\int a_k \cdot \nabla \psi = 0$, and $|\int (a_k - a) \cdot \nabla \psi| \leq \|a_k - a\|_2 \|\nabla \psi\|_2 \rightarrow 0$. This proves the weak-divergence identity.

Conversely, suppose that $a \in L^2$ satisfies that identity. With the Fourier convention $\widehat{f}(\xi) = \int e^{-2\pi i x \cdot \xi} f(x) dx$, for $\delta > 0$ define a vector potential A_δ and its curl b_δ by

$$\widehat{A}_\delta(\xi) = \frac{i \xi \times \widehat{a}(\xi)}{2\pi(|\xi|^2 + \delta^2)}, \quad \widehat{b}_\delta(\xi) = \frac{|\xi|^2}{|\xi|^2 + \delta^2} \widehat{a}(\xi). \quad (147)$$

The weak divergence identity gives $\xi \cdot \widehat{a}(\xi) = 0$ almost everywhere, so $b_\delta = \operatorname{curl} A_\delta$. The multipliers show that A_δ and its first weak derivatives belong to L^2 . Spatially mollifying this H^1 potential gives smooth L^2 potentials whose curls converge to b_δ in L^2 ; cutting off each smooth potential on expanding balls shows that its curl belongs to J . Thus $b_\delta \in J$.

For $\delta_n = 1/(n+1)$ the curl multiplier is bounded by one and converges to one almost everywhere. Plancherel and dominated convergence therefore give

$$\|b_{\delta_n} - a\|_2 \rightarrow 0. \quad (148)$$

The closedness of J in L^2 now gives $a \in J$. This route regularizes the Fourier potential and passes to its curl in L^2 ; it does not require an L^6 potential estimate or a prescribed cutoff-curl sequence for a . $\square$

Leray–Hopf solutions. The velocity of a Leray–Hopf solution is a function defined at every time, and it is this function, not an equivalence class, that the definition constrains at every time. The same function is used as the velocity of a suitable weak solution.

Definition 7.2 (Leray–Hopf solution). Let $0 < T < \infty$ and $a \in J$. The pair (u, Du) is a *Leray–Hopf solution* on $\mathbb{R}^3 \times [0, T]$ with *initial datum* a if the following hold.

- (LH1) *Energy class*, [6, (1.3)]. u and Du are measurable on $\mathbb{R}^3 \times (0, T)$; $\operatorname{ess\,sup}_{t \in (0, T)} \int |u(x, t)|^2 dx < \infty$; $\iint_{\mathbb{R}^3 \times (0, T)} |u|^2 + |Du|^2 < \infty$; and for a.e. $t \in (0, T)$, $Du(\cdot, t)$ is the weak gradient of $u(\cdot, t)$ on $\mathbb{R}^3$ and $\int u(x, t) \cdot \nabla \psi(x) dx = 0$ for every $\psi \in C_c^\infty(\mathbb{R}^3)$.
- (LH2) *Weak continuity*, [6, (1.4)]. For every $w \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ the map $t \mapsto \int u(x, t) \cdot w(x) dx$ is continuous on $[0, T]$.
- (LH3) *Weak equation*, [6, (1.5)]. For every $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$ with $\operatorname{div} \varphi = 0$,

$$\iint \left[-u \cdot \partial_t \varphi - u_i u_j \partial_j \varphi_i + Du_{ij} \partial_j \varphi_i \right] = 0.$$

- (LH4) *Energy inequality*, [6, (1.6)]. For every $t_0 \in [0, T]$,

$$\frac{1}{2} \int |u(x, t_0)|^2 dx + \int_0^{t_0} \int |Du|^2 dx dt \leq \frac{1}{2} \int |a|^2 dx.$$

(LH5) *Initial datum*, [6, (1.7)]. $\int |u(x, t) - a(x)|^2 dx \rightarrow 0$ as $t \downarrow 0$.

A *global Leray–Hopf solution* with datum a is a pair (u, Du) that is a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ for every $T > 0$.

Remark 7.3 (The global notion). [6] define the global notion by replacing $[0, T]$ with $[0, \infty)$ in (1.3)–(1.7). Read literally, (1.3) would then require $u \in L^2(0, \infty; W^{1,2})$, that is, $\int_0^\infty \int |u|^2 < \infty$, which Leray’s solutions need not satisfy; the energy inequality controls only $\sup_t \int |u|^2$ and $\int_0^\infty \int |\nabla u|^2$. We therefore define global solutions through every finite interval, which is the reading under which [6, Theorem 1.1] holds.

Main results. The first theorem is Leray’s existence theorem, with the constructed solution suitable in the sense of Definition 1.5.

Theorem 7.4 (Existence). *For every $a \in J$ there are u, Du and p such that (u, Du) is a global Leray–Hopf solution with datum a , and $(u, Du, p, 0)$ is a suitable weak solution on $\mathbb{R}^3 \times (0, \infty)$ for every $q > \frac{5}{2}$.*

Combined with the Caffarelli–Kohn–Nirenberg theorem, Theorem 2.10, this gives the headline of [3].

Corollary 7.5. *For every $a \in J$ there is a global Leray–Hopf solution (u, Du) with datum a whose singular set in $\mathbb{R}^3 \times (0, \infty)$ has zero one-dimensional parabolic Hausdorff measure.*

Every Leray–Hopf solution has a pressure.

Theorem 7.6 (Associated pressure). *Let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum $a \in J$. There is $p \in L^{5/3}(\mathbb{R}^3 \times (0, T))$ such that (S3) of Definition 1.5 holds on $\mathbb{R}^3 \times (0, T)$ with $f = 0$. If in addition $\text{ess sup}_{t \in (0, T)} \|u(\cdot, t)\|_{L^3(\mathbb{R}^3)} < \infty$, then this p also belongs to $L^\infty((0, T); L^{3/2}(\mathbb{R}^3))$.*

Forced Leray–Hopf solutions. The same all-time velocity representative is used in the forced equation, the work term, and the initial trace.

Definition 7.7 (Forced Leray–Hopf solution). Let $0 < T < \infty$, $a \in J$, and $f \in L^2(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$. The pair (u, Du) is a *forced Leray–Hopf solution* on $\mathbb{R}^3 \times [0, T]$ with datum a and force f if it satisfies (LH1), (LH2), and (LH5) of Definition 7.2, and also the following two clauses:

(FLH1) For every $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$ with $\text{div } \varphi = 0$,

$$\iint \left[-u \cdot \partial_t \varphi - u_i u_j \partial_j \varphi_i + Du_{ij} \partial_j \varphi_i - f \cdot \varphi \right] dx dt = 0.$$

(FLH2) For every $t \in [0, T]$,

$$\frac{1}{2} \int_{\mathbb{R}^3} |u(x, t)|^2 dx + \int_0^t \int_{\mathbb{R}^3} |Du|^2 dx ds \leq \frac{1}{2} \int_{\mathbb{R}^3} |a|^2 dx + \int_0^t \int_{\mathbb{R}^3} f \cdot u dx ds.$$

A *global forced Leray–Hopf solution* is one pair (u, Du) that is a forced Leray–Hopf solution on every finite interval, with the same representative and force.

The force work is an ordinary Lebesgue integral: on every finite slab, $u \in L_t^\infty L_x^2$ and $f \in L_{t,x}^2$, so $f \cdot u \in L_{t,x}^1$.

Theorem 7.8 (Forced suitable existence). *Let $a \in J$ and let $f: \mathbb{R}^3 \times (0, \infty) \rightarrow \mathbb{R}^3$ satisfy $f \in L^2(\mathbb{R}^3 \times (0, T))$ for every finite T and $f \in L_{\text{loc}}^{q_0}(\mathbb{R}^3 \times (0, \infty))$ for some $q_0 > 5/2$. There are*

one velocity u , weak gradient Du , and pressure p such that (u, Du) is a global forced Leray–Hopf solution with datum a . For every $q > 5/2$ for which $f \in L^q_{\text{loc}}(\mathbb{R}^3 \times (0, \infty))$, the same quadruple (u, Du, p, f) is a suitable weak solution on $\mathbb{R}^3 \times (0, \infty)$ in the sense of Definition 1.5. In particular, its local energy inequality contains the term $2(f \cdot u)\psi$.

Theorem 7.9 (Forced singular-set conclusion). *Under the hypotheses of Theorem 7.8, for every $q > 5/2$ for which $f \in L^q_{\text{loc}}$, the witness there satisfies*

$$\mathcal{P}^1(S(u)) = 0.$$

Theorem 7.10 (Forced associated pressure). *Let (u, Du) be a forced Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum $a \in J$ and $f \in L^2(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$. Let $p_N = P[u \otimes u]$ and let p_f be the force pressure of Lemma 13.1. Then*

$$p_N \in L^{5/3}(\mathbb{R}^3 \times (0, T)), \quad p_f \in L^2((0, T); L^6(\mathbb{R}^3)), \quad p = p_N + p_f \in L^{3/2}_{\text{loc}}(\mathbb{R}^3 \times (0, T)).$$

For every $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$,

$$\iint \left[-u \cdot \partial_t \varphi - u_i u_j \partial_j \varphi_i + Du_{ij} \partial_j \varphi_i - p \operatorname{div} \varphi - f \cdot \varphi \right] dx dt = 0.$$

When $f = 0$, this pressure is the pressure of Theorem 7.6 and has all of its stated global and conditional integrability properties.

8. RIESZ TRANSFORMS AND THE ASSOCIATED PRESSURE

The weak equation in the Leray–Hopf definition is tested only against solenoidal fields. We first construct the whole-space pressure that extends that equation to arbitrary compactly supported vector tests. Its sign is fixed by the convention that the pressure term in (S3) is $-p \operatorname{div} \varphi$.

Definition 8.1 (Whole-space pressure). For $1 < r < \infty$, on Schwartz functions let $\mathcal{R}_j$ denote the Fourier multiplier $i\xi_j/|\xi|$. The product $\mathcal{R}_i \mathcal{R}_j$ has a bounded extension to $L^r(\mathbb{R}^3)$. For a jointly measurable tensor field $F = (F_{ij})$ whose slices $F_{ij}(\cdot, t)$ belong to $L^r(\mathbb{R}^3)$ for almost every t , define

$$P[F](\cdot, t) = \sum_{i,j=1}^3 \mathcal{R}_i \mathcal{R}_j F_{ij}(\cdot, t). \quad (149)$$

The extensions for different exponents agree on their common domains, so this defines one pressure operator. Indeed, bounded compactly supported inputs belong to $L^r \cap L^s \cap L^2$ for any two finite exponents r, s ; the extensions of External Input A.1 agree there. Truncating an input in $L^r \cap L^s$ by balls and bounded values gives convergence in both norms, so the two outputs agree as distributions. With this convention

$$-\Delta P[F] = \sum_{i,j=1}^3 \partial_i \partial_j F_{ij} \quad (150)$$

in the spatial distribution sense for almost every t . Here $\mathcal{R}_i \mathcal{R}_j$ is the operator $R_i R_j$ of External Input A.1, the bounded extension of $\partial_i \partial_j (-\Delta)^{-1}$.

There is a jointly measurable representative of $P[F]$, and for almost every t ,

$$\|P[F](\cdot, t)\|_{L^r(\mathbb{R}^3)} \leq C_r \sum_{i,j=1}^3 \|F_{ij}(\cdot, t)\|_{L^r(\mathbb{R}^3)}. \quad (151)$$

Here and below C_r depends only on r in dimension three. If $F \in L^r(\mathbb{R}^3 \times (0, T))$, then

$$\|P[F]\|_{L^r(\mathbb{R}^3 \times (0, T))} \leq C_r \sum_{i,j=1}^3 \|F_{ij}\|_{L^r(\mathbb{R}^3 \times (0, T))}. \quad (152)$$

The bounded extensions are supplied by External Input A.1. For joint measurability, set $E_m = \{t : \sum_{i,j} \|F_{ij}(\cdot, t)\|_r \leq m\}$. Fubini's theorem gives a strongly measurable $L^r(\mathbb{R}^3)$ -valued slice map on each E_m , since F restricted to $\mathbb{R}^3 \times E_m$ belongs to the product L^r space. Applying the bounded operator to simple-function approximations gives a strongly measurable pressure map there, and hence a jointly measurable representative. The sets E_m cover almost every time. The slice estimate follows from the nine operator bounds; integrating its r th power gives the space-time estimate. The sign in (150) follows from the multiplier $-\xi_i \xi_j / |\xi|^2$ of $\mathcal{R}_i \mathcal{R}_j$.

Lemma 8.2 (Riesz duality). *Let $1 < r < \infty$, $r' = r/(r-1)$, $F \in L^r(\mathbb{R}^3 \times (0, T))$, and $g \in L^{r'}(\mathbb{R}^3 \times (0, T))$. Then*

$$\iint P[F]g = \sum_{i,j=1}^3 \iint F_{ij} \mathcal{R}_j \mathcal{R}_i g. \quad (153)$$

In particular, suppose that $F \in L^{3/2}(\mathbb{R}^3 \times (0, T))$ and that $\psi \in C^\infty(\mathbb{R}^3 \times (0, T))$ is supported in a compact time interval and satisfies, for every spatial multi-index α with $|\alpha| \leq 2$,

$$|\partial_x^\alpha \psi(x, t)| \leq C_\alpha (1 + |x|)^{-2-|\alpha|} \quad \text{for all } (x, t), \quad (154)$$

with constants independent of t . Then

$$\iint P[F] \Delta \psi = - \sum_{i,j=1}^3 \iint F_{ij} \partial_i \partial_j \psi. \quad (155)$$

Proof. On Schwartz functions, the multiplier $-\xi_i \xi_j / |\xi|^2$ is real and even, so $\mathcal{R}_i \mathcal{R}_j$ is self-adjoint and commutes with $\mathcal{R}_j \mathcal{R}_i$. This proves the scalar duality on Schwartz pairs. Density in L^r and $L^{r'}$, together with the operator bounds and Hölder's inequality, extends it to each pair of L^r and $L^{r'}$ functions. Apply this at almost every time and integrate; Hölder's inequality on space-time makes both sides integrable and proves (153).

For the particular identity, first take ψ smooth and compactly supported in space. The multiplier identity $\mathcal{R}_j \mathcal{R}_i \Delta \psi = -\partial_i \partial_j \psi$ gives (155). For a ψ satisfying (154), choose $\chi \in C_c^\infty(\mathbb{R}^3)$ equal to one on B_1 , supported in B_2 , and put $\chi_R(x) = \chi(x/R)$. The decay in (154) implies $\chi_R \psi \rightarrow \psi$ in L^3 together with all spatial derivatives of order at most two: the cubed tail bounds decay respectively as $(1 + |x|)^{-6}$, $(1 + |x|)^{-9}$, and $(1 + |x|)^{-12}$, each integrable on $\mathbb{R}^3$. Terms where a derivative falls on χ_R are supported in $R \leq |x| \leq 2R$; a term of total derivative order $k \leq 2$ is bounded there by CR^{-2-k} , so its cubed integral is $O(R^{-3-3k})$. Thus $\Delta(\chi_R \psi) \rightarrow \Delta \psi$ and $\partial_i \partial_j(\chi_R \psi) \rightarrow \partial_i \partial_j \psi$ in $L^3(\mathbb{R}^3 \times (0, T))$. Apply the compactly supported identity to $\chi_R \psi$ and pass to the limit by Hölder's inequality with $P[F]$, $F \in L^{3/2}$. This proves (155). $\square$

Lemma 8.3 (Helmholtz decomposition of a test field). *Let $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$. Define*

$$\psi(x, t) = -\frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{\operatorname{div} \varphi(y, t)}{|x - y|} dy, \quad \varphi^\sigma = \varphi - \nabla \psi. \quad (156)$$

Then $\Delta \psi = \operatorname{div} \varphi$, $\operatorname{div} \varphi^\sigma = 0$, and $\varphi^\sigma = \operatorname{curl} A$, where

$$A(x, t) = \frac{1}{4\pi} \int_{\mathbb{R}^3} \frac{\operatorname{curl} \varphi(y, t)}{|x - y|} dy. \quad (157)$$

For every integer $k \geq 0$, there are constants depending only on φ and k such that $|D_x^k \psi(x, t)| + |D_x^k \partial_t \psi(x, t)| \leq C_k(1 + |x|)^{-2-k}$ and $|D_x^k A(x, t)| + |D_x^k \partial_t A(x, t)| \leq C_k(1 + |x|)^{-2-k}$. Let χ be as in the proof of Lemma 8.2, set $\chi_R(x) = \chi(x/R)$, and define $w_R = \text{curl}(\chi_R A)$. Then w_R is smooth, compactly supported in space, divergence-free, and has the same compact time support as φ . As $R \rightarrow \infty$,

$$\|\partial_t(w_R - \varphi^\sigma)\|_{L_t^1 L_x^2} + \|\nabla(w_R - \varphi^\sigma)\|_{L_{t,x}^2} + \|\nabla(w_R - \varphi^\sigma)\|_{L_t^1 L_x^\infty} \longrightarrow 0. \quad (158)$$

These are the norms needed to pass all three terms of (LH3) under (LH1).

Proof. The spatial supports of φ and all its time derivatives lie in one fixed ball. Each component of $\text{div } \varphi$ and $\text{curl } \varphi$ has integral zero, by integration of a compactly supported derivative. For a compactly supported smooth function h with integral zero, differentiating its Newtonian potential and subtracting $D^k(|x|^{-1}) \int h = 0$ gives $|D^k(|\cdot|^{-1} * h)(x)| \leq C_k(1 + |x|)^{-2-k}$ outside twice the support; inside that ball smoothness gives the same bound after increasing C_k . Apply this argument to $\text{div } \varphi$, $\text{curl } \varphi$, and their time derivatives to obtain the stated estimates. The fundamental solution identities give $\Delta \psi = \text{div } \varphi$ and $-\Delta A = \text{curl } \varphi$. Since $\text{curl curl } \varphi = \nabla \text{div } \varphi - \Delta \varphi$,

$$\text{curl } A = (-\Delta)^{-1} \text{curl curl } \varphi = (-\Delta)^{-1} \nabla \text{div } \varphi + (-\Delta)^{-1} (-\Delta \varphi) = -\nabla \psi + \varphi.$$

Here $(-\Delta)^{-1} \text{div } \varphi = -\psi$ by $\Delta \psi = \text{div } \varphi$; the decay at infinity fixes the Newtonian representatives in these identities. This proves $\text{curl } A = \varphi^\sigma$, and $\text{div } \varphi^\sigma = 0$ follows from $\Delta \psi = \text{div } \varphi$.

The cutoff field w_R is compactly supported and smooth, and its divergence vanishes because it is a curl. The identity $w_R - \varphi^\sigma = \text{curl}((\chi_R - 1)A)$ and the derivative bounds on A show, for $R \geq 1$, that

$$\|\partial_t(w_R - \varphi^\sigma)\|_{L_t^1 L_x^2} + \|\nabla(w_R - \varphi^\sigma)\|_{L_{t,x}^2} \leq C_\varphi R^{-3/2}, \quad \|\nabla(w_R - \varphi^\sigma)\|_{L_t^1 L_x^\infty} \leq C_\varphi R^{-3}. \quad (159)$$

Indeed, outside B_R the time derivative error is bounded by a constant times $(1 + |x|)^{-3}$; the gradient error is bounded by a constant times $(1 + |x|)^{-4}$, including the annular terms where a derivative lands on χ_R . Their L^2 tails and spatial suprema give (159). The time support has finite measure. This proves (158).

For the passage in (LH3), (LH1) gives $u \in L_t^\infty L_x^2$, $Du \in L_{t,x}^2$, and $u \otimes u \in L_t^\infty L_x^1$. Thus the errors in the time, gradient, and quadratic terms are bounded respectively by $\|u\|_{L_t^\infty L_x^2} \|\partial_t(w_R - \varphi^\sigma)\|_{L_t^1 L_x^2}$, $\|Du\|_{L_{t,x}^2} \|\nabla(w_R - \varphi^\sigma)\|_{L_{t,x}^2}$, and $\|u \otimes u\|_{L_t^\infty L_x^1} \|\nabla(w_R - \varphi^\sigma)\|_{L_t^1 L_x^\infty}$. Each tends to zero by (159). $\square$

Lemma 8.4 (Energy interpolation). *If (u, Du) satisfies (LH1) and (LH4) on $\mathbb{R}^3 \times [0, T]$, then*

$$\iint_{\mathbb{R}^3 \times (0, T)} |u|^{10/3} \leq C \left(\text{ess sup}_{t \in (0, T)} \|u(\cdot, t)\|_2 \right)^{4/3} \iint_{\mathbb{R}^3 \times (0, T)} |Du|^2 \leq C \|a\|_2^{10/3}. \quad (160)$$

Consequently $u \otimes u \in L^{5/3}(\mathbb{R}^3 \times (0, T))$. Also $u \in L^3(\mathbb{R}^3 \times (0, T))$ and

$$\iint_{\mathbb{R}^3 \times (0, T)} |u|^3 \leq CT^{1/4} \|a\|_2^3. \quad (161)$$

Proof. For almost every t , (LH1) gives $u(\cdot, t) \in H^1(\mathbb{R}^3)$ with weak gradient $Du(\cdot, t)$. Interpolation between L^2 and L^6 in space, with interpolation weight $3/5$ on L^6 , and the Sobolev inequality give

$$\|u(\cdot, t)\|_{10/3} \leq \|u(\cdot, t)\|_2^{2/5} \|u(\cdot, t)\|_6^{3/5} \leq C \|u(\cdot, t)\|_2^{2/5} \|Du(\cdot, t)\|_2^{3/5}. \quad (162)$$

Raising (162) to the power $10/3$ gives the integrand bound with exponents $4/3$ and 2 . Integrating and using (LH4) at $t_0 = T$ proves the first inequality in (160). The same energy inequality gives $\text{ess sup}_t \|u(\cdot, t)\|_2 \leq \|a\|_2$ and $\iint |Du|^2 \leq \frac{1}{2} \|a\|_2^2$, proving the second. Since the Frobenius norm satisfies $|u \otimes u| = |u|^2$, $\|u \otimes u\|_{5/3} = \|u\|_{10/3}^2$.

Interpolation between L^2 and L^6 with equal weights, followed by the Sobolev inequality on $\mathbb{R}^3$, gives $\|u(\cdot, t)\|_3 \leq \|u(\cdot, t)\|_2^{1/2} \|u(\cdot, t)\|_6^{1/2} \leq C \|u(\cdot, t)\|_2^{1/2} \|Du(\cdot, t)\|_2^{1/2}$. Cubing this estimate and applying Hölder's inequality in time yields

$$\int_0^T \|Du(\cdot, t)\|_2^{3/2} dt \leq T^{1/4} \left(\int_0^T \|Du(\cdot, t)\|_2^2 dt \right)^{3/4}.$$

Using (LH4) once more proves (161). $\square$

Proof of Theorem 7.6. Set $F_{ij} = u_i u_j$ and $p = P[F]$. Lemma 8.4 and (152) at $r = 5/3$ give $p \in L^{5/3}(\mathbb{R}^3 \times (0, T))$. For almost every t , the Sobolev inequality and (LH1) give $u(\cdot, t) \in L^6 \cap L^2$; interpolation gives $u(\cdot, t) \in L^4$, so $F(\cdot, t) \in L^2$. Thus the $r = 3/2$ pressure used in Lemma 8.2 and the $r = 5/3$ pressure just defined agree almost everywhere by their common L^2 realization.

Fix $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$ and take its Helmholtz potentials A, ψ from Lemma 8.3, so $\varphi = \text{curl } A + \nabla \psi$. Let χ_n be the expanding spatial cutoffs used there, and set

$$V_n = \text{curl}(\chi_n A), \quad g_n = \chi_n \psi, \quad \Phi_n = V_n + \nabla g_n.$$

These are compactly supported smooth tests, $\text{div } V_n = 0$, and $\Phi_n \rightarrow \varphi$ pointwise with their first derivatives. Applying (LH3) to V_n removes its pressure term. For the compact gradient test ∇g_n , its contribution to (S3) is

$$\iint [-u_i \partial_t \partial_i g_n - F_{ij} \partial_j \partial_i g_n + Du_{ij} \partial_j \partial_i g_n - p \Delta g_n]. \quad (163)$$

The time term is zero by weak divergence-freeness in (LH1), applied to $\partial_t g_n$. The weak-gradient identity and then (LH1), both with compactly supported scalar tests, give

$$\iint Du_{ij} \partial_j \partial_i g_n = - \iint u_i \partial_i \Delta g_n = 0. \quad (164)$$

Compact-test Riesz duality gives

$$\iint p \Delta g_n = - \iint F_{ij} \partial_i \partial_j g_n, \quad (165)$$

so the nonlinear and pressure terms in (163) cancel. Thus the momentum functional vanishes on each Φ_n .

It remains to pass to the given test in one limit. If $J \Subset (0, T)$ contains the time support of φ , put $S(x, t) = \mathbf{1}_J(t)(1 + |x|)^{-2}$. The decay estimates for the Newtonian potentials and the cutoff derivative bounds give a common bound $C_\varphi S$ for every first derivative of Φ_n ; these derivatives converge pointwise to those of φ . Therefore the absolute value of the entire momentum integrand for Φ_n is bounded by the single integrable function

$$C_\varphi S(x, t)(|u| + |F| + |Du| + |p|).$$

Indeed, $S \in L^2 \cap L^{5/2}$, while $u, Du \in L^2$ and $F, p \in L^{5/3}$. Dominated convergence applied once to the full sum of terms proves (S3) for φ , establishing the first assertion.

The second assertion, for the same $p = P[u \otimes u]$, is proved in Lemma 8.6; that lemma uses only the first assertion and the bounds of Definition 8.1. $\square$

Remark 8.5 (Departure from [6]). The Leray–Hopf weak equation (1.5) of [6, p. 212] is stated only for solenoidal tests. Here we identify the canonical whole-space pressure by a double Riesz transform and prove the corresponding identity for arbitrary compactly supported vector tests. The Helmholtz decomposition and the noncompact-potential cutoff argument supply this pressure extension and fix its sign relative to (S3).

Lemma 8.6 (Pressure bound under the critical velocity hypothesis). *Let (u, Du) be a Leray–Hopf solution on $\mathbb{R}^3 \times [0, T]$ with datum $a \in J$, and suppose that $\text{ess sup}_{t \in (0, T)} \int_{\mathbb{R}^3} |u(x, t)|^3 dx < \infty$. Let $p_* = P[u \otimes u]$ be the pressure witness selected in the proof of Theorem 7.6. Then $p_* \in L^{5/3}(\mathbb{R}^3 \times (0, T))$, the identity (S3) of Definition 1.5 holds with $p = p_*$ and $f = 0$, and $p_* \in L^\infty((0, T); L^{3/2}(\mathbb{R}^3))$. There is a constant C depending only on the dimension and the $L^{3/2}$ double-Riesz-transform bound such that*

$$\text{ess sup}_{t \in (0, T)} \|p_*(\cdot, t)\|_{L^{3/2}(\mathbb{R}^3)} \leq C \left(\text{ess sup}_{t \in (0, T)} \|u(\cdot, t)\|_{L^3(\mathbb{R}^3)} \right)^2.$$

Proof. The proof of Theorem 7.6 takes $F_{ij} = u_i u_j$ and sets $p_* = P[F]$. Lemma 8.4 gives $F \in L^{5/3}(\mathbb{R}^3 \times (0, T))$; the proof there uses Lemma 8.2 to verify the gradient part of (S3). Thus the existential theorem’s chosen witness is exactly $P[u \otimes u]$, with the stated global integrability and (S3) identity.

For almost every t , the hypothesis gives $u(\cdot, t) \in L^3(\mathbb{R}^3)$. The slice estimate in Definition 8.1, applied with $r = 3/2$, yields

$$\|p_*(\cdot, t)\|_{L^{3/2}} \leq C_{3/2} \sum_{i,j=1}^3 \|u_i(\cdot, t) u_j(\cdot, t)\|_{L^{3/2}} \leq 9C_{3/2} \|u(\cdot, t)\|_{L^3}^2.$$

Taking the essential supremum proves the estimate. $\square$

9. THE LERAY-REGULARIZED EQUATIONS

The mollified velocity in the transport term suppresses the derivatives that would otherwise fall on the advecting field. This gives a local contraction in the energy norm; the same divergence-free transport cancellation then prevents higher spatial derivatives from growing faster than a linear Gronwall bound. We use the Fourier transform on $\mathbb{R}^3$ with phase $e^{-2\pi i x \cdot \xi}$. In this convention the heat multiplier is $e^{-4\pi^2 t |\xi|^2}$, and the derivative of the heat flow followed by the Leray projection has an integrable $(t - s)^{-1/2}$ bound in the mild formula.

Fix a radial function $\rho \in C_c^\infty(B_1)$ such that $\rho \geq 0$ and $\int_{\mathbb{R}^3} \rho = 1$. For $\varepsilon > 0$ set $\rho_\varepsilon(x) = \varepsilon^{-3} \rho(x/\varepsilon)$ and, componentwise,

$$(J_\varepsilon f)(x) = \int_{\mathbb{R}^3} \rho_\varepsilon(x - y) f(y) dy. \quad (166)$$

We use the pressure $P[F]$ of Definition 8.1; its normalization is $-\Delta P[F] = \sum_{i,j} \partial_i \partial_j F_{ij}$.

Fourier multipliers. For a Schwartz function f write $\widehat{f}(\xi) = \int_{\mathbb{R}^3} e^{-2\pi i x \cdot \xi} f(x) dx$ and extend this transform to L^2 by Plancherel. Vector and tensor transforms are taken componentwise. Thus $\widehat{\partial_j f} = 2\pi i \xi_j \widehat{f}$ and the heat semigroup $S(t) = e^{t\Delta}$ has symbol $e^{-4\pi^2 t |\xi|^2}$. For $\xi \neq 0$ let

$$\Pi(\xi) = I - \frac{\xi \otimes \xi}{|\xi|^2}, \quad (167)$$

and set $\Pi(0) = I$. This is the orthogonal projection onto $\xi^\perp$. For a tensor $F = (F_{ij})$ put

$$\widehat{K(t)F}(\xi) = e^{-4\pi^2 t |\xi|^2} \Pi(\xi) \left(2\pi i \sum_{i=1}^3 \xi_i \widehat{F}_{ij}(\xi) \right)_{j=1}^3, \quad (168)$$

where the vector in parentheses has components indexed by j . For $\xi \neq 0$ the pressure multiplier is $\widehat{\mathcal{R}_i \mathcal{R}_j g}(\xi) = -\xi_i \xi_j |\xi|^{-2} \widehat{g}(\xi)$. The value assigned at $\xi = 0$ does not affect any of these L^2 operators.

Lemma 9.1 (Multiplier bounds). *For $t > 0$, $v \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ and $F \in L^2(\mathbb{R}^3; \mathbb{R}^{3 \times 3})$,*

$$\|S(t)v\|_2 \leq \|v\|_2, \quad \|K(t)F\|_2 \leq \frac{1}{\sqrt{2e}t} \|F\|_2, \quad \|P[F]\|_2 \leq 3 \|F\|_2. \quad (169)$$

Moreover $S(t)$ and Π preserve weak divergence freedom, and $\operatorname{div} K(t)F = 0$.

Proof. Plancherel gives the first estimate because $|e^{-4\pi^2 t |\xi|^2}| \leq 1$. Since $\Pi(\xi)$ has operator norm at most one and $|F(\xi)^T \xi| \leq |F(\xi)|_F |\xi|$, the norm of the symbol in (168) is at most $2\pi |\xi| e^{-4\pi^2 t |\xi|^2}$. Its maximum over $|\xi| \geq 0$ is $(2et)^{-1/2}$, proving the second estimate. Each scalar double Riesz multiplier has modulus at most one, so Plancherel gives $\|\mathcal{R}_i \mathcal{R}_j F_{ij}\|_2 \leq \|F_{ij}\|_2$. Summing the nine terms and applying Cauchy–Schwarz gives the last estimate. The identities $\xi \cdot \Pi(\xi)z = 0$ and $\Pi(\xi)^2 = \Pi(\xi)$ give the divergence claims. The distributional Fourier identity extends from Schwartz fields to the stated spaces by density and the displayed bounds. $\square$

Lemma 9.2 (Mollifier bounds). *For every scalar or vector field $f \in L^2(\mathbb{R}^3)$ and every multi-index α ,*

$$\|\partial^\alpha J_\varepsilon f\|_{L^\infty} \leq \varepsilon^{-3/2-|\alpha|} \|\partial^\alpha \rho\|_{L^2} \|f\|_2, \quad \|J_\varepsilon f\|_2 \leq \|f\|_2. \quad (170)$$

Convolution commutes with weak spatial derivatives and preserves weak divergence freedom.

Proof. For smooth compactly supported f , differentiation under the integral in (166) and Cauchy–Schwarz yield $|\partial^\alpha J_\varepsilon f(x)| \leq \|\partial^\alpha \rho_\varepsilon\|_2 \|f\|_2$. The change of variables $y = \varepsilon z$ gives $\|\partial^\alpha \rho_\varepsilon\|_2 = \varepsilon^{-3/2-|\alpha|} \|\partial^\alpha \rho\|_2$. Density of smooth compactly supported functions in L^2 extends the pointwise convolution formula and the uniform bound to every $f \in L^2$; the same argument applies to each weak derivative. Young’s inequality and $\|\rho_\varepsilon\|_1 = 1$ give the L^2 contraction. Since radially implies $\rho_\varepsilon(x) = \rho_\varepsilon(-x)$, Fubini’s theorem gives $\int (J_\varepsilon f) \cdot \nabla \psi = \int f \cdot J_\varepsilon(\nabla \psi)$ for $\psi \in C_c^\infty(\mathbb{R}^3)$. The right side vanishes if f is weakly divergence free. All Fubini integrals are absolutely convergent by Cauchy–Schwarz and compact support of the test and mollifier. This proves the divergence claim; the weak-derivative commutation follows by applying the same identity to derivatives of the test function. $\square$

The local mild solution. For $v : [0, T] \rightarrow J$ define $F(v)_{ij}(s) = (J_\varepsilon v(s))_i v_j(s)$. The divergence-free condition gives $\sum_i \partial_i F(v)_{ij} = ((J_\varepsilon v) \cdot \nabla) v_j$ in distributions. The mild equation for the regularized system is

$$u(t) = S(t)b - \int_0^t K(t-s)F(u)(s) ds. \quad (171)$$

where $b \in J$ is the initial state. The integral is a Bochner integral in L^2 ; only its pointwise regular representative will be used in the theorem below.

Lemma 9.3 (Local mild solution). *For $b \in J$ and $\varepsilon > 0$, set*

$$M = 1 + \|b\|_2, \quad A_\varepsilon = \frac{\|\rho\|_2}{\sqrt{2e}} \varepsilon^{-3/2}, \quad T_0 = (16A_\varepsilon M)^{-2}. \quad (172)$$

There is a unique $u \in C([0, T_0]; J)$ satisfying (171). Uniqueness holds among all continuous L^2 mild solutions on this interval, and $u(0) = b$.

Proof. The space $C([0, T]; J)$ with the supremum norm is complete because J is closed in L^2 . For $v, w \in J$, Lemma 9.2 gives

$$\|F(v)\|_2 \leq \|\rho\|_2 \varepsilon^{-3/2} \|v\|_2^2, \quad \|F(v) - F(w)\|_2 \leq \|\rho\|_2 \varepsilon^{-3/2} (\|v\|_2 + \|w\|_2) \|v - w\|_2. \quad (173)$$

Here the tensor norm is the L^2 norm of its Frobenius norm. The second bound follows from $F(v) - F(w) = J_\varepsilon(v - w) \otimes v + J_\varepsilon w \otimes (v - w)$ and Hölder's inequality in space. Lemma 9.1 therefore shows that for continuous $F : [0, T] \rightarrow L^2$, the integrand in (171) is strongly measurable and has norm at most $(2e)^{-1/2} (t - s)^{-1/2} \sup_{r \in [0, T]} \|F(r)\|_2$. Since $\int_0^t (t - s)^{-1/2} ds = 2\sqrt{t}$, it is Bochner integrable. Its integral is continuous in t : on $t - s < \eta$ its norm is at most $2(2e)^{-1/2} \sqrt{\eta} \sup_r \|F(r)\|_2$; on the complement, strong continuity of $K(\tau)$ for $\tau > 0$, which follows from its Fourier symbol, and dominated convergence apply. Letting $\eta \downarrow 0$ proves continuity, including at $t = 0$.

Let Φ denote the right side of (171) and take the closed ball of radius $2M$ in $C([0, T_0]; J)$. The first estimate in (173) gives $\|\Phi(v) - S(\cdot)b\|_{C_t L^2} \leq 2A_\varepsilon (2M)^2 \sqrt{T_0} = M/2$. Also $\|S(t)b\|_2 \leq \|b\|_2 < M$, so Φ maps the ball into itself. The second estimate gives $\|\Phi(v) - \Phi(w)\|_{C_t L^2} \leq 8A_\varepsilon M \sqrt{T_0} \|v - w\|_{C_t L^2} = \frac{1}{2} \|v - w\|_{C_t L^2}$. The contraction theorem supplies a fixed point. The map takes values in J because both the heat flow and $K(t)$ have divergence-free range.

If u, v are any two continuous L^2 solutions of (171), their difference $d(t) = \|u(t) - v(t)\|_2$ satisfies $d(t) \leq A_\varepsilon (\|u\|_{C_t L^2} + \|v\|_{C_t L^2}) \int_0^t (t - s)^{-1/2} d(s) ds$. Partition $[0, T_0]$ into intervals of length h such that $2A_\varepsilon (\|u\|_{C_t L^2} + \|v\|_{C_t L^2}) \sqrt{h} < 1$. On the first interval this inequality forces $d = 0$ by taking its supremum. On each next interval the earlier difference is zero, and the same estimate applies after translating the lower time endpoint. Induction over the finite partition proves uniqueness. $\square$

Global regularity. The theorem's initial state is $b_0 = J_\varepsilon a$. It belongs to J : if $a_k \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ are divergence-free with $a_k \rightarrow a$ in L^2 , then each $J_\varepsilon a_k$ is smooth, compactly supported and divergence-free, since derivatives commute with convolution, and $\|J_\varepsilon a_k - J_\varepsilon a\|_2 \leq \|a_k - a\|_2 \rightarrow 0$ because $\rho \geq 0$ and $\int \rho = 1$. It also belongs to every integer Sobolev space. More precisely, Lemma 9.2 gives $\|\partial^\alpha J_\varepsilon a\|_{L^2} \leq \|\partial^\alpha \rho_\varepsilon\|_1 \|a\|_2$ for every α ; the derivative kernel is integrable because ρ is smooth and compactly supported. For each integer $k \geq 0$, give H^k the norm $\|v\|_{H^k}^2 = \sum_{|\alpha| \leq k} \|\partial^\alpha v\|_2^2$. The product rule and Lemma 9.2 imply, for a constant $C_{\varepsilon, k}$ depending only on ε, k, ρ ,

$$\|F(v)\|_{H^k} \leq C_{\varepsilon, k} \|v\|_2 \|v\|_{H^k}. \quad (174)$$

Indeed each derivative is a finite sum $\partial^\alpha F(v)_{ij} = \sum_{\beta \leq \alpha} \binom{\alpha}{\beta} (\partial^\beta J_\varepsilon v)_i \partial^{\alpha-\beta} v_j$, and the first factor is bounded in L^∞ by $\|\partial^\beta \rho_\varepsilon\|_2 \|v\|_2$.

For any $b \in H^k(\mathbb{R}^3; \mathbb{R}^3) \cap J$, spatial derivatives commute with $S(t)$ and $K(t)$, and the multiplier bound in Lemma 9.1 holds with the H^k norms on both sides. Thus the same mild map is locally contractive in $C([0, T_k]; H^k)$ on a ball whose radius is twice $1 + \|b\|_{H^k}$, for a positive time T_k depending only on $\varepsilon, k, \rho, \|b\|_2$ and $\|b\|_{H^k}$. The difference estimate follows by applying the product rule to $J_\varepsilon(v - w) \otimes v + J_\varepsilon w \otimes (v - w)$; on that ball each term is bounded in H^k by a constant times the ball radius times $\|v - w\|_{H^k}$. The Abel integral contributes the factor $2\sqrt{T_k}$, so choosing T_k small enough gives both invariance and contraction. The result agrees with the L^2 solution by the uniqueness just proved.

For this local interval, the construction gives

$$u \in C([0, T_k]; H^k(\mathbb{R}^3; \mathbb{R}^3)). \quad (175)$$

This local H^k construction supplies the spatial regularity used to identify the equation on its interval of existence. After the global L^2 energy bound is established, uniform Bessel-potential restarts will give the higher-regularity path on every finite interval.

The pointwise representative and pressure. Fix the local interval $[0, T_4]$ given by the preceding H^4 construction. If $s > m + 3/2$, Cauchy–Schwarz gives

$$\int_{\mathbb{R}^3} |\xi|^m |\widehat{f}(\xi)| d\xi \leq \left(\int_{\mathbb{R}^3} (1 + |\xi|^2)^s |\widehat{f}(\xi)|^2 d\xi \right)^{1/2} \left(\int_{\mathbb{R}^3} |\xi|^{2m} (1 + |\xi|^2)^{-s} d\xi \right)^{1/2} < \infty. \quad (176)$$

The last integral is finite exactly when $s > m + 3/2$. For $f \in H^s$, the inverse Fourier integral of its L^1 Fourier representative defines a bounded continuous function. Testing against Schwartz functions identifies it with f as a tempered distribution, hence almost everywhere. Applying the same estimate to derivatives of order at most m permits differentiation under the inverse integral by dominated convergence. In particular, $H^4(\mathbb{R}^3)$ embeds into $C_b^2(\mathbb{R}^3)$.

For each $t \in [0, T_4]$, set $p(t) = P[F(u(t))]$. The double Riesz multipliers are bounded on H^4 and commute with spatial derivatives. The path $u \in C([0, T_4]; H^4)$ and continuity of J_ε from L^2 into each $W^{m, \infty}$ imply that the tensor $F(u(t))$ and the canonical pressure $p(t)$ depend continuously on t in H^4 . The slice formula agrees almost everywhere in time with the jointly measurable pressure of Definition 8.1, and fixes the Riesz representative at every time. Fourier inversion gives continuous spatial derivatives of u through order two and of p through order one, jointly with time on this local interval. These fields are uniformly bounded there. The formula for $\partial_t u$ has right side $\Delta u - (J_\varepsilon u \cdot \nabla)u - \nabla p$, which is continuous and bounded on compact time strips; the same formula gives a continuous spatial L^2 path for $\partial_t u$.

Let $\mathcal{L}F$ have Fourier multiplier $\Pi(\xi)(2\pi i \xi_i \widehat{F}_{ij})_j$. The mild equation gives $\partial_t u - \Delta u + \mathcal{L}F(u) = 0$ in distributions. The Fourier symbols give $\mathcal{L}F = \operatorname{div} F + \nabla p$: the gradient part of $\operatorname{div} F$ has symbol $(I - \Pi)(2\pi i \xi_i \widehat{F}_{ij})_j$, while ∇p has its negative by the multiplier $-\xi_i \xi_j |\xi|^{-2}$. Thus, for every compactly supported test field φ , the pairing $G_\varphi(t) = \int u(x, t) \cdot \varphi(x) dx$ has distributional derivative

$$G'_\varphi(t) = \int u(x, t) \cdot \Delta \varphi(x) dx + \int u_i(x, t) (J_\varepsilon u(x, t))_j \partial_j \varphi_i(x) dx + \int p(x, t) \operatorname{div} \varphi(x) dx.$$

The right side is continuous in t . The one-dimensional fundamental theorem of calculus for weak derivatives gives, for $0 < s < t < T_4$,

$$G_\varphi(t) - G_\varphi(s) = \int_s^t G'_\varphi(\tau) d\tau.$$

Since this holds for every compactly supported spatial test and all fields in the identity are jointly continuous, test separation gives, for each x and $0 < s < t < T_4$,

$$u(x, t) - u(x, s) = \int_s^t (\Delta u - (J_\varepsilon u \cdot \nabla)u - \nabla p)(x, \tau) d\tau.$$

Differentiating this identity in time shows that $\partial_t u$ exists and is jointly continuous. Therefore

$$\partial_t u - \Delta u + \operatorname{div} F(u) + \nabla p = 0. \quad (177)$$

This is the pointwise equation on the local interval. Since $\operatorname{div}(J_\varepsilon u) = 0$, $\operatorname{div} F(u) = ((J_\varepsilon u) \cdot \nabla)u$.

Theorem 9.4 (Global Leray-regularized solution). *Let $a \in J$ and $\varepsilon > 0$. There are $u_\varepsilon: \mathbb{R}^3 \times [0, \infty) \rightarrow \mathbb{R}^3$ and $p_\varepsilon: \mathbb{R}^3 \times (0, \infty) \rightarrow \mathbb{R}$ with the following properties.*

(R1) *The map $t \mapsto u_\varepsilon(\cdot, t)$ is continuous from $[0, \infty)$ to $L^2(\mathbb{R}^3)$, $u_\varepsilon(\cdot, 0) = J_\varepsilon a$, and $u_\varepsilon(\cdot, t)$ is weakly divergence free for every $t \geq 0$ (hence classically divergence free for $t > 0$).*

- (R2) On $\mathbb{R}^3 \times (0, \infty)$, u_ε , its first and second spatial derivatives, its time derivative, p_ε , and the first spatial derivatives of p_ε are continuous. For every $0 < \delta < T < \infty$ these fields are bounded on $\mathbb{R}^3 \times [\delta, T]$, and $u_\varepsilon, \nabla u_\varepsilon, \nabla^2 u_\varepsilon, \partial_t u_\varepsilon \in L^2(\mathbb{R}^3 \times (\delta, T))$ and $p_\varepsilon \in L^2(\mathbb{R}^3 \times (\delta, T))$.
- (R3) The pointwise equation $\partial_t u_\varepsilon - \Delta u_\varepsilon + ((J_\varepsilon u_\varepsilon) \cdot \nabla) u_\varepsilon + \nabla p_\varepsilon = 0$ holds on $\mathbb{R}^3 \times (0, \infty)$.
- (R4) For every $t > 0$, $p_\varepsilon(\cdot, t) = P[(J_\varepsilon u_\varepsilon)_{i u_{\varepsilon, j}}](\cdot, t)$, with the sum over i, j understood and the sign convention of Definition 8.1; in particular, $-\Delta p_\varepsilon = \sum_{i, j} \partial_i \partial_j ((J_\varepsilon u_\varepsilon)_i u_{\varepsilon, j})$.
- (R5) For every $t \geq 0$,

$$\|u_\varepsilon(t)\|_2^2 + 2 \int_0^t \|\nabla u_\varepsilon(s)\|_2^2 ds = \|J_\varepsilon a\|_2^2. \quad (178)$$

The solution is unique among continuous L^2 solutions of the projected mild equation (171).

Proof. Lemma 9.3 gives the unique local mild solution when applied with initial state $b_0 = J_\varepsilon a$. The Sobolev construction above gives, on its interval of existence, a solution in every H^k ; the projected equation and the Riesz pressure give the classical equation (177) for positive time. We first establish its energy equality on a positive-time slab.

For $0 < \delta < t$, multiply (177) by $2u$ and integrate over space with a smooth cutoff χ_R satisfying $0 \leq \chi_R \leq 1$, equal to one in B_R , zero outside B_{2R} , and $|\nabla \chi_R| \leq C/R$. The identities $u \cdot \partial_t u = \frac{1}{2} \partial_t |u|^2$, $-2u \cdot \Delta u = -\Delta |u|^2 + 2|\nabla u|^2$, and $2u \cdot ((J_\varepsilon u) \cdot \nabla) u = \operatorname{div}(|u|^2 J_\varepsilon u)$ give the localized balance. The pressure term is $2 \operatorname{div}(pu)$ because $\operatorname{div} u = 0$. After integration in time, every cutoff error is bounded by a constant times R^{-1} times one of $\|u\|_{L^2((\delta, t); L^2)} \|\nabla u\|_{L^2((\delta, t); L^2)}$, $\|J_\varepsilon u\|_{L^\infty((\delta, t) \times \mathbb{R}^3)} \|u\|_{L^2((\delta, t); L^2)}^2$, or $\|p\|_{L^2((\delta, t); L^2)} \|u\|_{L^2((\delta, t); L^2)}$. These quantities are finite by the Sobolev bounds and the pressure multiplier estimate. Thus the errors tend to zero as $R \rightarrow \infty$. Dominated convergence for the endpoint norms and for the dissipation gives

$$\|u(t)\|_2^2 + 2 \int_\delta^t \|\nabla u(s)\|_2^2 ds = \|u(\delta)\|_2^2. \quad (179)$$

As $\delta \downarrow 0$, continuity in L^2 gives $\|u(\delta)\|_2 \rightarrow \|J_\varepsilon a\|_2$. The integrals on the left increase to the integral over $(0, t)$, which is finite by (179). This proves (178) on the local interval, including $t=0$.

In particular $\|u(t)\|_2 \leq \|J_\varepsilon a\|_2 \leq \|a\|_2$. The lifespan in Lemma 9.3, applied at any later time to the value $u(t)$, is bounded below by $(16A_\varepsilon(1 + \|a\|_2))^{-2}$, independently of t . Repeating the local construction therefore gives a global mild solution, and uniqueness identifies the successive pieces. The positive-time energy argument on each piece gives (178) for every t .

For each k and finite T , apply the Bessel-potential mild estimates of order $2k$. The local continuation step depends on the L^2 bound from (R5), not on the current H^{2k} norm. That bound is uniform at every restart time, so one positive step length covers $[0, T]$ with finitely many pieces. Mild uniqueness identifies them with the global L^2 solution. The resulting continuous Bessel path and the bounded coordinate-derivative maps into L^2 give $u \in C([0, T]; H^k)$ for every k . This gives the asserted higher regularity globally.

The Fourier embedding, canonical pressure construction, and weak-to-classical fundamental-theorem argument above now apply on every finite interval. They give the regularity clauses and the interval pointwise equation (R3) for all positive times, together with the every-time pressure identity (R4). The paths u , ∇u , $\nabla^2 u$, p , and ∇p are continuous in their spatial L^2 norms on compact positive-time intervals. The formula for $\partial_t u$ gives the same continuity for its spatial L^2 norm. These bounds yield the required space-time L^2 clauses by integration in time. This proves (R1)–(R5) and the stated uniqueness. $\square$

Lemma 9.5 (The regularized pressure in L^2). *For every $t > 0$,*

$$\|p_\varepsilon(t)\|_2 \leq C \|J_\varepsilon u_\varepsilon(t)\|_4 \|u_\varepsilon(t)\|_4, \quad (180)$$

where C is an absolute constant.

Proof. At each positive time the theorem gives $u_\varepsilon(t) \in H^1(\mathbb{R}^3)$. For completeness, the Sobolev bound used here follows first for $v \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ from the coordinate-line estimate $|g(x)| \leq \int_{\mathbb{R}} |\partial_i g(x_1, \dots, s, \dots, x_3)| ds$ applied to $g = |v|^4$. To see the resulting $W^{1,1}$ inequality explicitly, put $G_i(x_{\hat{i}}) = \int_{\mathbb{R}} |\partial_i g(x_1, \dots, s, \dots, x_3)| ds$, where $x_{\hat{i}}$ denotes the two coordinates other than x_i . Then $|g|^{3/2} \leq (G_1 G_2 G_3)^{1/2}$. Integrating first in x_1 and using Cauchy–Schwarz bounds the integral by

$$\int_{\mathbb{R}^2} G_1(x_2, x_3)^{1/2} \left(\int_{\mathbb{R}} G_2(x_1, x_3) dx_1 \right)^{1/2} \left(\int_{\mathbb{R}} G_3(x_1, x_2) dx_1 \right)^{1/2} dx_2 dx_3.$$

A second Cauchy–Schwarz inequality in (x_2, x_3) bounds this by $\prod_{i=1}^3 (\int_{\mathbb{R}^2} G_i)^{1/2}$. Since $\int G_i = \|\partial_i g\|_1$, taking the power $2/3$ gives $\|g\|_{3/2} \leq \prod_i \|\partial_i g\|_1^{1/3} \leq C \|\nabla g\|_1$. Since $|\nabla(|v|^4)| \leq 4|v|^3|\nabla v|$, Hölder yields $\|v\|_6^4 \leq C \|v\|_6^3 \|\nabla v\|_2$, hence $\|v\|_6 \leq C \|\nabla v\|_2$. Approximation in H^1 extends this inequality to every H^1 field. Interpolation gives $\|v\|_4 \leq \|v\|_2^{1/4} \|v\|_6^{3/4}$, so both $u_\varepsilon(t)$ and $J_\varepsilon u_\varepsilon(t)$ belong to L^4 . Indeed, Young’s inequality and commutation with weak derivatives give $\|J_\varepsilon u_\varepsilon(t)\|_2 + \|\nabla J_\varepsilon u_\varepsilon(t)\|_2 \leq \|u_\varepsilon(t)\|_2 + \|\nabla u_\varepsilon(t)\|_2$, so the same Sobolev and interpolation bounds apply.

Set $F_{ij} = (J_\varepsilon u_\varepsilon)_i u_{\varepsilon,j}$. Its Frobenius norm is $|F|_F = |J_\varepsilon u_\varepsilon| |u_\varepsilon|$, and Hölder gives $\|F\|_2 \leq \|J_\varepsilon u_\varepsilon\|_4 \|u_\varepsilon\|_4$. The $p=2$ Riesz bound in Definition 8.1, or equivalently Lemma 9.1, now proves (180) with an absolute constant in dimension three. $\square$

Lemma 9.6 (Local energy identity). *For every $\psi \in C_c^\infty(\mathbb{R}^3 \times (0, \infty))$,*

$$2 \iint |\nabla u_\varepsilon|^2 \psi = \iint |u_\varepsilon|^2 (\partial_t \psi + \Delta \psi) + (|u_\varepsilon|^2 J_\varepsilon u_\varepsilon + 2p_\varepsilon u_\varepsilon) \cdot \nabla \psi. \quad (181)$$

Proof. Multiply the pointwise equation in (R3) by $2u_\varepsilon \psi$ and integrate over the compact support of ψ . The time term is $\partial_t |u_\varepsilon|^2 \psi$; integration by parts in time gives $-|u_\varepsilon|^2 \partial_t \psi$. The Laplacian term is $-\Delta |u_\varepsilon|^2 \psi + 2|\nabla u_\varepsilon|^2 \psi$; moving $\Delta |u_\varepsilon|^2$ onto ψ gives the displayed $\Delta \psi$ term. Since $\operatorname{div}(J_\varepsilon u_\varepsilon) = 0$, $2u_\varepsilon \cdot ((J_\varepsilon u_\varepsilon) \cdot \nabla) u_\varepsilon = \operatorname{div}(|u_\varepsilon|^2 J_\varepsilon u_\varepsilon)$. Since $\operatorname{div} u_\varepsilon = 0$, $2u_\varepsilon \cdot \nabla p_\varepsilon = 2 \operatorname{div}(p_\varepsilon u_\varepsilon)$. Spatial integration by parts moves both divergences onto ψ with a negative sign; the compact support eliminates boundary terms. Rearranging gives (181). All integrations are justified by the continuity in (R2) on the compact support. $\square$

Lemma 9.7 (Weak momentum identity). *For every $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, \infty); \mathbb{R}^3)$,*

$$\iint [-u_\varepsilon \cdot \partial_t \varphi - (J_\varepsilon u_\varepsilon)_j u_{\varepsilon,i} \partial_j \varphi_i + \partial_j u_{\varepsilon,i} \partial_j \varphi_i - p_\varepsilon \operatorname{div} \varphi] = 0. \quad (182)$$

Proof. Integrate the pointwise equation (R3) against φ . Integration by parts in time gives the first term of (182). For diffusion, one spatial integration by parts gives $\int \partial_j u_{\varepsilon,i} \partial_j \varphi_i$. For convection, write it as $\partial_j ((J_\varepsilon u_\varepsilon)_j u_{\varepsilon,i})$; this is valid because $\operatorname{div}(J_\varepsilon u_\varepsilon) = 0$. Integration by parts gives the negative transport term displayed. The pressure term gives $-p_\varepsilon \operatorname{div} \varphi$. Each test is compactly supported in the open space-time cylinder, so no spatial or temporal boundary term remains. The regularity (R2) makes every integral finite. The resulting identity is (S3) of Definition 1.5 with the advecting factor $(J_\varepsilon u_\varepsilon)_j$ and zero force. $\square$

Remark 9.8 (Departure from [18]). The regularized existence proof in [18, Theorem 4.2, p. 41] uses heat and Oseen kernels. Here the local construction uses the L^2 Fourier multiplier (168); Plancherel gives the $(t - s)^{-1/2}$ estimate directly. The weak identity is derived from the pointwise equation rather than copied from the source: its Definition 4.1 (printed p. 40) display has a sign and tensor-slot mismatch with the regularized equation. The Riesz formula fixes the canonical time-slice pressure, and the lifespan in (172) also covers zero initial data. These changes affect the proof, not the properties (R1)–(R5).

10. UNIFORM BOUNDS

For the solutions of Theorem 9.4, energy equality (R5) gives the following separate global bounds.

Lemma 10.1. *For every $0 < \varepsilon \leq 1$,*

$$\begin{aligned} \sup_{t \geq 0} \|u_\varepsilon(t)\|_2^2 &\leq \|a\|_2^2, \\ 2 \int_0^\infty \|\nabla u_\varepsilon(s)\|_2^2 ds &\leq \|a\|_2^2. \end{aligned} \tag{183}$$

Moreover, spatial convolution is contractive in L^2 : for every $t \geq 0$, $\|J_\varepsilon u_\varepsilon(t)\|_2 \leq \|u_\varepsilon(t)\|_2$, and also $\|J_\varepsilon a\|_2 \leq \|a\|_2$.

Proof. For every finite $T \geq 0$, (R5) and L^2 contraction give

$$\|u_\varepsilon(T)\|_2^2 + 2 \int_0^T \|\nabla u_\varepsilon(s)\|_2^2 ds = \|J_\varepsilon a\|_2^2 \leq \|a\|_2^2.$$

Each term on the left is nonnegative, so the first bound follows for every T , and monotone convergence as $T \rightarrow \infty$ gives the second. Young's inequality for spatial convolution gives both stated contraction bounds. $\square$

Lemma 10.2. *For every $T \in (0, \infty]$ and every $0 < \varepsilon \leq 1$,*

$$\|u_\varepsilon\|_{L^{10/3}(\mathbb{R}^3 \times (0, T))} + \|J_\varepsilon u_\varepsilon\|_{L^{10/3}(\mathbb{R}^3 \times (0, T))} \leq C \|a\|_2,$$

where C depends only on the dimension and the interpolation constant in Lemma 8.4.

Proof. By (R1), $u_\varepsilon(\cdot, t) \in L^2(\mathbb{R}^3)$ for every $t \geq 0$. By (R2), each positive-time slice is smooth in space and its classical gradient is its weak gradient. The bound in (R5) gives $\nabla u_\varepsilon \in L^2(\mathbb{R}^3 \times (0, \infty))$, so Fubini implies $\nabla u_\varepsilon(\cdot, t) \in L^2(\mathbb{R}^3)$ for almost every $t > 0$. Thus $u_\varepsilon(\cdot, t) \in H^1(\mathbb{R}^3)$ for almost every $t > 0$, and the spatial interpolation estimate of Lemma 8.4 gives, for those times,

$$\int_{\mathbb{R}^3} |u_\varepsilon(x, t)|^{10/3} dx \leq C \|u_\varepsilon(t)\|_2^{4/3} \|\nabla u_\varepsilon(t)\|_2^2.$$

Integrating in time and using (183) proves the bound for u_ε , with a constant independent of T . Since ρ_ε is nonnegative and has integral one, Young's convolution inequality gives

$$\|J_\varepsilon u_\varepsilon(\cdot, t)\|_{10/3} \leq \|u_\varepsilon(\cdot, t)\|_{10/3}$$

for almost every t , which proves the other bound. $\square$

Lemma 10.3. *Let p_ε be the pressure in (R4). For every $0 < \varepsilon \leq 1$,*

$$\|p_\varepsilon\|_{L^{5/3}(\mathbb{R}^3 \times (0, \infty))} \leq C \|a\|_2^2.$$

Moreover, for every $T > 0$,

$$\|p_\varepsilon\|_{L^{3/2}(\mathbb{R}^3 \times (0, T))} \leq C \|a\|_2^2 T^{1/6}.$$

The constants depend only on the dimension, the interpolation constant in Lemma 10.2, and the Calderón–Zygmund operator bounds at exponents 5/3 and 3/2.

Proof. Set $F_\varepsilon = (J_\varepsilon u_\varepsilon) \otimes u_\varepsilon$. By Lemma 10.2 and Hölder's inequality,

$$\|F_\varepsilon\|_{L^{5/3}(\mathbb{R}^3 \times (0, \infty))} \leq \|J_\varepsilon u_\varepsilon\|_{L^{10/3}} \|u_\varepsilon\|_{L^{10/3}} \leq C \|a\|_2^2.$$

The Calderón–Zygmund bound in Definition 8.1, applied at exponent 5/3, and (R4) imply the first assertion. For the second, spatial Hölder and (183) give

$$\|F_\varepsilon\|_{L^1(\mathbb{R}^3 \times (0, T))} \leq \int_0^T \|J_\varepsilon u_\varepsilon(t)\|_2 \|u_\varepsilon(t)\|_2 dt \leq T \|a\|_2^2.$$

Interpolation on $\mathbb{R}^3 \times (0, T)$ yields

$$\|F_\varepsilon\|_{L^{3/2}} \leq \|F_\varepsilon\|_{L^1}^{1/6} \|F_\varepsilon\|_{L^{5/3}}^{5/6} \leq C \|a\|_2^2 T^{1/6}.$$

Applying the same pressure bound at exponent 3/2 proves the claim. $\square$

Lemma 10.4. *There is a constant C , independent of a , ε , R_1 , R_2 and t , such that for $0 < \varepsilon \leq 1$, $0 < R_1 < R_2$ and $t \geq 0$,*

$$\int_{|x| > R_2} |u_\varepsilon(x, t)|^2 dx \leq \int_{|x| > R_1} |J_\varepsilon a(x)|^2 dx + \frac{C(\|a\|_2^2 t^{1/2} + \|a\|_2^3 t^{1/4})}{R_2 - R_1}. \quad (184)$$

In particular,

$$\int_{|x| > R_1} |J_\varepsilon a(x)|^2 dx \leq \int_{|x| > R_1 - 1} |a(x)|^2 dx,$$

where the right side is understood to be the integral over all of $\mathbb{R}^3$ when $R_1 - 1 < 0$. The constant C depends only on the dimension, the fixed cutoff profile, and the universal Calderón–Zygmund and Gagliardo–Nirenberg constants.

Proof. At $t = 0$, (R1) identifies $u_\varepsilon(0) = J_\varepsilon a$, and the inequality follows from $R_1 < R_2$. Thus assume $t > 0$ below. Write $A = \|a\|_2$ and $D = R_2 - R_1$. Choose a smooth function $\eta: \mathbb{R} \rightarrow [0, 1]$ which is zero on $(-\infty, 0]$, one on $[1, \infty)$, and has bounded derivative. The radial function

$$f(x) = \eta\left(\frac{|x| - R_1}{D}\right)$$

is smooth (it is constant near the origin), vanishes on B_{R_1} , equals one outside B_{R_2} , and satisfies $|\nabla f| \leq C/D$.

We first justify the localized energy identity without a boundary assumption at spatial infinity. Choose smooth cutoffs χ_N with $0 \leq \chi_N \leq 1$, equal to one on B_N , supported in B_{2N} , and $|\nabla \chi_N| \leq C/N$, and set $q_N = f \chi_N$. For $0 < \delta < t$, multiply (R3) by $2q_N u_\varepsilon$, sum over the components, and integrate over $\mathbb{R}^3 \times (\delta, t)$. The test field is compactly supported in space. Integration by parts has no spatial boundary terms; the convection term uses $\operatorname{div}(J_\varepsilon u_\varepsilon) = 0$, and the pressure term uses $\operatorname{div} u_\varepsilon = 0$. It gives

$$\begin{aligned} \int q_N |u_\varepsilon(t)|^2 + 2 \int_\delta^t \int q_N |\nabla u_\varepsilon|^2 &= \int q_N |u_\varepsilon(\delta)|^2 \\ &+ \int_\delta^t \int \left(-2(u_\varepsilon)_i \partial_k q_N \partial_k (u_\varepsilon)_i + |u_\varepsilon|^2 (J_\varepsilon u_\varepsilon)_k \partial_k q_N + 2p_\varepsilon(u_\varepsilon)_i \partial_i q_N \right). \end{aligned}$$

Here and below, repeated spatial indices are summed from 1 to 3.

To pass to $N \rightarrow \infty$, the endpoint terms converge by dominated convergence, since $u_\varepsilon(\cdot, t)$ and $u_\varepsilon(\cdot, \delta)$ are in L^2 ; the dissipation term converges by domination by $|\nabla u_\varepsilon|^2$. The terms

with $\chi_N \nabla f$ converge by dominated convergence as well. The terms with $f \nabla \chi_N$ vanish: their absolute values are bounded by C/N times the respective integrals of $|u_\varepsilon| |\nabla u_\varepsilon|$, $|J_\varepsilon u_\varepsilon| |u_\varepsilon|^2$, and $|p_\varepsilon| |u_\varepsilon|$. These are integrable on $\mathbb{R}^3 \times (\delta, t)$. Indeed, (R1) gives $u_\varepsilon(\cdot, s) \in L^2(\mathbb{R}^3)$ for every $s \geq 0$, and (R2) identifies the classical gradient with the weak gradient on each positive-time slice. By (R5) and Fubini, $u_\varepsilon(\cdot, s) \in H^1(\mathbb{R}^3)$ for almost every $s \in (0, t)$. For these times, set $g(s) = \|\nabla u_\varepsilon(s)\|_2$. The Gagliardo–Nirenberg inequality gives the first estimate below. Young’s inequality also gives $\|J_\varepsilon u_\varepsilon(s)\|_4 \leq \|u_\varepsilon(s)\|_4$, and (R4) with the Calderón–Zygmund bound at exponent 2 gives the pressure estimate.

$$\|u_\varepsilon(s)\|_4^2 \leq C \|u_\varepsilon(s)\|_2^{1/2} g(s)^{3/2}, \quad \|p_\varepsilon(s)\|_2 \leq C \|J_\varepsilon u_\varepsilon(s)\|_4 \|u_\varepsilon(s)\|_4 \leq C \|u_\varepsilon(s)\|_4^2.$$

Thus the second and third integrands are time integrable by $g \in L^2(0, t)$, and the first is integrable by Cauchy–Schwarz in time. Letting next $\delta \downarrow 0$ is valid: (R1) gives $u_\varepsilon(\delta) \rightarrow J_\varepsilon a$ in L^2 , the dissipation is integrable by (R5), and the error terms are integrable down to time zero by the same bounds. We obtain the identity with f in place of q_N and with initial time zero.

The initial-time error is bounded by $\int f |J_\varepsilon a|^2 \leq \int_{|x| > R_1} |J_\varepsilon a|^2$. To estimate the other terms, (R5), convolution contraction and Gagliardo–Nirenberg imply

$$\begin{aligned} \int_0^t \|u_\varepsilon(s)\|_2 g(s) ds &\leq C A^2 t^{1/2}, \\ \int_0^t \|J_\varepsilon u_\varepsilon(s)\|_2 \|u_\varepsilon(s)\|_4^2 ds &\leq C A^{3/2} \int_0^t g(s)^{3/2} ds \leq C A^3 t^{1/4}, \\ \int_0^t \|p_\varepsilon(s)\|_2 \|u_\varepsilon(s)\|_2 ds &\leq C A^{3/2} \int_0^t g(s)^{3/2} ds \leq C A^3 t^{1/4}. \end{aligned}$$

The first line bounds the term with $\nabla u_\varepsilon \nabla f$; the next two bound the convection and pressure terms with ∇f . Using $|\nabla f| \leq C/D$ in the localized identity therefore bounds its total error by $C(A^2 t^{1/2} + A^3 t^{1/4})/D$. Since these estimates use only the common energy bound $\|u_\varepsilon(s)\|_2 \leq A$ from (R5) and the pressure identity (R4), the same dimension, cutoff-profile, and operator constants work for every datum, every $0 < \varepsilon \leq 1$, and every $t \geq 0$. Since $f = 1$ on $\{|x| > R_2\}$, this proves (184).

Finally, Jensen’s inequality and the support condition $\text{supp } \rho_\varepsilon \subset B_\varepsilon$ give

$$\int_{|x| > R_1} |J_\varepsilon a(x)|^2 dx \leq \int \rho_\varepsilon(z) \int_{|x+z| > R_1} |a(x)|^2 dx dz \leq \int_{|x| > R_1 - \varepsilon} |a(x)|^2 dx \leq \int_{|x| > R_1 - 1} |a(x)|^2 dx.$$

□

Remark 10.5 (Repair of Lemma 4.3 in [18]). The tail estimate is the separation-of-energy estimate of Lemma 4.3 of [18]. In place of its Newton-potential estimate for the pressure, the proof above uses (R4), the Calderón–Zygmund estimate in L^2 , and the Gagliardo–Nirenberg inequality. It also supplies the spatial cutoff limit and the limit to the initial time.

Lemma 10.6. *For every $w \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, every $0 < \varepsilon \leq 1$, and every $0 \leq s < t < \infty$,*

$$\left| \int_{\mathbb{R}^3} (u_\varepsilon(x, t) - u_\varepsilon(x, s)) \cdot w(x) dx \right| \leq C(A) N(w) |t - s|^{2/5},$$

where

$$N(w) = \|w\|_2 + \|\Delta w\|_2 + \|\nabla w\|_{L^\infty(\mathbb{R}^3)} + \|\nabla w\|_{L^{5/2}(\mathbb{R}^3)}, \quad A = \|a\|_2, \quad C(A) = C_0(A + A^2).$$

The constant C_0 depends only on the dimension and the fixed Calderón–Zygmund operator bounds.

Proof. For $w \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, the weak momentum equation gives the distributional derivative of $G_w(\tau) = \int u_\varepsilon(x, \tau) \cdot w(x) dx$ on $(0, \infty)$:

$$G'_w(\tau) = \int u_\varepsilon \cdot \Delta w + \int (u_\varepsilon)_i (J_\varepsilon u_\varepsilon)_j \partial_j w_i + \int p_\varepsilon \operatorname{div} w.$$

The integrations by parts have no spatial boundary terms because w is compactly supported; the convection term uses $\operatorname{div}(J_\varepsilon u_\varepsilon) = 0$. The first two terms are bounded in absolute value by $A \|\Delta w\|_2 + A^2 \|\nabla w\|_{L^\infty}$. Lemma 10.3 and Hölder's inequality give, for every $h > 0$,

$$\int_0^h \left| \int p_\varepsilon \operatorname{div} w \right| d\tau \leq C A^2 \|\operatorname{div} w\|_{L^{5/2}} h^{2/5}.$$

Thus G'_w is integrable down to zero. The weak-derivative fundamental theorem first gives

$$G_w(t) - G_w(0) = \int_0^t G'_w(\tau) d\tau, \quad t > 0,$$

where the endpoint value is justified by the L^2 continuity in (R1). Subtracting this identity at s from the identity at t yields the single increment formula

$$G_w(t) - G_w(s) = \int_s^t G'_w(\tau) d\tau \quad (0 \leq s < t).$$

The diffusion and convection terms on (s, t) are bounded by

$$A \|\Delta w\|_2 (t - s) + A^2 \|\nabla w\|_{L^\infty} (t - s),$$

and the pressure term by

$$C A^2 \|\operatorname{div} w\|_{L^{5/2}} (t - s)^{2/5} \leq C A^2 \|\nabla w\|_{L^{5/2}} (t - s)^{2/5}.$$

For $t - s \leq 1$, use $t - s \leq (t - s)^{2/5}$. For $t - s \geq 1$, (R1) and (183) give

$$|G_w(t) - G_w(s)| \leq 2A \|w\|_2 \leq 2A \|w\|_2 (t - s)^{2/5}.$$

This proves the result with a constant of the stated form, uniformly in ε . $\square$

11. COMPACTNESS

11.1. A compactness lemma. Spatial mollification separates the compactness argument into fixed-scale convergence and a small gradient error. At a fixed scale, pairings with every translate of the mollifier control the time variable, while the local L^2 bound controls spatial variation. The local gradient bound then makes the mollification error vanish in space-time as the scale tends to zero.

Lemma 11.1. *Let $U \subset \mathbb{R}^3$ be open and let $I \subset \mathbb{R}$ be an open interval. For each $n \in \mathbb{N}$, let $u_n : U \times I \rightarrow \mathbb{R}^3$ and $Du_n : U \times I \rightarrow \mathbb{R}^{3 \times 3}$ be measurable, with $Du_n(\cdot, t)$ the weak spatial gradient of $u_n(\cdot, t)$ for almost every $t \in I$. Suppose that for every compact $K \subset U$ and compact interval $J \subset I$ there are finite constants $M_{K,J}$ and $G_{K,J}$ such that*

$$\sup_n \sup_{t \in J} \int_K |u_n(x, t)|^2 dx \leq M_{K,J}^2, \quad \sup_n \int_J \int_K |Du_n(x, t)|^2 dx dt \leq G_{K,J}^2.$$

Suppose also that for every such K, J there are finite constants $A_{K,J}, P_{K,J}$ and an exponent $\theta_{K,J} \in (0, 1]$, independent of n, s, t, w , such that for all $n, s, t \in J$, and $w \in C_c^\infty(U; \mathbb{R}^3)$ with

$\text{supp } w \subset K$,

$$\left| \int_U (u_n(x, t) - u_n(x, s)) \cdot w(x) dx \right| \leq A_{K,J} |t - s| \left(\|\Delta w\|_{L^2(U)} + \|\nabla w\|_{L^\infty(U)} \right) \\ + P_{K,J} |t - s|^{\theta_{K,J}} \|\text{div } w\|_{L^\infty(U)}.$$

The constants may depend only on K, J and the uniform bounds and moduli assumed there, not on n . Then there is a subsequence (n_k) and a jointly measurable function $u: U \times I \rightarrow \mathbb{R}^3$, defined at every point, such that:

- (a) for every $t \in I$ and compact $K \subset U$, $u_{n_k}(\cdot, t) \rightharpoonup u(\cdot, t)$ weakly in $L^2(K)$, and for each t the function $u(\cdot, t)$ agrees almost everywhere on U with the resulting local weak limit;
- (b) $u_{n_k} \rightarrow u$ strongly in $L^2_{\text{loc}}(U \times I)$;
- (c) after passing to the same subsequence, $Du_{n_k} \rightharpoonup Du$ weakly in $L^2_{\text{loc}}(U \times I)$, where $Du(\cdot, t)$ is the weak spatial gradient of $u(\cdot, t)$ for almost every $t \in I$;
- (d) for every $w \in L^2(U; \mathbb{R}^3)$ with compact support, the map $t \mapsto \int_U u(x, t) \cdot w(x) dx$ is continuous on I ;
- (e) the local bounds pass to the limit: for every compact $K \subset U$ and compact interval $J \subset I$,

$$\sup_{t \in J} \int_K |u(x, t)|^2 dx \leq M_{K,J}^2, \quad \int_J \int_K |Du(x, t)|^2 dx dt \leq G_{K,J}^2.$$

Proof. Choose a nested countable exhaustion of U by relatively compact open sets V_j and, for each j , a countable family in $C_c^\infty(V_j; \mathbb{R}^3)$ dense in $L^2(V_j; \mathbb{R}^3)$. For each field in these families, the pairings with $u_n(\cdot, t)$ are uniformly bounded on compact time intervals by the slice bound on $\overline{V_j}$ and are equicontinuous by the assumed modulus. Arzelà–Ascoli and a diagonal subsequence give uniform convergence of these pairings on every compact time interval. At each fixed t , their limits extend to bounded linear functionals on each $L^2(V_j; \mathbb{R}^3)$, with norms bounded by the corresponding slice bound on $\overline{V_j}$. The Riesz representation theorem gives compatible limits $u_t \in L^2_{\text{loc}}(U; \mathbb{R}^3)$; compatibility follows because the limiting pairings agree on tests supported in intersections of the sets V_j . For any compact $K \subset U$ and $f \in L^2(K; \mathbb{R}^3)$, choose V_j containing K and approximate the zero extension of f in $L^2(V_j)$ by the dense family. The uniform slice bound then gives $u_{n_k}(\cdot, t) \rightharpoonup u_t$ in $L^2(K)$ for every t . The same uniform approximation shows that $t \mapsto \int_U u_t \cdot w$ is continuous for every compactly supported L^2 test w .

Fix a nonnegative smooth bump β with inner radius $1/2$ and outer radius 1 , equal to one on $B_{1/2}(0)$ and supported in $B_1(0)$. Normalize it by setting $\rho = \beta / \int_{\mathbb{R}^3} \beta(x) dx$, and set $\rho_\delta(x) = \delta^{-3} \rho(x/\delta)$. For each fixed $\delta = 1/m$, the mollified fields $u_{n_k} * \rho_\delta$ are uniformly bounded on compact subsets whose δ -neighborhood lies in U . Their spatial derivatives are bounded there by the slice L^2 bound. On a compact set of centers C for which the closed δ -neighborhood $K_{C,\delta} = \{y \in \mathbb{R}^3 : \text{dist}(y, C) \leq \delta\}$ is compactly contained in U , every translated kernel test is supported in the same compact $K_{C,\delta}$. The pairing hypothesis therefore gives the same constants $A_{K_{C,\delta},J}$, $P_{K_{C,\delta},J}$, and $\theta_{K_{C,\delta},J}$ for every center $x \in C$; the three test seminorms are invariant under translation. Arzelà–Ascoli on compact space-time sets and a diagonal extraction over m and a compact exhaustion therefore give locally uniform limits g_m of these mollified fields. In particular, for every x whose $1/m$ -ball lies in U and every $t \in I$, convergence of the translated-kernel pairings gives $g_m(x, t) = (u_t * \rho_{1/m})(x)$.

Extend each g_m by zero to the rest of $U \times I$, and define each component of $u(x, t)$ as $\limsup_{m \rightarrow \infty} g_m(x, t)$ when this is finite, and as zero otherwise. Every $x \in U$ is in the domain of g_m for all sufficiently large m . The local uniform limits g_m are measurable, so this pointwise definition is jointly measurable. For a fixed t , choose a countable exhaustion $W_j \Subset V_j \Subset U$ and functions $\chi_j \in C_c^\infty(U)$ equal to one on V_j . Extend $\chi_j u_t$ by zero to $\mathbb{R}^3$; this extension

is locally integrable on $\mathbb{R}^3$ because $u_t \in L^2_{\text{loc}}(U)$. Apply the Lebesgue differentiation theorem for approximate identities whose kernels have supports of comparable inner and outer radii componentwise to this extension and the scaled bumps with outer radius $1/m$ and inner radius $1/(2m)$. Their radius ratio is 2, and their integral-normalized kernels are $\rho_{1/m}$. The theorem gives a.e. spatial convergence for each j and component. For every $x \in W_j$, all sufficiently large kernels are supported in V_j , where $\chi_j = 1$, so their convolutions with the extension agree with $u_t * \rho_{1/m}$. Taking the countable union over j and the components proves convergence to u_t almost everywhere on U . The exceptional spatial null set may depend on the fixed time t , as allowed by the statement.

For a compact $K \subset U$, choose a relatively compact open neighborhood $V \Subset U$ containing the δ -neighborhood of K for all sufficiently small δ . For almost every t and each shift y in the kernel support, the Sobolev translation estimate on K follows for smooth fields by integrating Du_n along the segment from x to $x - y$; approximation in $W^{1,2}(V)$ gives the same estimate for the slices. Minkowski's inequality applied to the normalized kernel, followed by integration in t , gives

$$\|u_n - u_n * \rho_\delta\|_{L^2(K \times J)} \leq C_\rho \delta \|Du_n\|_{L^2(\bar{V} \times J)},$$

where C_ρ depends only on the fixed kernel profile. The convolution and every segment used in the estimate stay inside V , so no boundary extension enters the estimate. The right side tends to zero uniformly in n . At each fixed scale the mollified subsequence converges locally uniformly, hence in $L^2(K \times J)$. The translation estimate then makes (u_{n_k}) Cauchy in $L^2(K \times J)$, so it converges strongly there to some v . A further subsequence converges in $L^2(K)$ for almost every t ; its every-time weak limit is u_t , so $v(\cdot, t) = u_t$ for almost every t . The slice identification above and Fubini's theorem give $v = u$ almost everywhere in $K \times J$. This proves the asserted local strong convergence.

The gradient bounds make (Du_{n_k}) bounded in the separable Hilbert spaces $L^2(V_j \times J; \mathbb{R}^{3 \times 3})$ for the relatively compact open sets V_j in a countable exhaustion of U and compact time intervals J in a countable exhaustion of I . Weak sequential compactness and a diagonal subsequence give a local weak limit Du . Passing to the limit in the distributional identities defining Du_{n_k} is justified by strong convergence of u_{n_k} and weak convergence of Du_{n_k} ; it shows that Du is the space-time weak spatial gradient of u . For each j , choose a countable family $\mathcal{D}_j \subset C_c^\infty(V_j; \mathbb{R}^3)$ dense in $W_0^{1,2}(V_j; \mathbb{R}^3)$ with its $W^{1,2}$ norm. For each $\eta \in \mathcal{D}_j$, test the distributional identities with $\eta(x)\zeta(t)$ for every $\zeta \in C_c^\infty(I)$; Fubini gives the slice identity outside a null set that may depend on η . Since the families are countable, their null sets have a countable union, so one null-time set works for every test in every $\mathcal{D}_j$. At each remaining time the slices of u and Du are locally square integrable. Continuity of the distributional pairing in the $W^{1,2}$ norm therefore extends the identity to every test in $C_c^\infty(V_j; \mathbb{R}^3)$. Exhausting U by the V_j proves that $Du(\cdot, t)$ is the weak gradient of $u(\cdot, t)$ for almost every t . Weak lower semicontinuity gives the gradient bound in (e). For each fixed t , weak lower semicontinuity of the slice norm and (a) give the stated slice bound. The continuity assertion in (d) was proved first for the dense smooth family and extends to every compactly supported L^2 test by approximation, uniformly in t on compact intervals using the slice bound. $\square$

The compactness conclusion is local in space-time. In particular, it gives weak convergence of every time slice, not strong convergence of slice norms; the latter is not needed for the weak-pairing and lower-semicontinuity arguments used below.

Remark 11.2 (Departure from [6]). The local extraction here yields strong L^2 convergence in space-time and weak L^2 convergence of slices at every time. It does not assert the strong $C([a, b]; L^2_{\text{loc}})$ convergence in [6, (3.23), p. 225]. In the blow-up argument, uses of that stronger

topology must instead be handled with the all-time weak-pairing continuity and weak lower semicontinuity supplied by the lemma.

A uniform local $L^{10/3}$ bound upgrades this convergence below the exponent $10/3$.

Corollary 11.3. *In the setting of Lemma 11.1, suppose in addition that for every compact $Q \subset U \times I$ there is a finite constant C_Q such that*

$$\sup_n \|u_n\|_{L^{10/3}(Q)} \leq C_Q.$$

Then, along the subsequence in Lemma 11.1, $u_{n_k} \rightarrow u$ strongly in $L^q_{\text{loc}}(U \times I)$ for every $1 \leq q < 10/3$.

Proof. On each compact space-time set, strong L^2 convergence has a further subsequence converging almost everywhere. Fatou's lemma therefore gives the same local $L^{10/3}$ bound for u , and hence a uniform $L^{10/3}$ bound for $u_{n_k} - u$. For $1 \leq q < 2$, Hölder's inequality on finite-measure sets bounds the L^q norm by the L^2 norm. For $2 < q < 10/3$, choose $\lambda \in (0, 1)$ so that $1/q = \lambda/2 + (1 - \lambda)/(10/3)$. Interpolation gives

$$\|u_{n_k} - u\|_{L^q} \leq \|u_{n_k} - u\|_{L^2}^\lambda \|u_{n_k} - u\|_{L^{10/3}}^{1-\lambda},$$

which tends to zero. The case $q = 2$ is Lemma 11.1. $\square$

11.2. Stability of suitable weak solutions. The stability result below is the closure step used for the blow-up limit. Its uniform local energy hypothesis includes the essential-supremum bound in Definition 1.5; that bound is a hypothesis of the theorem and does not follow from the three strong or weak convergence statements alone.

Theorem 11.4 (Stability of suitable weak solutions). *Let $\Omega \subset \mathbb{R}^3$ be open, let $I \subset \mathbb{R}$ be an open interval, and let $q > 5/2$. For each $n \in \mathbb{N}$, suppose that $(u_n, Du_n, p_n, 0)$ is a suitable weak solution on $\Omega \times I$ in the sense of Definition 1.5. Let $u: \Omega \times I \rightarrow \mathbb{R}^3$, $Du: \Omega \times I \rightarrow \mathbb{R}^{3 \times 3}$, and $p: \Omega \times I \rightarrow \mathbb{R}$ be fields. Assume that for every open Ω' with compact closure contained in Ω and every interval J with compact closure contained in I , the following hold on $U = \Omega' \times J$:*

(i)

$$\sup_n \text{ess sup}_{t \in J} \int_{\Omega'} |u_n(x, t)|^2 dx < \infty, \quad \sup_n \int_J \int_{\Omega'} |Du_n(x, t)|^2 dx dt < \infty;$$

- (ii) $u_n \rightarrow u$ strongly in $L^3(U; \mathbb{R}^3)$;
- (iii) $Du_n \rightharpoonup Du$ weakly in $L^2(U; \mathbb{R}^{3 \times 3})$;
- (iv) $p_n \rightarrow p$ strongly in $L^{3/2}(U)$.

Then $(u, Du, p, 0)$ is a suitable weak solution on $\Omega \times I$ in the sense of Definition 1.5. In particular, the almost-every-time weak-gradient clause for Du follows from (ii) and (iii); it is not an additional hypothesis.

Proof. Fix a local box $U = \Omega' \times J$. The fields u, Du, p are already global in the theorem statement, and their restrictions to each such box are the same fixed fields. The strong L^3 convergence, weak L^2 convergence, and strong $L^{3/2}$ convergence identify these restrictions with measurable equivalence classes on U . A countable cover of $\Omega \times I$ therefore verifies global measurability of the given fields; no separately chosen local versions need to be patched on overlaps. The zero force is measurable and belongs to every required local L^q space.

The local energy of u is finite because $u \in L^3(U)$ and U has finite measure. Weak lower semicontinuity gives $Du \in L^2(U)$ and

$$\int_J \int_{\Omega'} |Du|^2 dx dt \leq \liminf_{n \rightarrow \infty} \int_J \int_{\Omega'} |Du_n|^2 dx dt.$$

Also $p \in L^{3/2}(U)$ by (iv). We prove the slice essential-supremum bound separately, since it does not follow by applying space-time lower semicontinuity to the essential supremum. Put

$$M = \sup_n \operatorname{ess\,sup}_{t \in J} \int_{\Omega'} |u_n(x, t)|^2 dx < \infty.$$

By (ii), choose a subsequence (n_k) such that $\sum_k \|u_{n_k} - u\|_{L^3(U)}^3 < \infty$. Tonelli's theorem then implies that $u_{n_k}(\cdot, t) \rightarrow u(\cdot, t)$ in $L^3(\Omega')$ for almost every $t \in J$. Since Ω' has finite measure, Hölder's inequality gives

$$\|g\|_{L^2(\Omega')} \leq |\Omega'|^{1/6} \|g\|_{L^3(\Omega')},$$

so the same subsequence converges in $L^2(\Omega')$ for almost every such t . For each k , the bound defining M holds outside a null set of times; the union of these countably many null sets is null. Passing to the slice limit outside that union yields

$$\operatorname{ess\,sup}_{t \in J} \int_{\Omega'} |u(x, t)|^2 dx \leq M.$$

This proves all the integrability and energy clauses of (S1), except its weak-gradient clause.

We derive that clause from (ii) and (iii). Fix components i, j , a function $\xi \in C_c^\infty(\Omega')$, and $\eta \in C_c^\infty(\operatorname{int} J)$. For each n , the slice weak-gradient property in (S1) and Fubini give

$$\int_J \eta(t) \int_{\Omega'} (u_{n,i}(x, t) \partial_j \xi(x) + (Du_n)_{ij}(x, t) \xi(x)) dx dt = 0.$$

The first term passes to the limit by strong L^3 convergence, and the second by weak L^2 convergence tested against $\eta(t)\xi(x)$, which is in $L^2(U)$. Thus for each fixed ξ , the function

$$G_{ij,\xi}(t) = \int_{\Omega'} (u_i(x, t) \partial_j \xi(x) + Du_{ij}(x, t) \xi(x)) dx$$

has zero pairing with every such η , and hence vanishes for almost every $t \in J$. To obtain one exceptional set for all spatial tests, choose compact sets $K_1 \subset K_2 \subset \dots$ with $K_m \subset \operatorname{int} K_{m+1}$, $K_{m+1} \subset \Omega'$ and $\bigcup_m K_m = \Omega'$. For each m choose a countable family $\mathcal{F}_m$ of smooth functions supported in K_{m+1} that is dense, in the C^1 norm, among the smooth functions supported in K_{m+1} ; every test supported in K_m is then a C^1 limit of members of $\mathcal{F}_m$, all supported in the fixed compact set K_{m+1} . Let N be the union of the null sets of times for the countable family $\bigcup_m \mathcal{F}_m$. For $t \notin N$ such that $u(\cdot, t)$ and $Du(\cdot, t)$ are locally integrable, the displayed pairing is continuous under C^1 convergence with supports in K_{m+1} , so the identity extends from $\mathcal{F}_m$ to every $\xi \in C_c^\infty(\Omega')$ supported in K_m , and every test is supported in some K_m . The weak-gradient identity therefore holds for every spatial test function for almost every $t \in J$. This is exactly the remaining clause of (S1).

For (S2), take $\psi \in C_c^\infty(\Omega \times I)$. Its support is compact, so it lies in a local box, and (ii) implies convergence of $\iint u_n \cdot \nabla \psi$ to $\iint u \cdot \nabla \psi$. The former integral is zero for every n ; hence the limit is zero.

For (S3), fix $\varphi \in C_c^\infty(\Omega \times I; \mathbb{R}^3)$ and choose a local box containing its support. Strong L^3 convergence passes the linear velocity term in L^1 . It also passes the quadratic term because

$$\|u_n \otimes u_n - u \otimes u\|_{L^{3/2}} \leq (\|u_n\|_{L^3} + \|u\|_{L^3}) \|u_n - u\|_{L^3} \rightarrow 0.$$

The weak gradient term passes by (iii), since $\nabla\varphi \in L^2$. Finally, (iv) passes the pressure term in L^1 against the bounded $\operatorname{div} \varphi$. Each u_n satisfies (S3) with force zero, so the limiting identity is (S3) for $(u, Du, p, 0)$.

For (S4), fix a nonnegative $\psi \in C_c^\infty(\Omega \times I)$ and take a local box U containing its support. Multiplication by the bounded function $\sqrt{\psi}$ preserves weak convergence in $L^2(U)$, so $\sqrt{\psi} Du_n \rightharpoonup \sqrt{\psi} Du$. Hilbert-space lower semicontinuity yields

$$\int_U |Du|^2 \psi \leq \liminf_{n \rightarrow \infty} \int_U |Du_n|^2 \psi.$$

On the right side of the local energy inequality, (ii) gives $|u_n|^2 \rightarrow |u|^2$ in $L^{3/2}(U)$. The cubic flux converges in L^1 :

$$\left\| |u_n|^2 u_n - |u|^2 u \right\|_{L^1(U)} \leq C(\|u_n\|_{L^3(U)}^2 + \|u\|_{L^3(U)}^2) \|u_n - u\|_{L^3(U)} \rightarrow 0.$$

The pressure flux also converges in L^1 , since

$$\|p_n u_n - p u\|_{L^1(U)} \leq \|p_n - p\|_{L^{3/2}(U)} \|u_n\|_{L^3(U)} + \|p\|_{L^{3/2}(U)} \|u_n - u\|_{L^3(U)} \rightarrow 0.$$

All derivatives of ψ are bounded and supported in U , so these convergences show that the right side of (S4) for u_n, p_n tends to its right side for u, p . Taking the lower limit in the suitable inequality for each n proves (S4) for $(u, Du, p, 0)$. Thus every clause (S1)–(S4) holds. $\square$

Lemma 11.5 (Restriction of suitable weak solutions). *Let $q > 5/2$, let $\Omega \subset \mathbb{R}^3$ be open, let $I \subset \mathbb{R}$ be an open interval, and suppose (u, Du, p, f) is a suitable weak solution on $\Omega \times I$ in the sense of Definition 1.5. If $\Omega_1 \subset \Omega$ is open and $I_1 \subset I$ is an open interval, then the restrictions of (u, Du, p, f) form a suitable weak solution on $\Omega_1 \times I_1$ with the same exponent q .*

Proof. Let $\Omega' \subset \Omega_1$ be open with compact closure in Ω_1 and let J be an interval with compact closure in I_1 . Then $\overline{\Omega'} \subset \Omega$ and $\overline{J} \subset I$, so $\Omega' \times J$ is also a local box for $\Omega \times I$. Every local measurability, integrability, essential-supremum, and slice weak-gradient clause in (S1) therefore restricts to this box. This proves (S1) for the smaller domain, including local L^q membership of f .

The test functions in (S2)–(S4) are globally defined smooth functions with compact support contained in the relevant space-time domain. A test function supported in $\Omega_1 \times I_1$ is consequently also an admissible test function for $\Omega \times I$. Its derivatives vanish outside its support, so every integrand in (S2) and (S3), and every term in (S4), has the same integral over $\Omega_1 \times I_1$ as over $\Omega \times I$. The original identities and inequality therefore give (S2)–(S4) on the smaller domain. $\square$

12. THE LIMIT IS A SUITABLE LERAY–HOPF SOLUTION

Spatial tails turn the local compactness of the regularized velocities into convergence on the whole space. The weak L^2 limit of each time slice is retained, because the Leray–Hopf energy inequality is required at every time. The dissipation passes through weak lower semicontinuity in L^2 ; no almost-everywhere convergence of the gradients is used.

Proposition 12.1. *Let $a \in J$, and let $\varepsilon_n \in (0, 1]$ tend to zero. There are a subsequence, still denoted ε_n , and measurable fields $u: \mathbb{R}^3 \times [0, \infty) \rightarrow \mathbb{R}^3$ and $Du: \mathbb{R}^3 \times (0, \infty) \rightarrow \mathbb{R}^{3 \times 3}$ with the following properties. Choose a measurable representative of a and set $u(x, 0) = a(x)$. For $t > 0$, take the jointly measurable representative given, componentwise on each finite slab by*

Lemma 11.1, using its fixed normalized bump profile ρ :

$$u_i(x, t) = \begin{cases} \limsup_{m \rightarrow \infty} \lim_{n \rightarrow \infty} (u_{\varepsilon_n}(\cdot, t) * \rho_{1/m})_i(x), & \text{if this value is finite,} \\ 0, & \text{otherwise,} \end{cases} \quad i = 1, 2, 3. \quad (185)$$

The diagonal subsequence can be chosen so that for every $T > 0$,

$$u_{\varepsilon_n} \longrightarrow u \quad \text{strongly in } L^2(\mathbb{R}^3 \times (0, T)), \quad (186)$$

$$Du_{\varepsilon_n} \rightharpoonup Du \quad \text{weakly in } L^2(\mathbb{R}^3 \times (0, T)), \quad (187)$$

$$u_{\varepsilon_n} \longrightarrow u \quad \text{strongly in } L^q(\mathbb{R}^3 \times (0, T)) \quad \text{for every } 2 \leq q < 10/3, \quad (188)$$

$$J_{\varepsilon_n} u_{\varepsilon_n} \longrightarrow u \quad \text{strongly in } L^3(\mathbb{R}^3 \times (0, T)). \quad (189)$$

For every $t > 0$, the slices $u_{\varepsilon_n}(\cdot, t)$ converge weakly to $u(\cdot, t)$ in $L^2(\mathbb{R}^3)$, and $Du(\cdot, t)$ is the weak spatial gradient of $u(\cdot, t)$ for almost every $t > 0$.

Proof. Apply Lemma 11.1 to u_ε on $\mathbb{R}^3 \times (0, m)$ for each positive integer m , and take a diagonal subsequence. The uniform slice and gradient bounds required there follow from Lemma 10.1; the translated-test time modulus is Lemma 10.6. The construction in that lemma gives (185), local strong L^2 convergence, and local weak convergence of the gradients. To track the representative, write the first diagonal subsequence as $\varepsilon_{\sigma(n)}$. If its mollified limit is taken along the same extraction once more, the resulting indices are $\sigma(\sigma(n))$. For each fixed spatial mollification scale, the inner pairings already converge along $\varepsilon_{\sigma(n)}$, so this further restriction leaves their limits unchanged. Thus the iterated representative in (185) is the same one selected by the compactness argument. On each finite slab the global energy bound also makes the gradients bounded in L^2 . A further diagonal weak extraction on the integer slabs therefore gives global weak convergence in (187); its limit agrees with the local weak gradient by uniqueness of distributional limits and does not change the velocity representative.

We next exhaust the exterior region uniformly in time and in the regularization parameter. Fix $T > 0$ and an integer $n \geq 1$, and set $R_1 = n + 1$ and $R_2 = 2(n + 1)$. For $0 < \varepsilon \leq 1$, positivity and support of the mollifier, Jensen's inequality, and Fubini give

$$\int_{|x| > R_1} |J_\varepsilon a(x)|^2 dx \leq \int_{|y| > R_1 - 1} |a(y)|^2 dy. \quad (190)$$

Since $R_1 - 1 = n$, Lemma 10.4 now yields, for every $t \in [0, T]$ and all $0 < \varepsilon \leq 1$,

$$\begin{aligned} \int_{|x| > 2(n+1)} |u_\varepsilon(x, t)|^2 dx &\leq \int_{|y| > n} |a(y)|^2 dy \\ &\quad + C \frac{\|a\|_2^2 T^{1/2} + \|a\|_2^3 T^{1/4}}{n+1} =: \eta_n(T). \end{aligned} \quad (191)$$

Both terms in $\eta_n(T)$ tend to zero as $n \rightarrow \infty$. Local weak convergence on every time slice and lower semicontinuity on bounded annuli imply, for each $S > 2(n + 1)$ and $t \in (0, T)$,

$$\int_{2(n+1) < |x| < S} |u(x, t)|^2 dx \leq \eta_n(T).$$

Letting $S \rightarrow \infty$ by monotone convergence gives the same tail bound for $u(\cdot, t)$. Thus the L^2 tails of both u_ε and u on $\mathbb{R}^3 \times (0, T)$ are uniformly small outside sufficiently large balls.

On each $\mathbb{R}^3 \times (\delta, T)$, spatial tail control and the local strong convergence from Lemma 11.1 give strong L^2 convergence. The initial strip is controlled by the uniform energy bound:

$$\int_0^\delta \int_{\mathbb{R}^3} |u_\varepsilon|^2 dx dt \leq \delta \|a\|_2^2.$$

A subsequence converges almost everywhere on $\mathbb{R}^3 \times (0, T)$ by local strong convergence on a countable exhaustion of the open slab. Fatou's lemma gives the same upper bound for u on $(0, \delta)$. Thus $\|u_\varepsilon - u\|_{L^2(\mathbb{R}^3 \times (0, \delta))}^2 \leq 4\delta \|a\|_2^2$. Letting $\delta \downarrow 0$ proves strong convergence on the full slab and hence (186).

Lemma 10.2 gives a uniform $L^{10/3}$ bound on every finite slab, and Fatou's lemma gives the same bound for u . Interpolating this bound with (186) proves (188) for $2 \leq q < 10/3$. In particular, $u_{\varepsilon_n} \rightarrow u$ in L^3 . Since spatial convolution is a contraction on L^3 and $J_{\varepsilon_n} u \rightarrow u$ in $L^3(\mathbb{R}^3 \times (0, T))$,

$$\|J_{\varepsilon_n} u_{\varepsilon_n} - u\|_{L^3} \leq \|u_{\varepsilon_n} - u\|_{L^3} + \|J_{\varepsilon_n} u - u\|_{L^3} \rightarrow 0.$$

The approximate-identity convergence follows first for finite sums of smooth compactly supported spatial functions times time indicators and then for all L^3 fields by density and contraction.

It remains to strengthen the slice convergence to global L^2 . The tail bound just proved makes the tails uniformly small for $u_{\varepsilon_n}(t)$ and $u(t)$ on every fixed slab, while Lemma 11.1 gives weak convergence in L^2 on each bounded spatial set for every $t > 0$. Approximating a global L^2 test by its restriction to a large ball proves $u_{\varepsilon_n}(\cdot, t) \rightharpoonup u(\cdot, t)$ in global L^2 . $\square$

Proposition 12.2. *For the fields in Proposition 12.1, and every $T > 0$, (u, Du) is a Leray-Hopf solution on $\mathbb{R}^3 \times [0, T]$ with initial datum a in the sense of Definition 7.2; that is, it satisfies (LH1)–(LH5).*

Proof. The local weak-gradient conclusion and measurability in (LH1) follow from Lemma 11.1; the global L^2 bounds and the global L^2 -integrability of Du follow from Lemma 10.1 and weak lower semicontinuity. For almost every t , the weak spatial gradient is therefore $Du(\cdot, t)$. Since every $u_{\varepsilon_n}(\cdot, t)$ is divergence-free, local weak convergence on each slice gives $\int u(x, t) \cdot \nabla \psi(x) dx = 0$ for every $\psi \in C_c^\infty(\mathbb{R}^3)$ and every $t > 0$.

For $w \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, Lemma 11.1 gives continuity of $t \mapsto \int u(x, t) \cdot w(x) dx$ on $(0, T)$. The initial value follows from the uniform modulus in Lemma 10.6: if $\omega_w(h) \rightarrow 0$ is its bound, then for every $t > 0$ the global weak slice convergence and $J_{\varepsilon_n} a \rightarrow a$ in L^2 give

$$\left| \int (u(x, t) - a(x)) \cdot w(x) dx \right| \leq \omega_w(t). \quad (192)$$

The convergence $J_{\varepsilon_n} a \rightarrow a$ follows from the approximate identity property in L^2 , proved by density of smooth compactly supported fields and L^2 contraction. The energy bound gives $\sup_{t \geq 0} \|u(t)\|_2 \leq \|a\|_2$. Approximating an arbitrary $w \in L^2(\mathbb{R}^3; \mathbb{R}^3)$ by smooth compactly supported fields therefore extends both the continuity and the limit at zero to every such w . Applying the compactness lemma on a slightly larger time interval also covers the right endpoint T . This proves (LH2) on $[0, T]$.

For the weak equation, use Lemma 9.7 with a solenoidal test $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$. The linear terms pass by strong L^2 convergence of u_{ε_n} , weak L^2 convergence of Du_{ε_n} , and compact support of the test. The transport tensors converge in $L^{3/2}$ because $J_{\varepsilon_n} u_{\varepsilon_n} \rightarrow u$ and $u_{\varepsilon_n} \rightarrow u$ in L^3 . Thus the limit is the weak identity (LH3); the pressure term vanishes since $\operatorname{div} \varphi = 0$.

Fix $t_0 \in (0, T]$. The global weak slice convergence in Proposition 12.1 gives weak L^2 convergence at t_0 , and the global weak gradient convergence restricts to $\mathbb{R}^3 \times (0, t_0)$. Lower

semicontinuity of the two Hilbert norms, applied to the energy equality (R5), gives

$$\frac{1}{2} \|u(t_0)\|_2^2 + \int_0^{t_0} \int_{\mathbb{R}^3} |Du|^2 \, dx \, dt \leq \liminf_{n \rightarrow \infty} \left(\frac{1}{2} \|u_{\varepsilon_n}(t_0)\|_2^2 + \int_0^{t_0} \int_{\mathbb{R}^3} |\nabla u_{\varepsilon_n}|^2 \, dx \, dt \right) = \frac{1}{2} \|a\|_2^2. \quad (193)$$

Here the equality on the right uses $J_{\varepsilon_n} a \rightarrow a$ in L^2 . This is exactly (LH4) at t_0 . At $t_0 = 0$, $u(0) = a$ gives equality in (LH4).

Finally, (LH2) implies $u(t) \rightharpoonup a$ in L^2 as $t \downarrow 0$, and (LH4) gives $\limsup_{t \downarrow 0} \|u(t)\|_2 \leq \|a\|_2$. The Hilbert-space identity $\|u(t) - a\|_2^2 = \|u(t)\|_2^2 + \|a\|_2^2 - 2\langle u(t), a \rangle$ then proves (LH5). In particular, weak lower semicontinuity in $L^2((0, t_0); L^2)$, rather than Fatou's lemma, supplies the dissipation term in (LH4). $\square$

Proposition 12.3. *For the fields in Proposition 12.1, define $p = P[u \otimes u]$ by Definition 8.1. For every $T > 0$,*

$$p_{\varepsilon_n} \longrightarrow p \quad \text{strongly in } L^{3/2}(\mathbb{R}^3 \times (0, T)). \quad (194)$$

Proof. By (R4), $p_{\varepsilon_n} = P[F_n]$ for $(F_n)_{ij} = (J_{\varepsilon_n} u_{\varepsilon_n})_i (u_{\varepsilon_n})_j$. The strong L^3 convergences in Proposition 12.1 and Hölder's inequality give

$$\|F_n - u \otimes u\|_{L^{3/2}} \leq \|J_{\varepsilon_n} u_{\varepsilon_n} - u\|_{L^3} \|u_{\varepsilon_n}\|_{L^3} + \|u\|_{L^3} \|u_{\varepsilon_n} - u\|_{L^3} \longrightarrow 0. \quad (195)$$

The pressure operator in Definition 8.1 is bounded on $L^{3/2}$, so applying it to (195) proves (194). This also identifies the limit with the canonical pressure, without a time-dependent additive constant. $\square$

Proof of Theorem 7.4. Fix $a \in J$ and take the sequence of regularized solutions with $\varepsilon_n \rightarrow 0$. Propositions 12.1–12.3 give a global Leray–Hopf pair (u, Du) and the pressure $p = P[u \otimes u]$. By Lemma 8.4, $u \otimes u \in L^{5/3}(\mathbb{R}^3 \times (0, T))$; the bounded $L^{5/3}$ extension in Definition 8.1 therefore gives $p \in L^{5/3}(\mathbb{R}^3 \times (0, T))$. This is the same canonical pressure used in Theorem 7.6. It remains to verify that the limit is suitable in the precise sense of Definition 1.5.

The regularity and weak-gradient passages in clause (S1), and the weighted weak lower semicontinuity used in (S4), are the corresponding clausewise arguments from the proof of Theorem 11.4. The approximating pairs here do not satisfy the unregularized Navier–Stokes equations: their transport field is $J_{\varepsilon} u_{\varepsilon}$. Thus Theorem 11.4 does not apply as a theorem to these pairs. We pass instead to the limit directly in Lemmas 9.6 and 9.7, which are the identities proved for the regularized pairs.

For (S1), measurability and the slice weak-gradient property follow from Proposition 12.1, and (LH1) supplies the local energy bounds. The pressure belongs to $L^{3/2}$ on every finite slab by Proposition 12.3. For (S2), the divergence-free identity was proved in (LH1).

Let $\varphi \in C_c^\infty(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$. Passing to the limit in Lemma 9.7 is justified by the linear convergences in Proposition 12.1, the strong pressure convergence in Proposition 12.3, and

$$\|u_{\varepsilon_n} \otimes (J_{\varepsilon_n} u_{\varepsilon_n}) - u \otimes u\|_{L^{3/2}} \leq \|u_{\varepsilon_n}\|_{L^3} \|J_{\varepsilon_n} u_{\varepsilon_n} - u\|_{L^3} + \|u\|_{L^3} \|u_{\varepsilon_n} - u\|_{L^3} \longrightarrow 0. \quad (196)$$

This gives (S3) for every vector test, with the pressure sign in Definition 1.5.

Take now a nonnegative $\psi \in C_c^\infty(\mathbb{R}^3 \times (0, T))$. In the right side of the identity of Lemma 9.6, strong L^3 convergence gives $|u_{\varepsilon_n}|^2 \rightarrow |u|^2$ in $L^{3/2}$, since

$$\||u_{\varepsilon_n}|^2 - |u|^2\|_{L^{3/2}} \leq (\|u_{\varepsilon_n}\|_{L^3} + \|u\|_{L^3}) \|u_{\varepsilon_n} - u\|_{L^3} \longrightarrow 0.$$

The additional mollified transport factor in Lemma 9.6 causes no loss:

$$\begin{aligned} \||u_{\varepsilon_n}|^2 J_{\varepsilon_n} u_{\varepsilon_n} - |u|^2 u\|_{L^1} &\leq \|u_{\varepsilon_n}\|_{L^3}^2 \|J_{\varepsilon_n} u_{\varepsilon_n} - u\|_{L^3} \\ &\quad + (\|u_{\varepsilon_n}\|_{L^3} + \|u\|_{L^3}) \|u_{\varepsilon_n} - u\|_{L^3} \|u\|_{L^3} \longrightarrow 0. \end{aligned} \quad (197)$$

Also,

$$\|p_{\varepsilon_n} u_{\varepsilon_n} - pu\|_{L^1} \leq \|p_{\varepsilon_n} - p\|_{L^{3/2}} \|u_{\varepsilon_n}\|_{L^3} + \|p\|_{L^{3/2}} \|u_{\varepsilon_n} - u\|_{L^3} \longrightarrow 0. \quad (198)$$

Multiplication by $\sqrt{\psi}$ is bounded on L^2 , so the weak gradient convergence gives

$$2 \iint |Du|^2 \psi \leq \liminf_{n \rightarrow \infty} 2 \iint |Du_{\varepsilon_n}|^2 \psi.$$

Passing to the limit in Lemma 9.6 now yields (S4) with $f = 0$. These verifications hold on every finite slab, and the same u, Du, p work for all $q > 5/2$. Thus $(u, Du, p, 0)$ is suitable on $\mathbb{R}^3 \times (0, \infty)$ for every such q , proving Theorem 7.4. $\square$

Proof of Corollary 7.5. Choose any $q > 5/2$, for example $q = 3$. Theorem 7.4 gives a suitable weak solution on $\mathbb{R}^3 \times (0, \infty)$ in the exact class of Definition 1.5. Apply the Caffarelli–Kohn–Nirenberg theorem, Theorem 2.10, to this solution with $\Omega = \mathbb{R}^3$, $I = (0, \infty)$, and $f = 0$. It follows that the singular set has zero one-dimensional parabolic Hausdorff measure, as asserted. $\square$

Remark 12.4 (Departure from [6]). Theorem 1.1 of [6, printed p. 212] states Leray’s existence result but does not supply the approximation and suitability proof used here. We obtain the solution from the regularized equations, retain weak limits of every time slice, and pass the regularized momentum and local energy identities directly to the limit. In the energy inequality, the dissipation is controlled by weak lower semicontinuity in L^2 ; the available gradient convergence is weak and does not justify a Fatou argument.

13. FORCED LERAY SOLUTIONS

The force enters the construction in two different ways. Its solenoidal part does work in the energy equation, while its gradient part is represented by a pressure. The latter pressure is local in the space-time integrability needed for suitability, but it need not have the global integrability of the quadratic pressure. We first fix this pressure split, then add the force to the regularized construction and pass to the limit.

13.1. The pressure of the force.

Lemma 13.1 (Force pressure). *Let $0 < T < \infty$ and $f \in L^2(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$. There is a unique $p_f \in L^2((0, T); L^6(\mathbb{R}^3))$ obtained from the Fourier multiplier*

$$\widehat{p_f}(\xi, t) = -\frac{i \xi \cdot \widehat{f}(\xi, t)}{2\pi |\xi|^2} \quad (\xi \neq 0), \quad (199)$$

such that, in spatial distributions for almost every t ,

$$\Delta p_f = \operatorname{div} f, \quad \nabla p_f = (I - \mathbb{P})f, \quad \|p_f\|_{L_t^2 L_x^6} \leq C \|f\|_{L_{t,x}^2}. \quad (200)$$

Here C depends only on the dimension. On every bounded cylinder $K \times J \subset \mathbb{R}^3 \times (0, T)$,

$$\|p_f\|_{L^{3/2}(K \times J)} \leq |K|^{1/2} |J|^{1/6} \|p_f\|_{L_t^2 L_x^6}. \quad (201)$$

Proof. For a smooth force whose spatial Fourier transform is supported in an annulus, the multiplier in (199) defines a function in $H^1(\mathbb{R}^3)$ at each time. The Fourier convention of Section 9 gives

$$\widehat{\nabla p_f}(\xi, t) = \frac{\xi \otimes \xi}{|\xi|^2} \widehat{f}(\xi, t), \quad \widehat{\Delta p_f}(\xi, t) = 2\pi i \xi \cdot \widehat{f}(\xi, t).$$

Plancherel therefore gives $\|\nabla p_f(t)\|_2 \leq \|f(t)\|_2$. The homogeneous Sobolev inequality used in the proof of Lemma 9.5 gives $\|p_f(t)\|_6 \leq C \|f(t)\|_2$. Approximation of f in $L_{t,x}^2$ by smooth

forces with annular Fourier cutoffs makes the corresponding p_f Cauchy in $L_t^2 L_x^6$. The limit has the stated multiplier and identities in distributions, and the same bound. The construction is a bounded linear map from $L_{t,x}^2$ to $L_t^2 L_x^6$, so it also gives a strongly measurable representative.

If two such pressures have the same Laplacian, their difference is harmonic in space for almost every time and belongs to $L^6(\mathbb{R}^3)$. The mean-value property on balls of radius R bounds its value at any point by $CR^{-1/2}$ times its global L^6 norm; letting $R \rightarrow \infty$ shows that the difference is zero. Finally, the finite-measure embeddings $L^6(K) \hookrightarrow L^{3/2}(K)$ and $L^2(J) \hookrightarrow L^{3/2}(J)$ give (201). $\square$

Proof of Theorem 7.10. Let $p_N = P[u \otimes u]$ and let p_f be given by Lemma 13.1. The forced energy class gives $u \in L_t^\infty L_x^2$ and $Du \in L_{t,x}^2$ on $(0, T)$. The interpolation calculation in the proof of Lemma 8.4, before its final use of the unforced energy bound, gives $u \in L_{t,x}^{10/3}$. Thus $u \otimes u \in L_{t,x}^{5/3}$ and the boundedness of P gives $p_N \in L_{t,x}^{5/3}$. Estimate (201) now implies $p = p_N + p_f \in L_{\text{loc}}^{3/2}$ on the cylinder.

Set $r = f - \nabla p_f$. The force-pressure multiplier gives $r = \mathbb{P}f \in L_{t,x}^2$ and $\text{div } r = 0$ in space for almost every time. For the solenoidal part of the Helmholtz decomposition, the forced weak identity applied to w_R therefore has source term $-\iint r \cdot w_R$: the gradient part of f pairs to zero with the divergence-free field w_R . The cutoff fields in Lemma 8.3 also satisfy

$$\|w_R - \varphi^\sigma\|_{L^2(\mathbb{R}^3 \times J)} \leq C_\varphi R^{-3/2}, \quad (202)$$

where J is the compact time support of φ . Indeed, $w_R - \varphi^\sigma = \text{curl}((\chi_R - 1)A)$; outside B_R the derivative of A is bounded by $C_\varphi(1 + |x|)^{-3}$, and the annular term $R^{-1}A$ is bounded by $C_\varphi R^{-3}$. Their squared integrals are $O(R^{-3})$.

For the gradient part, the velocity and p_N terms evaluated on $\nabla(\chi_R \psi)$ cancel by the gradient-test argument in the proof of Theorem 7.6; the residual work vanishes because $\text{div } r = 0$. The pointwise decay estimates in Lemma 8.3 dominate the vector cutoff tests w_R and $\nabla(\chi_R \psi)$ and their time derivatives by a common spatial profile $C_\varphi(1 + |x|)^{-3}$; their spatial derivatives paired with $u \otimes u$ and p_N decay faster. This profile belongs to $L^2(\mathbb{R}^3) \cap L^{5/2}(\mathbb{R}^3)$. Since $u, Du, r \in L_{t,x}^2$ and $u \otimes u, p_N \in L_{t,x}^{5/3}$, Hölder's inequality makes the corresponding majorants integrable on the finite time support. Dominated convergence therefore removes the Helmholtz cutoffs in the combined identity and gives the arbitrary-test identity with p_N and residual r . Finally, $\nabla p_f \in L_{t,x}^2$ and integration by parts against the compactly supported φ give $-\iint r \cdot \varphi = -\iint f \cdot \varphi - \iint p_f \text{div } \varphi$. The last term is integrable because $p_f \in L_t^2 L_x^6$. This is the asserted identity with pressure $p_N + p_f$ and force f .

When $f = 0$, the multiplier gives $p_f = 0$. The conclusion is then exactly Theorem 7.6, including its conditional $L_t^\infty L_x^{3/2}$ assertion. No such conditional bound is asserted for the total pressure when $f \neq 0$. $\square$

13.2. Regularization with a rough force. Write $\mathcal{P}(\xi) = I - \xi \otimes \xi / |\xi|^2$ for $\xi \neq 0$ and $\mathcal{P}(0) = I$. This is the Fourier multiplier of $\mathbb{P}$ on L^2 . For fixed $\varepsilon > 0$, the forced mild equation is

$$u_\varepsilon(t) = S(t)J_\varepsilon a - \int_0^t K(t-s)F_\varepsilon(u_\varepsilon(s))ds + \int_0^t S(t-s)\mathbb{P}f(s)ds, \quad (F_\varepsilon(v))_{ij} = (J_\varepsilon v)_{i,j}. \quad (203)$$

The last integral is an L^2 Bochner integral. Its norm on a time interval I of length at most h is bounded by $h^{1/2} \|f\|_{L^2(I; L^2)}$.

Lemma 13.2 (Forced regularized solution). *Let $a \in J$, $f \in L^2(\mathbb{R}^3 \times (0, T); \mathbb{R}^3)$, and $\varepsilon > 0$. There is a unique continuous L^2 solution of (203) on $[0, T]$. It is weakly divergence free at every time, has initial value $J_\varepsilon a$, and for every $t \in [0, T]$*

$$\|u_\varepsilon(t)\|_2^2 + 2 \int_0^t \|Du_\varepsilon(s)\|_2^2 ds \leq \|J_\varepsilon a\|_2^2 + 2 \int_0^t \int_{\mathbb{R}^3} f \cdot u_\varepsilon dx ds. \quad (204)$$

For every nonnegative $\psi \in C_c^\infty(\mathbb{R}^3 \times (0, T))$, with

$$p_\varepsilon = P[F_\varepsilon(u_\varepsilon)] + p_f,$$

the following local energy inequality holds:

$$\begin{aligned} 2 \iint |Du_\varepsilon|^2 \psi &\leq \iint |u_\varepsilon|^2 (\partial_t \psi + \Delta \psi) \\ &\quad + \iint (|u_\varepsilon|^2 J_\varepsilon u_\varepsilon + 2p_\varepsilon u_\varepsilon) \cdot \nabla \psi + 2 \iint (f \cdot u_\varepsilon) \psi. \end{aligned} \quad (205)$$

For every compactly supported smooth vector test φ on $\mathbb{R}^3 \times (0, T)$,

$$\iint [-u_\varepsilon \cdot \partial_t \varphi - (J_\varepsilon u_\varepsilon)_j (u_\varepsilon)_i \partial_j \varphi_i + (Du_\varepsilon)_{ij} \partial_j \varphi_i - p_\varepsilon \operatorname{div} \varphi - f \cdot \varphi] = 0. \quad (206)$$

All constants in the finite-time energy bounds below depend only on T , $\|a\|_2$, and $\|f\|_{L^2((0, T) \times \mathbb{R}^3)}$, uniformly in ε .

Proof. For smooth compactly supported f , add the Duhamel term in (203) to the contraction map in the proof of Lemma 9.3. The nonlinear difference estimate is unchanged. The force term is continuous in L^2 and has norm at most $h^{1/2} \|f\|_{L^2(I; L^2)}$ on an interval I of length h . Absolute continuity of the integral of $|f|^2$ makes this bound uniformly small on short intervals. The same contraction argument therefore gives local existence and uniqueness. Since the mollified initial datum and the smooth force belong to every spatial Sobolev space, the differentiated mild equation gives the regularity used in the proof of Theorem 9.4. In the H^k energy estimate, the force pairing is bounded by $\|f\|_{H^k}^2 + \|u_\varepsilon\|_{H^k}^2$; Gronwall therefore bounds each H^k norm on every finite time interval, with a constant allowed to depend on the smooth force and its derivatives. This prevents a finite-time loss of the regularity needed in the cutoff energy and local-energy calculations.

The Fourier identities used in the proof of Theorem 9.4 give $\mathcal{L}F_\varepsilon = \operatorname{div} F_\varepsilon + \nabla P[F_\varepsilon]$, and Lemma 13.1 gives $\mathbb{P}f = f - \nabla p_f$. Thus the projected mild equation has the unprojected form $\partial_t u_\varepsilon - \Delta u_\varepsilon + \operatorname{div} F_\varepsilon(u_\varepsilon) + \nabla p_\varepsilon = f$. Multiply this equation by $2u_\varepsilon$ on $B_R \times (\delta, t)$, using the spatial cutoff χ_R from that proof. The transport cancellation and the pressure divergence are unchanged, and the added term is $2f \cdot u_\varepsilon$. The quadratic-pressure cutoff errors vanish as $R \rightarrow \infty$ by the L^2 estimates used for (178). The additional force-pressure cutoff term is bounded, on the annulus $A_R = \{R < |x| < 2R\}$, by

$$\frac{C}{R} \int_\delta^t \int_{A_R} |p_f u_\varepsilon| \leq C \int_\delta^t \|p_f(s)\|_6 \|u_\varepsilon(s)\|_{L^2(A_R)} ds.$$

Here Hölder's inequality contributes $|A_R|^{1/3} = O(R)$, which cancels R^{-1} . On $[\delta, t]$, p_f is bounded in L^6 and the continuous L^2 curve $u_\varepsilon(s)$ has uniformly small tails, so this term tends to zero. The force pairing is integrable by Cauchy-Schwarz, and the cutoff force integrals converge by dominated convergence. Letting $\delta \downarrow 0$ using continuity in L^2 gives the energy

equality for this smooth force. In particular,

$$\|u_\varepsilon(t)\|_2^2 + 2 \int_0^t \|Du_\varepsilon(s)\|_2^2 ds \leq \|a\|_2^2 + \|f\|_{L^2((0,t) \times \mathbb{R}^3)}^2 + \int_0^t \|u_\varepsilon(s)\|_2^2 ds. \quad (207)$$

Gronwall bounds the L^2 norm on every finite interval, so the local solution can be restarted before any finite endpoint. This proves global existence for smooth f and uniqueness follows from the same interval contraction argument.

For a general $f \in L^2((0, T) \times \mathbb{R}^3)$, choose smooth compactly supported $f_m \rightarrow f$ strongly in L^2 and solve the smooth-force equation. Estimate (207) bounds these solutions in $C([0, T]; L^2)$ and their gradients in $L_{t,x}^2$, uniformly in m . On each short interval, subtracting the two mild equations and using the Volterra estimate of Lemma 9.3 shows convergence in $C_t L_x^2$ when f_m is Cauchy in $L_{t,x}^2$. A finite subdivision of $[0, T]$ gives convergence on the whole interval. The limit is the unique continuous mild solution; weak compactness of the gradients gives its gradient data.

The force work converges for every t because $f_m \rightarrow f$ in $L_{t,x}^2$ and $u_{\varepsilon,m} \rightarrow u_\varepsilon$ in $C_t L_x^2$. Weak lower semicontinuity of the gradient norm therefore gives (204) at every time. For fixed ε , the map $v \mapsto F_\varepsilon(v)$ is continuous from L^2 to L^2 on bounded sets: J_ε maps L^2 to L^∞ , so strong $C_t L_x^2$ convergence gives strong $L_{t,x}^2$ convergence of the tensors. The L^2 pressure bound in Lemma 9.1 then gives strong convergence of their quadratic pressures. The force pressures converge in $L_t^2 L_x^6$ by Lemma 13.1. Passing to the limit in the smooth weak momentum identities proves (206).

For each smooth-force solution, multiplication by $2u_{\varepsilon,m}\psi$ gives equality in (205). The velocity-square and transport terms converge locally by strong $C_t L_x^2$ convergence and the $L^2 \rightarrow L^\infty$ bound for J_ε . The force term converges by Cauchy-Schwarz. For the pressure flux, the quadratic pressure converges strongly in $L_{t,x}^2$ and the velocity strongly in L^2 locally; the force pressure converges in $L_t^2 L_x^6$, while the velocities converge in $L_t^2 L_x^{6/5}$ on the compact spatial support. Finally, $\sqrt{\psi} Du_{\varepsilon,m}$ converges weakly in L^2 , so its squared norm is lower semicontinuous. These facts give the inequality (205). The construction on successive finite intervals agrees by mild uniqueness, so it defines one solution on every finite time interval. $\square$

13.3. Bounds and spatial tails. For $T > 0$ put

$$E_T = e^T \left(\|a\|_2^2 + \|f\|_{L^2((0,T) \times \mathbb{R}^3)}^2 \right),$$

$$G_T^2 = \frac{1}{2} \left(\|a\|_2^2 + \|f\|_{L^2((0,T) \times \mathbb{R}^3)}^2 + T E_T \right).$$

These quantities are finite and independent of ε .

Lemma 13.3 (Forced energy bounds). *For the solutions of Lemma 13.2,*

$$\sup_{0 \leq t \leq T} \|u_\varepsilon(t)\|_2^2 \leq E_T, \quad \int_0^T \|Du_\varepsilon(t)\|_2^2 dt \leq G_T^2. \quad (208)$$

Consequently,

$$\|u_\varepsilon\|_{L^{10/3}(\mathbb{R}^3 \times (0,T))}^{10/3} + \|J_\varepsilon u_\varepsilon\|_{L^{10/3}(\mathbb{R}^3 \times (0,T))}^{10/3} \leq C E_T^{2/3} G_T^2, \quad (209)$$

and, with $p_{N,\varepsilon} = P[F_\varepsilon(u_\varepsilon)]$,

$$\|p_{N,\varepsilon}\|_{L^{5/3}(\mathbb{R}^3 \times (0,T))} + \|p_{N,\varepsilon}\|_{L^{3/2}(\mathbb{R}^3 \times (0,T))} \leq C_T. \quad (210)$$

Here C_T depends only on T , $\|a\|_2$, $\|f\|_{L^2((0,T) \times \mathbb{R}^3)}$, and the dimension-three interpolation and Riesz-transform constants.

Proof. Use $2f \cdot u_\varepsilon \leq |f|^2 + |u_\varepsilon|^2$ in (204). The resulting inequality and Gronwall give the first bound in (208). Substitution back into the same inequality gives

$$2 \int_0^T \|Du_\varepsilon(t)\|_2^2 dt \leq \|a\|_2^2 + \|f\|_{L^2((0,T) \times \mathbb{R}^3)}^2 + TE_T,$$

which is the second bound.

For almost every t , the weak gradient and the homogeneous Sobolev inequality give the spatial interpolation estimate

$$\|u_\varepsilon(t)\|_{10/3}^{10/3} \leq C \|u_\varepsilon(t)\|_2^{4/3} \|Du_\varepsilon(t)\|_2^2.$$

Integrating in time and using Young's convolution inequality for J_ε proves (209). Thus $F_\varepsilon(u_\varepsilon) \in L_{t,x}^{5/3}$, and the $L^{5/3}$ Riesz-transform bound in Definition 8.1 proves the first pressure bound. On the finite slab, the same tensor is in L^1 with

$$\|F_\varepsilon(u_\varepsilon)\|_{L_{t,x}^1} \leq TE_T.$$

Interpolation between L^1 and $L^{5/3}$ gives the $L^{3/2}$ bound in (210); the constants have the stated dependence. $\square$

Lemma 13.4 (Forced spatial tails). *There is a constant C depending only on the dimension, the cutoff profile, and the Riesz-transform bounds such that for $0 < R_1 < R_2$, $0 < \varepsilon \leq 1$, and $0 \leq t \leq T$,*

$$\begin{aligned} \int_{|x|>R_2} |u_\varepsilon(x,t)|^2 dx &\leq \int_{|x|>R_1} |J_\varepsilon a(x)|^2 dx + \frac{C(E_T^{1/2} G_T T^{1/2} + E_T^{3/4} G_T^{3/2} T^{1/4})}{R_2 - R_1} \\ &\quad + 2E_T^{1/2} T^{1/2} \|(\mathbb{P}f) \mathbf{1}_{\{|x|>R_1\}}\|_{L^2((0,T) \times \mathbb{R}^3)}. \end{aligned} \quad (211)$$

Also

$$\int_{|x|>R_1} |J_\varepsilon a(x)|^2 dx \leq \int_{|x|>R_1-1} |a(x)|^2 dx. \quad (212)$$

Proof. Set $D = R_2 - R_1$ and choose a smooth radial function ζ which is zero on B_{R_1} , one outside B_{R_2} , and satisfies $|\nabla \zeta| \leq C/D$. First take a smooth force approximation from the proof of Lemma 13.2, and multiply its classical equation by $2u_\varepsilon \zeta \chi_N$ on (δ, t) , where χ_N is one on B_N , zero outside B_{2N} , and $|\nabla \chi_N| \leq C/N$. Spatial integration by parts gives the usual viscous, transport, and quadratic-pressure fluxes from the proof of Lemma 10.4. The remaining force and force-pressure terms combine exactly: since $\nabla p_f = (I - \mathbb{P})f$ and the velocity is divergence free,

$$\int f \cdot u_\varepsilon q + \int p_f u_\varepsilon \cdot \nabla q = \int (\mathbb{P}f) \cdot u_\varepsilon q \quad (213)$$

for each compactly supported scalar cutoff q . The identity follows by integrating $\nabla p_f \cdot (u_\varepsilon q)$ by parts; there is no boundary term because q is compactly supported.

Let $N \rightarrow \infty$ in the cutoff identity. The endpoint terms and the dissipation converge by dominated convergence. The terms containing $\nabla \chi_N$ vanish: their absolute values are bounded by C/N times the space-time integrals of $|u_\varepsilon| |Du_\varepsilon|$, $|J_\varepsilon u_\varepsilon| |u_\varepsilon|^2$, and $|p_{N,\varepsilon}| |u_\varepsilon|$. These are finite on (δ, t) . For the last two, the Gagliardo–Nirenberg estimate in the proof of Lemma 10.4 gives

$$\|u_\varepsilon(s)\|_4^2 + \|p_{N,\varepsilon}(s)\|_2 \leq CE_T^{1/4} \|Du_\varepsilon(s)\|_2^{3/2};$$

the force pressure has already been removed by (213). Letting $\delta \downarrow 0$ is justified by the energy inequality and the same integrable bounds.

The initial term is at most the first term on the right of (211). For the cutoff-gradient errors, the viscous term is bounded by $CD^{-1}E_T^{1/2}G_T T^{1/2}$. The transport and quadratic-pressure terms are each bounded by $CD^{-1}E_T^{3/4}G_T^{3/2}T^{1/4}$, by Hölder's inequality in time and the preceding Gagliardo–Nirenberg bound. The remaining projected-force work, which carries the factor 2 from multiplying the local energy balance by $2u_\varepsilon$, is bounded by

$$2 \int_0^t \int_{|x|>R_1} |\mathbb{P}f| |u_\varepsilon| \leq 2T^{1/2}E_T^{1/2} \|(\mathbb{P}f)\mathbf{1}_{\{|x|>R_1\}}\|_{L^2((0,T)\times\mathbb{R}^3)}.$$

This proves (211) for smooth forces. For rough f , the smooth-force solutions converge in $C_t L_x^2$ on finite slabs, their gradients converge weakly in $L_{t,x}^2$, and $\mathbb{P}f_m \rightarrow \mathbb{P}f$ strongly in $L_{t,x}^2$. Lower semicontinuity and convergence of the other terms pass the estimate to the solution of Lemma 13.2.

Finally, Jensen's inequality and $\text{supp } \rho_\varepsilon \subset B_\varepsilon$ give

$$\int_{|x|>R_1} |J_\varepsilon a(x)|^2 dx \leq \int_{|y|>R_1-\varepsilon} |a(y)|^2 dy \leq \int_{|y|>R_1-1} |a(y)|^2 dy,$$

which proves (212). $\square$

Remark 13.5 (Spatial tail of the projected force). The force contribution in (211) is the tail of $\mathbb{P}f$. The pressure p_f cancels the gradient component of f in the localized energy identity, so the relevant field is its solenoidal projection. For a general L^2 force this projection is nonlocal; its tail tends to zero because $\mathbb{P}f \in L^2$, but it is not bounded by the tail of f with the same radius. This is the form of the estimate used below.

Lemma 13.6 (Forced time modulus). *Fix $T > 0$ and a compact set $K \subset \mathbb{R}^3$. There is a finite constant $C_{T,K}$, independent of $0 < \varepsilon \leq 1$, $s, t \in [0, T]$, and $w \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$ supported in K , such that*

$$\left| \int (u_\varepsilon(x, t) - u_\varepsilon(x, s)) \cdot w(x) dx \right| \leq C_{T,K} \left(\|\Delta w\|_2 + \|\nabla w\|_\infty + \|w\|_2 + \|\text{div } w\|_\infty \right) |t - s|^{2/5}. \quad (214)$$

The constant depends only on $T, K, \|a\|_2, \|f\|_{L^2((0,T)\times\mathbb{R}^3)}$, and the dimension-three operator bounds.

Proof. For $0 < s < t$, pair (206) with w between the two times. The heat and transport terms are bounded by $E_T^{1/2} \|\Delta w\|_2 (t - s)$ and $E_T \|\nabla w\|_\infty (t - s)$. The quadratic pressure term is bounded by $C_T |K|^{2/5} \|\text{div } w\|_\infty (t - s)^{2/5}$ using $p_{N,\varepsilon} \in L_{t,x}^{5/3}$. The force term satisfies

$$\left| \int_s^t \int f \cdot w \right| \leq |t - s|^{1/2} \|f\|_{L^2((0,T)\times\mathbb{R}^3)} \|w\|_2.$$

By Hölder's inequality in space and Lemma 13.1, the force-pressure term is bounded by

$$|t - s|^{1/2} \|p_f\|_{L^2((0,T);L^6)} \|\text{div } w\|_{L^{6/5}(K)} \leq |t - s|^{1/2} |K|^{5/6} \|p_f\|_{L_t^2 L_x^6} \|\text{div } w\|_\infty.$$

When $|t - s| \leq 1$, both $|t - s|$ and $|t - s|^{1/2}$ are at most $|t - s|^{2/5}$. When $|t - s| \geq 1$, the endpoint bound $2E_T^{1/2} \|w\|_2$ suffices. Continuity in L^2 handles $s = 0$. $\square$

13.4. Compactness and the forced limit.

Proposition 13.7 (Compactness with a fixed force). *Let $a \in J$, let $f \in L^2((0, T) \times \mathbb{R}^3)$ for every finite T , and let $\varepsilon_n \downarrow 0$. There are a subsequence and measurable fields $u: \mathbb{R}^3 \times [0, \infty) \rightarrow \mathbb{R}^3$ and $Du: \mathbb{R}^3 \times (0, \infty) \rightarrow \mathbb{R}^{3 \times 3}$ such that for every finite T ,*

$$u_{\varepsilon_n} \longrightarrow u \quad \text{strongly in } L^2(\mathbb{R}^3 \times (0, T)), \quad (215)$$

$$Du_{\varepsilon_n} \rightharpoonup Du \quad \text{weakly in } L^2(\mathbb{R}^3 \times (0, T)), \quad (216)$$

$$u_{\varepsilon_n} \longrightarrow u \quad \text{strongly in } L^r(\mathbb{R}^3 \times (0, T)) \quad (2 \leq r < 10/3), \quad (217)$$

$$J_{\varepsilon_n} u_{\varepsilon_n} \longrightarrow u \quad \text{strongly in } L^3(\mathbb{R}^3 \times (0, T)). \quad (218)$$

For every $t > 0$, $u_{\varepsilon_n}(\cdot, t) \rightharpoonup u(\cdot, t)$ in $L^2(\mathbb{R}^3)$, and $Du(\cdot, t)$ is the weak spatial gradient of $u(\cdot, t)$ for almost every $t > 0$.

Proof. Fix a finite T . Lemma 13.3 supplies the uniform slice and gradient bounds in Lemma 11.1. When its test field w is replaced by a translate of a fixed spatial mollifier, the proof of Lemma 13.6 gives a time modulus independent of the translate: all spatial norms of w are translation invariant, and on the fixed compact set of centers the $L^{6/5}$ and $L^{5/2}$ norms of $\operatorname{div} w$ are bounded by its supremum norm times a constant depending only on the set. The extra force terms carry the factor $\|w\|_2 |t - s|^{1/2}$ and obey the same uniform bound. The Arzelà–Ascoli and spatial-mollification proof of Lemma 11.1 therefore gives local strong L^2 convergence, weak convergence of every time slice, and local weak convergence of the gradients. A diagonal extraction over integer time intervals, followed by weak extraction in global L^2 for the gradients, gives one subsequence for all finite T .

We upgrade the velocity convergence using the per-solution exterior estimate in Lemma 13.4, whose constants are uniform for $0 < \varepsilon \leq 1$ and $0 \leq t \leq T$. Given $\eta > 0$, first take $R_1 > 1$ so that the L^2 tail of a outside B_{R_1-1} and the L^2 tail of $\mathbb{P}f$ outside B_{R_1} on $(0, T)$ are each small enough that their contributions in (211) sum to less than $\eta/2$. Such a choice is possible because both fields belong to L^2 . Next choose $R_2 > R_1$ so that the term with denominator $R_2 - R_1$ is less than $\eta/2$. Then

$$\sup_n \sup_{0 \leq t \leq T} \int_{|x| > R_2} |u_{\varepsilon_n}(x, t)|^2 dx \leq \eta.$$

Weak convergence on every time slice and lower semicontinuity on bounded annuli give the same tail bound for $u(\cdot, t)$, first on $R_2 < |x| < S$ and then on $|x| > R_2$ by monotone convergence. Thus the space-time L^2 tails of the approximants and the limit are uniformly small. Local strong convergence then gives strong convergence on $\mathbb{R}^3 \times (\delta, T)$ for every $\delta > 0$. On the initial strip the energy bound gives

$$\int_0^\delta \int_{\mathbb{R}^3} |u_{\varepsilon_n}|^2 dx dt \leq \delta E_T;$$

a subsequence converging almost everywhere and Fatou's lemma give the same bound for u . Sending $\delta \downarrow 0$ proves (215).

The $L^{10/3}$ bounds in Lemma 13.3 and Fatou's lemma give a uniform $L^{10/3}$ bound for the limit. Interpolation with (215) proves (217). In particular the convergence holds in L^3 . Spatial convolution is contractive on L^3 , and $J_{\varepsilon_n} u \rightarrow u$ strongly in L^3 by the approximate-identity theorem. Hence

$$\|J_{\varepsilon_n} u_{\varepsilon_n} - u\|_{L^3} \leq \|u_{\varepsilon_n} - u\|_{L^3} + \|J_{\varepsilon_n} u - u\|_{L^3} \longrightarrow 0.$$

Finally, local weak slice convergence and the uniform tails extend the slice convergence from each ball to every global L^2 test, by density. This also identifies the representative obtained

from iterated mollified limits. If σ denotes the compactness reindexing, then for every point, component, and fixed mollifier scale the corresponding pairing converges along σ by the all-time weak slice convergence just proved. Restricting that convergent sequence once more along σ leaves its limit unchanged, so the limit along $\sigma \circ \sigma$ agrees with the one along σ . Thus the iterated mollified-limit representative is the same u . This proves the proposition. $\square$

Proof of Theorem 7.8. Choose $\varepsilon_n \downarrow 0$ and take the fields from Proposition 13.7. Set $p_N = P[u \otimes u]$ and let p_f be the force pressure from Lemma 13.1; put $p = p_N + p_f$. We check the global Leray–Hopf clauses and local suitability.

The energy and gradient bounds in (LH1) follow from Lemma 13.3 and weak lower semi-continuity. The slice weak-gradient property follows from Proposition 13.7. Weak divergence freedom at each positive time follows by passing to the limit in the corresponding identity for $u_{\varepsilon_n}(t)$ on compact spatial tests. The same proposition gives weak convergence of each slice.

For $w \in C_c^\infty(\mathbb{R}^3; \mathbb{R}^3)$, the time-pairing estimate in the proof of Lemma 13.6 is uniform in ε and tends to zero with $|t - s|$. Passing to the limit gives continuity of $t \mapsto \int u(t) \cdot w$ on $(0, T)$. At zero, use $u_{\varepsilon_n}(0) = J_{\varepsilon_n} a$ and $J_{\varepsilon_n} a \rightarrow a$ in L^2 ; the same uniform modulus gives $u(t) \rightarrow a$ as $t \downarrow 0$. The uniform energy bound and density of compactly supported smooth tests extend weak continuity to every L^2 test and across both endpoints of $[0, T]$. This proves (LH2).

For an arbitrary compactly supported vector test φ , pass to the limit in (206). The time and viscous terms pass by strong L^2 convergence of u_{ε_n} and weak L^2 convergence of Du_{ε_n} . The force term is fixed and pairs continuously with the strong L^2 limit. For the transport term,

$$\|(J_{\varepsilon_n} u_{\varepsilon_n}) \otimes u_{\varepsilon_n} - u \otimes u\|_{L^{3/2}} \leq \|J_{\varepsilon_n} u_{\varepsilon_n}\|_{L^3} \|u_{\varepsilon_n} - u\|_{L^3} + \|J_{\varepsilon_n} u_{\varepsilon_n} - u\|_{L^3} \|u\|_{L^3} \rightarrow 0.$$

The same product estimate and boundedness of P give $p_{N, \varepsilon_n} \rightarrow p_N$ strongly in $L^{3/2}$ on each finite slab; p_f is fixed. Thus the pressure term also converges, and the limit is (S3) with force f . Restricting to solenoidal tests gives (FLH1).

Fix $t_0 \in (0, T]$. The energy inequality (204) holds at that time for every n . The left-hand side is lower semicontinuous under the weak slice convergence and weak gradient convergence on $(0, t_0)$. On the right,

$$\int_0^{t_0} \int f \cdot u_{\varepsilon_n} \rightarrow \int_0^{t_0} \int f \cdot u$$

by $f \in L^2_{t,x}$ and strong convergence in (215); also $J_{\varepsilon_n} a \rightarrow a$ in L^2 . Thus the every-time forced energy inequality holds. At $t_0 = 0$ it is an identity. The work term tends to zero as $t_0 \downarrow 0$, since

$$\left| \int_0^{t_0} \int f \cdot u \right| \leq \|f\|_{L^2((0, t_0) \times \mathbb{R}^3)} t_0^{1/2} \sup_{0 \leq t \leq T} \|u(t)\|_2 \rightarrow 0.$$

Weak convergence to a at zero and the energy inequality now give $\limsup_{t \downarrow 0} \|u(t)\|_2 \leq \|a\|_2$; the Hilbert-space identity for $\|u(t) - a\|_2^2$ proves strong attainment (LH5).

We now pass the local energy inequality. The quadratic pressures $p_{N, \varepsilon_n} = P[(J_{\varepsilon_n} u_{\varepsilon_n}) \otimes u_{\varepsilon_n}]$ converge strongly to p_N in $L^{3/2}$ by (218) and boundedness of P on $L^{3/2}$. The force pressure p_f is fixed. On any compact cylinder it belongs to L^2 , so its product with u_{ε_n} converges in L^1 by strong local L^2 convergence. The cubic transport flux converges in L^1 by the strong L^3 convergences in Proposition 13.7; the force work converges by $f \in L^2_{\text{loc}}$ and the same strong L^2 convergence. Finally, multiplication by $\sqrt{\psi}$ is bounded on L^2 , so weak convergence of the gradients gives lower semicontinuity of $\int |Du|^2 \psi$. Passing to the limit in (205) gives (S4), with precisely the term $2(f \cdot u)\psi$.

The remaining local clauses follow from the energy bounds, the pressure split, and the assumed local L^q integrability of f . The same fields u, Du, p were selected without reference

to q . Thus for every $q > 5/2$ for which $f \in L^q_{\text{loc}}$, the quadruple (u, Du, p, f) is suitable. This proves Theorem 7.8. $\square$

Proof of Theorem 7.9. Choose any $q > 5/2$ for which the force is locally in L^q . By Theorem 7.8, its witness is suitable in the exact class of Definition 1.5. Apply Theorem 2.10 on $\mathbb{R}^3 \times (0, \infty)$. It gives zero one-dimensional parabolic Hausdorff measure for the singular set. $\square$

Zero-force recovery. If $f = 0$, the forced weak equation and energy inequality in Definition 7.7 reduce to (LH3) and (LH4), while its other clauses are exactly (LH1), (LH2), and (LH5). Hence, for $f = 0$, the forced class and the Leray–Hopf class of Definition 7.2 agree with the same representative. The zero-force case of Theorem 7.8 therefore gives Theorem 7.4 with the same u, Du, p ; applying Theorem 2.10 with any $q > 5/2$ gives Corollary 7.5 and the zero-force singular-set conclusion. In Theorem 7.10, the multiplier gives $p_f = 0$, so the pressure is exactly $P[u \otimes u]$. Its global $L^{5/3}$ bound and, under the critical $L_t^\infty L_x^3$ hypothesis, the conditional $L_t^\infty L_x^{3/2}$ bound are those already proved in Theorem 7.6 and Lemma 8.6. Thus the forced pressure statement specializes to Theorem 7.6.

Remark 13.8 (Departure from [6]). Theorem 1.1 of [6, printed p. 212] is unforced. Here the force adds work to the every-time energy inequality and a local term to the suitability inequality; its gradient component also creates the pressure p_f . The force class here does not give a global $L^{5/3}$ bound for this pressure, so the forced theorem records p_f in $L_t^2 L_x^6$ and claims only local $L^{3/2}$ integrability for the total pressure.

Tsai’s suitable-existence theorem uses $L^2(0, T; (D_0^{1,2}(\Omega))') \cap L_t^{5/3} L_x^{15/14}$ and proves a global $L^{5/3}$ pressure bound; see Theorem 3.9 and displays (3.38)–(3.42), pp. 63–66 of [25]. The force class here gives global $L_{t,x}^2$ on finite slabs and local L^q for some $q > 5/2$. The order-one force pressure is therefore controlled here in $L_t^2 L_x^6$, which suffices locally for suitability but does not give global $L^{5/3}$ integrability. The unforced regularization, uniform-bound, and compactness arguments are those of Lemmas 9.3–11.1 and Propositions 12.1–12.3; the new terms are the force work, the projected-force tail, and the fixed force contribution in the local energy inequality.

APPENDIX A. EXTERNAL INPUTS

The analytic inputs and comparison results are collected below with their precise scope. Several inputs are proved internally in the formal development; their designation here does not mean that they are additional axioms. The Campanato comparison and Aubin–Lions compactness entries are not used by the direct A/B/C proof route. Constants named in this appendix are absolute unless a dependence is displayed.

External input A.1 (Calderón–Zygmund bounds for the Riesz transforms). For $i \in \{1, 2, 3\}$ let R_i denote the i -th Riesz transform on $\mathbb{R}^3$, that is, for the Fourier convention with kernel $e^{-ix \cdot \xi}$, the multiplier operator with symbol $-i\xi_i/|\xi|$; equivalently $R_i = -\partial_i(-\Delta)^{-1/2}$, so that $R_i R_j = \partial_i \partial_j (-\Delta)^{-1}$. For every $s \in (1, \infty)$ there is a constant $C_{11} = C_{11}(s)$ such that

$$\|R_i R_j g\|_{L^s(\mathbb{R}^3)} \leq C_{11}(s) \|g\|_{L^s(\mathbb{R}^3)} \quad \text{for all } g \in L^s(\mathbb{R}^3) \text{ and all } i, j.$$

For the pressure itself we write $C_{11} = C_{11}(3/2)$; the weak-gradient route also uses the fixed exponent $6/5$. These fixed-exponent constants are absolute. The operators are the bounded L^s extensions of the second Riesz transforms from a dense class, not absolutely convergent global integrals of the Hessian kernel. Literal kernel formulas are used only off the source support, or with their proper singular-integral interpretation. Interpolation is first performed on $L^2 \cap L^s$ and then passed to the completed L^s operator. Reference: [22, Ch. II–III].

External input A.2 (Hardy–Littlewood–Sobolev for the Riesz potential of order 1). *Let $N(y) = -\frac{1}{4\pi|y|}$, so that $|\nabla N(y)| = \frac{1}{4\pi|y|^2}$. There is an absolute finite constant C'_{11} such that, for every $g \in L^{5/2}(\mathbb{R}^3)$ and every $j \in \{1, 2, 3\}$, the literal convolution $\partial_j N * g$ is absolutely convergent for almost every x and*

$$\|\partial_j N * g\|_{L^{15}(\mathbb{R}^3)} \leq C'_{11} \|g\|_{L^{5/2}(\mathbb{R}^3)}.$$

This is the specialization $1/15 = 1/(5/2) - 1/3$ needed here; the full Hardy–Littlewood–Sobolev exponent family is not part of this input. No pointwise measurability or compact-support assumption is added to the $L^{5/2}$ datum. Reference: [22, Ch. V, Thm. 1].

External input A.3 (Newtonian representation of a compactly supported solution). *Let $N(y) = -\frac{1}{4\pi|y|}$, so that $\Delta N = \delta_0$ in $\mathcal{D}'(\mathbb{R}^3)$. Let $v \in L^{3/2}(\mathbb{R}^3)$ have compact support and suppose that in $\mathcal{D}'(\mathbb{R}^3)$*

$$\Delta v = \partial_i \partial_j G_{ij} + \partial_j H_j + K$$

where $G \in L^{3/2}(\mathbb{R}^3; \mathbb{R}^{3 \times 3})$, $H \in L^m(\mathbb{R}^3; \mathbb{R}^3)$ for some $m \in (1, 3)$, and $K \in L^{m'}(\mathbb{R}^3)$ for some $m' \in (1, 3/2)$, all with compact support. Then

$$v = -R_i R_j G_{ij} + \partial_j N * H_j + N * K \quad \text{almost everywhere in } \mathbb{R}^3.$$

*Reference: [8, Ch. 2 and Ch. 4] for the Newtonian potential and [22, Ch. III, V] for the identification $\partial_i \partial_j N * G_{ij} = -R_i R_j G_{ij}$; the uniqueness statement is Liouville's theorem applied to the difference, which is harmonic on $\mathbb{R}^3$ and tends to 0 in the $L^{3/2}$ -average sense at infinity.*

External input A.4 (Sobolev embeddings).

- (i) *There is an absolute constant such that $\|g\|_{L^6(\mathbb{R}^3)} \leq C \|\nabla g\|_{L^2(\mathbb{R}^3)}$ for all $g \in H^1(\mathbb{R}^3)$; consequently $\|g\|_{L^6(\mathbb{R}^3)} \leq C \|g\|_{H^1(\mathbb{R}^3)}$.*
- (ii) *There is an absolute constant C such that for every Euclidean ball $B_r(x)$, $r > 0$, and every $g \in H^1(B_r(x))$,*

$$\|g\|_{L^6(B_r(x))} \leq C (\|\nabla g\|_{L^2(B_r(x))} + r^{-1} \|g\|_{L^2(B_r(x))}).$$

In particular the unscaled H^1 estimate has an absolute constant on B_1 . No general Lipschitz-domain extension theorem is required here.

- (iii) *There is an absolute constant with $\|g\|_{L^\infty(\mathbb{R}^3)} \leq C \|g\|_{H^2(\mathbb{R}^3)}$ for all $g \in H^2(\mathbb{R}^3)$.*
- (iv) *Let J be an interval and $U = B_r(x)$, $r > 0$. If $g \in L^\infty(J; L^2(U)) \cap L^2(J; H^1(U))$, then $g \in L^{10/3}(U \times J)$ and, with one absolute constant C ,*

$$\|g\|_{L^{10/3}(U \times J)}^{10/3} \leq C (\|g\|_{L^\infty(J; L^2(U))}^{4/3} \|\nabla g\|_{L^2(U \times J)}^2 + r^{-2} \|g\|_{L^\infty(J; L^2(U))}^{10/3} |J|).$$

For unbounded J , the membership conclusion is asserted separately; an infinite right side is not a proof of membership. The displayed bound is (44) integrated in time with the slice L^2 mass bounded by the essential supremum. No embedding on an arbitrary bounded open set, or sharper public actual-mass estimate, is asserted here.

References: [8, Ch. 7], [22, Ch. V].

External input A.5 (Poincaré inequalities on the unit ball).

- (i) *There is an absolute constant with $\left\|g - [g]_{B_1}\right\|_{L^1(B_1)} \leq C \|\nabla g\|_{L^1(B_1)}$ for all $g \in W^{1,1}(B_1)$.*
- (ii) *There is an absolute constant with $\left\|g - [g]_{B_1}\right\|_{L^6(B_1)} \leq C \|\nabla g\|_{L^2(B_1)}$ for all $g \in H^1(B_1)$.*

- (iii) Additional classical statement. *There is an absolute constant with $\|g - [g]_{B_1}\|_{L^{3/2}(B_1)} \leq C \|\nabla g\|_{L^1(B_1)}$ for all $g \in W^{1,1}(B_1)$. Rescaling gives the same coefficient on any Euclidean ball, since both sides have scaling degree 2. This arbitrary $W^{1,1}$ version is recorded as classical paper material, not as an available formal export; the formal scalar estimate presently treats C^1 data.*

Reference: [8, Ch. 7].

The formalized Caccioppoli application uses a separate squared-velocity estimate: for $0 < r < R$, $B_R(x_0) \subset U$ with U open, and $u \in H^1(U; \mathbb{R}^3)$ with its weak gradient,

$$\| |u|^2 - [u]^2 \|_{L^{3/2}(B_r(x_0))} \leq C \|u\|_{L^2(B_r(x_0))} \|\nabla u\|_{L^2(B_r(x_0))}.$$

The constant is absolute. Approximation on nested balls proves this particular weak-slice estimate; it does not by itself export clause (iii) for arbitrary $W^{1,1}$ functions.

External input A.6 (Interior estimates for harmonic functions). *Let h be harmonic on an open $U \subset \mathbb{R}^3$.*

- (i) (Mean value property.) *If $\overline{B_s(x)} \subset U$ then $h(x) = |B_s(x)|^{-1} \int_{B_s(x)} h$.*
(ii) (Gradient estimate.) *If $\overline{B_s(x)} \subset U$ then $|\nabla h(x)| \leq \frac{3}{s} \sup_{B_s(x)} |h|$.*

Reference: [8, Ch. 2].

External input A.7 (Vitali covering lemma, $5r$ form). *Let (X, d) be a separable metric space and let $\mathcal{F}$ be a family of open balls $\mathfrak{B}_{r_a}(z_a)$, $a \in \mathcal{I}$, with $\sup_a r_a < \infty$. Then there is a countable set $\mathcal{J} \subset \mathcal{I}$ such that the balls $\{\mathfrak{B}_{r_j}(z_j)\}_{j \in \mathcal{J}}$ are pairwise disjoint and*

$$\bigcup_{a \in \mathcal{I}} \mathfrak{B}_{r_a}(z_a) \subset \bigcup_{j \in \mathcal{J}} \mathfrak{B}_{5r_j}(z_j).$$

In particular every $\mathfrak{B}_{r_a}(z_a)$ is contained in some $\mathfrak{B}_{5r_j}(z_j)$. Separability is needed for the subfamily to be countable: without it the standard Zorn's-lemma argument produces only a maximal disjoint subfamily, which may be uncountable. The open-ball version follows directly from the usual $5r$ argument. Choose $R > \sup_a r_a$ and successively choose a maximal disjoint family among balls of radii in $(2^{-n-1}R, 2^{-n}R]$ that avoid the earlier choices. Every unchosen ball meets a chosen ball of radius at least half its own, and is therefore contained in that ball's fivefold enlargement. The chosen balls are countable because each contains a distinct point of a fixed countable dense set. This also proves the individual containment assertion. For the covering argument in its standard form see [16, Thm. 2.1].

External input A.8 (Hausdorff measures of a metric space). *Let (X, d) be a metric space and $\alpha \geq 0$. The set functions $\mathcal{P}_\delta^\alpha$ and $\mathcal{P}^\alpha$ of Definition 2.4 are outer measures on X ; in particular $\mathcal{P}^\alpha$ is countably subadditive and monotone, and $\mathcal{P}^\alpha(E) = \lim_{\delta \rightarrow 0^+} \mathcal{P}_\delta^\alpha(E) = \sup_{\delta > 0} \mathcal{P}_\delta^\alpha(E)$, the family $\delta \mapsto \mathcal{P}_\delta^\alpha(E)$ being nonincreasing in δ . Reference: [16, Ch. 4].*

External input A.9 (Parabolic Campanato characterization of Hölder continuity). *Let $\alpha \in (0, 1)$, $R > 0$, $z_0 \in \mathbb{R}^3 \times \mathbb{R}$, and $w \in L^3(Q_{2R}(z_0); \mathbb{R}^m)$. Suppose*

$$\Lambda^3 := \sup \left\{ \frac{1}{r^{5+3\alpha}} \iint_{Q_r(z)} |w - (w)_{Q_r(z)}|^3 dz' : z \in Q_R(z_0), 0 < r \leq R \right\} < \infty.$$

Then w agrees almost everywhere on $Q_R(z_0)$ with a function $\bar{w} \in C^{\alpha, \alpha/2}(\overline{Q_R(z_0)}; \mathbb{R}^m)$, and there is a constant $C(\alpha)$ with

$$[\bar{w}]_{C^{\alpha, \alpha/2}(Q_R(z_0))} \leq C(\alpha) \Lambda, \quad \|\bar{w}\|_{L^\infty(Q_R(z_0))} \leq C(\alpha) \left(\Lambda R^\alpha + R^{-5/3} \|w\|_{L^3(Q_{2R}(z_0))} \right).$$

The mean $(w)_{Q_r(z)}$ is the genuine space-time average (5). Note that the normalization is $r^{-5-3\alpha}$, so that in terms of the excess (30) the hypothesis reads $\tilde{C}(z, r)^{1/3} \leq \Lambda r^{1+\alpha}$. Reference: [11, §4.2], [7, §III.1] for the elliptic model case, and [25, after (6.24)] for the form used here. This input is recorded for comparison with the blow-up route only: Section 5 uses instead the self-contained Lemma 5.11, and the exponent it delivers is $\gamma_0 = \min\{2 - 5/q, 1/5\}$ (Remark 5.33).

External input A.10 (Heat kernel). Let $G(x, \tau) = (4\pi\tau)^{-3/2} \exp(-|x|^2/(4\tau))$ for $x \in \mathbb{R}^3$, $\tau > 0$. Then $G \in C^\infty(\mathbb{R}^3 \times (0, \infty))$, $G > 0$, $\int_{\mathbb{R}^3} G(x, \tau) dx = 1$ for every $\tau > 0$, and $\partial_\tau G = \Delta G$ on $\mathbb{R}^3 \times (0, \infty)$. Consequently, for $r > 0$, the function $\psi_r(x, t) = r^2 G(x, r^2 - t)$ is smooth and positive on $\mathbb{R}^3 \times (-\infty, r^2)$ and satisfies $\partial_t \psi_r + \Delta \psi_r = 0$ there. Reference: [25, Ch. 2].

External input A.11 (Heat kernel: size and derivative bounds). Let

$$W_+(x, t) = \mathbf{1}_{t>0} (4\pi t)^{-3/2} e^{-|x|^2/(4t)}.$$

Then $W_+ \in L^1_{loc}(\mathbb{R}^3 \times \mathbb{R})$, $\text{supp } W_+ \subset \{t \geq 0\}$, $(\partial_t - \Delta)W_+ = \delta_{(0,0)}$ in $\mathcal{D}'(\mathbb{R}^3 \times \mathbb{R})$, and convolution with a compactly supported distribution is well-defined. Causal uniqueness, if used, must be restricted to tempered distributions: a tempered d supported in $\{t \geq c\}$ and satisfying $(\partial_t - \Delta)d = 0$ is zero. No such assertion holds for arbitrary distributions, and half-space support does not define a universal convolution algebra. The proof below uses the adjoint-test identity in Lemma 5.25, not this uniqueness assertion. There is an absolute constant C such that, with $\varrho = \max\{|x|, |t|^{1/2}\}$, for $t > 0$,

$$|W_+| \leq C\varrho^{-3}, \quad |\nabla_x W_+| \leq C\varrho^{-4}, \quad |\partial_t W_+| \leq C\varrho^{-5},$$

and, for every Fourier multiplier $\varsigma(D)$ with ς smooth on $\mathbb{R}^3 \setminus \{0\}$ and homogeneous of degree 1,

$$|\varsigma(D)W_+| \leq C_\varsigma \varrho^{-4}, \quad |\nabla_x \varsigma(D)W_+| \leq C_\varsigma \varrho^{-5}, \quad |\partial_t \varsigma(D)W_+| \leq C_\varsigma \varrho^{-6}.$$

The causal kernels vanish for $t < 0$. Their size estimates hold almost everywhere on spacetime, but no time derivative at $t = 0$ is asserted for the general multiplier kernels. Reference: [11, §IV.11], [13, Ch. 5].

External input A.12 (Aubin–Lions compactness). Let $X \subset\subset Y \subset Z$ be Banach spaces with X reflexive, the embedding $X \hookrightarrow Y$ compact and $Y \hookrightarrow Z$ continuous. Let J be a bounded interval, $1 < s_0 \leq \infty$, $1 < s_1 < \infty$, and let (g_k) be a sequence with $\sup_k (\|g_k\|_{L^{s_0}(J; X)} + \|\partial_t g_k\|_{L^{s_1}(J; Z)}) < \infty$. Then (g_k) is relatively compact in $L^{s_0}(J; Y)$ (and in $C(\bar{J}; Y)$ if $s_0 = \infty$ and $s_1 > 1$). For comparison with a compactness proof one may take $X = H^1(B)$, $Y = L^2(B)$, $Z = Z_B$ of (6), $s_0 = 2$, $s_1 = 3/2$. This input is not consumed in the direct proof here. Reference: [21, Corollary 4], including the time-continuity endpoint.

External input A.13 (Pettis measurability theorem). Let X be a separable Banach space, $J \subset \mathbb{R}$ an interval, and $w : J \rightarrow X$ such that $t \mapsto \langle w(t), \varphi \rangle$ is Lebesgue measurable for every $\varphi \in X^*$. Then w is strongly (Bochner) measurable, i.e. an a.e. limit of countably valued measurable functions. In particular, if X is a separable Hilbert space or the dual of a separable Hilbert space, weak measurability implies strong measurability. Reference: [5, Ch. II].

External input A.14 (Vector-valued du Bois-Reymond lemma). Let X be a Banach space, $J \subset \mathbb{R}$ an open interval and $w \in L^1_{loc}(J; X)$ with $\int_J w(t) \theta'(t) dt = 0$ in X for every $\theta \in C_c^\infty(J)$. Then there is $\xi \in X$ with $w(t) = \xi$ for almost every $t \in J$. Reference: [24, Ch. III, Lemma 1.1].

External input A.15 (Mollification and density). Let $\zeta \in C_c^\infty(B_1)$ with $\zeta \geq 0$ and $\int_{\mathbb{R}^3} \zeta = 1$, and $\zeta_\varepsilon(y) = \varepsilon^{-3} \zeta(y/\varepsilon)$. For $g \in L^s(\mathbb{R}^3)$, $1 \leq s < \infty$, one has $g * \zeta_\varepsilon \in C^\infty(\mathbb{R}^3)$, $\|g * \zeta_\varepsilon\|_{L^s} \leq \|g\|_{L^s}$ and $g * \zeta_\varepsilon \rightarrow g$ in $L^s(\mathbb{R}^3)$; the analogous statements hold in one time variable. Moreover

$C_c^\infty(U)$ is dense in $H_0^2(U)$ by definition of the latter, and $C_c^\infty(U)$ is dense in $L^s(U)$ for $1 \leq s < \infty$ and $U \subset \mathbb{R}^3$ open. Reference: [8, Ch. 7].

APPENDIX B. CORRECTIONS TO THE SOURCES OF THE EXISTENCE PART

This appendix records where the sources used in this part are corrected or completed by the arguments above. The corrections to the sources used only in the endpoint regularity theory are recorded in [2].

Escauriaza–Seregin–Šverák (2003).

- (1) **Unproved existence citation.** Theorem 1.1, printed p. 212, states Leray’s global existence result and cites prior work rather than proving it. The manuscript proves the regularized construction and carries out the suitability and Leray-limit passages in Proposition 12.1, Proposition 12.2, and Proposition 12.3, culminating in Theorem 7.4. This is an addition to the proof, not a correction to the theorem statement.
- (2) **Associated pressure normalization and mixed-norm notation.** Equations (3.2)–(3.4), printed p. 221, invoke coercive Stokes estimates and state the pressure equation and $L_{3/2,\infty}$ bound without an explicit whole-space Riesz formula or pressure normalization. Here $L_{3/2,\infty}$ means $L_t^\infty L_x^{3/2}$, not Lorentz $L^{3/2,\infty}$. Derivative estimates are interior to positive time even though $\delta_1 > 0$ is arbitrary. The manuscript fixes a canonical double-Riesz pressure and sign in Definition 8.1, proves the arbitrary-test extension in Theorem 7.6, and integrates the L^3 -solution pressure bound in Lemma 8.6.

Ożański–Pooley, arXiv:1708.09787v1.

- (1) **Young and convolution hypotheses.** Section 1.1, Lemma 1.1 and the integral Minkowski/Young statements, leaves measurability and integrability conditions implicit; its spatial Young inequality permits positive exponents although the norm inequality needs exponents at least one. The proof also takes a maximum over a closed time interval although the kernel is defined only for positive time. Appendix A.2 repeats this lemma and has a likely time-index typo in its continuity proof. These auxiliary results are not used in the manuscript.
- (2) **Mollifier range and weak derivatives.** Section 1.1, Lemma 1.2, states an inhomogeneous H^m estimate for every $\varepsilon > 0$ with an ε^{-m} factor; for large ε this needs a restriction such as $0 < \varepsilon \leq 1$, or a bound with $1 + \varepsilon^{-m}$. The weak-derivative clause also leaves its distributional interpretation implicit. The manuscript uses a fixed compactly supported mollifier with $0 < \varepsilon \leq 1$ in Lemma 9.2.
- (3) **Undifferentiated Newtonian potential on L^2 .** Section 1.1, Definition 1.2, flags that L^2 decay may not make its displayed potential integral pointwise defined. The later differentiated multiplier does not define that undifferentiated integral. The manuscript defines pressure by bounded Riesz extensions in Definition 8.1; it does not use that potential formula on arbitrary L^2 data.
- (4) **Multiplier scope and value at zero frequency.** Section 1.1, Lemma 1.4, proves only the L^2 multiplier bound, not general L^p Calderón–Zygmund boundedness, and leaves the multiplier value at $\xi = 0$ unspecified in the Sobolev extension. Lemma 1.3 states its estimate scalarly; a vector version must specify summation over components. The manuscript uses the all-exponent Riesz-transform extension of External Input A.1 and records its sign in Definition 8.1.
- (5) **Oseen kernel display and norms.** Section 1.2, the radial Oseen kernel definition, assigns $\tilde{P}(0) = 1$, although the displayed formula’s limit is $1/(4\pi^{3/2}\sqrt{t})$. The Oseen

- tensor estimates also leave the matrix norm implicit, and the proof's split at $x = y$ is degenerate unless that case is treated separately. This heat/Oseen route is not used in the manuscript's Fourier construction Theorem 9.4.
- (6) **Pressure variable typo.** Section 2, the pressure representation following Theorem 2.2, writes X immediately after defining the forcing F ; $X = F$ is intended. The manuscript supplies its own explicit tensor input in Definition 8.1.
 - (7) **Fourier exponential sign typo.** Appendix A.4.1, in the Fourier proof of Theorem 2.2, displays the transform of P with $e^{+4\pi^2 t |\xi|^2}$; $-\Delta P = \Phi$, the next derivative display, and the final Oseen multiplier require a negative exponent. This route is not used in Theorem 9.4.
 - (8) **Nonlinear forcing and weak-form sign/index conflict.** Section 2.2, equations (2.12)–(2.16), declares $F = -(Y \cdot \nabla)Y$, but its component formula, positive weak nonlinear term, and positive-gradient Oseen representation do not all match that forcing. The displayed pressure sign agrees with the negative divergence forcing. Corollaries 2.4 and 2.7 also use incompatible tensor slots; in Corollary 2.7 the displayed $F_i = -\partial_i(Y_i Z_k)$ repeats i and leaves k free. The same sign inconsistency propagates to Theorem 2.6 and later §3 formulas. The manuscript does not adopt these identities: Theorem 9.4 derives the weak identity from its pointwise equation, and Lemma 9.7 fixes the tensor order used here.
 - (9) **Pressure gauge.** Section 2, Theorem 2.5, establishes uniqueness of velocity, not literal uniqueness of pressure; pressure is determined only modulo a time-only function. Section 3, Corollary 3.3, claims pressure regularity for every pressure allowed by Definition 3.1, which is false after adding an arbitrary locally integrable $c(t)$. A canonical normalization is required. The manuscript makes this choice in Definition 8.1.
 - (10) **Unbounded-domain energy integrations.** Section 3, Theorem 3.4 and Lemma 3.5, suppress cutoff/density justifications for integration by parts on $\mathbb{R}^3$. The source integrability is enough to make a cutoff proof plausible, but the vanishing boundary terms are not shown. These comparison results are not used in the manuscript.
 - (11) **Mild representation time argument typo.** Section 3, equation (3.5), integrates over absolute $s \in [t_1, t_2]$ but prints kernel time $t_2 - t_1 - s$; the Duhamel time should be $t_2 - s$. The later formula (3.11) uses $t - s$. This mild representation is not used in Theorem 9.4.
 - (12) **Weak-divergence quantifier and pressure differentiability.** Section 3, Definition 3.1, leaves the time quantifier on weak divergence freedom implicit. Corollary 3.3's pressure regularity also omits weak time-differentiation and induction details, in addition to the gauge defect above. These source definitions are not used in the manuscript.
 - (13) **Strong-solution proof details.** Section 3, Theorem 3.2, sketches the regularity induction without a precise local-time bound; Lemma 3.6 does not claim convergence at $t = 0$ and leaves the nonnegative range of its scalar function implicit; Lemma 3.8's "claim follows" omits separate energy and comparison limit details. These routes are not used in the manuscript.
 - (14) **Picard existence sign and zero-data defects.** Section 3, Theorem 3.9, combines the standard-sign Oseen iteration with an opposite-sign Stokes equation and weak identity; the displayed proof therefore does not establish its stated Definition 3.7 solution. Its lifespan divides by $\|u_0\|_\infty^2$, leaving zero data undefined, and the invariant contraction set is not fully stated. The manuscript uses the Fourier- L^2 route in Theorem 9.4 instead.
 - (15) **Regularized weak-form mismatch.** Section 4.1, equation (4.2), has a nonlinear sign and tensor-order mismatch with the regularized PDE (4.1) and its later time-sliced

- identity. For an initial-plus-integral identity with tests supported before T , the sign is opposite as well. The manuscript derives its pressure-inclusive weak identity from the pointwise regularized equation in Lemma 9.7.
- (16) **Regularized existence typos and zero data.** Section 4.1, Theorem 4.2 proof, uses $u(s)$ where the argument requires $u_\varepsilon(s)$, and prints $\varepsilon^{+3/2}$ where the mollifier estimate and subsequent coefficient require $\varepsilon^{-3/2}$. Its lifespan $C/\|u_0\|_\infty^2$ also leaves $u_0 = 0$ undefined. The manuscript's Fourier construction Theorem 9.4 repairs the proof route and uses a lifespan formula defined for zero data.
 - (17) **Separation-of-energy proof gaps.** Section 4.1, Lemma 4.3, has a missing terminal t in its localized balance; the following display indicates the intended endpoint. Whole-space cutoff testing and the Newton-potential integration by parts lack truncation and decay arguments, and the theorem's regularity/data context is not restated. Its positive-time regularity hypothesis also requires a limit argument at the initial endpoint. The tail estimate of Lemma 10.4 replaces the Newton-potential step by an L^2 double-Riesz estimate and supplies the cutoff and initial-time limits.
 - (18) **Global energy-space overclaim.** Section 4.2, Corollary 4.6, claims global $L_t^2 H_x^1$ from the energy inequality. The proof also obtains weak continuity using density of admissible tests in L^2 without spelling out that step. The inequality controls u in $L_t^\infty L_x^2$ and ∇u in $L_{t,x}^2$, but does not make $\int_0^\infty \|u(t)\|_2^2 dt$ finite. This corollary is not used.
 - (19) **Global-existence proof display defects.** Section 4.2, Theorem 4.7, has an index typo in its Helly step (" $k \rightarrow \infty$ " instead of the sequence index), and its energy-limit display uses $\int_0^t$ while equating to the norm at a later start s ; the lower-semicontinuity step requires $\int_s^t$. The simple-function density and time-uniform continuity arguments are also compressed. Definition 4.5 uses an all-time representative, with an a.e.-time gradient, but does not state strong L^2 attainment at zero; Theorem 4.7 proves that trace. Theorem 4.7 itself gives no pressure, suitability, or local energy inequality; its all-space strong convergence uses the tail estimate. The manuscript gives its own compactness and every-time energy passage in Lemma 11.1, Proposition 12.1, and Proposition 12.2; it does not rely on Theorem 4.7.
 - (20) **A.7 gradient sign and decay hypotheses.** Appendix A.7, printed pp. 75–76, alternative pressure-bound proof, prints $\nabla_x |x - y|^{-1} = \nabla_y |x - y|^{-1}$; the correct identity has a minus sign, on which the subsequent integration by parts depends. The triple-integral steps also need decay/approximation assumptions on g . This alternative proof is not used.
 - (21) **Volterra induction exponent typo.** Appendix A.10, printed pp. 73–75, proof of Theorem A.10, mixes $M\|\phi\|C^k$ in the induction hypothesis with $\|\phi\|M^k$ in the target and IC^2M^k in the resulting estimate. The written induction is not algebraically aligned; this theorem is not used.
 - (22) **Approximation exponent mismatch.** Appendix A.12, printed pp. 76–78, Lemma A.12, is stated for general p but its conclusion is in L^2 , while its proof estimates L^p . It needs either an L^p conclusion or restriction to $p = 2$; its cited uses are the latter. This lemma is not used.
 - (23) **Continuity, trace, and Bochner conventions.** Appendix A.1–A.5 and A.7 leave some measurability/Bochner hypotheses implicit; the pointwise heat convergence in Lemma A.1 requires a representative continuous at the point, and "classical" heat/Stokes solutions are only positive-time regular with an L^2 initial trace. Appendix A.4.3 uses a bound up to time zero although its stated Hölder regularity is only for positive times.

Appendix A.4.4 leaves growth/cutoff hypotheses for mollified pressure and Fubini implicit and writes an $\mathbb{R}^n$ test in an $\mathbb{R}^3$ argument. Appendix A.5 and Corollary A.7 use convolution-continuity and measurability assumptions not fully stated. Appendix A.5 Corollary A.6 is justified only by “similar,” with no proof. These auxiliary routes are not used.

Tsai.

- (1) Theorem 3.9, printed p. 66, introduces the local energy identity by assertion without deriving the integrations by parts or stating their required regularity/integrability; it also asserts, rather than proves, uniqueness of the harmonic pressure part. This source serves only for comparison; the manuscript proves the identity in Lemma 9.6 for its regularized smooth pair.
- (2) Theorem 3.9 uses Lemma 3.7 for strong local compactness in Step 3, despite the accompanying route description suggesting otherwise; Lemma 3.7 is bounded-domain compactness, and Lemma 3.8 is the bounded-domain pressure branch, not the whole-space branch. The manuscript uses its own compactness result Lemma 11.1 and carries out the whole-space tail and Leray-limit arguments in Lemma 10.4 and Proposition 12.1.
- (3) Lemma 3.4, printed pp. 57–58, first gives an a.e.-time equivalence class; the modified representative is essential for fixed-time assertions. Definition 7.2 retains every-time slices; the manuscript carries out the limit identification in Proposition 12.1.

Robinson–Rodrigo–Sadowski.

- (1) Equation (14.5), printed pp. 264–265, is labeled formal for weak Navier–Stokes solutions. For the regularized pair, the omitted regularity/integrability and most smooth-pair integrations by parts must be supplied; the source spells out only the nonlinear transport integration.
- (2) Theorem 14.2, printed pp. 268–272, is a torus suitability proof. Its invocation of whole-space Theorem B.6 needs the periodic multiplier version, and its pressure convergence in $L^{3/2}(\mathbb{T}^3)$ cannot be transplanted as a whole-space pressure argument. Theorem 4.4 and the strong energy theorem 4.6 are likewise stated for periodic or bounded domains, not as whole-space construction results. These are not used; the manuscript provides the whole-space pressure and identity in Definition 8.1, Lemma 9.6, and Theorem 7.6.
- (3) Exercise 14.2 is left as the weak-equation convergence step in Theorem 14.2. Section 14.4, printed pp. 272–273, also needs a nonnegative cutoff test and yields an a.e.-time endpoint after a time approximate identity and subsequence, not an every-time assertion. The manuscript states the space-time local identity in Lemma 9.6 and carries out the every-time suitable-limit passage in Proposition 12.1.

Lemarié–Rieusset.

- (1) Theorem 13.6, printed p. 416, calls for a nonnegative $\phi \in \mathcal{D}'$; a distribution has no pointwise nonnegative test-function value. The intended test appears to be a nonnegative compactly supported smooth function. The pressure passage through the nonlocal Riesz operator is compressed into “Similarly.” This theorem serves only for comparison and is not used for the pressure of Definition 8.1.
- (2) Theorem 12.2, printed pp. 352–355, spatial-mollification argument serves only for comparison; it is not a pressure or suitability proof. The manuscript supplies its pressure extension in Theorem 7.6 and carries out the suitable Leray limit in Proposition 12.1.

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